Solving Motion Tasks
Hybrid Approaches using Iterative Learning Control in Motion Planning

Master thesis

of

Ferdinand Otto Werner Campe

born

17.01.1996 in Braunschweig, Germany

ferdinand@familie-campe.de (PGP)
f.campe@tu-berlin.de (PGP)
Matriculation Number: 365932
Study Program: Electrical Engineering
Specialization: Automation

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Supervisor:	M.Sc. Dustin Marlon Lehmann M.Sc. Michael Meindl
Examiner:	Prof. Dr.-Ing. Jörg Raisch Control Systems Group Department of Energy and Automation Technology Faculty IV - Electrical Engineering and Computer Science Technische Universität Berlin
Co-Examiner:	Prof. Dr.-Ing. Clemens Gühmann Chair of Electronic Measurement and Diagnostic Technology Department of Energy and Automation Technology Faculty IV - Electrical Engineering and Computer Science Technische Universität Berlin

Eigenständigkeitserklärung

Hiermit erkläre ich, dass ich die vorliegende Arbeit selbstständig und eigenhändig sowie ohne unerlaubte fremde Hilfe und ausschließlich unter Verwendung der aufgeführten Quellen und Hilfsmittel angefertigt habe.

Berlin, den 12. Januar 2023

Ferdinand Campe

Ferdinand Otto Werner Campe

Kurzfassung

Die vorliegende Arbeit widmet sich der Fragestellung, wie ein autonomer Agent ein Hindernis mittels eines hochdynamischen Manövers überwinden kann. Eine besondere Schwierigkeit ist dabei, dass der Roboter unteraktuiert ist und somit beliebige Trajektorien nicht verfolgt werden können. Als praktische Anwendung wird ein zweirädriges invertiertes Pendel untersucht. Für die Planung steht nur ein lineares Modell zur Verfügung, das das tatsächliche Verhalten nur grob wiedergibt. Um mit diesem Problem um zu gehen, werden Methoden der Iterativ Lernenden Regelung eingesetzt, um die kritischen Bereiche der geplanten Trajektorie gezielt zu verfolgen.

Zunächst wird gezeigt, wie mit Hilfe von sampling basierten Methoden, das Planungsproblem gelöst wird. Um die geplante Trajektorie zu realisieren wird eine Regelung benötigt, die mit Hilfe von Iterativ lernender Regelung implementiert wird. Zunächst wird diese Methode vorgestellt. Da es sich allerdings um ein Mehrgrößen System handelt, werden ergänzende theoretische Betrachtungen angestellt.

Die Kombination aus iterativ lernender Regelung und dem Rapidliy expanding Random Tree (RRT) stellt eine Methode dar, das Bewegungsproblem zu lösen. Die Method wird hybrid genannt, da sie die Teile der Planung aus der Offline-Phase in die Online-Phase verschiebt, sie also in die Regelschleife aufnimmt.

Die Gewichte der iterativ lernenden Regelung werden schlussendlich so gewählt, dass nur ausgewählte Teile der Referenz verfolgt werden. Dadurch wird erreicht, dass sich tortz der Unteraktuierung und eines ungenauen Systemodells die Punkte oder auch Teile der Referenz sehr präzise verfolgen lassen. Die Methoden werden in Simulation und Experiment verglichen, und es wird untersucht, ob sie die erwarteten Eigenschaften zeigen.

Darauf aufbauend werden Methoden entwikelt und verglichen, die auf Basis von Abständen der Trajectorie zum Hindernis, wichtige Teile der Referenz extrahieren.

Ferner wird experimentell gezeigt, dass ein Roboter eine Bewegungsaufgabe auch mit weniger Wissen lösen kann. So wird in einem Experiment ein einzelner Wegpunkt sowie das Ziel und in einem weitere in einem weiteren nur das Ziel dem Regler übergeben. Auch in diesen Fällen löst der Regler das Problem.

Zum besseren Problem Verständis wurde der Roboter mit Hilfe von analytischer Mechanik untersucht und dabei die dynamische Zwangsbedingung isoliert. Ferner lassen sich allgemeine Aussagen über den Einfluss von Reibung auf die beobachten Trajektorien machen.

Die entwickelte Software ist in einem git-Repsitory im zugänglich.

Abstract

This thesis deals with the problem how an agent can autonomously pass an obstacle with a highly dynamic maneuver. A special challenge is that the robot is under-actuated, and therefore arbitrary trajectories can not be tracked. The practical application is the Two Wheeled Inverted Pendulum Robot (TWIPR). The methods for planning and control must rely on a linear model only, even though the problem is non-linear and prone to imperfections like friction. To concur this task first sampling-based motion planning is applied to get a movement that is realized and optimized with Iterative Learning Control (ILC) methods.

It is shown first how a solution for the motion task is obtained using sampling-based motion planning. This results in a trajectory that needs to be controlled using ILC. As the system is multiple output, additional theory has to be discussed.

This combination is a new method to solve motion tasks. The methods are called hybrid as the motion optimization is done online in in the iterative learning control loop.

The weights of the ILC are chosen so that only important parts of the trajectory are controlled. Thus, also in the light of under-actuation and model imperfections precise tracking is achieved. The methods are compared with a low amplitude reference in simulation and experiment. It is shown if the results reflect the desired properties.

This leads to the development of four methods to realize planned trajectories, of whom three extract the important part of the trajectory. Those are also compared in the simulation and experiment. All those methods solved the problem in some cases but also lacked in robustness.

Furthermore, it is shown that the robot can also solve the motion problem with less information. An experiment is conducted that achives the arrival at a desired position in an agile maneuver. Also, an experiment is done that only considers a single waypoint to pass the obstacle.

To better understand the planning problem for the present robot, analytical mechanics are applied and the dynamic constraint is characterized. An important result therin is that the actuator friction is irrelevant for planning purposes.

The developed software is delivered in a git repository.

Keywords: motion planning, learning control, under actuated robotics

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List of Acronyms

ILC Iterative Learning Control

NO-ILC Norm Optimal Iterative Learning Control

P2P-ILC Point-to-Point Iterative Learning Control

CILC Colective Iterative Learning Control

TWIPR Two Wheeled Inverted Pendulum Robot

P2P-ILC Point-to-Point Iterative Learning Control

CSG Control Systems Group at TU Berlin

AI Artificial Intelligence

LTI Linear Time Invariant

SI Single Input

MIMO Multiple Input Multiple Output

SISO Single Input Single Output

SIMO Single Input Multiple Output

LMP Local Motion Planning

PSPACE-hard at lest decidable using polynomial memory

RRT Rapidliy expanding Random Tree

ODE Ordinary Differential Equation

DoF Degrees of freedom

DuT Device under test

DuT Device under test

w. l. o. g. without loss of generality

TUB Technical Univercity Berlin

DLR Deutsche Zentrum für Luft- und Raumfahrt

PD-Controler Proportional-Differential-Controler

MIP Mixed Integer Program

C-Space Configuration Space

ODE Ordinary Differential Equations

DAE Differential Algebraic Equations

BVP Boundary Value Problem

UDP User Datagram Protocol

cog. center of gravity

QP Quadratic Program

w.o.l.g. without loss of generality

TCP Transmission Control Protocol

UDP User Datagram Protocol

FTP File Transfer Protocol

FIR Finite Impulse Response

IIR Infinite Impulse Response

LQR Linear Quadratic Regulator

PDF Partially Differentially Flat

FCL flexible collision checking library

ROS Robotic Operating System

SVD Singular Value Decomposition

FD Finite Differences

RMSE Root Mean Square Error

LTV Linear Time Varriant

Notations

In this thesis the following conventions are chosen:

- scalar values are represented by lowercase symbols x, φ.
- vectors are represented by lowercase bold symbols $\boldsymbol{x}, \boldsymbol{\varphi}$
- matrices are represented by uppercase bold symbols $\boldsymbol{X}, \boldsymbol{\Phi}$
- sets are represented by caligraphic letters $\mathcal{S}$
- the algebraic core sets are given by $\mathbb{N}, \mathbb{Z}, \mathbb{R}, \mathbb{C}$
- the Laplace transform of a quantity is given by $\mathcal{L}\{\boldsymbol{X}\} = \boldsymbol{\mathfrak{X}}, \mathcal{L}\{\boldsymbol{x}\} = \boldsymbol{\mathfrak{x}}$
- a rotation matrix represents a orientation from the special orthogonal group $R \in$
- Construct a matrix from specific elements $\left[a_{ij}\right]^{i,j} = \boldsymbol{A}$
- Take one element out of a matrix $[\boldsymbol{A}]_{i,j} = a_{ij}$
- Indicator matrix $\left[\boldsymbol{I}_{i_1 j_1}\right]_{i,j} = \begin{cases} 1 & i_1 = i \wedge j_1 = j \\ 0 & \text{else} \end{cases}$
- Pseudo inverse $\boldsymbol{A}^{\ddagger}$
- Pseudo inverse of a matrix with full row rank $\boldsymbol{A}^{+}$
- Pseudo inverse of a matrix with full column rank $\boldsymbol{A}^{\dagger}$
- Kronecker Product $\otimes$ (Apendix A.5.2)
- State space system $\left[\begin{array}{c|c} \boldsymbol{A} & \boldsymbol{B} \\ \hline \boldsymbol{C} & \boldsymbol{D} \end{array}\right]$ represents $\begin{array}{ll} \boldsymbol{x}(k+1) & = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{u} \\ \boldsymbol{y}(k) & = \boldsymbol{C}\boldsymbol{x} + \boldsymbol{D}\boldsymbol{u} \end{array}$ (Notation is adopted to [5, p. 68]

List of Symbols

t	time
k	time index
i	sequence index
j	iteration index
m	input dimension
n	system order
z	point
σ	trajectory
ρ	path
κ	pathlength
ι	interpolation
p	output dimension
γ	system delay
$\mathcal{I}$	index set
$\mathcal{Y}$	output space
$\mathcal{X}$	state evolution space
$\boldsymbol{u}$	input
u	scalar input
$\boldsymbol{y}$	output
$\boldsymbol{x}$	state
$\boldsymbol{P}$	markov parameter matrix
$\boldsymbol{p}$	markov parameter vector
$\boldsymbol{r}$	reference signal
$\boldsymbol{x}_0$	initial state
$\boldsymbol{x}_\mathrm{f}$	final state
$\boldsymbol{A}$	system matrix
$\boldsymbol{B}$	input matrix
$\boldsymbol{C}$	output matrix
$\boldsymbol{D}$	feedthrough matrix
$\boldsymbol{v}$	state noise
$\boldsymbol{f}$	system function
$\boldsymbol{P}$	lifted system dynmaics
$\boldsymbol{u}$	lifted system input
$\boldsymbol{u}_\mathrm{anc}$	lifted ancillary input
$\boldsymbol{u}_\mathrm{P}$	lifted primal input

$\boldsymbol{y}$	lifted system output
$\boldsymbol{d}$	lifted system disturbance
$\boldsymbol{r}$	lifted reference signal
$\boldsymbol{e}$	lifted system error sequence
$\boldsymbol{Q}$	Q-filter
$\boldsymbol{L}$	leaning rule
$\boldsymbol{L}$	learing rule matrix
$\boldsymbol{\Phi}$	lifted selection matrix
$\boldsymbol{P}_{\Phi}$	lifted system dynamics selected
$\boldsymbol{d}_{\Phi}$	lifted system disturbance selected
$\boldsymbol{y}_{\Phi}$	lifted system output selected
$\boldsymbol{r}_{\Phi}$	lifted system reference selected
$\boldsymbol{e}_{\Phi}$	lifted system error selected
n_{b}	batch size
n_{c}	way point clearence
c	clearence
n_{Φ}	number of waypointssym
n_{ILC}	number of trials
$\mathbf{0}$	zero matrix
λ	eignevalue
ρ	spectral radius
p	markov paramter
$\tilde{d}$	disturbance sequence
q	time shift operator
$\boldsymbol{R}_{\mathrm{W}}$	input weights
$\boldsymbol{Q}_{\mathrm{W}}$	state weights
$\boldsymbol{S}_{\mathrm{W}}$	update weights
$\boldsymbol{P}$	covariance matrix
τ	tory duration
$\mathcal{N}$	Normaly distributed Random Variable
$\mathcal{K}$	Controlability Matrix
J	Cost Functional
l_{cg}	lever arm center of gravity
r_{W}	wheel radius
w_{W}	wheel width
d	wheel spacing
h_{b}	body hegiht
l_{b}	body height under axel
w_{b}	body width
d_{b}	body depth
φ	pitch
ϑ	wheel angle
ϑ_{Δ}	wheel angle diffrernce
ψ	headding
s_{W}	lateral position of the axis
s_{cg}	lateral position of the center of gravity

x_{C}	lateral position x coordinate
y_{C}	lateral position y coordinate
$\boldsymbol{K}_{\mathrm{FB}}$	controlerfb
$\boldsymbol{G}$	plant
$\boldsymbol{M}$	Mass matrix
$\boldsymbol{C}$	Coriolis matrix
$\boldsymbol{c}$	generelized coriolis and centripetal forces
$\boldsymbol{g}$	generelized potential forces
$\boldsymbol{\tau}$	generelized forces
τ	generelized forces
q	generelized coordinates
$\boldsymbol{q}$	generelized coordinates
$\boldsymbol{\chi}$	endeffector workspace coordinates
$\boldsymbol{\omega}$	rotational velocity
$\boldsymbol{v}$	velocity
$\boldsymbol{q}$	robot configuration
$\mathcal{C}$	configspace
$\boldsymbol{\Phi}$	selection operator
y	output signal
$\boldsymbol{E}$	identity matrix
$\boldsymbol{E}$	lifted indentity matrix
$\mathbf{0}$	lifted zero matrix
Λ	politopic constraint linear operator
b	politopic constraints affine offset
$\mathcal{D}$	confidence set for the output update
$\boldsymbol{I}$	indicator matrix
δ	kronecker delta
j	imaginary unit
e	euler number
L	Lagrange Function
T	Energy of translation
V	Potential Energy
J	Rotational Momentum of Inertia
m	Mass
g	gravitational acceleration
T	time contant
h	sampling period
s	laplace varriable
ϵ	convergence rate
F	flatten operator

Chapter 1

Introduction

1.1 Motivation

In the past decades Artificial Intelligence (AI) and Controls of robotic system has gained a lot of interest. In the industrial domain, robotic systems are well known and have become standard in production lines. Therefore, a huge knowledge base exists. But there is a big drawback regarding these applications. They are embedded in a well-known predefined environment, the production line, where all environmental circumstances can be designed carefully. So, nowadays, the main challenge in robotics is build robots that are capable to act in unknown environments. This topic involves a variety of challenges e. g., computer vision, image segmentation, design, motion planning and feedback control.

This thesis will consider an approach that combines the problem of motion planning and learning control of highly dynamic under-actuated robots.

The field of control has evolved over decades, originating in the operation of steam engines to large industrial processes, high-end electronics, complex systems and many more applications. The methods developed in control are widely applicable and are not limited to fully actuated systems. Recent hot topics in underactuated control is the realization of complex movements like walking of humanoid robots. In research, it turned out that kinetical design is still a key problem when realizing control systems, e. g.Collins has shown that careful design realizes a very efficient walking motion [6], unfortunately this property conflicts with active robotic control as it operates close to singularities. Therefore, most humanoids do not walk energy efficiently but more controllable an example is the atlas of Boston Dynamics and also a humanoid operated at Deutsche Zentrum für Luft- und Raumfahrt (DLR) that can be considered as a standard research humanoid [7]. Recent lead research suggests the introduction of soft materials into the domain of robotics. This paradigm coincides with the philosophy of intelligence by design. An illustrative result is the work of Pall [8] that analysis the grasping of goods with soft robotic actuators without any feedback controller. This can be considered as a paradigm shift in robotics as the most other end efectors that consist of rigid bodies that are controlled carefully.

To understand the aim of this thesis, the essence of control research must be characterized. The overall control research is both analyses of possibly very complex systems called plants and building up on analysis the synthesis of possibly many auxiliary systems called controllers to modify the behaviour of the original system. The combination of plant and controller is called closed loop system. The process involves a conflict of aims between performance and robustness. Performance is the magnitude of modification that is realizable with the controller. The robustness it the ability of the closed loop system to tolerate unforeseen impacts e. g.unmodeled forces [9].

In past research impressive results have been made by using this model-based controller design to do very complex maneuvers. Those involved very careful controller design e. g.Moore [10] synthesised a controller for a post-stall landing on a perch upon the models discovered by Tedrake [11]. This example shows the timespan needed in the engineering cycle which in this case is at least 5 years.

The research group around Thomas Seel, where this thesis contributes, explores methods that are either more applicable and accessible for engineers who face control tasks in their everyday life and need to budget their effort in development. Also, research is done on which control methods need to be used to also facilitate the related robotics advances like soft robotics and intelligence by design. One major turnout is that the method of Iterative Learning Control (ILC) is a very robust well, performing control strategy shown in many experiments as Seel shows in his disertation [12]. The problems considered are mostly from the biomedical domain, precisely the application of functional electrical stimulation, that involves problems that are similar to those in soft robotics. Furthermore, the group explores how the necessity of tuning can be reduced e. g.[13].

This thesis aims to explore how those methods can be extended to solve more general tasks than tracking, where a reference trajectory is given. In contrast to the known literature we shift from careful planning to ILC. To perform this task the interface to the domain of motion planning is characterized and ILC algorithms are applied. To make the consideration, expressive experiments in the existing testbed will be done for validation.

1.2 The Robotic Testbench

This thesis project is embedded in a broader scope project to synthesize intelligent behavior of agile autonomous robots exploiting cooperation embedded in the Science of Intelligence cluster. The robotic test bench incorporates the task of doing highly agile maneuvers, e. g.a dive under an obstacle (Fig. 1-1). The reviewed methods shall include cooperation to synthesize intelligent behavior. In this matter Iterative Learning Control (ILC) turned out to be promising as the cooperation can be exploited using Colective Iterative Learning Control (CILC) [14] or transfer learning. Furthermore, the robots are under-actuated, forming a particular class of control problems in robotics.

The results obtained in this thesis do not cover this cooperative issue but serve as a basis for solving tasks as they strip the hand-constructed reference trajectories out. Thus, the robots are enabled to solve motion tasks in the presence of obstacles autonomously even if only few model information

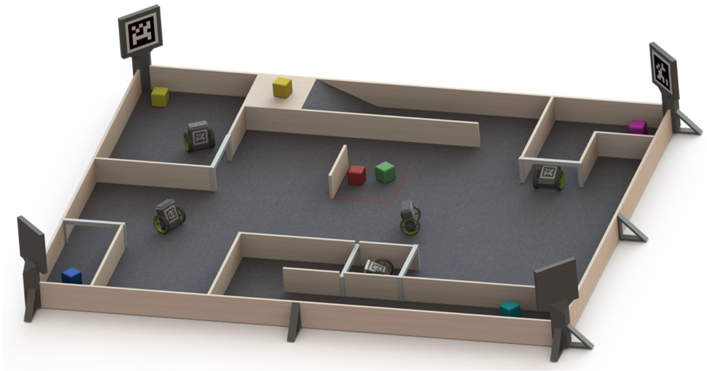


Figure 1-1. Target state of the robotic test bench in the project 11 of SCIoI at Technical Univercity Berlin (TUB). The test bench enables testing algorithms for distributed control and intelligent behaviour. The sample robot is the Two Wheeled Inverted Pendulum Robot (TWIPR) combining being a complex system that learning methods may show to be superior. The robots perform complex maneuvers as diving or pick up events.

is available. More specifically, they serve to solve motion problems and not higher-level tasks like exploration and cooperative problem-solving. Furthermore, the system is analyzed in a level of detail to gain an understanding of the physical trajectories and the impact of imperfections. Due to the state of development in the overall project, the experiments restrict to the case omitting the planning problem in the yaw ore the direction. But it will be discussed how the problem extends, including the yaw motion, and that the methods still apply. Therefore, it can be considered for future strategic decisions.

1.3 Case and Problem statement for the reviewed case

In this section, the case that is reviewed is introduced. Furthermore, a problem statement is given that defines the interfaces and information that are delivered from the outside of this project.

1.3.1 Case

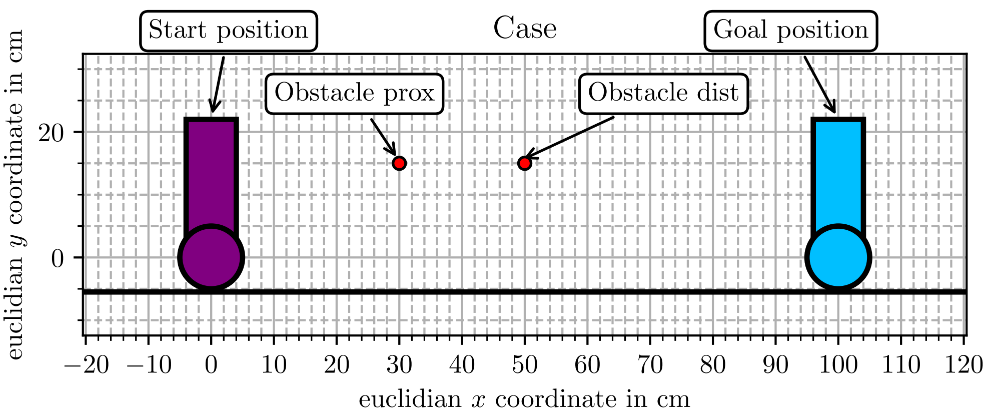


Figure 1-2. Illustration of the test case.

The case that is experimentally reviewed is the dive under an obstacle for the planar robot, i. e. the robot only moves forward and backward and does not change its heading. The reason for this decision is that it does not require additional hardware development, introducing a project risk. Furthermore, it introduces an additional actuated degree of freedom that is of no importance for the research aim, to especially the under-actuated part of the system.

The case is illustrated in Figure 1-2. The two robots show the start and the goal position of the robot. The red circles represent the obstacles that the robot must dive under to reach the goal.

There are two cases reviewed that differ in position of the obstacle one is with the proximal obstacle and one with a distant one.

1.3.2 Problem statement

This section introduces all information that is used to solve the motion task.

- A geometric representation of the robot and obstacle is given. (As it is given in Figure 1-2)
- A start and a goal position is given.
- A linear discrete time model is given as an approximation of the robot's dynamics. It has the form $\boldsymbol{x}(k+1) = \boldsymbol{A}\boldsymbol{x}(k) + \boldsymbol{B}\boldsymbol{u}(k)$
- The algorithm is required to traverse the robot from the start to the goal.
- The algorithm is allowed to learn the movement.This means the robot is allowed to perform movements that collide with the obstacle for the purpose of training.
- As the movement is learned it must not collide with the obstacle.

1.4 State of Research

A main challenge in modern robotics it motion planning, as in traditional applications, the movement can be planned with any amount of time, whereas in an adoptive new environment, they need to be planned and optimized jet in place, which leads to direct cost in the scene of delay or the risk of not finding a feasible jet solution for a given problem, addressed by the narrow passage problem. Furthermore, it is known from theory that the general motion planning problems are likely at lest decidable using polynomial memory (PSPACE-hard) [15, pp. 249] which is considered as a not reasonably solvable problem. Therefore the development of generic algorithms is for motion planning not very promising.

A currently published dissertation presents the control design of an automated humanoid robot designed by the Deutsche Zentrum für Luft- und Raumfahrt (DLR) that tries to enable the robot to perform classical mounting tasks in airplane construction [7]. But it has also shown that an economical application is far from being possible as the robot lags in speed and reliance.

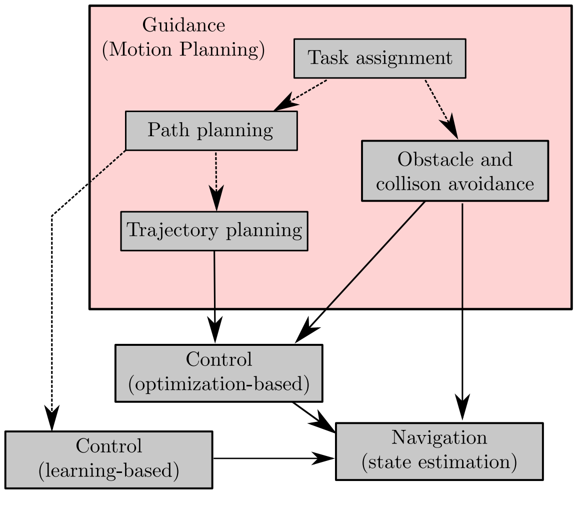
Furthermore, the new approaches in robotics address new challenges in control as traditional applications in mechanics are deterministic and modeled well. Even simple approaches like exactly linearized plants with Proportional-Differential-Controler (PD-Controler) lead to satisfying results. This does not hold for soft robotics non. The less applicable road maps serving the needs of the industry may come up in the next years [16, 17, 18] and sophisticated applications like in medicine exist e.g. [19].

The problem of motion planning is an active field of research for the past 30 years. The monograph that seems impactful and is also used in out group is the one written by LaValle [15], who sets up the research framework for motion planning applications. Many publications focus on the application of well-known techniques to real-world applications [20] or on important design choices like sampling strategies. The leading paradigm is the sampling-based approach that is dominating research and application. Non the less, the direct optimization techniques are also reviewed with recent research in Riemann manifold optimization that is jet active in mathematics [21]. The methodology seems to be very involved, though, and is not as easily accessible for the author and tent to be very abstract. Nevertheless, it is promising as it convexifies problems or at least transforms them to a low order Mixed Integer Program (MIP) and in theory, simplifies the obstacle representation a lot. Presentations for simple geometries are released as open source [22].

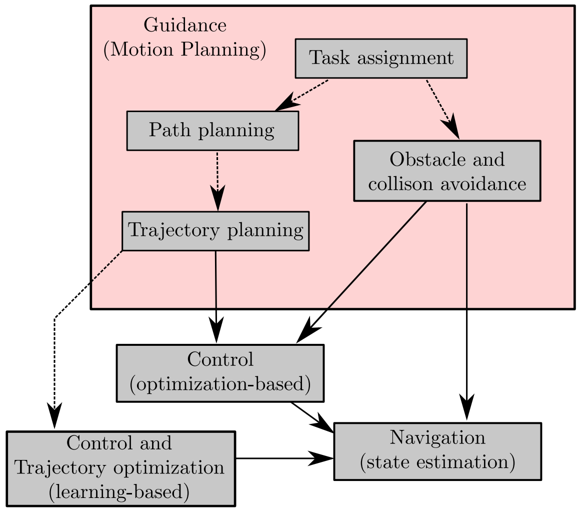
Finally, a glimpse on the research in the domain of Iterative Learning Control (ILC) is given. The methods are not new and are formalized in textbooks like [23, 24, 25, 26] and one very application oriented survey paper by Brisow [2] exist. Furthermore, there exists a course at Technical Univercity Berlin (TUB) [27] that introduces the method in an application-oriented manner. The method is proven to be useful in many applications and naturally in production systems that perform the same operations thousand times a day. The range of applications starts in the nearly impossible-to-model biomedical motor system to the optimal production of microchip bonding [28, 29] to name a few. Naturally, the past research focused on implementing easy algorithms realizable on easy hardware by digital filtering of sequences. Nowadays, in light of vast computing power, one can also consider powerful optimizers to handle constraints more easily [3]. In the author's point of view, the methods leg in the uncertainty quantification techniques applied allow save operation even and extreme utilization of the robot's capabilities. The combination of motion planning and ILC has be shown to be beneficial on a soft robotic application [30].

1.5 Solution approach

The reviewed method found throughout the project work can be embedded in the general motion planning as it is presented in [1] as, it illustrated in Figure 1-3. The first (Figure 1.3(a)) one omits trajectory generation and just relies on paths, who are generated model free. Those method is applied to a fully actuated robot in simulation (Section **??**). This method failed for the TWIPR due to difficulty to assign the waypoins to the right time. The methods that have been applied successfully are characterized to the diagram given in Figure 1.3(b) that uses the trajectory as initial guess and optimizes the trajectory within the control loop. This leads to optimal trajectories realized by the physical system in contrast to trajectory relying on a model as in offline trajectory generation. This enables the robot to perform the task robustly in contrast to classical tracking methods.



(a) Bypassing trajectory generation.



(b) Online Trajectory optimization.

Figure 1-3. Motion Planning Framework [1] extended by ILC.

As throughout the work on the thesis project it turned out that the trajectory optimization approach is more promising and also an improvement to the present methods, it focuses on this branch.

1.6 Outline

To reach this goal, the first chapter introduces sampling based motion planning fundamentals and the used algorithm. In the next step Iterative Learning Control (ILC) is introduced, and it is shown how the method is combined with the sampling based planing. And simulations are carried out. In the fourth chapter the robot is presented and method form analytical mechanics are carried out to improve the problem understanding. The simulation model is improved so that it represent the planing specific challenges, by the introduction of a friction model. Finally, in the fifth chapter results of experiments are presented.

Chapter 2

Motion Planning

This chapter introduces the methods of motion planning used in this work. Methods present in literature spilt up in two branches. One is the sampling-based method built up from robotics research that is meant to explore unknown environments with universal robots modeled as kinematic chains that are relatively easy to control but have multiple actuators and complex geometry. The other branch is flatness based motion planning that comes from the control perspective. It enables under-actuated robots to quickly perform precise maneuvers in relatively simple environments, mostly without constraints. In the present work, an under-actuated robot that moves in complicated environments is investigated. Therefore, both methods are reviewed and applied. Furthermore, it is highly desirable to perform movements even with loose model information.

Therefore, the present chapter builds up as follows. Firstly, the fundamental sampling-based algorithm of Rapidliy expanding Random Tree (RRT) is presented and applied to the setup under review (Section 2.1). To extend the setup so that it takes the dynamics of the TWIPR into account, the algorithm is extended to the kinodynaic RRT (Section 2.2.1). This is acchived introducing a local motion planning method that relies on a linear state space representation (Section 2.2.1)).

The motion planning problem is, generally speaking, the task to find a movement or strategy that enables a subject to transition from one state to another, that fulfill some task specific requirements e. g.clearance of obstacles, desired way-points etc., enabling a robot to perform a task as grasping, moving, taking a gesture, autonomously [31, pp. ix]

As the problem is quite abstract and does not involve control problems in the first place the abstraction of the configuration space is used.

Definition 1 (Configuration space). *The set of all configurations $\mathcal{C}$, called configuration space, is a manifold that can be portioned in two sets disjoint, the free space $\mathcal{C}_{\text{free}}$ representing all configurations for those collisions are absent and the obstacle region $\mathcal{C}_{\text{obs}}$ representing configurations that are not valid due to collisions with of the robot either any surrounding object or two links of the robot.*

The underling idea is that robots and obstacles define subsets of the euclidean space that contain all points that are occupied by the object. As the configuration is given, determining the robots pose, the state of collision can be uniquely queried [15, pp. 127].

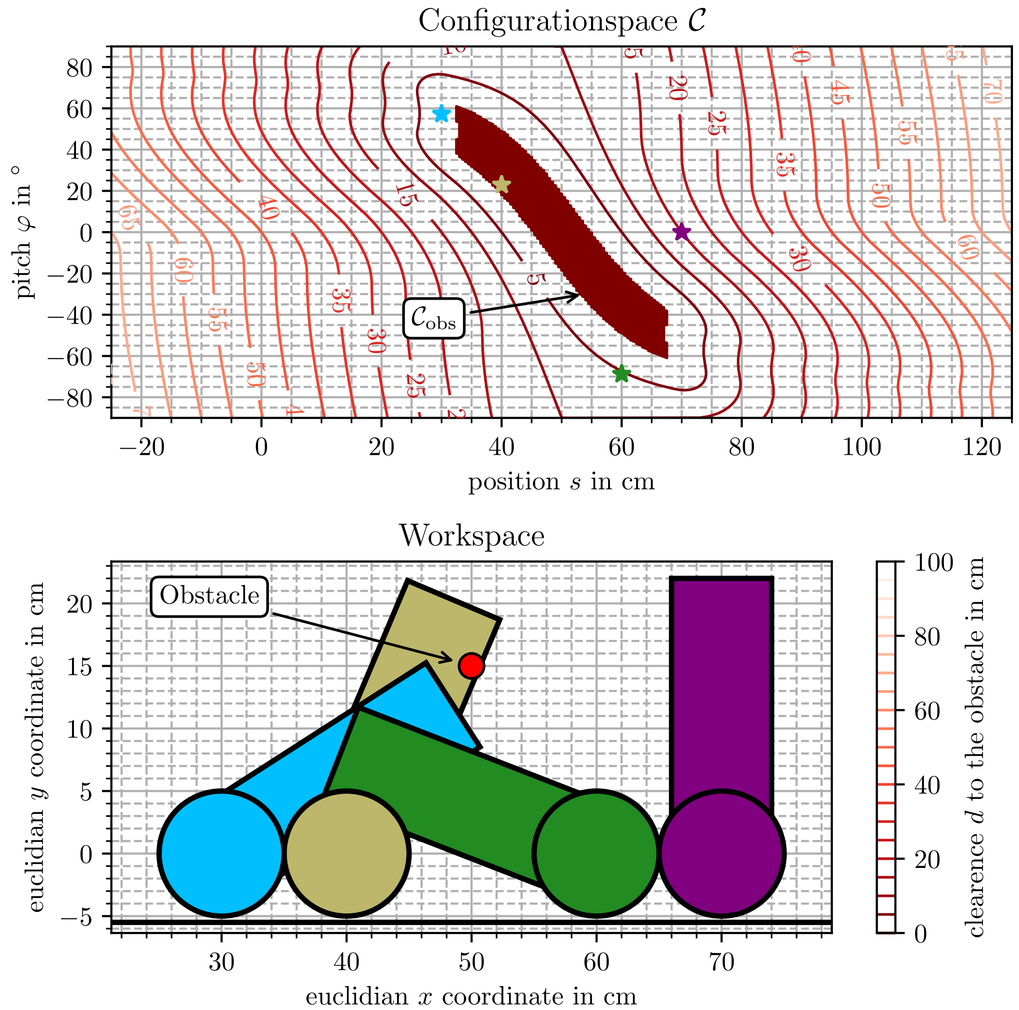


Figure 2-1. A valid configuration space of the planar TWIPR model. In the bottom the workspace representation of the configuration marked by crosses in the below are displayed. In the top the configuration space is presented. An approximation of the is obtained by a grid sampling of the configuration space. This is only feasible for low dimensional configuration spaces. The obstacle space is indicated by a negative distance vlaue. `code_to_generate_result`

Example. *A valid configuration space definition of the planar Two Wheeled Inverted Pendulum Robot (TWIPR) (considering no yaw movement) is given by its pitch angle* φ *and its lateral position* s_w*. It is sufficient to detemine the state of collision with parts of the robot or known environmental obstacles. The realization is illustrated in Figure 2-1. Obviously the evaluation it the robot is colliding can be done using available graphics software as the environment is known. Furthermore, in this special example the Obstacle space* $\mathcal{C}_{\text{obs}}$ *can be approximated.*

In the Figure 2-1 the before mentioned configuration space is illustrated. In the lower plot some robot configurations are given in the above those are marked in the configuration space. Furthermore, the contourlines are the equidistant lines indicating the minimum distance to the obstacle called *clearance* in the following. It can be observed that the geometry of the set of all colliding configurations is complicated even though the geometry of the robot and the obstacle is very simple. The difficulty to represent the set of colliding configurations in closed form is described in [15] in detail. The author points out that it is only possible for very few joints and geometries so that no reasonable implementation can be accomplished.

The collision checking problem arises frequently in robotic tasks, so that open source implementations exist. The most promising one found is the *flexible collision checking library* [32] that is used in *MoveIt* [33] being the motion planning node for ROS. Therefore, it is assumed to be stable and reliant.

The Figure 2-1 is generated using the *felxible collision checkin library* in python. The library consist represents basic geometries as boxes modeling the TWIPR body and cylinders modelling the obstacle. The contour plot was generated by sampling the distance query of flexible collision checking library (FCL) on a grid.

In the following it is described how the configuration space relates to the state space that is the fundamental tool in control theory.

Definition 2 (State Space)**.** *The statespace is a set $\mathcal{X}$ that fulfills the markov property. An element is called state.*

The markov property says that knowlege of state at a time contains the information about the future evolution that is encode in the systems past. So any information about the systems past is obsolete for prediction as the state is known. Therefore, the state space must also contain the configuration of the robot as it encodes the subspace that robot occupies in the euclidean. This leads to

Remark 1 (Dimension.)**.** *The configuration space conditionality is most likely smaller but never larger the dimension of the state space.*

In the remainder of this introductory part, the planning problem is defined using the configuration space abstraction.

Definition 3 (Trajectory)**.** *A trajectory σ is a continuous mapping that maps a finite time domain to the configuration space.*

$$\sigma : [0,\tau] \to \mathcal{C}$$

Definition 4 (Path)**.** *A path ρ of length κ is a finite sequence of points in configuration space.*

$$(z_i \in \mathcal{C})_{i\in\{1,\dots,\kappa\}}$$

Definition 5 (Interpolation method)**.** *A interpolation method ι is a mapping that maps a path to trajectory,*

$$\iota : (z_i \in \mathcal{C})_{i\in\{1,\dots,\kappa\}} \to \{\sigma : [0,\tau] \to \mathcal{C}\}$$

so that all points are traversed and their sequence is maintained.

$$\exists (t_i \in [0,\tau])_{i\in\{1,\ldots,\kappa\}} \quad \forall i \quad z_i = \iota(t_i)$$

$$t_1 = 0 \quad t_\kappa = \tau \quad \forall i \quad t_i < t_{i+1}$$

Definition 6 (Collision free path)**.** *A path ρ is called collision free given an interpolation method ι if the trajectory generated by the interpolation method lies entirely in the freespace $\mathcal{C}_{\text{free}}$.*

$$\forall t \in \mathrm{Def}\big(\iota(\rho)\big) \quad \iota(\rho)(t) \in \mathcal{C}_{\text{free}}$$

Problem Statement 1 (Planning problem)**.** *Find a contentious trajectory $\boldsymbol{q}_{\text{trj}} : [0,1] \to \mathcal{C}_{\text{free}}$ that connects $\boldsymbol{q}_{\text{init}}$ and $\boldsymbol{q}_{\text{final}}$, i. e.$\boldsymbol{q}_{\text{trj}}(0) = \boldsymbol{q}_{\text{init}}$ and $\boldsymbol{q}_{\text{path}}(1) = \boldsymbol{q}_{\text{final}}$*

2.1 Sampling based motion planning Rapidliy expanding Random Tree (RRT)

In this section the RRT is introduced that is the fundamental strategy for motion planning.

The RRT algorithm explores the configuration space structure by constructing a search tree to connect two configurations, namely the initial configuration $\boldsymbol{q}_{\text{init}}$ and the final configuration $\boldsymbol{q}_{\text{final}}$. Therefore, two steps are necessary *sampling* and *expanding*. Sampling is the selection of a configuration that one attempts to add to the tree. Expanding is the procedure of adding a sample to the tree.

Algorithm 1 Fundamental RRT. [34]

Require: $\boldsymbol{q}_{\text{init}}, \boldsymbol{q}_{\text{final}}$
  $\mathcal{G}$ = Graph($\boldsymbol{q}_{\text{init}}$)
  loop
    $\boldsymbol{q} \leftarrow$ sampleCspace()
    $\boldsymbol{q}_{\text{n}} \leftarrow \mathcal{G}$.getNearest($\boldsymbol{q}$)
    if isInterpolationCollisionFree($\boldsymbol{q}_{\text{n}}, \boldsymbol{q}$) **then**
      $e \leftarrow$ interpolate($\boldsymbol{q}_{\text{n}}, \boldsymbol{q}$)
      $\mathcal{G}$.addVertex($\boldsymbol{q}$)
      $\mathcal{G}$.addEdge(e)
      if random() $\leq p_{\text{complete}}$ **then**
        if isInterpolationCollisionFree($\boldsymbol{q}, \boldsymbol{q}_{\text{final}}$) **then**
          $e \leftarrow$ interpolate($\boldsymbol{q}, \boldsymbol{q}_{\text{final}}$)
          $\mathcal{G}$.addVertex($\boldsymbol{q}_{\text{final}}$)
          $\mathcal{G}$.addEdge(e)
          return $\mathcal{G}$.path_to($\boldsymbol{q}_{\text{final}}$)
        end if
      end if
    end if
  end loop

The fundamental application of the motion planning algorithm is the of linear interpolation technique and uniform sampling in the configuration space. The sampling strategy and the interpolation method are the parameters of the planning algorithm.

$$\text{interpolate}: \mathcal{C} \times \mathcal{C} \to ([0,1] \to \mathcal{C}); \boldsymbol{q}_1, \boldsymbol{q}_2 \mapsto f(s) = \boldsymbol{q}_1(1-s) + \boldsymbol{q}_2 s \tag{2.1}$$

The base sampling strategy is a uniform sampling of the configuration space.

This method will be referred to as pure geometric planning. The basic method is presented in Algorithm 1 as fundamental algorithm [15, pp. 232]. The method is a single query motion planner as the tree is a directed graph that can only find paths that start in the tree's root node. This contrasts to multiple query methods that build bidirectional graphs, which generate roadmaps that are also used for future queries.

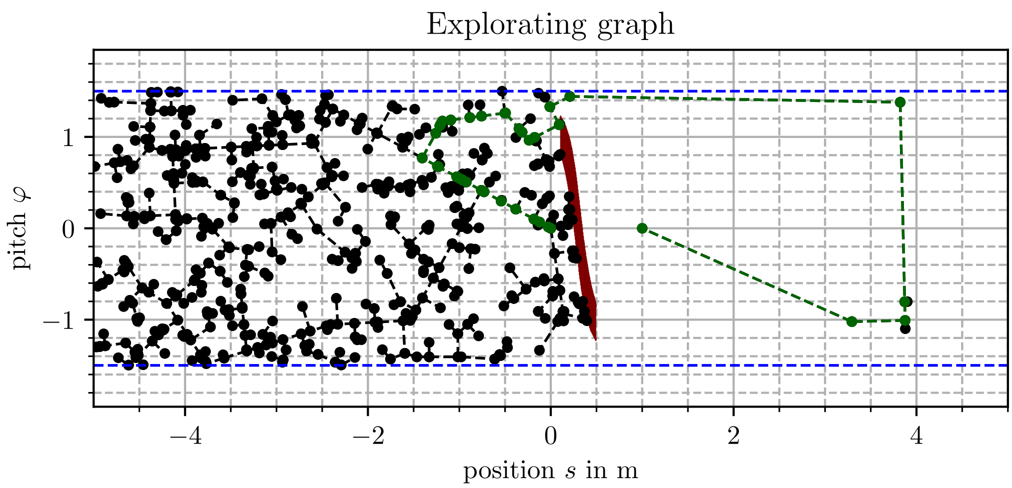


Figure 2-2. RRT

Figure 2-2 shows the result of a RRT query. The typical structure of the search tree is replicated [15, p. 230]. The solution trajectory build by the algorithms is presented in green.

2.2 Kinodynamic planning

The extension to kinodynamic planning requires the state trajectory to fulfill equations of motion, which need to be known. This introduces constraints on the trajectory in the sense it is given the definitions before. The trajectory must fulfill the dynamic constraints.

Definition 7. *Dynamic trajectory constraints.*

$$\begin{aligned} \boldsymbol{f}(\ddot{\boldsymbol{q}}, \dot{\boldsymbol{q}}, \boldsymbol{q}) &= 0 \\ \boldsymbol{g}(\ddot{\boldsymbol{q}}, \dot{\boldsymbol{q}}, \boldsymbol{q}) &\leq 0 \end{aligned} \tag{2.2}$$

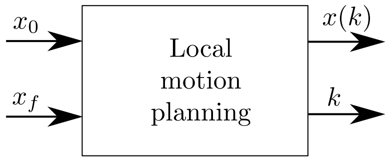


Figure 2-3. Local Motion Planner. [34, p. 817]

LaValle [15, p. 792] introduces the term kinodynamic planning to be unique for second-order differential constraints. He also distinguishes the term non-holonomic that captures non integrability of first-order constraints. Nonholonomic means that even though all points in the configuration space are reachable, the movement is bound to a subspace [15, pp. 893]. If a constraint would be integrable, then not all configurations are reachable. This property is especially true for underactuated robots as they involve fewer actuators than possible movements.

For typical mechanic systems, as mentioned in Equation A.89, the dynamic constraints can be represented by an actuated generalized coordinate portion and an unactuated portion. Thus the acuated coordinate introduces the inequality constraint, while the unauctuated coordinate introduces the actuated portion.

Problem Statement 2 (Kinodynamic planning problem)**.** *Find a contentious trajectory* $\boldsymbol{q}_{\text{trj}} : [0,1] \rightarrow \mathcal{C}_{\text{free}}$ *that connects* $\boldsymbol{q}_{\text{init}}$ *and* $\boldsymbol{q}_{\text{final}}$*, i. e.*$\boldsymbol{q}_{\text{trj}}(0) = \boldsymbol{q}_{\text{init}}$ *and* $\boldsymbol{q}_{\text{path}}(1) = \boldsymbol{q}_{\text{final}}$*, and satisfies the kinodynamic constraints.*

$$\begin{aligned} \boldsymbol{f}(\ddot{\boldsymbol{q}}, \dot{\boldsymbol{q}}, \boldsymbol{q}) &= 0 \\ \boldsymbol{g}(\ddot{\boldsymbol{q}}, \dot{\boldsymbol{q}}, \boldsymbol{q}) &\leq 0 \end{aligned} \tag{2.3}$$

2.2.1 Kinodynamic RRT

The rapidly exploring random tree applies to the kinodynamic problem when it is extended by a Local Motion Planning (LMP). In contrast to the basic Rapidliy expanding Random Tree (RRT) (Algorithm 1) that assumes that the configuration space can be traversed arbitrarily in the kinodynamic case, the movement is constrained. To produce trajectories that are feasible solutions of a boundary value problem is required to find the trajectory realizing a path (Figures 2-3). Therefore, the LMP is a modified interpolation strategy. Furthermore, the sampling must also consider the velocities, which significantly enlarges the computational effort as the space sampled form has a dimension twice the geometric model. This is for robots with more than three joints, a pretty severe drawback. Having those changes named the algorithm 1 can be directly adopted by modifying the interpolation strategy. The interpolation is then also called Local Motion Planning (LMP) that solves the boundary value problem to connect two given states (Figure 2-3) [34, pp. 816].

Local motion planning for a discrete time state space system using minimum input energy

Now it is explained how the LMP is solved for a Boundary Value Problem (BVP) relying on a linear time discrete state space model.

Therefore, recover

Definition 8 (Dynamics of a state space system)**.** *A discrete time state space system is governed by the following dynamics.*

$$\begin{aligned} \boldsymbol{x}(k+1) &= \boldsymbol{A}\boldsymbol{x}(k) + \boldsymbol{B}\boldsymbol{u}(k) \\ \boldsymbol{x}(0) &= \boldsymbol{x}_0 \end{aligned} \tag{2.4}$$

For a fixed time interval the state trajectory can be represented by:

$$\boldsymbol{x}(k) = \sum_{i=0}^{k-1} \boldsymbol{A}^{k-i-1}\boldsymbol{B}\boldsymbol{u}(i) + \boldsymbol{A}^k \boldsymbol{x}_0 \tag{2.5}$$

Therefore, the final state can be represented in the following form.

$$\begin{aligned} \boldsymbol{x}(k) &= \begin{bmatrix} \boldsymbol{A}^{k-1}\boldsymbol{B} & \boldsymbol{A}^{k-2}\boldsymbol{B} & \dots & \boldsymbol{A}^{1}\boldsymbol{B} & \boldsymbol{B} \end{bmatrix} \cdot \begin{bmatrix} \boldsymbol{u}(0) \\ \boldsymbol{u}(1) \\ \vdots \\ \boldsymbol{u}(k-2) \\ \boldsymbol{u}(k-1) \end{bmatrix} + \boldsymbol{A}^k \boldsymbol{x}_0 \\ &= \underbrace{\begin{bmatrix} \boldsymbol{A}^{k-1}\boldsymbol{B} & \boldsymbol{A}^{k-2}\boldsymbol{B} & \dots & \boldsymbol{A}^{1}\boldsymbol{B} & \boldsymbol{B} \end{bmatrix}}_{\mathcal{K}_k} \cdot \boldsymbol{u} + \boldsymbol{A}^k \boldsymbol{x}_0 \end{aligned}$$

Thus, the LMP can be solved to optimallity regarding the cost Function $J = \sum_{i=0}^{k-1} ||\boldsymbol{u}||^2$ using the pseudo inverse as derived in Section 2.2.2.

$$\boldsymbol{u} = \begin{bmatrix} \boldsymbol{A}^{k-1}\boldsymbol{B} & \boldsymbol{A}^{k-2}\boldsymbol{B} & \dots & \boldsymbol{A}^{1}\boldsymbol{B} & \boldsymbol{B} \end{bmatrix}^{+} \left(\boldsymbol{x}(k) - \boldsymbol{A}^k \boldsymbol{x}_0\right) = \mathcal{K}_k{}^{+} \left(\boldsymbol{x}_{\mathrm{f}} - \boldsymbol{A}^k \boldsymbol{x}_0\right) \tag{2.6}$$

As this is a relatively easy-to-implement and numerically efficient method, it is feasible to do a grid search on the trajectory duration. The implementation of the local motion planning attempts to connect two states by a trajectory of the shortest duration and checks if the constraints are fulfilled, if they are not, the attempt is repeated with a longer duration. The shortest timespan that guarantees a solution of the problem $\mathcal{K}_k$ must be full row rank. This only becomes true if the system is controllable and specifically for single input systems for $k \geq n$. Thus the local motion planner solves the optimization problem.

$$\min_{k,\boldsymbol{u}} \quad k \tag{2.7}$$

$$\begin{aligned} \text{s.t.} \quad & \min \frac{1}{2}\boldsymbol{u}^{\mathrm{T}}\boldsymbol{u} \\ & \boldsymbol{x}_{\mathrm{f}} = \mathcal{K}_k \boldsymbol{u} + \boldsymbol{A}^k \boldsymbol{x}_0 \\ & \left[\sum_{i=0}^{j-1} \boldsymbol{A}^{j-i-1}\boldsymbol{B}\boldsymbol{u}(i) + \boldsymbol{A}^j \boldsymbol{x}_0 \right] \in \mathcal{X} \quad \forall j \in \{1,\dots,k\} \end{aligned} \tag{2.8}$$

The set $\mathcal{X}$ constrains the trajectory to the state space.

To obtain the solution quickly, matrix potentials $\boldsymbol{A}^k$ and $\mathcal{K}_k{}^{\ddagger}$ are precomputed.

2.2.2 Derivation of the Solution

This section shows how the before mentioned solution is obtained by fundamental optimization strategies.

That the pseudo inverse solves the equation can be shown by solving the optimization problem.

$$\begin{aligned} \min \quad & \frac{1}{2}\boldsymbol{u}^{\mathrm{T}}\boldsymbol{u} \\ \text{s.t.} \quad & \boldsymbol{x}_{\mathrm{f}} = \mathcal{K}_k\boldsymbol{u} + \boldsymbol{A}^k\boldsymbol{x}_0 \end{aligned}$$

The solution of the constraint optimization is obtained by formulation of the Lagrangian ans solving it to optimality.

$$\begin{aligned}
& & L(\boldsymbol{u},\boldsymbol{\lambda}) &= \frac{1}{2}\boldsymbol{u}^{\mathrm{T}}\boldsymbol{u} + \boldsymbol{\lambda}^{\mathrm{T}}\left(\mathcal{K}_k\boldsymbol{u} + \boldsymbol{A}^k\boldsymbol{x}_0 - \boldsymbol{x}_{\mathrm{f}}\right) \\
& & &\min_{\boldsymbol{u}}\max_{\boldsymbol{\lambda}} L(\boldsymbol{u},\boldsymbol{\lambda}) \\
\text{I:} & & \frac{\partial L}{\partial \boldsymbol{\lambda}} &= \boldsymbol{x}_{\mathrm{f}} - \mathcal{K}_k\boldsymbol{u} - \boldsymbol{A}^k\boldsymbol{x}_0 \overset{!}{=} 0_n \\
\text{II:} & & \frac{\partial L}{\partial \boldsymbol{u}} &= \boldsymbol{u} - \mathcal{K}_k{}^{\mathrm{T}}\boldsymbol{\lambda} \overset{!}{=} 0_{mk} \\
\text{II:} & & \boldsymbol{u} &= \mathcal{K}_k{}^{\mathrm{T}}\boldsymbol{\lambda} \\
\text{II} \to \text{I} = \text{III:} & & 0_n &= \boldsymbol{x}_{\mathrm{f}} - \mathcal{K}_k\mathcal{K}_k{}^{\mathrm{T}}\boldsymbol{\lambda} - \boldsymbol{A}^k\boldsymbol{x}_0 \\
\text{III:} & & \mathcal{K}_k\mathcal{K}_k{}^{\mathrm{T}}\boldsymbol{\lambda} &= \boldsymbol{x}_{\mathrm{f}} - \boldsymbol{A}^k\boldsymbol{x}_0 \\
\text{III:} & & \boldsymbol{\lambda} &= \left(\mathcal{K}_k\mathcal{K}_k{}^{\mathrm{T}}\right)^{-1}\left(\boldsymbol{x}_{\mathrm{f}} - \boldsymbol{A}^k\boldsymbol{x}_0\right) \\
\text{I} \to \text{III:} & & \boldsymbol{u} &= \mathcal{K}_k{}^{\mathrm{T}}\left(\mathcal{K}_k\mathcal{K}_k{}^{\mathrm{T}}\right)^{-1}\left(\boldsymbol{x}_{\mathrm{f}} - \boldsymbol{A}^k\boldsymbol{x}_0\right)
\end{aligned}$$

The cost of the optimal solution is

$$\begin{aligned}
\boldsymbol{u}^{\mathrm{T}}\boldsymbol{u} &= \left(\boldsymbol{x}_{\mathrm{f}} - \boldsymbol{A}^k\boldsymbol{x}_0\right)^{\mathrm{T}}\left(\mathcal{K}_k{}^{\dagger}\right)^{\mathrm{T}}\mathcal{K}_k{}^{\dagger}\left(\boldsymbol{x}_{\mathrm{f}} - \boldsymbol{A}^k\boldsymbol{x}_0\right) && (2.9) \\
&= \left(\boldsymbol{x}_{\mathrm{f}} - \boldsymbol{A}^k\boldsymbol{x}_0\right)^{\mathrm{T}}\left(\mathcal{K}_k{}^{\dagger}\right)^{\mathrm{T}}\mathcal{K}_k{}^{\dagger}\left(\boldsymbol{x}_{\mathrm{f}} - \boldsymbol{A}^k\boldsymbol{x}_0\right) && (2.10) \\
&= \left(\boldsymbol{x}_{\mathrm{f}} - \boldsymbol{A}^k\boldsymbol{x}_0\right)^{\mathrm{T}}\left(\mathcal{K}_k\mathcal{K}_k{}^{\mathrm{T}}\right)^{-1}\left(\boldsymbol{x}_{\mathrm{f}} - \boldsymbol{A}^k\boldsymbol{x}_0\right) && (2.11)
\end{aligned}$$

that decays montonically with the time index k [35, Theorem 2]. Therefore, the local motion planning method as is proposed that aims to minimize the time that is needed, that will also lead to a high controller input as the input trajectory decays monotonically with the trajectory duration of the LMP.

2.3 Simulation results of the kinodynamic RRT

The kinodynamic RRT was implemented in python and tested for the setup. The Model that has been used is the linerized model without frition (4.56) derived in Chapter 4. The state is constrained to $s_{\mathrm{W}} \in [-2, 4]$, $\varphi \in [-50°, 70°]$, $|\dot{s_{\mathrm{W}}}| \leq 2\,\mathrm{m\,s^{-1}}$ and $|\dot{\varphi}| \leq 2\,\mathrm{s^{-1}}$ They define the states that samples are drawn from and set$\mathcal{X}$that is checked in the Local Motion Planning (LMP). The code that has lead to the results is composed in a jupyter-notebook `9_Python_Motion_Planning/kinodynamic_motion_planning.ipynb` in the git repository. The underlying system dynamics are taken from Chapter 4 Eq. (4.56) and discretized using the `c2d` function in matlab [36].

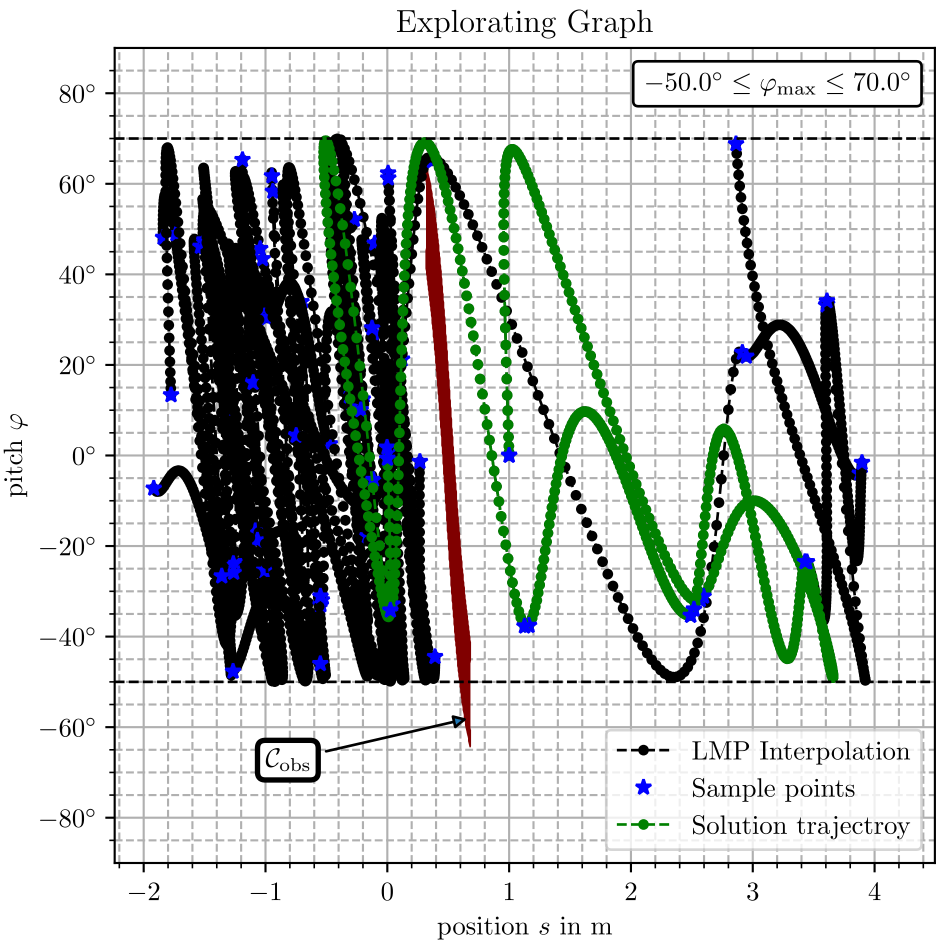


Figure 2-4. Kinodynamic RRT applied to the TWIPR dynamics in the Configuration space. The Solution trajectory reflects the made assumptions as a high amplitude due to time optimization. Furthermore, in the given setup the solution trajectories are often similar.

Figure 2-4 presents the algorithm's result trajectory and the exploring graph. The initial configuration is located in the origin, and the goal configuration is at 1 m and a pitch of 0°. The vertices are given in blue, and interpolation trajectories in green when they are part of the solution trajectory and black otherwise. It is observed that the green trajectory solves the planning problem as it does not intersect the obstacle region $\mathcal{C}_{\mathrm{obs}}$, fulfills the dynamics by construction, and connects the initial configuration $\boldsymbol{q}_{\mathrm{init}}$ and the goal configuration $\boldsymbol{q}_{\mathrm{goal}}$. Furthermore, all interpolations satisfy the constraint in the pitch and the position.

The graph's appearance is characteristic for this problem, i. e. most other attempts have a similar appearance. In the earlier phase the graph explores the left side of the obstacle region until a node is sampled on the right. As this happend the two parts are connected and the algorithm quickly approaches the goal configuration. This also leads to the observation that many vertices are on the left, while only a few vertices appear on the right.

2.4 Applying planned motions

The planned motion relies on the reference model eventually. Therefore, the motion can not be executed precisely. Three main problems lead to tracking imperfections. The first is the linearization of the model. This is done at some point and time instant leading to an error apart from this particular point. Secondly, models used in controls are usually simplified and only model the most significant dynamics e. g.controlling the chetha is done only considering the trunk but not the legs [37]. Finally, there are environmental influences that can not be foreseen. They may disturb the state but also change the dynamics. In fully actuated robotic tasks, this is a minor problem though. Since all set-points are reachable, an arbitrarily given trajectory can be traversed to a predefined confidence set. If the robot is under-actuated, these properties do no longer hold. A trajectory is not feasible as the model is imperfect. Therefore, the control algorithm has to tolerate a deviation from the original trajectory. How those deviations behave is not uniquely given by the system's properties but can be manipulated using feedback and feedforward controllers. As Iterative Learning Control (ILC) enables learning a feedforward control with rough system information [2], it is promising to combine these methods.

2.5 Conclusions and Takeaways

Investigating the Motion planning methods, it has been found that the planning problem for the Two Wheeled Inverted Pendulum Robot (TWIPR) is relatively easy to solve in planar cases. This leads to trajectories that reflect the properties of sampling-based motion planning. In planning literature, many extensions exist, focusing on the optimization of the algorithm to mitigate problems in high dimensional cluttered environments [34]. Those extensions are not promising in the thesis context as they optimize execution time while the characteristics of the graph remain. Therefore, in the following as the thesis focuses on adopting the Iterative Learning Control (ILC) methods, no further extensions are introduced.

Chapter 3

Iterative Learning Control

In this chapter, the paradigm of Iterative Learning Control (ILC) is introduced. The method solves the tracking problem for repeated processes. Therefore a feedback loop from iteration to iteration is used to obtain a feedforward control signal. Generally speaking, ILC are improving the performance using the information in past trials. Thus, the control loop improves the input trajectory from one trial to the next based on the observations. Doing so, the feedforward control signal converges to the optimal one that may also achieve zero tracking error, which is unique in control algorithms [23, 38].

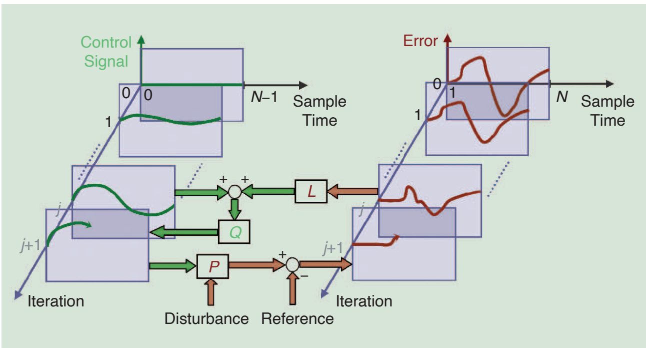


Figure 3-1. Basic ILC principle [2].

An example application is made on the TWIPR in the master thesis of Meindl [4], who applied ILC to solve the reference tracking problem of the pitch angle. To solve the motion problems, a reference trajectory was designed by hand. In doing so, agile maneuvering is possible in the absence of precise model information that was needed for pure feedback solutions. As this work serves as a preliminary for this thesis project, in the following, the problem is taken care of more generally considering also control problems that are over or under-actuated (Section 3.2), and the assumption of reference tracking will be relaxed by introducing the point to point problem (Section 3.4). Furthermore, the DoF that are not necessary to solve the point to point problem will be optimized by an ancillary ILC controller (Section 3.5). As preliminaries, it is discussed how general control problems are augmented by ILC controllers to form an ILC system (Section 3.1), the design technique of Norm

Optimal Iterative Learning Control (NO-ILC) (Section 3.3) and how constraints are introduced using optimization (Section 3.3).

The methods available divide into two main branches which both are model based. The first uses time or frequency domain methods synthesizing resource efficient algorithms, meaning easy implementation even in the light of limited computing power and controller memory, thus suitable for higher frequency applications. An introductory tutorial can be found [39]. The second main branch is part of the optimal control algorithms that aim to minimize a cost function by applying the control update. The latter gains interest as the computational capabilities are constantly growing and enabling the solution of mathematical programs (optimization problems) even in embedded devices. Those methods may be extended by nonlinear stochastic models which is not practical in the time or frequency domain analysis and synthesis.

In the scope of this thesis ILC is applied to problems that motion planners have formally solved. Therefore, three algorithms are tested and compared. They are rising in complexity and thus discussed consecutively. The first is NO-ILC for Single Input Single Output (SISO) systems forming the baseline and can also be considered as a standard algorithm. It is a special case of the Single Input Multiple Output (SIMO) case. As the SIMO case is under-actuated by definition, arbitrary output trajectories can not be realized. Thus this method likely required carefully planned input trajectories. This problem is overcome in the second step, which is restricted to special waypoints on the trajectory. Thus, the input trajectory has enough Degrees of freedom (DoF) to realize exact tracking again. In case of a sparse selection of waypoints, the optimal solution is not unique, i. e. there exist a lot of input trajectories solving the control problem to optimally. In the fourth step, an ancillary cost is introduced that is embedded in the update rule leading to a unique optimum. Throughout the experiments done in this thesis project, two additional problems are observed, namely time allocation and robust constraint satisfaction. Promising methods facing this issue have not been found by the author. The system representation common for ILC literature is introduced as a preliminary.

3.1 System representation

In this section, it is discussed how systems need to be represented to apply the theory that is derived in this thesis. Therefore, an ILC-plant representation is introduced first. Then this slightly more general representation is compared to standard ILC lifted dynamics representation in the literature adopted to the $\mathcal{H}_\infty$-methods. This enables the application of constraints to quantities that are not the controlled output and is closely related to the matlab implementation. It is presented by an example of how a given control loop can be represented in this notation to apply Iterative Learning Control (ILC) analysis and synthesis. Finally, the ILC control loop is introduced to be equipped for the analysis and synthesis sections. The decision to model the system in this slightly more general way is that is reflect the software that has been developed in the project.

Commonly, control theory assumes linear systems [9]. This also holds true for ILC controller design. Therefore, consider the exemplary control loop depicted in Figure 3-2. The system controlled consists of a plant model $\boldsymbol{G}(q)$ and a stabilizing controller $\boldsymbol{K}_{\mathrm{FB}}(q)$ in a standard control loop. The ILC controller is augmented to this loop by governing an additional input signal to control the error trajectory of the standard control loop to the reference given to the ILC-Controller.

In the following, it is emphasized that the structure of the possibly controlled system does not matter for the ILC design. Therefore, the system shall be represented by a system as given in Definition 9.

Definition 9 (Discrete-time MIMO LTI system with an observed output)**.** *The tuple of equations*

$$
\begin{aligned}
\boldsymbol{x}(k+1) &= \boldsymbol{A}\boldsymbol{x}(k) + \boldsymbol{B}_{\mathrm{c}}\boldsymbol{u}_{\mathrm{c}}(k) + \boldsymbol{B}_{\mathrm{e}}\boldsymbol{u}_{\mathrm{e}}(k)\\
\boldsymbol{y}_{\mathrm{c}}(k) &= \boldsymbol{C}_{\mathrm{c}}\boldsymbol{x}(k) + \boldsymbol{D}_{\mathrm{cc}}\boldsymbol{u}_{\mathrm{c}}(k) + \boldsymbol{D}_{\mathrm{ce}}\boldsymbol{u}_{\mathrm{e}}(k)\\
\boldsymbol{y}_{\mathrm{o}}(k) &= \boldsymbol{C}_{\mathrm{o}}\boldsymbol{x}(k) + \boldsymbol{D}_{\mathrm{oc}}\boldsymbol{u}_{\mathrm{c}}(k) + \boldsymbol{D}_{\mathrm{eo}}\boldsymbol{u}_{\mathrm{e}}(k)\\
\boldsymbol{x}(0) &= \boldsymbol{x}_0
\end{aligned}
\tag{3.1}
$$

Where $\boldsymbol{x} \in \mathbb{R}^n$ ist the state, $\boldsymbol{u}_{\mathrm{c}} \in \mathbb{R}^{m_{\mathrm{c}}}$ is the input governed by the ILC-Controller, $\boldsymbol{y}_{\mathrm{c}} \in \mathbb{R}^{p_{\mathrm{c}}}$ is the controlled output, $\boldsymbol{y}_{\mathrm{o}} \in \mathbb{R}^{p_{\mathrm{o}}}$ is the constrained output and $\boldsymbol{u}_{\mathrm{e}} \in \mathbb{R}^{m_{\mathrm{e}}}$ an exogenous input. The corresponding sampling period is denoted h.

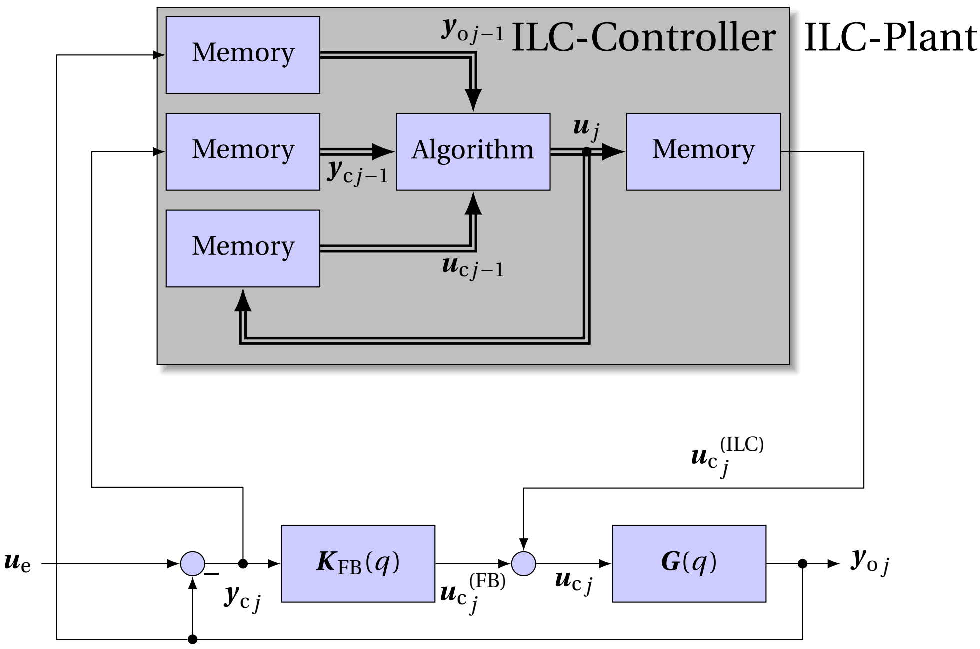


Figure 3-2. Standard NO-ILC algorithm. The algorithm of the gray ILC controller is executed as a trial is completed. Throughout a trail the calculated input is applied to the ILC plant. Double lines represent lifted qualities and thin sequential data.

This definition is a basic discrete-time LTI system that divides its inputs and outputs. The outputs serves two purposes, the controlled output $\boldsymbol{y}_{\mathrm{c}}$ is to be controlled (i. e. governed to a reference), the observed $\boldsymbol{y}_{\mathrm{o}}$ is subject to constrained. Regarding the system in Figure 3-2 everything outside the gray box is the ILC-Plant, inside the box is the ILC-Controller.

In the followin the ILC-Plant shown in Figure 3-2 is described compatible with the Definition 9. Therefore, consider that the systems of the inner loop are represented by state space systems themselves $\boldsymbol{G} \cong \left[\begin{array}{c|c} \boldsymbol{A}_\mathrm{G} & \boldsymbol{B}_\mathrm{G} \\ \hline \boldsymbol{C}_\mathrm{G} & \boldsymbol{0} \end{array}\right]$ and $\boldsymbol{K}_\mathrm{FB} \cong \left[\begin{array}{c|c} \boldsymbol{A}_\mathrm{K} & \boldsymbol{B}_\mathrm{K} \\ \hline \boldsymbol{C}_\mathrm{K} & \boldsymbol{D}_\mathrm{K} \end{array}\right]$.

This lead to the following representation according to Definition 9.

$$
\begin{aligned}
\boldsymbol{A} &= \begin{bmatrix} \boldsymbol{A}_\mathrm{G} - \boldsymbol{B}_\mathrm{G}\boldsymbol{D}_\mathrm{K}\boldsymbol{C}_\mathrm{G} & \boldsymbol{B}_\mathrm{G}\boldsymbol{C}_\mathrm{K} \\ -\boldsymbol{B}_\mathrm{K}\boldsymbol{C}_\mathrm{G} & \boldsymbol{A}_\mathrm{K} \end{bmatrix} & \boldsymbol{B}_\mathrm{c} &= \begin{bmatrix} \boldsymbol{B}_\mathrm{G} \\ \boldsymbol{0} \end{bmatrix} & \boldsymbol{B}_\mathrm{e} &= \begin{bmatrix} \boldsymbol{B}_\mathrm{G}\boldsymbol{D}_\mathrm{K} \\ \boldsymbol{B}_\mathrm{K} \end{bmatrix} \\
\boldsymbol{C}_\mathrm{c} &= \begin{bmatrix} -\boldsymbol{C}_\mathrm{G} & \boldsymbol{0} \end{bmatrix} & \boldsymbol{D}_\mathrm{cc} &= \begin{bmatrix} \boldsymbol{0} \end{bmatrix} & \boldsymbol{D}_\mathrm{ce} &= \begin{bmatrix} \boldsymbol{E} \end{bmatrix} \\
\boldsymbol{C}_\mathrm{o} &= \begin{bmatrix} \boldsymbol{C}_\mathrm{G} & \boldsymbol{0} \end{bmatrix} & \boldsymbol{D}_\mathrm{oc} &= [\boldsymbol{0}] & \boldsymbol{D}_\mathrm{eo} &= [\boldsymbol{0}]
\end{aligned}
$$

The system is a distcte time Linear Time Invariant (LTI) system and can be described by the Markov parameters applying a convolution sum. The solution for the representation is given by:

$$
\begin{aligned}
\boldsymbol{y}_\mathrm{c}(k) &= \sum_{i=0}^{k} \boldsymbol{P}_\mathrm{cc}(k-i)\boldsymbol{u}_\mathrm{c}(i) + \sum_{i=0}^{k} \boldsymbol{P}_\mathrm{ec}(k-i)\boldsymbol{u}_\mathrm{e}(i) + \boldsymbol{C}_\mathrm{c}\boldsymbol{A}^k\boldsymbol{x}_0 \\
\boldsymbol{y}_\mathrm{o}(k) &= \sum_{i=0}^{k} \boldsymbol{P}_\mathrm{co}(k-i)\boldsymbol{u}_\mathrm{c}(i) + \sum_{i=0}^{k} \boldsymbol{P}_\mathrm{eo}(k-i)\boldsymbol{u}_\mathrm{e}(i) + \boldsymbol{C}_\mathrm{o}\boldsymbol{A}^k\boldsymbol{x}_0
\end{aligned} \tag{3.2}
$$

Where the matrices $\boldsymbol{P}(k)$ are the markov parameters of the system $\left[\begin{array}{c|c} \boldsymbol{A} & \boldsymbol{B} \\ \hline \boldsymbol{C} & \boldsymbol{D} \end{array}\right]$ and are defined by

$$
\boldsymbol{P}(k) = \begin{cases} \boldsymbol{0} & k < 0 \\ \boldsymbol{D} & k = 0 \\ \boldsymbol{C}\boldsymbol{A}^{k-1}\boldsymbol{B} & k > 0 \end{cases}. \tag{3.3}
$$

As the time horizon is finite the signals become finite dimensional and the convolution sum can be represented by a matrix called the lifted dynamics representation. Furthermore, the disturbance by the initial state is arranged in lifted vectors.

$$
\boldsymbol{P} = \begin{bmatrix} \boldsymbol{P}(0) & \boldsymbol{0} & \dots & \boldsymbol{0} \\ \boldsymbol{P}(1) & \boldsymbol{P}(0) & \dots & \boldsymbol{0} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{P}(n_\mathrm{b}-1) & \boldsymbol{P}(n_\mathrm{b}-2) & \dots & \boldsymbol{P}(0) \end{bmatrix} \qquad \boldsymbol{d}_0 = \begin{bmatrix} \boldsymbol{C}\boldsymbol{x}_0 \\ \boldsymbol{C}\boldsymbol{A}\boldsymbol{x}_0 \\ \vdots \\ \boldsymbol{C}\boldsymbol{A}^{n_\mathrm{b}-1}\boldsymbol{x}_0 \end{bmatrix} \tag{3.4}
$$

$$
\boldsymbol{u} = \begin{bmatrix} \boldsymbol{u}(0) \\ \boldsymbol{u}(1) \\ \vdots \\ \boldsymbol{u}(n_\mathrm{b}-1) \end{bmatrix} \qquad \boldsymbol{y} = \begin{bmatrix} \boldsymbol{y}(0) \\ \boldsymbol{y}(1) \\ \vdots \\ \boldsymbol{y}(n_\mathrm{b}-1) \end{bmatrix} \tag{3.5}
$$

Thus, the whole system can be represented by an overall lifted dynamics representation

$$\boldsymbol{y}_{\mathrm{c}} = \boldsymbol{P}_{\mathrm{cc}}\boldsymbol{u}_{\mathrm{c}} + \boldsymbol{P}_{\mathrm{ec}}\boldsymbol{u}_{\mathrm{e}} + \boldsymbol{d}_{0\mathrm{c}} \tag{3.6}$$
$$\boldsymbol{y}_{\mathrm{o}} = \boldsymbol{P}_{\mathrm{co}}\boldsymbol{u}_{\mathrm{c}} + \boldsymbol{P}_{\mathrm{eo}}\boldsymbol{u}_{\mathrm{e}} + \boldsymbol{d}_{0\mathrm{o}} \tag{3.7}$$

The four matices are obtained by the four permutations of the inputs and outputs. E. g. the parameter $\boldsymbol{P}_{\mathrm{eo}}$ represents the state space system $\left[\begin{array}{c|c} \boldsymbol{A} & \boldsymbol{B}_{\mathrm{e}} \\ \hline \boldsymbol{C}_{\mathrm{o}} & \boldsymbol{D}_{\mathrm{eo}} \end{array}\right]$.

Now the learning controller is introduced. The learning controller is a discrete-time controller that acts in the trail domain $j \in \mathbb{N}$. One trail is one execution of an experiment. Therefore inputs and outputs of the former system are augmented by trail indexes j. The trail domain does not correspond to a smapling period. An index corresponds to a full experiment called *trail*. As the disturbance introduced by the initial state $\boldsymbol{d}_0$ are assumed to not change from trail to trail, as indexed in the trail domain, the assumption of the same initial state is done implicitly (3.4).

In ILC literature exogenous inputs and observed outputs are not considered as they are not important in analysis. They will be important in the design process considered later on. Namely, the controlled output is governed to a reference while the observed is constrained. The exogenous input is in contrast another input trajectory, e. g.the reference signal fed to feedback control as indicated in Figure 3-2, that needs to be trail invariant. It, therefore, can be absorbed to the lifted disturbance $\boldsymbol{d}$. For the analysis purpose the standard lifted dynamics representation is considered:

$$\boldsymbol{y}_j = \boldsymbol{P}\boldsymbol{u}_j + \boldsymbol{d} \quad [\boldsymbol{P} = \boldsymbol{P}_{\mathrm{cc}} \quad \boldsymbol{d} = \boldsymbol{P}_{\mathrm{ec}}\boldsymbol{u}_{\mathrm{e}} + \boldsymbol{d}_{0\mathrm{c}}]\,. \tag{3.8}$$

Therefore, the system representation is complete and the standard ILC control loop is introduced. First, the standard ILC controller is defined.

Definition 10 (Linear ILC update)**.** *The tuple $(\boldsymbol{Q}, \boldsymbol{L})$ is called linear ILC update law. The input of the next iteration is computed by*

$$\boldsymbol{u}_{j+1} = \boldsymbol{Q}(\boldsymbol{u}_j - \boldsymbol{L}(\boldsymbol{y}_j - \boldsymbol{r})). \tag{3.9}$$
$$\boldsymbol{u}_0 = \boldsymbol{0} \quad \textit{w.o.l.g.}$$

The matrix $\boldsymbol{Q}$ is called Q-filter and $\boldsymbol{L}$ learn rule.

This linear ILC update might also be presented by LTV systems and is discussed in teaching material [27].

Finnaly an ILC system is defined.

Definition 11 (ILC system)**.** *The composition of equations (3.8) and (3.9) is standard ILC system.*

3.1.1 Rank and invertability of the lifted dynamics

Most analysis techniques used later rely on the full rank of the lifted dynamics. Therefore, this section argues why this is the case for almost every application. This is also done in most publications for SISO systems [14, 4]. In this case the Markov parameters become scalar and the one can simply shift the sequence by the systems delay to assure that $p(0) \neq 0$ guaranteeing that the lifted dynamics matrix is invertible as it is lower triangular and holds only non zero values on its diagonal. In the case of MIMO systems, this is not possible as the matrix is block lower triangular. The matrices on the diagonal must be full rank to let the matrix be full rank as well.

The following thoughts are closely related to [23, pp. 92].

To reason about this question consider the lifted dynamics column wise and assure that the zeroth element is non zero and compose them to yet another Markov parameter matrix.

Definition 12 (Single Input (SI) Markov paramter vector sequence, time delay and disturbance sequence)**.** *Each input l of a discrete-time LTI system has an unique sequence of markov parameter column vectors and an associated delay. The Markov parameters are defined by*

$$\boldsymbol{p}_l(k) := \begin{cases} \boldsymbol{C}_\mathrm{c}\boldsymbol{A}^{k+\gamma_l-1}\,{}^\mathrm{C}[\boldsymbol{B}_\mathrm{c}]_l & k > -\gamma_l \\ {}^\mathrm{C}[\boldsymbol{D}_\mathrm{cc}]_l & k = -\gamma_l \\ \boldsymbol{0}_{p\times 1} & else \end{cases} \tag{3.10}$$

so that $\boldsymbol{p}_l(0) \neq \boldsymbol{0}_{p\times 1}$ and for $k < 0$ also $p_l(k) = \boldsymbol{0}_{p\times 1}$ hold. The output disturbance induced by the initial state is given by

$$\tilde{d}(k) = \boldsymbol{C}_\mathrm{c}\boldsymbol{A}^k\boldsymbol{x}_0 + \boldsymbol{D}_\mathrm{ce}\boldsymbol{u}_\mathrm{e}(k) + \sum_{i=1}^{k+\gamma} \boldsymbol{C}_\mathrm{c}\boldsymbol{A}^{k-i-1}\boldsymbol{B}_\mathrm{e}\boldsymbol{u}_\mathrm{e}(i). \tag{3.11}$$

Definition 13 (Markov parameter matrix sequence and delay vector)**.** *Each discrete-time LTI system has an unique sequence of Markov parameter matrices and an associated delay vector. The time delay vector is $\gamma = \left(\gamma_1 \dots \gamma_p\right)^\mathrm{T}$ a composition of the input delays and the markov parameter matrices $\boldsymbol{P}(k) = \left[\boldsymbol{p}_1 \dots \boldsymbol{p}_p\right]$ are the composition of the Markov parameter row vectors.*

Alternatively, the system can be represented by an equivalent system without delays that uses the delayed input as input. The system uses a modified version of the feed through and input matrices. (adopted to [23, pp. 92])

$$\tilde{\boldsymbol{B}} = \left[\boldsymbol{A}^{\gamma_1}\,{}^\mathrm{C}[\boldsymbol{B}_\mathrm{c}]_1 \quad \dots \quad \boldsymbol{A}^{\gamma_m}\,{}^\mathrm{C}[\boldsymbol{B}_\mathrm{c}]_m\right]$$

$$\tilde{\boldsymbol{D}} = \left[\boldsymbol{C}_\mathrm{c}\boldsymbol{A}^{\gamma_1-1}\,{}^\mathrm{C}[\boldsymbol{B}_\mathrm{c}]_1 \quad \dots \quad \boldsymbol{C}_\mathrm{c}\boldsymbol{A}^{\gamma_m-1}\,{}^\mathrm{C}[\boldsymbol{B}_\mathrm{c}]_m\right]$$

As the matrix is block diagonal (3.4) its invertability is equivalent to the invertability of the first markov parameter matrix $\boldsymbol{P}(0) = \tilde{\boldsymbol{D}}$. If the matrix would loose rank one linear dependent output could be omitted. Using this derivation it is justified to assume that the lifted dynamics representation has full rank, as it is done in the remainder of this thesis.

3.2 Analysis of ILC system

This section presents the ILC analysis techniques used later. The analysis methods rely on the ILC control system (3-3) as representation of the control loop to derive system properties. The most important properties that are discussed in this section are stability, steady-state error and monotonic convergence. Furthermore, it is discussed how the state of actuation effects the fundamental properties. It is shown that overactuation in the absence of a Q-filter maintains the most important property, a vanishing steady-state error, while it needs another stability criterion than the other cases. Considering a Q-filter the convergence rate can not be approximated. Contrarily, in the underactuated case, perfect tracking is not achievable generally, but the properties of the solution is still controllable by the learn rule. The results are summarized in Table 3.1.

The most important property of a control system is asymptotic stability. As the ILC system is defined in Definition 11 the input trajectory is considered as a state in the iteration domain. Therefore, as the input becomes stationary so does the whole system. To make this clear the system is written down as a discrete-time $n_\mathrm{b} \cdot m$ dimensional state space system in the iteration domain, meaning that the time index is the iteration index j.

Definition 14 (Asympthotic Stability and steady state error)**.** *[2] The standard ILC control loop is asymptotically stable if the limits*

$$\boldsymbol{e}_\infty = \lim_{j\to\infty} \boldsymbol{e}_j \qquad \boldsymbol{u}_\infty = \lim_{j\to\infty} \boldsymbol{u}_j \tag{3.12}$$

do exist. The value $\boldsymbol{e}_\infty$ is called the steady state error.

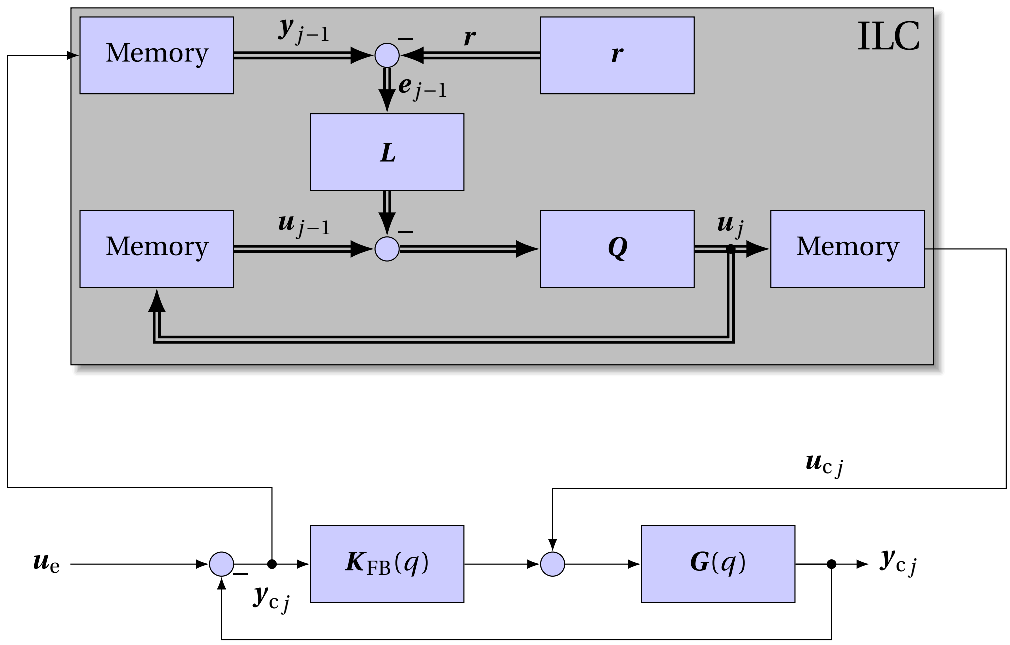


Figure 3-3. Standard ILC system using the linear ILC update rule.

The definition of stability includes the error as it is the design goal of the ILC system (3.8), (3.9) and defines the steady state error. Conversely, as the discrete-time ILC system is finite dimensional with the lifted input being the state, the existence of the limit of the input is already sufficient independent of the state of actuation.

An other property that is commonly analyzed is monotonic convergence [2]. If this property is fulfilled the variation of the output trajectory can be upper bounded, e. g. considering the maximum norm the error trajectory will not exceed an interval as big as the maximum initial error.

Definition 15 (Monothonic convergence)**.** *[2] The ILC standard control loop is monotonically convergent under a norm* $||\cdot||$ *if*

$$\left|\left|\boldsymbol{e}_{j+1} - \boldsymbol{e}_{\infty}\right|\right| = \epsilon \left|\left|\boldsymbol{e}_j - \boldsymbol{e}_{\infty}\right|\right| \tag{3.13}$$

where $0 \leq \epsilon \leq 1$ *is the convergence rate.*

In most applications a review of SISO systems is sufficient. The methods used in the thesis project will also deal with over- and under actuated systems. The state of actuation is the ability to control the outputs to given states using the inputs governed by the controller. This property is characterized by the lifted dynamics matrix rank. The state of under actuation describes the lack of the ability to control the output arbitrarily and the state of over actuation abandon the uniqueness of the input for a given output.

Definition 16 (Under actuation)**.** *A batch process is under actuated as its lifted dynamics representation does not have full row rank.*

$$\operatorname{rank}[\boldsymbol{P}] < n_b m \tag{3.14}$$

Definition 17 (Over actuation)**.** *A batch process is over actuated as its lifted dynamics representation does not have full column rank.*

$$\operatorname{rank}[\boldsymbol{P}] < n_b p \tag{3.15}$$

Definition 18 (Uniquely actuated)**.** *A batch process is fully actuated if it is neither under nor over actuated.*

As discussed in the previous section the lifted dynamics can be assumed to have full rank, so that the state of over or under actuation is equivalent to the dimension lifted dynamics or equivalently the dimension of the controlled and the governed variable. Furthermore, the state of under, over and unique actuation is exclusive under this assumption.

As under, over and unique actuation are exclusive, their states need to be investigated. The state of unique actuation leads to the simplest system behavior which is discussed in the beginning, afterwards it is discussed how the solutions change as unique actuation is dropped.

Unique actuation

Unique actuation implies the ivertability of the lifted dynamics matrix and the exact coupling of an input to an output trajectory. Regarding the asymptotic stability it is sufficient to proof the existence of one of the both limits [2].

The evolution of the input trajectory is given by the closed loop dynamics obtained by inserting (3.8) in (3.9).

$$\begin{aligned}
\boldsymbol{u}_{j+1} &= \boldsymbol{Q}\left(\boldsymbol{u}_j - \boldsymbol{L}\left(\boldsymbol{P}\boldsymbol{u}_j + \boldsymbol{d} - \boldsymbol{r}\right)\right) \\
\boldsymbol{u}_0 &= \boldsymbol{0} \\
\boldsymbol{u}_{j+1} &= \boldsymbol{Q}\left(\boldsymbol{E} - \boldsymbol{LP}\right)\boldsymbol{u}_j - \boldsymbol{QL}\left(\boldsymbol{d} - \boldsymbol{r}\right)
\end{aligned}$$

This evolution leads to the following sequence that defines the steady state input:

$$\boldsymbol{u}_\infty = \sum_{i=0}^{\infty} -\boldsymbol{Q}\left(\boldsymbol{E} - \boldsymbol{LP}\right)^i \boldsymbol{QL}\left(\boldsymbol{d} - \boldsymbol{r}\right) \overset{\text{Neumann series}}{=} -\left(\boldsymbol{E} - \boldsymbol{Q}\left(\boldsymbol{E} - \boldsymbol{LP}\right)\right)^{-1} \boldsymbol{QL}\left(\boldsymbol{d} - \boldsymbol{r}\right) \tag{3.16}$$

The term contains the Neumann series [40, p. 20] whose convergence criterion leads to the convergence criterion of uniquely actuated ILC systems [2].

$$\rho\left(\boldsymbol{Q}\left(\boldsymbol{E} - \boldsymbol{LP}\right)\right) < 1 \tag{3.17}$$

The same investigation can be done for the error trajectory whose evolution is derived in the Appendix A.6.2.

$$\boldsymbol{e}_{j+1} = \boldsymbol{PQ}\left(\boldsymbol{E} - \boldsymbol{LP}\right)\boldsymbol{P}^{-1}\boldsymbol{e}_j + \left(\boldsymbol{E} - \boldsymbol{PQP}^{-1}\right)\left(\boldsymbol{d} - \boldsymbol{r}\right) \tag{3.18}$$

$$\begin{aligned}
\boldsymbol{e}_\infty &= \boldsymbol{P}\sum_{i=0}^{\infty}\left[\boldsymbol{Q}\left(\boldsymbol{E} - \boldsymbol{LP}\right)\right]^i \boldsymbol{P}^{-1}\left(\boldsymbol{E} - \boldsymbol{PQP}^{-1}\right)\left(\boldsymbol{d} - \boldsymbol{r}\right) + \boldsymbol{P}\left[\boldsymbol{Q}\left(\boldsymbol{E} - \boldsymbol{LP}\right)\right]^\infty \boldsymbol{P}^{-1}\boldsymbol{e}_0 \\
&= \boldsymbol{P}\sum_{i=0}^{\infty}\left[\boldsymbol{Q}\left(\boldsymbol{E} - \boldsymbol{LP}\right)\right]\left(\boldsymbol{E} - \boldsymbol{Q}\right)\boldsymbol{P}^{-1}\left(\boldsymbol{d} - \boldsymbol{r}\right)
\end{aligned} \tag{3.19}$$

Therefore, the steady state error is found.

$$\boldsymbol{e}_\infty = \boldsymbol{P}(\boldsymbol{E} - \boldsymbol{Q}\left(\boldsymbol{E} - \boldsymbol{LP}\right))^{-1}\left(\boldsymbol{E} - \boldsymbol{Q}\right)\boldsymbol{P}^{-1}\left(\boldsymbol{d} - \boldsymbol{r}\right) \tag{3.20}$$

It is observed that the steady state error vanishes if and only if the Q-filter is the identity.

Lastly, the convergence behavior (3.18) is investigated to obtain a criterion for monotonicity, therefore, the submulitplicativity of operator norms is exploited to find an upper bound of the convergence rate.

$$\left|\left|\boldsymbol{e}_{j+1} - \boldsymbol{e}_\infty\right|\right| \leq \underbrace{\left|\left|\boldsymbol{PQ}\left(\boldsymbol{E} - \boldsymbol{LP}\right)\boldsymbol{P}^{-1}\right|\right|}_{\geq\epsilon}\left|\left|\left(\boldsymbol{e}_j - \boldsymbol{e}_\infty\right)\right|\right| \tag{3.21}$$

Under actuation

In the under actuated case the convergence criterion and the steady state input do not differ from those of the unique actuated case. Contrarily, the steady state error has a fundamental difference to the earlier results as a vanishing error trajectory is not guaranteed. In the absence of a $\boldsymbol{Q}$ the error trajectory depends on the learn rule if it is not realizable.

The evolution of the input trajectory coincides with the previous results and thus maintains its properties.

$$\boldsymbol{u}_{j+1} = \boldsymbol{Q}\left(\boldsymbol{E}-\boldsymbol{L}\boldsymbol{P}\right)\boldsymbol{u}_j - \boldsymbol{Q}\boldsymbol{L}\left(\boldsymbol{d}-\boldsymbol{r}\right)$$

So the criterion of the existence of the limit of the input is maintained.

$$\rho\left(\boldsymbol{Q}\left(\boldsymbol{E}-\boldsymbol{L}\boldsymbol{P}\right)\right) < 1$$

Due to its under actuation the system's output and error evolution differ as derived in Appendix A.6.3.

$$\boldsymbol{e}_{j+1} = \boldsymbol{P}\boldsymbol{Q}\left(\boldsymbol{E}-\boldsymbol{L}\boldsymbol{P}\right)\boldsymbol{P}^{\dagger}\boldsymbol{e}_j - \boldsymbol{P}\boldsymbol{Q}\boldsymbol{L}\left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{P}^{\dagger}\right)\left(\boldsymbol{d}-\boldsymbol{r}\right) + \left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{Q}\boldsymbol{P}^{\dagger}\right)\left(\boldsymbol{d}-\boldsymbol{r}\right) \tag{3.22}$$

$$\begin{aligned}
\boldsymbol{e}_{\infty} &= \boldsymbol{P}\underbrace{\left[\boldsymbol{Q}\left(\boldsymbol{E}-\boldsymbol{L}\boldsymbol{P}\right)\right]^{\infty}}_{\boldsymbol{0}}\boldsymbol{P}^{\dagger}\boldsymbol{e}_0 + \left(\sum_{i=0}^{\infty}\left[\boldsymbol{P}\boldsymbol{Q}\left(\boldsymbol{E}-\boldsymbol{L}\boldsymbol{P}\right)\boldsymbol{P}^{\dagger}\right]^i\right)\left(\left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{Q}\boldsymbol{P}^{\dagger}\right) - \boldsymbol{P}\boldsymbol{Q}\boldsymbol{L}\left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{P}^{\dagger}\right)\right)\left(\boldsymbol{d}-\boldsymbol{r}\right)\\
\boldsymbol{e}_{\infty} &= \left(\left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{P}^{\dagger}\right) + \boldsymbol{P}\sum_{i=0}^{\infty}\left[\boldsymbol{Q}\left(\boldsymbol{E}-\boldsymbol{L}\boldsymbol{P}\right)\right]^i\boldsymbol{P}^{\dagger}\right)\left(\left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{Q}\boldsymbol{P}^{\dagger}\right) - \boldsymbol{P}\boldsymbol{Q}\boldsymbol{L}\left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{P}^{\dagger}\right)\right)\left(\boldsymbol{d}-\boldsymbol{r}\right)\\
\boldsymbol{e}_{\infty} &= \boldsymbol{P}\sum_{i=0}^{\infty}\left[\boldsymbol{Q}\left(\boldsymbol{E}-\boldsymbol{L}\boldsymbol{P}\right)\right]^i\left(\left(\boldsymbol{E}-\boldsymbol{Q}\right)\boldsymbol{P}^{\dagger} - \boldsymbol{Q}\boldsymbol{L}\left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{P}^{\dagger}\right)\right)\left(\boldsymbol{d}-\boldsymbol{r}\right)\\
&\quad + \left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{P}^{\dagger}\right)\left(\left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{Q}\boldsymbol{P}^{\dagger}\right) - \boldsymbol{P}\boldsymbol{Q}\boldsymbol{L}\left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{P}^{\dagger}\right)\right)\left(\boldsymbol{d}-\boldsymbol{r}\right)\\
\boldsymbol{e}_{\infty} &= \boldsymbol{P}\left(\boldsymbol{E}-\boldsymbol{Q}\left(\boldsymbol{E}-\boldsymbol{L}\boldsymbol{P}\right)\right)^{-1}\left(\left(\boldsymbol{E}-\boldsymbol{Q}\right)\boldsymbol{P}^{\dagger} - \boldsymbol{Q}\boldsymbol{L}\left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{P}^{\dagger}\right)\right)\left(\boldsymbol{d}-\boldsymbol{r}\right) + \left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{P}^{\dagger}\right)\left(\boldsymbol{d}-\boldsymbol{r}\right) && (3.23)
\end{aligned}$$

From now on assume that $\boldsymbol{Q} = \boldsymbol{E}$.

$$\begin{aligned}
\boldsymbol{e}_{\infty} &= -\boldsymbol{P}\sum_{i=0}^{\infty}\left(\boldsymbol{E}-\boldsymbol{L}\boldsymbol{P}\right)^i\boldsymbol{L}\left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{P}^{\dagger}\right)\left(\boldsymbol{d}-\boldsymbol{r}\right) + \left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{P}^{\dagger}\right)\left(\boldsymbol{d}-\boldsymbol{r}\right)\\
\boldsymbol{e}_{\infty} &= -\boldsymbol{P}(\boldsymbol{L}\boldsymbol{P})^{-1}\boldsymbol{L}\left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{P}^{\dagger}\right)\left(\boldsymbol{d}-\boldsymbol{r}\right) + \left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{P}^{\dagger}\right)\left(\boldsymbol{d}-\boldsymbol{r}\right)\\
\boldsymbol{e}_{\infty} &= \left(\boldsymbol{E}-\boldsymbol{P}(\boldsymbol{L}\boldsymbol{P})^{-1}\boldsymbol{L}\right)\left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{P}^{\dagger}\right)\left(\boldsymbol{d}-\boldsymbol{r}\right)\\
\boldsymbol{e}_{\infty} &= \left(\boldsymbol{E}-\boldsymbol{P}(\boldsymbol{L}\boldsymbol{P})^{-1}\boldsymbol{L}\right)\left(\boldsymbol{d}-\boldsymbol{r}\right) && (3.24)
\end{aligned}$$

In contrast to the unique actuation the absence of an Q-filter does not imply zero steady state error, as not all trajectories can be realized. But if the trajectory matches the system's abilities zero steady state error is achieved. As it is shown in Remark 2. This information is encoded in the term $\left(\boldsymbol{E}-\boldsymbol{P}(\boldsymbol{L}\boldsymbol{P})^{-1}\boldsymbol{L}\right)$ that represents a projection to the space of all trajectories cut of the image of the lifted dynamics. Therefore, $\left(\boldsymbol{E}-\boldsymbol{P}(\boldsymbol{L}\boldsymbol{P})^{-1}\boldsymbol{L}\right)\left(\boldsymbol{d}-\boldsymbol{r}\right)$ vanishes as $\left(\boldsymbol{d}-\boldsymbol{r}\right)$ lies in the range of $\boldsymbol{P}$. In contrast to the former case in the absence of a Q-filter the steady state error is dependent on the learn rule $\boldsymbol{L}$. Thus, the steady state error can be manipulated using the learn rule as the reference is not realizable.

Remark 2. *If the reference trajectory is realizable and $\boldsymbol{Q}$ is chosen to be identity, the error vanishes as the iteration proceed to infinity. Therefore, consider there exists an input $\boldsymbol{u}$ so that $\boldsymbol{r} = \boldsymbol{P}\boldsymbol{u} + \boldsymbol{d}$ holds.*

$$\begin{aligned}
\boldsymbol{e}_{\infty} &= -\left(\boldsymbol{E}-\boldsymbol{P}(\boldsymbol{L}\boldsymbol{P})^{-1}\boldsymbol{L}\right)\boldsymbol{P}\boldsymbol{u}\\
&= -\left(\boldsymbol{P}-\boldsymbol{P}(\boldsymbol{L}\boldsymbol{P})^{-1}\boldsymbol{L}\boldsymbol{P}\right)\boldsymbol{u}\\
&= -\left(\boldsymbol{P}-\boldsymbol{P}\right)\boldsymbol{u}\\
&= \boldsymbol{0} && (3.25)
\end{aligned}$$

In this case the input trajectory does not have additional Degrees of freedom (DoF) and therefore the error trajectory determines the input trajectory leading to an approximation of the convergence rate similar to the one found in the uniquely actuated case (Appendix A.6.3).

$$\begin{aligned} \boldsymbol{e}_{j+1} - \boldsymbol{e}_\infty &= \boldsymbol{PQ}\left(\boldsymbol{E} - \boldsymbol{LP}\right)\boldsymbol{P}^\dagger\left(\boldsymbol{e}_j - \boldsymbol{e}_\infty\right) \\ \left|\left|\boldsymbol{e}_{j+1} - \boldsymbol{e}_\infty\right|\right| &\leq \underbrace{\left|\left|\boldsymbol{PQ}\left(\boldsymbol{E} - \boldsymbol{LP}\right)\boldsymbol{P}^\dagger\right|\right|}_{\geq \epsilon}\left|\left|\boldsymbol{e}_j - \boldsymbol{e}_\infty\right|\right| \end{aligned} \tag{3.26}$$

Overactuation

In the case of over-actuation all properties are maintained in the presence of a Q-filter. In its absence, stability is not reachable since the DoF in the null space are not stabilized. This problem is of minor importance as those DoF are controller states that do not have any physical meaning or importance. Furthermore, they are not exited by design of the standard ILC system so they maintain their initial state.

As in the previous cases the input trajectory follows the same dynamics.

$$\boldsymbol{u}_{j+1} = \boldsymbol{Q}\left(\boldsymbol{E} - \boldsymbol{LP}\right)\boldsymbol{u}_j - \boldsymbol{QL}\left(\boldsymbol{d} - \boldsymbol{r}\right)$$

Thus, the steady state error can be found using the same series representation.

$$\boldsymbol{u}_\infty = \sum_{i=0}^{\infty} - \left[\boldsymbol{Q}\left(\boldsymbol{E} - \boldsymbol{LP}\right)\right]^i \boldsymbol{QL}\left(\boldsymbol{d} - \boldsymbol{r}\right) \overset{\text{Neumann series}}{=} -\left(\boldsymbol{E} - \boldsymbol{Q}\left(\boldsymbol{E} - \boldsymbol{LP}\right)\right)^{-1}\boldsymbol{QL}\left(\boldsymbol{d} - \boldsymbol{r}\right)$$

In the state of under actuation the convergence criterion of the Neumann series can not be matched using $\boldsymbol{Q} = \boldsymbol{E}$. Since, this case is highly desirable to obtain prefect tracking in the steady state, this case must be investigated either. Therefore, assume that $\boldsymbol{Q}$ to be identity and the following fundamental matrix identy.

$$\left(\boldsymbol{E} - \boldsymbol{AB}\right)\left(\boldsymbol{E} - \boldsymbol{AB}\right)\boldsymbol{A} = \left(\boldsymbol{E} - \boldsymbol{AB}\right)\boldsymbol{A}\left(\boldsymbol{E} - \boldsymbol{BA}\right) = \boldsymbol{A}\left(\boldsymbol{E} - \boldsymbol{BA}\right)\left(\boldsymbol{E} - \boldsymbol{BA}\right)$$

Doing so the steady state input trajectory is found to be:

$$\begin{aligned} \boldsymbol{u}_\infty &= \sum_{i=0}^{\infty} - \left(\boldsymbol{E} - \boldsymbol{LP}\right)^i \boldsymbol{L}\left(\boldsymbol{d} - \boldsymbol{r}\right) \\ &= \sum_{i=0}^{\infty} -\boldsymbol{L}\left(\boldsymbol{E} - \boldsymbol{PL}\right)^i\left(\boldsymbol{d} - \boldsymbol{r}\right) \overset{\text{Neumann series}}{=} -\boldsymbol{L}\left(\boldsymbol{PL}\right)^{-1}\left(\boldsymbol{d} - \boldsymbol{r}\right) \\ &= \boldsymbol{L}\left(\boldsymbol{P}^{\dagger}\boldsymbol{PL}\right)^\ddagger\left(\boldsymbol{PLL}^\dagger\right)^\ddagger \end{aligned} \tag{3.27}$$

Therefore, in the state of over actuation a special stability criterion is used in the case of the $\boldsymbol{Q}$ being identity.

$$\begin{aligned} \rho\left(\boldsymbol{Q}\left(\boldsymbol{E} - \boldsymbol{LP}\right)\right) < 1 \quad & \boldsymbol{Q} \neq \boldsymbol{E} \\ \rho\left(\boldsymbol{E} - \boldsymbol{PL}\right) < 1 \quad & \boldsymbol{Q} = \boldsymbol{E} \end{aligned} \tag{3.28}$$

Now the monotonicity is reviewed. The evolution of the error trajectory is governed by:

$$\boldsymbol{e}_{j+1} = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{\dagger}\boldsymbol{e}_j + \boldsymbol{PQ}\left(\boldsymbol{E}-\boldsymbol{P}^{\dagger}\boldsymbol{P}\right)\boldsymbol{u}_j + \left(\boldsymbol{E}-\boldsymbol{PQP}^{\dagger}\right)(\boldsymbol{d}-\boldsymbol{r}) \tag{3.29}$$

In contrast to the previous cases the error evolution depends on the evolution of the input trajectory (3.31). Therefore, an upper bound of the convergence rate is not found similar to the previous cases. The upper bound is found, if an upper bound for $\frac{||\boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{P}^{\dagger}\boldsymbol{P})(\boldsymbol{u}_j-\boldsymbol{u}_\infty)||}{||(\boldsymbol{e}_j-\boldsymbol{e}_\infty)||}$ is known (3.32). Nevertheless, the concept still applies in the absence of a Q-filter as the influence of the input vanishes (3.30). Furthermore, the monotonic convergence of the system of inputs and errors can be checked using the overall dynamics (3.33).

Nevertheless, the convergence criterion can be used, when the $\boldsymbol{Q}$ is chosen to be identity, as the term $\boldsymbol{PQ}\left(\boldsymbol{E}-\boldsymbol{P}^{\dagger}\boldsymbol{P}\right)$ in (3.30) vanishes. In this special case the convergence rate in the sense of Definition 15 can be upper bounded $\epsilon \leq \left|\left|\boldsymbol{P}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{\dagger}\right|\right|$.

$$\begin{aligned}
\boldsymbol{u}_{j+1}-\boldsymbol{u}_\infty &= \boldsymbol{Q}(\boldsymbol{E}-\boldsymbol{LP})\left(\boldsymbol{u}_j-\boldsymbol{u}_\infty\right)\\
\boldsymbol{e}_{j+1}-\boldsymbol{e}_\infty &= \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{\dagger}\left(\boldsymbol{e}_j-\boldsymbol{e}_\infty\right) + \boldsymbol{PQ}\left(\boldsymbol{E}-\boldsymbol{P}^{\dagger}\boldsymbol{P}\right)\left(\boldsymbol{u}_j-\boldsymbol{u}_\infty\right)
\end{aligned} \tag{3.30}$$

$$\left|\left|\boldsymbol{e}_{j+1}-\boldsymbol{e}_\infty\right|\right| \leq \left|\left|\boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{\dagger}\right|\right|\,\left|\left|(\boldsymbol{e}_j-\boldsymbol{e}_\infty)\right|\right| + \left|\left|\boldsymbol{PQ}\left(\boldsymbol{E}-\boldsymbol{P}^{\dagger}\boldsymbol{P}\right)\left(\boldsymbol{u}_j-\boldsymbol{u}_\infty\right)\right|\right| \tag{3.31}$$

$$\left|\left|\boldsymbol{e}_{j+1}-\boldsymbol{e}_\infty\right|\right| \leq \left(\left|\left|\boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{\dagger}\right|\right| + \frac{\left|\left|\boldsymbol{PQ}\left(\boldsymbol{E}-\boldsymbol{P}^{\dagger}\boldsymbol{P}\right)\left(\boldsymbol{u}_j-\boldsymbol{u}_\infty\right)\right|\right|}{\left|\left|(\boldsymbol{e}_j-\boldsymbol{e}_\infty)\right|\right|}\right)\left|\left|(\boldsymbol{e}_j-\boldsymbol{e}_\infty)\right|\right| \tag{3.32}$$

$$\left|\left|\begin{matrix}\boldsymbol{u}_{j+1}-\boldsymbol{u}_\infty\\ \boldsymbol{e}_{j+1}-\boldsymbol{e}_\infty\end{matrix}\right|\right| \leq \underbrace{\left|\left|\begin{matrix}\boldsymbol{Q}(\boldsymbol{E}-\boldsymbol{LP}) & \boldsymbol{0}\\ \boldsymbol{PQ}\left(\boldsymbol{E}-\boldsymbol{P}^{\dagger}\boldsymbol{P}\right) & \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{PL})\boldsymbol{P}^{\dagger}\end{matrix}\right|\right|}_{\geq\epsilon_{\mathrm{a}}}\left|\left|\begin{matrix}\boldsymbol{u}_j-\boldsymbol{u}_\infty\\ \boldsymbol{e}_j-\boldsymbol{e}_\infty\end{matrix}\right|\right| \tag{3.33}$$

Definition 19 (Augmented monothonic convergence)**.** *The over actuated ILC standard control loop is augmented monotonically convergent under a norm* $||\cdot||$ *if*

$$\left|\left|\begin{matrix}\boldsymbol{u}_{j+1}-\boldsymbol{u}_\infty\\ \boldsymbol{e}_{j+1}-\boldsymbol{e}_\infty\end{matrix}\right|\right| \leq \epsilon_{\mathrm{a}}\left|\left|\begin{matrix}\boldsymbol{u}_j-\boldsymbol{u}_\infty\\ \boldsymbol{e}_j-\boldsymbol{e}_\infty\end{matrix}\right|\right| \tag{3.34}$$

where $0 \leq \epsilon_{\mathrm{a}} \leq 1$ *is the convergence rate.*

The steady state input trajectory does not change apart the image space of the learnrule if the Q-filter is identity, otherwise the result coincides with those found previously.

$$\boldsymbol{u}_\infty = \begin{cases} -(\boldsymbol{E}-\boldsymbol{Q}(\boldsymbol{E}-\boldsymbol{LP}))^{-1}\boldsymbol{QL}(\boldsymbol{d}-\boldsymbol{r}) & \boldsymbol{Q}\neq\boldsymbol{E}\\ \left(\boldsymbol{E}-\boldsymbol{LL}^{\dagger}\right)\boldsymbol{u}_0-\boldsymbol{L}(\boldsymbol{PL})^{-1}(\boldsymbol{d}-\boldsymbol{r}) & \boldsymbol{Q}=\boldsymbol{E}\end{cases} \tag{3.35}$$

Lastly it is observed that the steady state error vanishes as the steady Q-filter is chosen to be identity. But in contrast to the previous cases the absence of a Q-filter is not necessary for a vanishing steady state error as it is shown in the following.

$$\boldsymbol{e}_\infty = \begin{cases} \left(\boldsymbol{E}-\boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{\dagger}\right)^{-1}\left(\boldsymbol{PQ}\left(\boldsymbol{E}-\boldsymbol{P}^{\dagger}\boldsymbol{P}\right)\boldsymbol{u}_\infty+\left(\boldsymbol{E}-\boldsymbol{PQP}^{\dagger}\right)(\boldsymbol{d}-\boldsymbol{r})\right) & \boldsymbol{Q}\neq\boldsymbol{E}\\ \boldsymbol{0} & \boldsymbol{Q}=\boldsymbol{E}\end{cases} \tag{3.36}$$

Convergence of overactuated ILC systems in presence of a Q-filter

If the Q-filter is diagonal and has a spectral radius of one, the input can be decomposed in a part that is not damped by the filter and one part that is undamped. Therefore, the undamped part might lead to no tracking error.

This can be shown. So, consider the orthonormal transformation $\boldsymbol{T}\boldsymbol{u} = [\bar{\boldsymbol{u}} \quad \tilde{\boldsymbol{u}}]^{\mathrm{T}} = [\bar{\boldsymbol{T}} \quad \tilde{\boldsymbol{T}}]^{\mathrm{T}}\boldsymbol{u}$ so that $\boldsymbol{T}\boldsymbol{Q}\boldsymbol{T}^{\mathrm{T}} = \mathrm{Diag}\left[\boldsymbol{E}, \tilde{\boldsymbol{Q}}\right]$ with $\rho\left(\tilde{\boldsymbol{Q}}\right) < 1$. Then the overall ILC system can be written as:

$$\bar{\boldsymbol{u}}_{j+1} = \bar{\boldsymbol{u}}_j - \bar{\boldsymbol{T}}\boldsymbol{L}\boldsymbol{e}_j \tag{3.37}$$

$$\tilde{\boldsymbol{u}}_{j+1} = \tilde{\boldsymbol{Q}}\left(\tilde{\boldsymbol{u}}_j - \tilde{\boldsymbol{T}}\boldsymbol{L}\boldsymbol{e}_j\right) \tag{3.38}$$

$$\boldsymbol{y}_{j+1} = \boldsymbol{P}\left(\bar{\boldsymbol{T}}^{\mathrm{T}}\bar{\boldsymbol{u}}_{j+1} + \tilde{\boldsymbol{T}}^{\mathrm{T}}\tilde{\boldsymbol{u}}_{j+1}\right) + \boldsymbol{d} \tag{3.39}$$

$$\boldsymbol{e}_{j+1} = \boldsymbol{y}_{j+1} - \boldsymbol{r} \tag{3.40}$$

Here $\boldsymbol{e}_{\mathrm{f}} = \boldsymbol{0}, \bar{\boldsymbol{u}}_{\mathrm{f}} = \left(\boldsymbol{P}\bar{\boldsymbol{T}}^{\mathrm{T}}\right)^{+}(\boldsymbol{d} - \boldsymbol{r}), \tilde{\boldsymbol{u}}_{\mathrm{f}} = \boldsymbol{0}$ is a fixed point. If $\mathrm{rank}\left[\bar{\boldsymbol{T}}\boldsymbol{L}\right] = \dim \boldsymbol{e}$ holds, the vanishing error is assured as the necessary condition for stationarity $\boldsymbol{0} = \bar{\boldsymbol{T}}\boldsymbol{L}\boldsymbol{e}_{\infty}$ (3.37) is only satisfied if $\boldsymbol{e}_{\infty} = \boldsymbol{0}$. This also implies that $\tilde{\boldsymbol{u}}_{\infty}$ vanishes (3.38). Therefore, the condition on (3.36) shall be modified.

This concludes the presentation of the analysis methods. The criteria are summarized in Table 3.1.

Actuation	Stability criterion
Unique	$\rho\left(\boldsymbol{Q}\left(\boldsymbol{E}-\boldsymbol{L}\boldsymbol{P}\right)\right) < 1$
Over	$\rho\left(\boldsymbol{Q}\left(\boldsymbol{E}-\boldsymbol{L}\boldsymbol{P}\right)\right) < 1 \quad \boldsymbol{Q} \neq \boldsymbol{E}$ $\rho\left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{L}\right) < 1 \quad \boldsymbol{Q} = \boldsymbol{E}$
Under	$\rho\left(\boldsymbol{Q}\left(\boldsymbol{E}-\boldsymbol{L}\boldsymbol{P}\right)\right) < 1$
	convergence rate ϵ **Def. 15**
Unique	$\epsilon \leq \left\lVert \boldsymbol{P}\boldsymbol{Q}\left(\boldsymbol{E}-\boldsymbol{L}\boldsymbol{P}\right)\boldsymbol{P}^{-1} \right\rVert$
Over	$\epsilon \leq \left\lVert \boldsymbol{P}\boldsymbol{Q}\left(\boldsymbol{E}-\boldsymbol{L}\boldsymbol{P}\right)\boldsymbol{P}^{\ddagger} \right\rVert + \frac{\left\lVert \boldsymbol{P}\boldsymbol{Q}\left(\boldsymbol{E}-\boldsymbol{P}^{\ddagger}\boldsymbol{P}\right)\left(\boldsymbol{u}_j-\boldsymbol{u}_\infty\right)\right\rVert}{\left\lVert\left(\boldsymbol{e}_j-\boldsymbol{e}_\infty\right)\right\rVert} \quad \boldsymbol{Q} \neq \boldsymbol{E}$ $\epsilon_{\mathrm{a}} \leq \left\lVert \begin{matrix} \boldsymbol{Q}\left(\boldsymbol{E}-\boldsymbol{L}\boldsymbol{P}\right) & \boldsymbol{0} \\ \boldsymbol{P}\boldsymbol{Q}\left(\boldsymbol{E}-\boldsymbol{P}^{\ddagger}\boldsymbol{P}\right) & \boldsymbol{P}\boldsymbol{Q}\left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{L}\right)\boldsymbol{P}^{\ddagger} \end{matrix} \right\rVert$ **Def. 19** $\quad \boldsymbol{Q} \neq \boldsymbol{E}$ $\epsilon \leq \left\lVert \boldsymbol{P}\boldsymbol{Q}\left(\boldsymbol{E}-\boldsymbol{L}\boldsymbol{P}\right)\boldsymbol{P}^{\ddagger} \right\rVert = \lVert \boldsymbol{E}-\boldsymbol{P}\boldsymbol{L} \rVert \quad \boldsymbol{Q} = \boldsymbol{E}$
Under	$\epsilon \leq \left\lVert \boldsymbol{P}\boldsymbol{Q}\left(\boldsymbol{E}-\boldsymbol{L}\boldsymbol{P}\right)\boldsymbol{P}^{\dagger} \right\rVert$
	Steady state error $\boldsymbol{e}_\infty$
Unique	$\boldsymbol{P}\left(\boldsymbol{E}-\boldsymbol{Q}\left(\boldsymbol{E}-\boldsymbol{L}\boldsymbol{P}\right)\right)^{-1}\left(\boldsymbol{E}-\boldsymbol{Q}\right)\boldsymbol{P}^{-1}\left(\boldsymbol{d}-\boldsymbol{r}\right) \quad \boldsymbol{Q} \neq \boldsymbol{E}$ $\boldsymbol{0} \quad \boldsymbol{Q} = \boldsymbol{E}$
Over	$\left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{Q}\left(\boldsymbol{E}-\boldsymbol{L}\boldsymbol{P}\right)\boldsymbol{P}^{\ddagger}\right)^{-1}\left(\boldsymbol{P}\boldsymbol{Q}\left(\boldsymbol{E}-\boldsymbol{P}^{\ddagger}\boldsymbol{P}\right)\boldsymbol{u}_\infty + \left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{Q}\boldsymbol{P}^{\ddagger}\right)\left(\boldsymbol{d}-\boldsymbol{r}\right)\right) \quad \mathrm{rank}\left[\bar{\boldsymbol{T}}\boldsymbol{L}\right] < \dim \boldsymbol{y}$ $\boldsymbol{0} \quad \mathrm{rank}\left[\bar{\boldsymbol{T}}\boldsymbol{L}\right] = \dim \boldsymbol{y}$
Under	$\boldsymbol{P}\left(\boldsymbol{E}-\boldsymbol{Q}\left(\boldsymbol{E}-\boldsymbol{L}\boldsymbol{P}\right)\right)^{-1}\left(\left(\boldsymbol{E}-\boldsymbol{Q}\right)\boldsymbol{P}^{\dagger}-\boldsymbol{Q}\boldsymbol{L}\left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{P}^{\dagger}\right)\right)\left(\boldsymbol{d}-\boldsymbol{r}\right) \quad \boldsymbol{Q} \neq \boldsymbol{E}$ $\left(\boldsymbol{E}-\boldsymbol{P}\left(\boldsymbol{L}\boldsymbol{P}\right)^{-1}\boldsymbol{L}\right)\left(\boldsymbol{d}-\boldsymbol{r}\right) \quad \boldsymbol{Q} = \boldsymbol{E}$
	Steady state input $\boldsymbol{u}_\infty$
Unique	$-\left(\boldsymbol{E}-\boldsymbol{Q}\left(\boldsymbol{E}-\boldsymbol{L}\boldsymbol{P}\right)\right)^{-1}\boldsymbol{Q}\boldsymbol{L}\left(\boldsymbol{d}-\boldsymbol{r}\right) \quad \boldsymbol{Q} \neq \boldsymbol{E}$ $-\boldsymbol{P}^{-1}\left(\boldsymbol{d}-\boldsymbol{r}\right) \quad \boldsymbol{Q} = \boldsymbol{E}$
Over	$-\left(\boldsymbol{E}-\boldsymbol{Q}\left(\boldsymbol{E}-\boldsymbol{L}\boldsymbol{P}\right)\right)^{-1}\boldsymbol{Q}\boldsymbol{L}\left(\boldsymbol{d}-\boldsymbol{r}\right) \quad \boldsymbol{Q} \neq \boldsymbol{E}$ $-\boldsymbol{L}\left(\boldsymbol{P}\boldsymbol{L}\right)^{-1}\left(\boldsymbol{d}-\boldsymbol{r}\right) \quad \boldsymbol{Q} = \boldsymbol{E}$
Under	$-\left(\boldsymbol{E}-\boldsymbol{Q}\left(\boldsymbol{E}-\boldsymbol{L}\boldsymbol{P}\right)\right)^{-1}\boldsymbol{Q}\boldsymbol{L}\left(\boldsymbol{d}-\boldsymbol{r}\right) \quad \boldsymbol{Q} \neq \boldsymbol{E}$ $-\left(\boldsymbol{L}\boldsymbol{P}\right)^{-1}\boldsymbol{L}\left(\boldsymbol{d}-\boldsymbol{r}\right) \quad \boldsymbol{Q} = \boldsymbol{E}$

Table 3.1. Summary of the analysis quantities slit up by the state of actuation.

3.3 Norm Optimal Iterative Learning Control (NO-ILC)

The norm optimal ILC approach is a design technique that uses optimization to compute the update of the ILC input trajectory [23, pp. 233]. In the unconstrained case, the optimal solution can be computed explicitly, leading to a linear ILC update (3.9). This observation is, on the one hand side useful to generate a save convergence, on the other maintains the applicability of the analysis tools discussed in Section 3.2. This section starts with the formulation of the problem and the resulting linear ILC update. Using this the properties of the algorithm are analyzed. Then it is presented how the cost function is represented as a quadratic program. Finally, it is shown how to integrate equality and inequality constraints based on the one-step-ahead predictor.

A comon seleciton is the usage of a weighted norm $\left\lVert \boldsymbol{y} \right\rVert_{\boldsymbol{Q}_{\mathrm{W}}} = \boldsymbol{y}^{\mathrm{T}}\boldsymbol{Q}_{\mathrm{W}}\boldsymbol{y}$ [2].

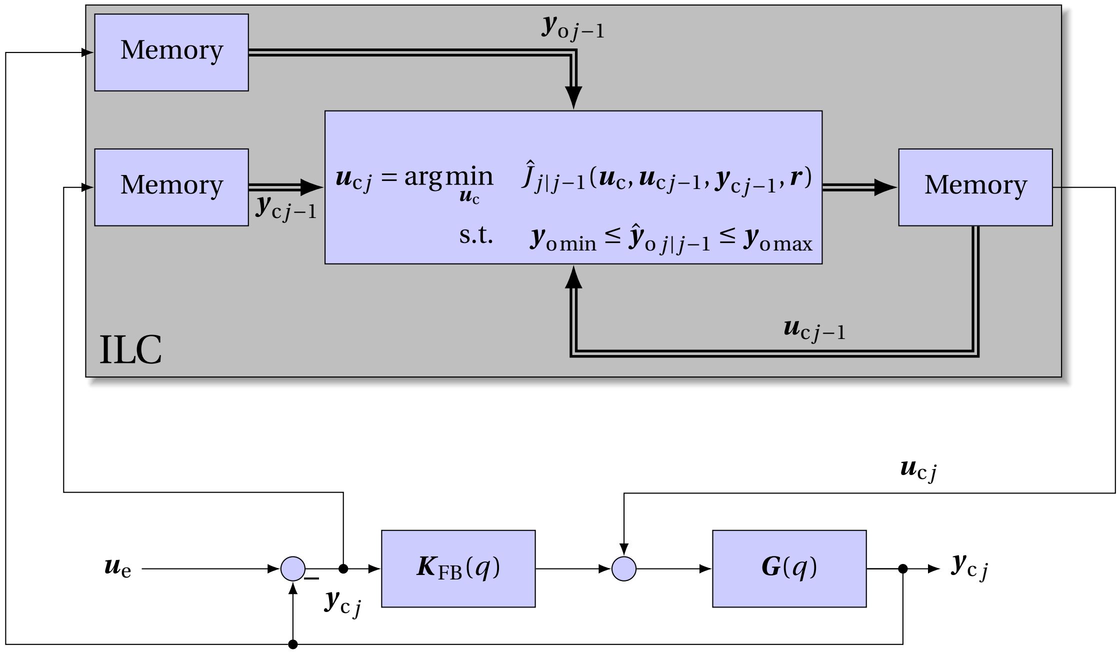


Figure 3-4. Concept of Norm optimal ILC.

$$J_j(\boldsymbol{u}_j, \boldsymbol{u}_{j-1}, \boldsymbol{y}_j, \boldsymbol{r}) = \left\|\boldsymbol{y}_j - \boldsymbol{r}\right\|_{\boldsymbol{Q}_{\mathrm{W}}} + \left\|\boldsymbol{u}_j - \boldsymbol{u}_{j-1}\right\|_{\boldsymbol{S}_{\mathrm{W}}} + \left\|\boldsymbol{u}_j\right\|_{\boldsymbol{R}_{\mathrm{W}}} \tag{3.41}$$

The weighting matrices are in general required to be positive definite to meat all norm properties. In this work this property relaxed requiring the matrices to be positive semidefinete.

The norm optimal ILC relies on the one step ahead predictor of the cost function to compute the optimal input.

$$\boldsymbol{y}_{j+1|j} = \boldsymbol{y}_j + \boldsymbol{P}\left(\boldsymbol{u}_{j+1} - \boldsymbol{u}_j\right) \tag{3.42}$$

Inserting the one step ahead predictor into the cost yields predicted cost that is optimized in the algorithm.

$$J_{j+1|j}(\boldsymbol{u}_{j+1}, \boldsymbol{u}_j, \boldsymbol{y}_j, \boldsymbol{r}) = \left\|\boldsymbol{y}_j + \boldsymbol{P}\left(\boldsymbol{u}_{j+1} - \boldsymbol{u}_j\right) - \boldsymbol{r}\right\|_{\boldsymbol{Q}_{\mathrm{W}}} + \left\|\boldsymbol{u}_{j+1} - \boldsymbol{u}_j\right\|_{\boldsymbol{S}_{\mathrm{W}}} + \left\|\boldsymbol{u}_{j+1}\right\|_{\boldsymbol{R}_{\mathrm{W}}} \tag{3.43}$$

If the quadratic program (3.43) is unconstrained, it can be solved to optimality using a linear ILC update (3.9). The learn rule is then represented by the matrices: [2]

$$\begin{aligned} \boldsymbol{L} &= \left(\boldsymbol{P}^{\mathrm{T}}\boldsymbol{Q}_{\mathrm{W}}\boldsymbol{P} + \boldsymbol{S}_{\mathrm{W}}\right)^{-1}\boldsymbol{P}^{\mathrm{T}}\boldsymbol{Q}_{\mathrm{W}} \\ \boldsymbol{Q} &= \left(\boldsymbol{P}^{\mathrm{T}}\boldsymbol{Q}_{\mathrm{W}}\boldsymbol{P} + \boldsymbol{S}_{\mathrm{W}} + \boldsymbol{R}_{\mathrm{W}}\right)^{-1}\left(\boldsymbol{P}^{\mathrm{T}}\boldsymbol{Q}_{\mathrm{W}}\boldsymbol{P} + \boldsymbol{S}_{\mathrm{W}}\right) \end{aligned} \tag{3.44}$$

Two observations are commonly done regarding this design equation (3.44). First, the Q-filter becomes identity if and only if the weights on the input trajectory $\boldsymbol{R}_\mathrm{W}$ are zero. Second, the solution becomes a plant inversion if and only if the weights on the input trajectories update are chosen to be zero. Plant inversion is only possible if the system is not over-actuated. The solution provides the weighted least squares input trajectory [41, p. 29]. [4]

$$\begin{aligned}\boldsymbol{Q}|_{\boldsymbol{R}_\mathrm{W}=\boldsymbol{0}} &= \boldsymbol{E}\\ \boldsymbol{L}|_{\boldsymbol{S}_\mathrm{W}=\boldsymbol{0}} &= \left(\boldsymbol{P}^\mathrm{T}\boldsymbol{Q}_\mathrm{W}\boldsymbol{P}\right)^{-1}\boldsymbol{P}^\mathrm{T}\boldsymbol{Q}_\mathrm{W}\end{aligned} \tag{3.45}$$

For the special case of unique acutuation the learnrule is a system inversion:

$$\boldsymbol{L}|_{\boldsymbol{S}_\mathrm{W}=\boldsymbol{0}} = \boldsymbol{P}^{-1}\left(\boldsymbol{P}^\mathrm{T}\boldsymbol{Q}_\mathrm{W}\right)^{-1}\boldsymbol{P}^\mathrm{T}\boldsymbol{Q}_\mathrm{W} = \boldsymbol{P}^{-1} \tag{3.46}$$

An advantage of NO-ILC is that the algorithm can incorporate constraint. To apply the algorithm the cost function need to be expressed in a way that allows the application of optimizer. This is discussed in the following.

In the light of constraints within the Iterative Learning Control (ILC) problem, a linear ILC update is no longer available. The problem can than be solved using a quadratic program solver. Then the one step ahead predictor cost function (3.43) is reformulated as it is desired for the software. The standard formulation of a Quadratic Program (QP) objective is given by $J(\boldsymbol{u}) = \alpha\left(\frac{1}{2}\boldsymbol{u}^\mathrm{T}\boldsymbol{H}\boldsymbol{u} + \boldsymbol{f}^\mathrm{T}\boldsymbol{u}\right) + \beta$ with arbitrary $\alpha \in \mathbb{R}^+$ and $\beta \in \mathbb{R}$ [42]. In the following the matrices are derived on the basis of the cost function and the one step ahead predictor.

$$\begin{aligned}J\left(\boldsymbol{u}_{j+1}\right) = \hat{J}_{j+1|j}\left(\boldsymbol{u}_{j+1}, \boldsymbol{u}_j, \boldsymbol{y}_j, \boldsymbol{r}\right) &= \left|\left|\boldsymbol{y}_j + \boldsymbol{P}\cdot\left(\boldsymbol{u}_{j+1} - \boldsymbol{u}_j\right) - \boldsymbol{r}\right|\right|_{\boldsymbol{Q}_\mathrm{W}} + \left|\left|\boldsymbol{u}_{j+1} - \boldsymbol{u}_j\right|\right|_{\boldsymbol{S}_\mathrm{W}} + \left|\left|\boldsymbol{u}_{j+1}\right|\right|_{\boldsymbol{R}_\mathrm{W}}\\ &= \left(\boldsymbol{y}_j + \boldsymbol{P}\cdot\left(\boldsymbol{u}_{j+1} - \boldsymbol{u}_j\right) - \boldsymbol{r}\right)^\mathrm{T}\boldsymbol{Q}_\mathrm{W}\left(\boldsymbol{y}_j + \boldsymbol{P}\cdot\left(\boldsymbol{u}_{j+1} - \boldsymbol{u}_j\right) - \boldsymbol{r}\right)\\ &\quad + \left(\boldsymbol{u}_{j+1} - \boldsymbol{u}_j\right)^\mathrm{T}\boldsymbol{S}_\mathrm{W}\left(\boldsymbol{u}_{j+1} - \boldsymbol{u}_j\right) + \boldsymbol{u}_{j+1}^\mathrm{T}\boldsymbol{R}_\mathrm{W}\boldsymbol{u}_{j+1}\\ &= \boldsymbol{u}_{j+1}^\mathrm{T}\left(\boldsymbol{P}^\mathrm{T}\boldsymbol{Q}_\mathrm{W}\boldsymbol{P} + \boldsymbol{S}_\mathrm{W} + \boldsymbol{R}_\mathrm{W}\right)\boldsymbol{u}_{j+1}\\ &\quad + 2\cdot\boldsymbol{u}_{j+1}^\mathrm{T}\left(\boldsymbol{P}^\mathrm{T}\boldsymbol{Q}_\mathrm{W}\left(\boldsymbol{y}_j - \boldsymbol{r} - \boldsymbol{P}\boldsymbol{u}_j\right) - \boldsymbol{S}_\mathrm{W}\boldsymbol{u}_j\right)\\ &\quad + \left(\boldsymbol{y}_j - \boldsymbol{P}\boldsymbol{u}_j - \boldsymbol{r}\right)^\mathrm{T}\boldsymbol{Q}_\mathrm{W}\left(\boldsymbol{y}_j - \boldsymbol{P}\boldsymbol{u}_j - \boldsymbol{r}\right) + \boldsymbol{u}_j^\mathrm{T}\boldsymbol{S}_\mathrm{W}\boldsymbol{u}_j\\ \boldsymbol{H} = \boldsymbol{P}^\mathrm{T}\boldsymbol{Q}_\mathrm{W}\boldsymbol{P} + \boldsymbol{S}_\mathrm{W} + \boldsymbol{R}_\mathrm{W} \qquad & \boldsymbol{f} = \boldsymbol{P}^\mathrm{T}\boldsymbol{Q}_\mathrm{W}\left(\boldsymbol{y}_j - \boldsymbol{r} - \boldsymbol{P}\boldsymbol{u}_j\right) - \boldsymbol{S}_\mathrm{W}\boldsymbol{u}_j\end{aligned} \tag{3.47}$$

An alternative formulation only considers the update on the input.

$$
\begin{aligned}
J(\Delta \boldsymbol{u}) &= \hat{J}_{j+1|j}(\Delta \boldsymbol{u} + \boldsymbol{u}_j, \boldsymbol{u}_j, \boldsymbol{y}_j, \boldsymbol{r}) \\
&= \left|\left|\boldsymbol{y}_j + \boldsymbol{P} \cdot \Delta \boldsymbol{u} - \boldsymbol{r}\right|\right|_{\boldsymbol{Q}_\mathrm{W}} + ||\Delta \boldsymbol{u}||_{\boldsymbol{S}_\mathrm{W}} + \left|\left|\Delta \boldsymbol{u} + \boldsymbol{u}_j\right|\right|_{\boldsymbol{R}_\mathrm{W}} \\
&= \left(\boldsymbol{y}_j + \boldsymbol{P} \cdot \Delta \boldsymbol{u} - \boldsymbol{r}\right)^\mathrm{T} \boldsymbol{Q}_\mathrm{W} \left(\boldsymbol{y}_j + \boldsymbol{P} \cdot \Delta \boldsymbol{u} - \boldsymbol{r}\right) + \Delta \boldsymbol{u}^\mathrm{T} \boldsymbol{S}_\mathrm{W} \Delta \boldsymbol{u} + \left(\Delta \boldsymbol{u} + \boldsymbol{u}_j\right)^\mathrm{T} \boldsymbol{R}_\mathrm{W} (\Delta \boldsymbol{u} + \boldsymbol{u}) \\
&= \Delta \boldsymbol{u}^\mathrm{T} \left(\boldsymbol{P}^\mathrm{T} \boldsymbol{Q}_\mathrm{W} \boldsymbol{P} + \boldsymbol{S}_\mathrm{W} + \boldsymbol{R}_\mathrm{W}\right) \Delta \boldsymbol{u} \\
&\quad + 2 \cdot \Delta \boldsymbol{u}^\mathrm{T} \left(\boldsymbol{P}^\mathrm{T} \boldsymbol{Q}_\mathrm{W} \left(\boldsymbol{y}_j - \boldsymbol{r}\right) + \boldsymbol{R}_\mathrm{W} \boldsymbol{u}_j\right) + \left(\boldsymbol{y}_j - \boldsymbol{r}\right)^\mathrm{T} \boldsymbol{Q}_\mathrm{W} \left(\boldsymbol{y}_j - \boldsymbol{r}\right) + \boldsymbol{u}_j^\mathrm{T} \boldsymbol{R}_\mathrm{W} \boldsymbol{u}_j \\
\boldsymbol{H} &= \boldsymbol{P}^\mathrm{T} \boldsymbol{Q}_\mathrm{W} \boldsymbol{P} + \boldsymbol{S}_\mathrm{W} + \boldsymbol{R}_\mathrm{W} \qquad \boldsymbol{f} = \boldsymbol{P}^\mathrm{T} \boldsymbol{Q}_\mathrm{W} \left(\boldsymbol{y}_j - \boldsymbol{r}\right) + \boldsymbol{R}_\mathrm{W} \boldsymbol{u}_j
\end{aligned}
\tag{3.48}
$$

In the latter it is also necessary to bind the input to a certain subspace of $\boldsymbol{u}$, i. e.one way to formulate an equality constraint. This is done by the introduction of a lower dimensional vector $\boldsymbol{u}_\mathrm{anc}$ and a basis $\boldsymbol{F}$ forming the subspace. Doing so the optimization problem is equivalent with transformed matrices as it is shown is (3.49).

$$
\begin{aligned}
J\left(\boldsymbol{z}_{j+1}\right) &= \hat{J}_{j+1|j}\left(\boldsymbol{F}\boldsymbol{z}_{j+1}, \boldsymbol{F}\boldsymbol{z}_j, \boldsymbol{y}_j, \boldsymbol{r}\right) = \left|\left|\boldsymbol{y}_j + \boldsymbol{P}\boldsymbol{F} \cdot \left(\boldsymbol{z}_{j+1} - \boldsymbol{z}_j\right) - \boldsymbol{r}\right|\right|_{\boldsymbol{Q}_\mathrm{W}} + \left|\left|\boldsymbol{F}\left(\boldsymbol{z}_{j+1} - \boldsymbol{z}_j\right)\right|\right|_{\boldsymbol{S}_\mathrm{W}} + \left|\left|\boldsymbol{F}\boldsymbol{z}_{j+1}\right|\right|_{\boldsymbol{R}_\mathrm{W}} \\
&= \left|\left|\boldsymbol{y}_j + \underbrace{\boldsymbol{P}\boldsymbol{F}}_{\tilde{\boldsymbol{P}}} \cdot \left(\boldsymbol{z}_{j+1} - \boldsymbol{z}_j\right) - \boldsymbol{r}\right|\right|_{\boldsymbol{Q}_\mathrm{W}} + \left|\left|\boldsymbol{z}_{j+1} - \boldsymbol{z}_j\right|\right|_{\boldsymbol{F}^\mathrm{T}\boldsymbol{S}_\mathrm{W}\boldsymbol{F} = \tilde{\boldsymbol{S}_\mathrm{W}}} + \left|\left|\boldsymbol{z}_{j+1}\right|\right|_{\boldsymbol{F}^\mathrm{T}\boldsymbol{R}_\mathrm{W}\boldsymbol{F} = \tilde{\boldsymbol{R}_\mathrm{W}}}
\end{aligned}
\tag{3.49}
$$

This works similar for the update formulation.

$$
\begin{aligned}
J(\Delta \boldsymbol{z}) &= \hat{J}_{j+1|j}(\Delta \boldsymbol{z} + \boldsymbol{z}_j, \boldsymbol{z}_j, \boldsymbol{y}_j, \boldsymbol{r}) \\
&= \left|\left|\boldsymbol{y}_j + \boldsymbol{P}\boldsymbol{F}\Delta \boldsymbol{z} - \boldsymbol{r}\right|\right|_{\boldsymbol{Q}_\mathrm{W}} + ||\boldsymbol{F}\Delta \boldsymbol{z}||_{\boldsymbol{S}_\mathrm{W}} + \left|\left|\boldsymbol{F}\Delta \boldsymbol{z} + \boldsymbol{u}_j\right|\right|_{\boldsymbol{R}_\mathrm{W}} \\
&= \left|\left|\boldsymbol{y}_j + \underbrace{\boldsymbol{P}\boldsymbol{F}}_{\tilde{\boldsymbol{P}}} \Delta \boldsymbol{z} - \boldsymbol{r}\right|\right|_{\boldsymbol{Q}_\mathrm{W}} + ||\boldsymbol{F}\Delta \boldsymbol{z}||_{\boldsymbol{F}^\mathrm{T}\boldsymbol{S}_\mathrm{W}\boldsymbol{F} = \tilde{\boldsymbol{S}_\mathrm{W}}} + \left|\left|\Delta \boldsymbol{z} + \boldsymbol{z}_j\right|\right|_{\boldsymbol{F}^\mathrm{T}\boldsymbol{R}_\mathrm{W}\boldsymbol{F} = \tilde{\boldsymbol{R}_\mathrm{W}}}
\end{aligned}
\tag{3.50}
$$

Lastly, the incorporation of constraints is discussed. Therefore, suppose a constraint on the observed output of the form

$$
\boldsymbol{y}_{\mathrm{o,min}} \leq \boldsymbol{y}_\mathrm{o} \leq \boldsymbol{y}_{\mathrm{o,max}}. \tag{3.51}
$$

Those are included in the optimization problem using the one step ahead predictor (3.42).

$$\begin{aligned}
\boldsymbol{y}_{\text{o,min}} &\leq \boldsymbol{y}_{\text{o},j+1|j} \leq \boldsymbol{y}_{\text{o,max}} \\
\boldsymbol{y}_{\text{o,min}} &\leq \boldsymbol{y}_{\text{o},j} + \boldsymbol{P}_{\text{co}} \left(\boldsymbol{u}_{j+1} - \boldsymbol{u}_j\right) \leq \boldsymbol{y}_{\text{o,max}} \\
\boldsymbol{y}_{\text{o,min}} &\leq \boldsymbol{y}_{\text{o},j} + \boldsymbol{P}_{\text{co}} \Delta \boldsymbol{u} \leq \boldsymbol{y}_{\text{o,max}}
\end{aligned}$$

The constraints can be reformulated to the standard form suitable for a quadratic program when optimizing the input

$$\begin{aligned}
\Lambda \boldsymbol{u}_{j+1} &\leq b \\
\begin{bmatrix} \boldsymbol{P}_{\text{co}} \\ -\boldsymbol{P}_{\text{co}} \end{bmatrix} \boldsymbol{u}_j &\leq \begin{bmatrix} \boldsymbol{y}_{\text{o,max}} - \boldsymbol{y}_{\text{o},j} + \boldsymbol{P}_{\text{co}} \boldsymbol{u}_j \\ -\boldsymbol{y}_{\text{o,min}} + \boldsymbol{y}_{\text{o},j} - \boldsymbol{P}_{\text{co}} \boldsymbol{u}_j \end{bmatrix}
\end{aligned} \tag{3.52}$$

Or when optimizing the update of the input

$$\begin{aligned}
\Lambda \Delta \boldsymbol{u} &\leq b \\
\begin{bmatrix} \boldsymbol{P}_{\text{co}} \\ -\boldsymbol{P}_{\text{co}} \end{bmatrix} \Delta \boldsymbol{u} &\leq \begin{bmatrix} \boldsymbol{y}_{\text{o,max}} - \boldsymbol{y}_{\text{o},j} \\ -\boldsymbol{y}_{\text{o,min}} + \boldsymbol{y}_{\text{o},j} \end{bmatrix}.
\end{aligned} \tag{3.53}$$

Properties of the solution

This section deals with the question how the remaining Degrees of freedom (DoF) evolve in under actuated systems using the NO-ILC design strategy, as in this case the optimizer $\boldsymbol{u}_\infty$ of the problem is not unique.

A useful result present in literature is found by Owens [23, p.250, Theorem 9.3] who reviewed system without input weights and has proven that the optimizer solves the optimization problem

$$\begin{aligned}
\min_{\boldsymbol{u}_\infty} \quad & ||\boldsymbol{u}_\infty - \boldsymbol{u}_0||_{\boldsymbol{S}_\text{W}} \\
\text{s.t.} \quad & \boldsymbol{e}_\infty = 0
\end{aligned}$$

under the assumption of perfect plant knowledge.

An natural extension to the case in presence of input weights is

$$\begin{aligned}
\min_{\boldsymbol{u}_\infty} \quad & ||\boldsymbol{u}_\infty - \boldsymbol{u}_0||_{\boldsymbol{S}_\text{W}} \\
\text{s.t.} \, & J(\boldsymbol{u}_\infty, \boldsymbol{u}_\infty, \boldsymbol{y}, \boldsymbol{r}) = \min_{\boldsymbol{u}} J(\boldsymbol{u}, \boldsymbol{u}, \boldsymbol{y}, \boldsymbol{r})
\end{aligned}$$

Design requirements for NO-ILC

The design of a NO-ILC requires the following specifications:

- A cost on the output lifted dynamics vector $\boldsymbol{Q}_\text{W}$
- A cost on the input update $\boldsymbol{S}_\text{W}$
- A cost on the final input $\boldsymbol{R}_\text{W}$
- Specification of constraints on the observed outputs $\boldsymbol{y}_{\text{o,max}}$ and $\boldsymbol{y}_{\text{o,min}}$ (optional).

3.4 Transition to the Point-to-Point Problem

To relax the reference tracking assumption, as precise tracking of the whole trajectory might not be the goal of the control task, but some specialized points need to be tracked, Point-to-Point Iterative Learning Control (P2P-ILC) is introduced. In the example of the TWIPR it is desirable to reach the goal state and maintain clearance to obstacles but the intermediate speed is of minor interest. Therefore, the lifted dynamics are augmented by linear mapping, the selection matrix $\boldsymbol{\Phi}$, that selects the points of interest in the output trajectory. Thus, the system output dimension is reduced, and the state of actuation might be changed, e. g.the formally under-actuated TWIPR that can not track the output trajectory composed of pitch and position exactly can track some of the waypoints exactly as it becomes over actuated applying the selection. In literature, the point-to-point problem is introduced in the teaching book of Owens as an intermediate point problem [23, pp. 286] and also in fellow publications as [3]

To accomplish the task, the point-to-point reference is defined. To be general, besides reference values $\boldsymbol{r}_i$ and times k_i a linear mapping $\boldsymbol{\Phi}_i$ is also introduced. This allows the designer to choose special values of the output e. g.in a task where at one point, the velocity of traversal is important while it is not of interest in another one.

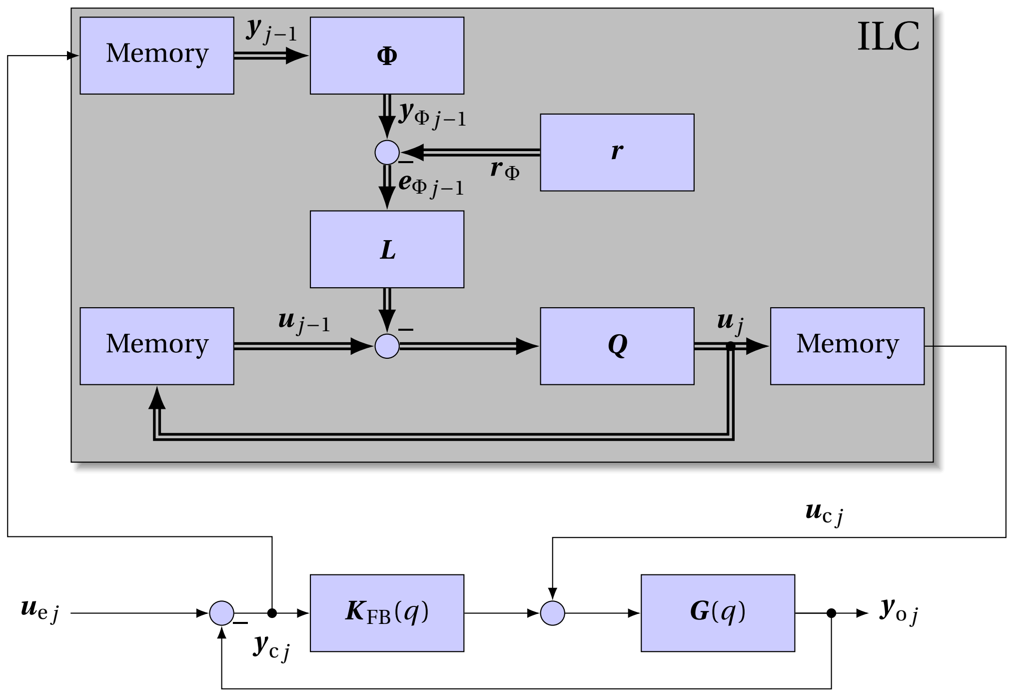


Figure 3-5. Illustration of the Point to Point controller.

Definition 20 (Point-to-Point reference and selector.). *The reference for a Point-to-Point Iterative Learning Control (ILC) is given by a sequence of n_{Φ} tuples $(k_i, \boldsymbol{r}_i, \boldsymbol{\Phi}_i) \in (\mathcal{I}, \mathbb{R}^{p_i}, \mathbb{R}^{p_i \times p})$, with* rank$[\boldsymbol{\Phi}_i] = p_i$. *Such that the ILC system is required to reach the following tracking behaviour.*

$$\boldsymbol{r}_i = \boldsymbol{\Phi}_i y(k_i) \qquad \forall i \tag{3.54}$$

This reformulation leads to a modified lifted dynamics representation.

$$\boldsymbol{r}_{\Phi} = \begin{bmatrix} \boldsymbol{r}_1 & \dots & \boldsymbol{r}_{n_\Phi} \end{bmatrix}^{\mathrm{T}} \tag{3.55}$$

$$\boldsymbol{\Phi} = \begin{array}{c} \begin{array}{cccccccccc} 0 & \dots & k_1 & \dots & k_i & \dots & k_{n_\Phi} & \dots & n_{\mathrm{b}} & \end{array} \\ \left[\begin{array}{ccccccccc} \mathbf{0}_{p_1\times p} & \dots & \boldsymbol{\Phi}_1 & \dots & \mathbf{0}_{p_1\times p} & \dots & \mathbf{0}_{p_1\times p} & \dots & \mathbf{0}_{p_1\times p} \\ \vdots & & \vdots & \ddots & \vdots & & \vdots & & \vdots \\ \mathbf{0}_{p_i\times p} & \dots & \mathbf{0}_{p_i\times p} & \dots & \boldsymbol{\Phi}_i & \dots & \mathbf{0}_{p_i\times p} & \dots & \mathbf{0}_{p_i\times p} \\ \vdots & & \vdots & & \vdots & \ddots & \vdots & & \vdots \\ \mathbf{0}_{p_{n_\Phi}\times p} & \dots & \mathbf{0}_{p_{n_\Phi}\times p} & \dots & \mathbf{0}_{p_{n_\Phi}\times p} & \dots & \boldsymbol{\Phi}_{n_\Phi} & \dots & \mathbf{0}_{p_{n_\Phi}\times p} \end{array}\right] \begin{array}{c} 1 \\ \vdots \\ i \\ \vdots \\ n_\Phi \end{array} \end{array} \tag{3.56}$$

The selection matrix $\boldsymbol{\Phi}$ has full row rank by definition, therefore it is possible that the modified dynamics $\boldsymbol{P}_\Phi$ has full row rank either. Non the less this is not necessarily true, as the selected outputs of the original system might depend on each other. A selection of linearily dependent outputs is not desirable an in such a case it is advisable to change the selection.

The output is been made conform to full output trajectory.

$$\boldsymbol{y}_\Phi = \boldsymbol{P}_\Phi \boldsymbol{u} + \boldsymbol{d}_\Phi = \boldsymbol{\Phi}\boldsymbol{P}\boldsymbol{u} + \boldsymbol{\Phi}\boldsymbol{d} \tag{3.57}$$

This notation is adopted to [23, p. 294].

This transition is no structural change to the lifted dynamics representation (Equation 3.57). Therefore, most properties from the earlier sections are maintained. A fundamental property lost is that the Matrix is no longer Block Toeplitz as $\boldsymbol{\Phi}$ is not. Furthermore, the method of PD-learning that only requires very rough model information is no longer applicable, and the Norm Optimal Iterative Learning Control (NO-ILC) or gradient-based as described in [23, pp. 165] approaches are available of whom all require a more detailed model.

Application Note 1. *Model free design of Point to Point controllers.*

This note discusses a model-free design method for point-to-point problems based on input shaping. This also enhances that the point-to-point method tends to be robust. As such a robust design technique exists. This is motivated by the model-free PD learning for the fully actuated case. For simplicity, the method is introduced for SISO systems but can be easily extended for arbitrary MIMO systems.

Therefore, recover the stability criterion in the absence of a $\boldsymbol{Q}$-Filter, i. e.$\rho\left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{L}\right) < 1$. In the case $\boldsymbol{\Phi}^{\mathrm{T}}\boldsymbol{L}$ is lower triangular, so is $\boldsymbol{E}-\boldsymbol{P}\boldsymbol{L}$. This especially holds true for the following structure (Single Input Single Output (SISO) systems):

$$
\boldsymbol{L} = \begin{bmatrix} l_{k-1-1} & 0 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ l_0 & 0 & \dots & 0 \\ 0 & l_{k_2-k_1-1} & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & l_0 & \dots & 0 \\ \vdots & \vdots & & \vdots \end{bmatrix} \tag{3.58}
$$

Then $\boldsymbol{PL}$ is lower triangular and the diagonal entries can be bounded by $\min_{k\in\{\min_i k_i-k_{i-1},\dots\}} \sum_{i=0}^{k-1} p_i l_i \le d \le \max_{k\in\{\min_i k_i-k_{i-1},\dots\}} \sum_{i=0}^{k-1} p_i l_i$. *Thus, the ILC system becomes stable if* $0 < \sum_{i=0}^{k} p_i l_i < 2$ *for all* $k \ge \min_i k_i$. *Intuitively, this leads to a design strategy, i. e.the designer can select a prototype signal* $\tilde{u}$ *of a duration* $\tilde{k}$ *limited to* $\min_i k_i$, *that is able to steer the system in the desired direction and does not overshoot, i. e.*$0 < \sum_{i=0}^{\tilde{k}} p_i \tilde{u}_{\tilde{k}-i} < 2$. *If no model information is available, this quantity can be obtained in an experiment. Then construct the learn rule so that* $l_i = \tilde{u}_{\tilde{k}-i}$. *This method could be extended to the under-actuated case where the matrices become block diagonal. One must assure* $0 < \lambda_i \left(\sum_{i=0}^{\tilde{k}} \boldsymbol{p}_{\Phi i} \tilde{\boldsymbol{u}}_{\tilde{k}-i}^{\mathrm{T}}\right) < 2$ *for all eigenvalues.*

Remark 3. *To the author this method appeared natural and improved understanding. If it was known in literature, has not been checked for the matter of time.*

$$
\boldsymbol{\Phi P} = \begin{array}{c} \begin{array}{cccccccc} \dots & k_{i-1}-1 & k_{i-1} & k_{i-1}+1 & \dots & k_i-1 & k_i & k_i+1 \quad \dots \end{array} \\ \begin{bmatrix} & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots & \\ \dots & \boldsymbol{0}_{p_{i-2}\times p} & \boldsymbol{0}_{p_{i-2}\times p} & \boldsymbol{0}_{p_{i-2}\times p} & \dots & \boldsymbol{0}_{p_{i-2}\times p} & \boldsymbol{0}_{p_{i-2}\times p} & \boldsymbol{0}_{p_{i-2}\times p} & \dots \\ \dots & \boldsymbol{\Phi}_{i-1}\boldsymbol{P}_1 & \boldsymbol{\Phi}_{i-1}\boldsymbol{P}_0 & \boldsymbol{0}_{p_{i-1}\times p} & \dots & \boldsymbol{0}_{p_{i-1}\times p} & \boldsymbol{0}_{p_{i-1}\times p} & \boldsymbol{0}_{p_{i-1}\times p} & \dots \\ \dots & \boldsymbol{\Phi}_i\boldsymbol{P}_{k_i-k_{i-1}} & \boldsymbol{\Phi}_i\boldsymbol{P}_{k_i-k_{i-1}-1} & \boldsymbol{\Phi}_i\boldsymbol{P}_{k_i-k_{i-1}-2} & \dots & \boldsymbol{\Phi}_i\boldsymbol{P}_1 & \boldsymbol{\Phi}_i\boldsymbol{P}_0 & \boldsymbol{0}_{p_i\times p} & \dots \\ & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots & \end{bmatrix} \begin{array}{c} \\ i-2 \\ i-1 \\ i \\ \\ \end{array} \end{array} \tag{3.59}
$$

It can be observed that (3.59) has a lower block diagonal structure. The size of the diagonal blocks depends on the period of two consecutive waypoints and the output dimension. The i-th block forms as follows.

$$
\boldsymbol{P}_{\Phi i} := \begin{bmatrix} \boldsymbol{\Phi}_i\boldsymbol{P}_{k_i-k_{i-1}-2} & \dots & \boldsymbol{\Phi}_i\boldsymbol{P}_1 & \boldsymbol{\Phi}_i\boldsymbol{P}_0 \end{bmatrix} \tag{3.60}
$$

As all this blocks are full row rank, so the selected lifted dynamics $\boldsymbol{P}_\Phi$ is.

This leads to a requirement for a system that has fewer inputs than outputs and the minimum clearance of waypoints. The Markov parameter matrices dimension reflects the number of in- and outputs. To observe full row-rank, the matrix must be square or broad, i.e. have at least one row per column. To achieve this property, it is required to introduce a minimum period of two consecutive samples. To assure that $\boldsymbol{P}_{\Phi i}$ is at least square, $k_i - k_{i-1} \ge \frac{p_i}{m}$ must hold for all i.

Remark 4. *Minimum clearance of waypoints in point to point problems. To assure no under actuation a minimum clearance of consecutive way points $k_i - k_{i-1} \geq \frac{p_i}{m}$ shall be assured.*

Design requirement for a point to point problem.

To specify a point to point iterative leaning control problem in the basis of a given ILC plant, a point to point reference as defined in Definition 20 need to be specified.

3.5 Ancillary cost functions

In this part, it will be explained how multiple cost functions can be considered in ILC problems. I. e. consider the case of a quantity optimized that does not require all Degrees of freedom (DoF) of the input signal. In this case, one might define an ancillary cost that shall be optimized on the solution manifold of the primary optimization problem. The group of Freeman introduced a method to stage several cost functions [3]. The main advantage of their method is that one maintains the ability to choose the design method. That is the major improvement in contrast to the results found by Owens that are restricted to the NO-ILC case [23, pp. 323]. Another improvement of the former is that the ancillary cost is optimized in a feedback loop, while the latter only modifies the operator used as an update and, therefore, only applies to feed-forward control for the ancillary problem. Furthermore, the method could be continued arbitrarily by defining another selection operator. A major drawback is that the method relies on the model information to decouple the outputs.

This section is structured as follows. The algorithm is introduced, showing how the cost functions' hierarchical behavior is generated. It is shown that the overall algorithm itself can be represented as a linear ILC update (3.9) of an over-actuated ILC plant, meaning that the analysis tools derived also apply to this algorithm. It is also shown that the Q-filter can be designed separately for the original input signal domain, therefore enabling the designer to specify the input's bandwidth. This design technique will also maintain the dimension of the ILC controller state. Furthermore, an extension is introduced that is to utilize the one-step ahead predictor in the ancillary controller. Up to this point, the argument focus on the linear Iterative Learning Control (ILC). To point out how constraints are treated, the constrained optimization problem is introduced finally.

The fundamental idea of [3] is to decouple the output behavior by the null space projection. Therefore, suppose $B^{\perp}_{\boldsymbol{P}_{\Phi}}$ forms a basis of the nullspace of $\boldsymbol{P}_{\Phi}$. Another input is introduced that does not affect the selected outputs but all the others. Thus, the primary ILC system is not affected by the ancillary. The newly introduced input $\boldsymbol{u}_{\mathrm{anc}}$ is of lower dimension, i. e.the dimension coincides with the dimension of the nullspace of $\boldsymbol{P}_{\Phi}$.

$$\boldsymbol{u} = \boldsymbol{u}_{\mathrm{p}} + B^{\perp}_{\boldsymbol{P}_{\Phi}} \boldsymbol{u}_{\mathrm{anc}} \tag{3.61}$$

Now it is analyzed how the two control loops are coupled. Therefore, suppose each input to be computed using a linear ILC update rule, the primary as $(\boldsymbol{Q}_{\mathrm{P}}, \boldsymbol{L}_{\mathrm{P}})$ and the ancillary as $(\boldsymbol{Q}_{\mathrm{anc}}, \boldsymbol{L}_{\mathrm{anc}})$, the overall system builds up as follows.

$$\boldsymbol{u}_{\mathrm{P}\,j+1} = \boldsymbol{Q}_{\mathrm{P}}\left(\boldsymbol{u}_{\mathrm{P}\,j} + \boldsymbol{L}_{\mathrm{P}}\left(\boldsymbol{\Phi}\boldsymbol{y}_j - \boldsymbol{r}_{\Phi}\right)\right) \tag{3.62}$$

$$\boldsymbol{u}_{\mathrm{anc}\,j+1} = \boldsymbol{Q}_{\mathrm{anc}}\left(\boldsymbol{u}_{\mathrm{anc}\,j} + \boldsymbol{L}_{\mathrm{anc}}\left(\boldsymbol{y}_j - \boldsymbol{r}\right)\right) \tag{3.63}$$

$$\begin{aligned} \boldsymbol{u}_{j+1} &= \boldsymbol{u}_{\mathrm{P}\,j+1} + B^{\perp}_{\boldsymbol{P}_{\Phi}}\boldsymbol{u}_{\mathrm{anc}\,j+1} \\ \boldsymbol{y}_j &= \boldsymbol{P}\boldsymbol{u}_j + \boldsymbol{d} \end{aligned}$$

That the primary ILC system is not affected by the second in the nominal case is obvious by design. Therefore, observe that the error $\boldsymbol{\Phi}\boldsymbol{y}_j - \boldsymbol{r}_{\Phi}$ in (3.62) does only depend on the primary input in the nominal case.

$$\begin{aligned} \boldsymbol{y}_{\Phi\,j} = \boldsymbol{\Phi}\boldsymbol{y}_j &= \boldsymbol{P}_{\Phi}\left(\boldsymbol{u}_{\mathrm{P}\,j} + B^{\perp}_{\boldsymbol{P}_{\Phi}}\boldsymbol{u}_{\mathrm{anc}\,j}\right) + \boldsymbol{d}_{\Phi} \\ \boldsymbol{y}_{\Phi\,j} &= \boldsymbol{P}_{\Phi}\boldsymbol{u}_{\mathrm{P}\,j} + \boldsymbol{P}_{\Phi}B^{\perp}_{\boldsymbol{P}_{\Phi}}\boldsymbol{u}_{\mathrm{anc}\,j} + \boldsymbol{d}_{\Phi} \\ \boldsymbol{y}_{\Phi\,j} &= \boldsymbol{P}_{\Phi}\boldsymbol{u}_{\mathrm{P}\,j} + \boldsymbol{d}_{\Phi} \end{aligned}$$

The ancillary is not decoupled of the primary as it is shown in the following.

$$\boldsymbol{y}_j = \boldsymbol{P}\left(\boldsymbol{u}_{\mathrm{P}\,j} + B^{\perp}_{\boldsymbol{P}_{\Phi}}\boldsymbol{u}_{\mathrm{anc}\,j}\right) + \boldsymbol{d} \tag{3.64}$$

$$\boldsymbol{y}_j = \boldsymbol{P}B^{\perp}_{\boldsymbol{P}_{\Phi}}\boldsymbol{u}_{\mathrm{anc}\,j} + \boldsymbol{P}\boldsymbol{u}_{\mathrm{P}\,j} + \boldsymbol{d} \tag{3.65}$$

$$\boldsymbol{u}_{\mathrm{anc}\,j+1} = \boldsymbol{Q}_{\mathrm{aux}}\left(\boldsymbol{u}_{\mathrm{anc}\,j} + \boldsymbol{L}_{\mathrm{anc}}\left(\boldsymbol{P}B^{\perp}_{\boldsymbol{P}_{\Phi}}\boldsymbol{u}_{\mathrm{anc}\,j} + \boldsymbol{P}\boldsymbol{u}_{\mathrm{P}\,j} + \boldsymbol{d} - \boldsymbol{r}\right)\right) \tag{3.66}$$

Thus, one observes an additional disturbance $\boldsymbol{P}\boldsymbol{u}_{\mathrm{P}\,j}$ that is time varying. Fortunately, the primary control converges to a stable trajectory and therefore becomes a constant disturbance eventually. Furthermore, the influence of an input to the output trajectory by the primary update can be estimated and the update shall be modified.

The change of the output trajectory takes the following form.

$$\begin{aligned} \Delta\boldsymbol{y}_{j+1} = \boldsymbol{y}_{j+1} - \boldsymbol{y}_j &= \boldsymbol{P}B^{\perp}_{\boldsymbol{P}_{\Phi}}\boldsymbol{u}_{\mathrm{anc}\,j+1} + \boldsymbol{P}\boldsymbol{u}_{\mathrm{P}\,j+1} + \boldsymbol{d} - \boldsymbol{P}B^{\perp}_{\boldsymbol{P}_{\Phi}}\boldsymbol{u}_{\mathrm{anc}\,j} - \boldsymbol{P}\boldsymbol{u}_{\mathrm{P}\,j} - \boldsymbol{d} \\ &= \boldsymbol{P}B^{\perp}_{\boldsymbol{P}_{\Phi}}\boldsymbol{u}_{\mathrm{anc}\,j+1} - \boldsymbol{P}B^{\perp}_{\boldsymbol{P}_{\Phi}}\boldsymbol{u}_{\mathrm{anc}\,j} + \boldsymbol{P}\boldsymbol{u}_{\mathrm{P}\,j+1} - \boldsymbol{P}\boldsymbol{u}_{\mathrm{P}\,j} \\ &= \boldsymbol{P}B^{\perp}_{\boldsymbol{P}_{\Phi}}\Delta\boldsymbol{u}_{\mathrm{anc}\,j+1} + \boldsymbol{P}\Delta\boldsymbol{u}_{\mathrm{P}\,j+1} \end{aligned} \tag{3.67}$$

To utilize the prediction to the output in the ancillary ILC problem modify the observation fed to the ancillary controller.

$$\boldsymbol{y}_{\mathrm{anc}\,j} = \boldsymbol{y}_j + \boldsymbol{P}\Delta\boldsymbol{u}_{\mathrm{P}\,j+1} \tag{3.68}$$

The compensated ILC update therefore builds up as follows.

$$\boldsymbol{u}_{\mathrm{anc}\,j+1} = \boldsymbol{Q}_{\mathrm{anc}}\left(\boldsymbol{u}_{\mathrm{anc}\,j} - \boldsymbol{L}_{\mathrm{anc}}\left(\boldsymbol{e}_j + \boldsymbol{P}\Delta\boldsymbol{u}_{\mathrm{P}\,j+1}\right) + \boldsymbol{d}\right) \tag{3.69}$$

An ILC system can be designed for both parts. The ancillary ILC system has fewer inputs than the primary, so that is does not influence the points in the primary problem. Also note that the selected lifted dynamics need to be overactuated as, otherwise, kernel of $P^{\perp}_{\boldsymbol{P}_\Phi}$ is empty. If the system has originally been square, the ancillary is underactuated. The principle is illustrated in Figure 3-6. It is important to analyze the stability of the overall system. To do so, consider the stored inputs as the state of the ILC controller. To do so the (3.69) is written down explicitly by substituting $\Delta\boldsymbol{u}_{\mathrm{P}j+1}$.

$$\begin{aligned}
\Delta\boldsymbol{u}_{\mathrm{P}j+1} &= \boldsymbol{Q}_{\mathrm{P}}\left(\boldsymbol{u}_{\mathrm{P}j} - \boldsymbol{L}_{\mathrm{P}}\boldsymbol{e}_{\Phi j}\right) - \boldsymbol{u}_{\mathrm{P}j}\\
&= \left(\boldsymbol{Q}_{\mathrm{P}} - \boldsymbol{E}\right)\boldsymbol{u}_{\mathrm{P}j} - \boldsymbol{Q}_{\mathrm{P}}\boldsymbol{L}_{\mathrm{P}}\boldsymbol{e}_{\Phi}\\
\boldsymbol{u}_{\mathrm{anc}j+1} &= \boldsymbol{Q}_{\mathrm{anc}}\left(\boldsymbol{u}_{\mathrm{anc}j} - \boldsymbol{L}_{\mathrm{anc}}\left(\boldsymbol{e}_j + \boldsymbol{P}\left(\boldsymbol{Q}_{\mathrm{P}} - \boldsymbol{E}\right)\boldsymbol{u}_{\mathrm{P}j} - \boldsymbol{Q}_{\mathrm{P}}\boldsymbol{L}_{\mathrm{P}}\boldsymbol{e}_{\Phi}\right)\right)\\
&= \boldsymbol{Q}_{\mathrm{anc}}\left(-\boldsymbol{L}_{\mathrm{anc}}\boldsymbol{P}\left(\boldsymbol{Q}_{\mathrm{P}} - \boldsymbol{E}\right)\boldsymbol{u}_{\mathrm{P}j} + \boldsymbol{u}_{\mathrm{anc}j} - \boldsymbol{L}_{\mathrm{anc}}\boldsymbol{e}_j + \boldsymbol{L}_{\mathrm{anc}}\boldsymbol{P}\boldsymbol{Q}_{\mathrm{P}}\boldsymbol{L}_{\mathrm{P}}\boldsymbol{e}_{\Phi}\right)
\end{aligned}$$

In the next step, the overall ILC algorithm is represented in matrix notation and rearranged to form a linear ILC update. Therefore, the overall input vector is assumed to be the composition of the primary and ancillary input $\boldsymbol{u} = \begin{bmatrix}\boldsymbol{u}_{\mathrm{P}}^{\mathrm{T}} & \boldsymbol{u}_{\mathrm{anc}}^{\mathrm{T}}\end{bmatrix}^{\mathrm{T}}$. Furthermore, the ancillary reference is consistent with the point-to-point reference, i. e.$\boldsymbol{r}_{\Phi} = \boldsymbol{\Phi}\boldsymbol{r}$, so the error trajectories are also consistent $\boldsymbol{e}_{\Phi} = \boldsymbol{\Phi}\boldsymbol{e}$.

$$\begin{aligned}
\begin{bmatrix}\boldsymbol{u}_{\mathrm{P}j+1}\\ \boldsymbol{u}_{\mathrm{anc}j+1}\end{bmatrix} &= \begin{bmatrix}\boldsymbol{Q}_{\mathrm{P}} & \boldsymbol{0}\\ \boldsymbol{0} & \boldsymbol{Q}_{\mathrm{anc}}\end{bmatrix}\left(\begin{bmatrix}\boldsymbol{E} & \boldsymbol{0}\\ -\boldsymbol{L}_{\mathrm{anc}}\boldsymbol{P}\left(\boldsymbol{Q}_{\mathrm{P}} - \boldsymbol{E}\right) & \boldsymbol{E}\end{bmatrix}\begin{bmatrix}\boldsymbol{u}_{\mathrm{P}j}\\ \boldsymbol{u}_{\mathrm{anc}j}\end{bmatrix} - \begin{bmatrix}\boldsymbol{L}_{\mathrm{P}} & \boldsymbol{0}\\ -\boldsymbol{L}_{\mathrm{anc}}\boldsymbol{P}\boldsymbol{Q}_{\mathrm{P}}\boldsymbol{L}_{\mathrm{P}} & \boldsymbol{L}_{\mathrm{anc}}\end{bmatrix}\begin{bmatrix}\boldsymbol{e}_{\Phi j}\\ \boldsymbol{e}_j\end{bmatrix}\right)\\
\begin{bmatrix}\boldsymbol{u}_{\mathrm{P}j+1}\\ \boldsymbol{u}_{\mathrm{anc}j+1}\end{bmatrix} &= \begin{bmatrix}\boldsymbol{Q}_{\mathrm{P}} & \boldsymbol{0}\\ \boldsymbol{0} & \boldsymbol{Q}_{\mathrm{anc}}\end{bmatrix}\begin{bmatrix}\boldsymbol{E} & \boldsymbol{0}\\ -\boldsymbol{L}_{\mathrm{anc}}\boldsymbol{P}\left(\boldsymbol{Q}_{\mathrm{P}} - \boldsymbol{E}\right) & \boldsymbol{E}\end{bmatrix}\\
&\quad\cdot\left(\begin{bmatrix}\boldsymbol{u}_{\mathrm{P}j}\\ \boldsymbol{u}_{\mathrm{anc}j}\end{bmatrix} - \begin{bmatrix}\boldsymbol{E} & \boldsymbol{0}\\ -\boldsymbol{L}_{\mathrm{anc}}\boldsymbol{P}\left(\boldsymbol{Q}_{\mathrm{P}} - \boldsymbol{E}\right) & \boldsymbol{E}\end{bmatrix}^{-1}\begin{bmatrix}\boldsymbol{L}_{\mathrm{P}} & \boldsymbol{0}\\ -\boldsymbol{L}_{\mathrm{anc}}\boldsymbol{P}\boldsymbol{Q}_{\mathrm{P}}\boldsymbol{L}_{\mathrm{P}} & \boldsymbol{L}_{\mathrm{anc}}\end{bmatrix}\begin{bmatrix}\boldsymbol{e}_{\Phi j}\\ \boldsymbol{e}_j\end{bmatrix}\right)\\
\begin{bmatrix}\boldsymbol{u}_{\mathrm{P}j+1}\\ \boldsymbol{u}_{\mathrm{anc}j+1}\end{bmatrix} &= \begin{bmatrix}\boldsymbol{Q}_{\mathrm{P}} & \boldsymbol{0}\\ -\boldsymbol{Q}_{\mathrm{anc}}\boldsymbol{L}_{\mathrm{anc}}\boldsymbol{P}\left(\boldsymbol{Q}_{\mathrm{P}} - \boldsymbol{E}\right) & \boldsymbol{Q}_{\mathrm{anc}}\end{bmatrix}\\
&\quad\cdot\left(\begin{bmatrix}\boldsymbol{u}_{\mathrm{P}j}\\ \boldsymbol{u}_{\mathrm{anc}j}\end{bmatrix} - \begin{bmatrix}\boldsymbol{E} & \boldsymbol{0}\\ \boldsymbol{L}_{\mathrm{anc}}\boldsymbol{P}\left(\boldsymbol{Q}_{\mathrm{P}} - \boldsymbol{E}\right) & \boldsymbol{E}\end{bmatrix}\begin{bmatrix}\boldsymbol{L}_{\mathrm{P}} & \boldsymbol{0}\\ -\boldsymbol{L}_{\mathrm{anc}}\boldsymbol{P}\boldsymbol{Q}_{\mathrm{P}}\boldsymbol{L}_{\mathrm{P}} & \boldsymbol{L}_{\mathrm{anc}}\end{bmatrix}\begin{bmatrix}\boldsymbol{e}_{\Phi j}\\ \boldsymbol{e}_j\end{bmatrix}\right)\\
\begin{bmatrix}\boldsymbol{u}_{\mathrm{P}j+1}\\ \boldsymbol{u}_{\mathrm{anc}j+1}\end{bmatrix} &= \begin{bmatrix}\boldsymbol{Q}_{\mathrm{P}} & \boldsymbol{0}\\ -\boldsymbol{Q}_{\mathrm{anc}}\boldsymbol{L}_{\mathrm{anc}}\boldsymbol{P}\left(\boldsymbol{Q}_{\mathrm{P}} - \boldsymbol{E}\right) & \boldsymbol{Q}_{\mathrm{anc}}\end{bmatrix}\left(\begin{bmatrix}\boldsymbol{u}_{\mathrm{P}j}\\ \boldsymbol{u}_{\mathrm{anc}j}\end{bmatrix} - \begin{bmatrix}\boldsymbol{L}_{\mathrm{P}} & \boldsymbol{0}\\ -\boldsymbol{L}_{\mathrm{anc}}\boldsymbol{P}\boldsymbol{L}_{\mathrm{P}} & \boldsymbol{L}_{\mathrm{anc}}\end{bmatrix}\begin{bmatrix}\boldsymbol{\Phi}\\ \boldsymbol{E}\end{bmatrix}\boldsymbol{e}_j\right)\\
\begin{bmatrix}\boldsymbol{u}_{\mathrm{P}j+1}\\ \boldsymbol{u}_{\mathrm{anc}j+1}\end{bmatrix} &= \begin{bmatrix}\boldsymbol{Q}_{\mathrm{P}} & \boldsymbol{0}\\ -\boldsymbol{Q}_{\mathrm{anc}}\boldsymbol{L}_{\mathrm{anc}}\boldsymbol{P}\left(\boldsymbol{Q}_{\mathrm{P}} - \boldsymbol{E}\right) & \boldsymbol{Q}_{\mathrm{anc}}\end{bmatrix}\left(\begin{bmatrix}\boldsymbol{u}_{\mathrm{P}j}\\ \boldsymbol{u}_{\mathrm{anc}j}\end{bmatrix} - \begin{bmatrix}\boldsymbol{L}_{\mathrm{P}}\boldsymbol{\Phi}\\ \boldsymbol{L}_{\mathrm{anc}} - \boldsymbol{L}_{\mathrm{anc}}\boldsymbol{P}\boldsymbol{L}_{\mathrm{P}}\boldsymbol{\Phi}\end{bmatrix}\boldsymbol{e}_j\right)
\end{aligned} \tag{3.70}$$

Therefore, a iterative learning control system with ancillary cost function and the update compensation as it is proposed can be presented by a linear ILC update (3.9), if the primary control problem is to track a specified set of way points of the referecnce given two the ancillary controller. In this case the Q-filter and the learnrule $\left(\boldsymbol{Q}_{\mathrm{c}}, \boldsymbol{L}_{\mathrm{c}}\right)$ are given by:

$$\boldsymbol{Q}_{\mathrm{c}} = \begin{bmatrix} \boldsymbol{Q}_{\mathrm{P}} & \boldsymbol{0} \\ -\boldsymbol{Q}_{\mathrm{anc}}\boldsymbol{L}_{\mathrm{anc}}\boldsymbol{P}\left(\boldsymbol{Q}_{\mathrm{P}}-\boldsymbol{E}\right) & \boldsymbol{Q}_{\mathrm{anc}} \end{bmatrix} \qquad \boldsymbol{L}_{\mathrm{c}} = \begin{bmatrix} \boldsymbol{L}_{\mathrm{P}}\boldsymbol{\Phi} \\ \boldsymbol{L}_{\mathrm{anc}}\left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{L}_{\mathrm{P}}\boldsymbol{\Phi}\right) \end{bmatrix} \tag{3.71}$$

In the uncompensated case (3.62) and (3.63) extraction of the combined update $\left(\boldsymbol{Q}_{\mathrm{u}}, \boldsymbol{L}_{\mathrm{u}}\right)$ is trivial.

$$\boldsymbol{Q}_{\mathrm{u}} = \begin{bmatrix} \boldsymbol{Q}_{\mathrm{P}} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{Q}_{\mathrm{anc}} \end{bmatrix} \qquad \boldsymbol{L}_{\mathrm{u}} = \begin{bmatrix} \boldsymbol{L}_{\mathrm{P}}\boldsymbol{\Phi} \\ \boldsymbol{L}_{\mathrm{anc}} \end{bmatrix} \tag{3.72}$$

In the next step the closed loop dynamics are derived to analyze the influence of the compensation term (3.68).

To use this quantities in analysis it is also required to have a lifted dynamics representation of the overall plant, being a matrix representation of (3.61).

$$\boldsymbol{P}_{\mathrm{comb}} = \boldsymbol{P}\begin{bmatrix} \boldsymbol{E} & B^{\perp}_{\boldsymbol{P}_{\Phi}} \end{bmatrix} \tag{3.73}$$

$$\begin{aligned}
\boldsymbol{L}_{\mathrm{c}} \cdot \boldsymbol{P}_{\mathrm{comb}} &= \begin{bmatrix} \boldsymbol{L}_{\mathrm{P}}\boldsymbol{\Phi}\boldsymbol{P} & \boldsymbol{L}_{\mathrm{P}}\boldsymbol{\Phi}\boldsymbol{P}B^{\perp}_{\boldsymbol{P}_{\Phi}} \\ \boldsymbol{L}_{\mathrm{anc}}\left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{L}_{\mathrm{P}}\boldsymbol{\Phi}\right)\boldsymbol{P} & \boldsymbol{L}_{\mathrm{anc}}\left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{L}_{\mathrm{P}}\boldsymbol{\Phi}\right)\boldsymbol{P}B^{\perp}_{\boldsymbol{P}_{\Phi}} \end{bmatrix} \\
&= \begin{bmatrix} \boldsymbol{L}_{\mathrm{P}}\boldsymbol{P}_{\Phi} & \boldsymbol{0} \\ \boldsymbol{L}_{\mathrm{anc}}\left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{L}_{\mathrm{P}}\boldsymbol{\Phi}\right)\boldsymbol{P} & \boldsymbol{L}_{\mathrm{anc}}\boldsymbol{P}B^{\perp}_{\boldsymbol{P}_{\Phi}} \end{bmatrix}
\end{aligned} \tag{3.74}$$

$$\begin{aligned}
\boldsymbol{Q}_{\mathrm{c}}\left(\boldsymbol{E}-\boldsymbol{L}_{\mathrm{c}} \cdot \boldsymbol{P}_{\mathrm{comb}}\right) &= \begin{bmatrix} \boldsymbol{Q}_{\mathrm{P}} & \boldsymbol{0} \\ -\boldsymbol{Q}_{\mathrm{anc}}\boldsymbol{L}_{\mathrm{anc}}\boldsymbol{P}\left(\boldsymbol{Q}_{\mathrm{P}}-\boldsymbol{E}\right) & \boldsymbol{Q}_{\mathrm{anc}} \end{bmatrix} \begin{bmatrix} \boldsymbol{E}-\boldsymbol{L}_{\mathrm{P}}\boldsymbol{P}_{\Phi} & \boldsymbol{0} \\ -\boldsymbol{L}_{\mathrm{anc}}\left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{L}_{\mathrm{P}}\boldsymbol{\Phi}\right)\boldsymbol{P} & \boldsymbol{E}-\boldsymbol{L}_{\mathrm{anc}}\boldsymbol{P}B^{\perp}_{\boldsymbol{P}_{\Phi}} \end{bmatrix} \\
&= \begin{bmatrix} \boldsymbol{Q}_{\mathrm{P}}\left(\boldsymbol{E}-\boldsymbol{L}_{\mathrm{P}}\boldsymbol{P}_{\Phi}\right) & \boldsymbol{0} \\ -\boldsymbol{Q}_{\mathrm{anc}}\boldsymbol{L}_{\mathrm{anc}}\boldsymbol{P}\left(\boldsymbol{Q}_{\mathrm{P}}-\boldsymbol{E}\right)\left(\boldsymbol{E}-\boldsymbol{L}_{\mathrm{P}}\boldsymbol{P}_{\Phi}\right)-\boldsymbol{Q}_{\mathrm{anc}}\boldsymbol{L}_{\mathrm{anc}}\left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{L}_{\mathrm{P}}\boldsymbol{\Phi}\right)\boldsymbol{P} & \boldsymbol{Q}_{\mathrm{anc}}\left(\boldsymbol{E}-\boldsymbol{L}_{\mathrm{anc}}\boldsymbol{P}B^{\perp}_{\boldsymbol{P}_{\Phi}}\right) \end{bmatrix} \\
&= \begin{bmatrix} \boldsymbol{Q}_{\mathrm{P}}\left(\boldsymbol{E}-\boldsymbol{L}_{\mathrm{P}}\boldsymbol{P}_{\Phi}\right) & \boldsymbol{0} \\ -\boldsymbol{Q}_{\mathrm{anc}}\boldsymbol{L}_{\mathrm{anc}}\boldsymbol{P}\left(\boldsymbol{Q}_{\mathrm{P}}-\boldsymbol{E}\right)\left(\boldsymbol{E}-\boldsymbol{L}_{\mathrm{P}}\boldsymbol{P}_{\Phi}\right)-\boldsymbol{Q}_{\mathrm{anc}}\boldsymbol{L}_{\mathrm{anc}}\boldsymbol{P}\left(\boldsymbol{E}-\boldsymbol{L}_{\mathrm{P}}\boldsymbol{P}_{\Phi}\right) & \boldsymbol{Q}_{\mathrm{anc}}\left(\boldsymbol{E}-\boldsymbol{L}_{\mathrm{anc}}\boldsymbol{P}B^{\perp}_{\boldsymbol{P}_{\Phi}}\right) \end{bmatrix} \\
&= \begin{bmatrix} \boldsymbol{Q}_{\mathrm{P}}\left(\boldsymbol{E}-\boldsymbol{L}_{\mathrm{P}}\boldsymbol{P}_{\Phi}\right) & \boldsymbol{0} \\ -\boldsymbol{Q}_{\mathrm{anc}}\boldsymbol{L}_{\mathrm{anc}}\boldsymbol{P}\boldsymbol{Q}_{\mathrm{P}}\left(\boldsymbol{E}-\boldsymbol{L}_{\mathrm{P}}\boldsymbol{P}_{\Phi}\right) & \boldsymbol{Q}_{\mathrm{anc}}\left(\boldsymbol{E}-\boldsymbol{L}_{\mathrm{anc}}\boldsymbol{P}B^{\perp}_{\boldsymbol{P}_{\Phi}}\right) \end{bmatrix}
\end{aligned} \tag{3.75}$$

The closed loop dynamics of a compensated ILC system with ancillary cost optimization is given by (3.75).

$$\begin{aligned}
\boldsymbol{L}_\mathrm{u} \cdot \boldsymbol{P}_\mathrm{comb} &= \begin{bmatrix} \boldsymbol{L}_\mathrm{P}\boldsymbol{\Phi}\boldsymbol{P} & \boldsymbol{L}_\mathrm{P}\boldsymbol{\Phi}\boldsymbol{P}B^{\perp}_{\boldsymbol{P}_\Phi} \\ \boldsymbol{L}_\mathrm{anc}\boldsymbol{P} & \boldsymbol{L}_\mathrm{anc}\boldsymbol{P}B^{\perp}_{\boldsymbol{P}_\Phi} \end{bmatrix} \\
&= \begin{bmatrix} \boldsymbol{L}_\mathrm{P}\boldsymbol{P}_\Phi & \boldsymbol{0} \\ \boldsymbol{L}_\mathrm{anc}\boldsymbol{P} & \boldsymbol{L}_\mathrm{anc}\boldsymbol{P}B^{\perp}_{\boldsymbol{P}_\Phi} \end{bmatrix}
\end{aligned} \tag{3.76}$$

$$\boldsymbol{Q}_\mathrm{u}\left(\boldsymbol{E} - \boldsymbol{L}_\mathrm{u} \cdot \boldsymbol{P}_\mathrm{comb}\right) = \begin{bmatrix} \boldsymbol{Q}_\mathrm{P}\left(\boldsymbol{E} - \boldsymbol{L}_\mathrm{P}\boldsymbol{\Phi}\boldsymbol{P}\right) & \boldsymbol{0} \\ -\boldsymbol{Q}_\mathrm{anc}\boldsymbol{L}_\mathrm{anc}\boldsymbol{P} & \boldsymbol{Q}_\mathrm{anc}\left(\boldsymbol{E} - \boldsymbol{L}_\mathrm{anc}\boldsymbol{P}B^{\perp}_{\boldsymbol{P}_\Phi}\right) \end{bmatrix} \tag{3.77}$$

The uncompensated systems closed loop dynamics are given by (3.77). Now analyse the closed loop properties.

Stability of the overall systems $\left(\boldsymbol{Q}_\mathrm{c}, \boldsymbol{L}_\mathrm{c}, \boldsymbol{P}_\mathrm{comb}, \boldsymbol{d}\right)$ and $\left(\boldsymbol{Q}_\mathrm{u}, \boldsymbol{L}_\mathrm{u}, \boldsymbol{P}_\mathrm{comb}, \boldsymbol{d}\right)$ is equivalent to the stability of both the primary $\left(\boldsymbol{Q}_\mathrm{p}, \boldsymbol{L}_\mathrm{p}, \boldsymbol{P}_\Phi, \boldsymbol{d}_\Phi\right)$ and ancillary ILC system, $\left(\boldsymbol{Q}_\mathrm{p}, \boldsymbol{L}_\mathrm{p}, \boldsymbol{P}_\Phi, \boldsymbol{d}_\Phi\right)$ and $\left(\boldsymbol{Q}_\mathrm{anc}, \boldsymbol{L}_\mathrm{anc}, \boldsymbol{P}_\Phi B^{\perp}_{\boldsymbol{P}_\Phi}, \boldsymbol{d}_\Phi\right)$, as the spectral radius of block lower triangular matrices (3.75) and (3.77) is the maximum of the spectral radii of the matrix blocks on the diagonal.

In many applications in the research at the Control Systems Group at TU Berlin (CSG), it turned out that the ILC system is very robust if the learn rule is designed to be norm optimal while the Q-filter is designed using standard filtering strategies. This is also possible by using the ancillary controller algorithm. An approach to do so is discussed in the following.

To apply classical filtering, the state of the ILC controller needs to be represented as the input applied to the plant. This is not doable since the transformation (3.61) that is proposed is not invertible. But, in the case the design of the learn rules $\boldsymbol{L}_\mathrm{P}$ and $\boldsymbol{L}_\mathrm{anc}$ is done so that the Q-filters $\boldsymbol{Q}_\mathrm{Q}$ and $\boldsymbol{Q}_\mathrm{anc}$ become identity, e. g.using Norm Optimal Iterative Learning Control (NO-ILC) with no input weights, the input of the primary problem $\boldsymbol{u}_\mathrm{P}$ is restricted to the the image of its learnrule $\boldsymbol{L}_\mathrm{P}$ as discussed in Section 3.2. The primary input can, therefore, be represented by $\boldsymbol{u}_{\mathrm{P}\,j} = \boldsymbol{L}_\mathrm{P}\tilde{\boldsymbol{u}}_{\mathrm{P}\,j}$. Using this, the input fusion of the states to the real input can be reformulated.

$$\boldsymbol{u}_{j+1} = \boldsymbol{L}_\mathrm{P}\tilde{\boldsymbol{u}}_{\mathrm{P}\,j+1} + B^{\perp}_{\boldsymbol{P}_\Phi}\boldsymbol{u}_{\mathrm{anc}\,j+1} \tag{3.78}$$

The reformulation of the input fusion is potentially invertible as the dimension of $\begin{bmatrix}\tilde{\boldsymbol{u}}_{\mathrm{P}\,j+1} & \boldsymbol{u}_{\mathrm{anc}\,j+1}\end{bmatrix}^\mathrm{T}$ coincides with the actual input. It is also surely invertible as the learnrule $\boldsymbol{L}_\mathrm{P}$ is designed to independently manipulate the selected output, i. e.$\boldsymbol{P}_\Phi\boldsymbol{L}_\mathrm{P}$ has full rank. Conversely, $\boldsymbol{P}_\Phi B^{\perp}_{\boldsymbol{P}}$ vanishes by definition. So the image of $\begin{bmatrix}\boldsymbol{L}_\mathrm{P} & B^{\perp}_{\boldsymbol{P}_\Phi}\end{bmatrix}$ is the full domain $\boldsymbol{P}_\Phi$, namely the space input trajectories, implying its invertibility. To use this observation, consider the composed linear ILC update, obtained by the combination of (3.70), (3.78) and a custom Q-filter $\boldsymbol{Q}_\mathrm{C}$ that is designed for the input trajectory.

$$\begin{bmatrix} \tilde{\boldsymbol{u}}_{\mathrm{P}j+1} \\ \boldsymbol{u}_{\mathrm{anc}j+1} \end{bmatrix} = \begin{bmatrix} \boldsymbol{L}_{\mathrm{P}} & B^{\perp}_{\boldsymbol{P}_{\Phi}} \end{bmatrix}^{-1} \boldsymbol{Q}_{\mathrm{C}} \begin{bmatrix} \boldsymbol{L}_{\mathrm{P}} & B^{\perp}_{\boldsymbol{P}_{\Phi}} \end{bmatrix} \left(\begin{bmatrix} \tilde{\boldsymbol{u}}_{\mathrm{P}j} \\ \boldsymbol{u}_{\mathrm{anc}j} \end{bmatrix} - \begin{bmatrix} \boldsymbol{\Phi} \\ \boldsymbol{L}_{\mathrm{anc}} - \boldsymbol{L}_{\mathrm{anc}} \boldsymbol{P} \boldsymbol{L}_{\mathrm{P}} \boldsymbol{\Phi} \end{bmatrix} \boldsymbol{e}_j \right)$$

$$\begin{bmatrix} \boldsymbol{L}_{\mathrm{P}} & B^{\perp}_{\boldsymbol{P}_{\Phi}} \end{bmatrix} \begin{bmatrix} \tilde{\boldsymbol{u}}_{\mathrm{P}j+1} \\ \boldsymbol{u}_{\mathrm{anc}j+1} \end{bmatrix} = \boldsymbol{Q}_{\mathrm{C}} \left(\begin{bmatrix} \boldsymbol{L}_{\mathrm{P}} & B^{\perp}_{\boldsymbol{P}_{\Phi}} \end{bmatrix} \begin{bmatrix} \tilde{\boldsymbol{u}}_{\mathrm{P}j} \\ \boldsymbol{u}_{\mathrm{anc}j} \end{bmatrix} - \begin{bmatrix} \boldsymbol{L}_{\mathrm{P}} & B^{\perp}_{\boldsymbol{P}_{\Phi}} \end{bmatrix} \begin{bmatrix} \boldsymbol{\Phi} \\ \boldsymbol{L}_{\mathrm{anc}} - \boldsymbol{L}_{\mathrm{anc}} \boldsymbol{P} \boldsymbol{L}_{\mathrm{P}} \boldsymbol{\Phi} \end{bmatrix} \boldsymbol{e}_j \right)$$

$$\boldsymbol{u}_{j+1} = \boldsymbol{Q}_{\mathrm{C}} \left(\boldsymbol{u}_j - \left(\boldsymbol{L}_{\mathrm{P}} \boldsymbol{\Phi} + B^{\perp}_{\boldsymbol{P}_{\Phi}} (\boldsymbol{L}_{\mathrm{anc}} - \boldsymbol{L}_{\mathrm{anc}} \boldsymbol{P} \boldsymbol{L}_{\mathrm{P}} \boldsymbol{\Phi}) \right) \boldsymbol{e}_j \right) \tag{3.79}$$

If the compensation term is dropped one the update is:

$$\boldsymbol{u}_{j+1} = \boldsymbol{Q}_{\mathrm{C}} \left(\boldsymbol{u}_j - \left(\boldsymbol{L}_{\mathrm{P}} \boldsymbol{\Phi} - B^{\perp}_{\boldsymbol{P}_{\Phi}} \boldsymbol{L}_{\mathrm{anc}} \right) \boldsymbol{e}_j \right) \tag{3.80}$$

Insert:

$$\begin{bmatrix} \boldsymbol{u}_{\mathrm{P}j+1} \\ \boldsymbol{u}_{\mathrm{anc}j+1} \end{bmatrix} = \begin{bmatrix} \boldsymbol{Q}_{\mathrm{P}} & \boldsymbol{0} \\ -\boldsymbol{Q}_{\mathrm{anc}} \boldsymbol{L}_{\mathrm{anc}} \boldsymbol{P} \boldsymbol{Q}_{\mathrm{P}} & \boldsymbol{Q}_{\mathrm{anc}} \end{bmatrix} \left(\begin{bmatrix} \boldsymbol{u}_{\mathrm{P}j} \\ \boldsymbol{u}_{\mathrm{anc}j} \end{bmatrix} - \begin{bmatrix} \boldsymbol{L}_{\mathrm{P}} \boldsymbol{\Phi} \\ \boldsymbol{L}_{\mathrm{anc}} \end{bmatrix} \left(\boldsymbol{P}_{\mathrm{Real}} \left(\begin{bmatrix} \boldsymbol{E} & B^{\perp}_{\boldsymbol{P}_{\Phi}} \end{bmatrix} \begin{bmatrix} \boldsymbol{u}_{\mathrm{P}j} \\ \boldsymbol{u}_{\mathrm{anc}j} \end{bmatrix} \right) - \boldsymbol{r} \right) \right)$$

$$\begin{bmatrix} \boldsymbol{u}_{\mathrm{P}j+1} \\ \boldsymbol{u}_{\mathrm{anc}j+1} \end{bmatrix} = \begin{bmatrix} \boldsymbol{Q}_{\mathrm{P}} & \boldsymbol{0} \\ -\boldsymbol{Q}_{\mathrm{anc}} \boldsymbol{L}_{\mathrm{anc}} \boldsymbol{P} \boldsymbol{Q}_{\mathrm{P}} & \boldsymbol{Q}_{\mathrm{anc}} \end{bmatrix} \left(\begin{bmatrix} \boldsymbol{u}_{\mathrm{P}j} \\ \boldsymbol{u}_{\mathrm{anc}j} \end{bmatrix} - \begin{bmatrix} \boldsymbol{L}_{\mathrm{P}} \boldsymbol{\Phi} \\ \boldsymbol{L}_{\mathrm{anc}} \end{bmatrix} \boldsymbol{P}_{\mathrm{Real}} \begin{bmatrix} \boldsymbol{E} & B^{\perp}_{\boldsymbol{P}_{\Phi}} \end{bmatrix} \begin{bmatrix} \boldsymbol{u}_{\mathrm{P}j} \\ \boldsymbol{u}_{\mathrm{anc}j} \end{bmatrix} - \begin{bmatrix} \boldsymbol{L}_{\mathrm{P}} \boldsymbol{\Phi} \\ \boldsymbol{L}_{\mathrm{anc}} \end{bmatrix} \boldsymbol{r} \right)$$

$$\begin{bmatrix} \boldsymbol{u}_{\mathrm{P}j+1} \\ \boldsymbol{u}_{\mathrm{anc}j+1} \end{bmatrix} = \begin{bmatrix} \boldsymbol{Q}_{\mathrm{P}} & \boldsymbol{0} \\ -\boldsymbol{Q}_{\mathrm{anc}} \boldsymbol{L}_{\mathrm{anc}} \boldsymbol{P} \boldsymbol{Q}_{\mathrm{P}} & \boldsymbol{Q}_{\mathrm{anc}} \end{bmatrix} \left(\begin{bmatrix} \boldsymbol{u}_{\mathrm{P}j} \\ \boldsymbol{u}_{\mathrm{anc}j} \end{bmatrix} - \begin{bmatrix} \boldsymbol{L}_{\mathrm{P}} \boldsymbol{\Phi} \boldsymbol{P}_{\mathrm{Real}} & \boldsymbol{L}_{\mathrm{P}} \boldsymbol{\Phi} \boldsymbol{P}_{\mathrm{Real}} B^{\perp}_{\boldsymbol{P}_{\Phi}} \\ \boldsymbol{L}_{\mathrm{anc}} \boldsymbol{P}_{\mathrm{Real}} & \boldsymbol{L}_{\mathrm{anc}} \boldsymbol{P}_{\mathrm{Real}} B^{\perp}_{\boldsymbol{P}_{\Phi}} \end{bmatrix} \begin{bmatrix} \boldsymbol{u}_{\mathrm{P}j} \\ \boldsymbol{u}_{\mathrm{anc}j} \end{bmatrix} - \begin{bmatrix} \boldsymbol{L}_{\mathrm{P}} \boldsymbol{\Phi} \\ \boldsymbol{L}_{\mathrm{anc}} \end{bmatrix} \boldsymbol{r} \right)$$

This concludes the analysis discussion. In the following, the implementation is given for the constrained NO-ILC case. In contrast to the pure point-to-point case, the ancillary optimization algorithm requires the solution of two optimization problems to achieve the hierarchical behavior.

The structure is similar to the one given in Figure 3-6. In contrast, both ILC updates are computed implicitly using optimization as derived in Section 3.3.

$$\begin{aligned} \boldsymbol{u}_{\mathrm{P}j+1} = \arg\min_{\boldsymbol{u}_{\mathrm{P}}} \quad & \hat{J}_{\mathrm{P}j+1|j} \left(\boldsymbol{u}_{\mathrm{P}}, \boldsymbol{u}_{\mathrm{P}j}, \boldsymbol{y}_{\Phi j}, \boldsymbol{r}_{\Phi} \right) && (3.81) \\ \text{s.t.} \quad & \boldsymbol{y}_{\mathrm{o,min}} \le \boldsymbol{y}_{\mathrm{o}} + \boldsymbol{P}_{\mathrm{oc}} \left(\boldsymbol{u}_{\mathrm{P}} - \boldsymbol{u}_{\mathrm{P}j} \right) \le \boldsymbol{y}_{\mathrm{o,max}} \\ \boldsymbol{u}_{\mathrm{anc}j+1} = \arg\min_{\boldsymbol{u}_{\mathrm{anc}}} \quad & \hat{J}_{\mathrm{anc}j+1|j} \left(B^{\perp}_{\boldsymbol{P}_{\Phi}} \boldsymbol{u}_{\mathrm{anc}}, B^{\perp}_{\boldsymbol{P}_{\Phi}} \boldsymbol{u}_{\mathrm{anc}j}, \boldsymbol{y}_j + \boldsymbol{P}_{\mathrm{cc}} \left(\boldsymbol{u}_{\mathrm{P}j+1} - \boldsymbol{u}_{\mathrm{P}j} \right), \boldsymbol{r} \right) && (3.82) \\ \text{s.t.} \quad & \boldsymbol{y}_{\mathrm{o,min}} \le \boldsymbol{y}_{\mathrm{o}} + \boldsymbol{P}_{\mathrm{oc}} \left(\boldsymbol{u}_{\mathrm{Pj+1}} - \boldsymbol{u}_{\mathrm{P}j} + B^{\perp}_{\boldsymbol{P}_{\Phi}} \boldsymbol{u}_{\mathrm{anc}} - B^{\perp}_{\boldsymbol{P}_{\Phi}} \boldsymbol{u}_{\mathrm{anc}j} \right) \le \boldsymbol{y}_{\mathrm{o,max}} \end{aligned}$$

In this setup, the primary ILC controller is given higher priority as the constraints can exploit the whole constraint variable, while the ancillary must act on the unused part. Contrarily, omitting the constraint, there is no reason to stage the optimization problems leading to an ordinary NO-ILC.

$$\begin{bmatrix} \boldsymbol{u}_{\mathrm{P}j+1} \\ \boldsymbol{u}_{\mathrm{anc}j+1} \end{bmatrix} = \arg \min_{\begin{bmatrix} \boldsymbol{u}_{\mathrm{P}} \\ \boldsymbol{u}_{\mathrm{anc}} \end{bmatrix}} \hat{J}_{\mathrm{P}j+1|j} \left(\boldsymbol{u}_{\mathrm{P}}, \boldsymbol{u}_{\mathrm{P}j}, \boldsymbol{y}_{\Phi j}, \boldsymbol{r}_{\Phi} \right) + \hat{J}_{\mathrm{anc}j+1|j} \left(B^{\perp}_{\boldsymbol{P}_{\Phi}} \boldsymbol{u}_{\mathrm{anc}}, B^{\perp}_{\boldsymbol{P}_{\Phi}} \boldsymbol{u}_{\mathrm{anc}j}, \boldsymbol{y}_j + \boldsymbol{P}_{\mathrm{cc}} \left(\boldsymbol{u}_{\mathrm{P}} - \boldsymbol{u}_{\mathrm{P}j} \right), \boldsymbol{r} \right) \tag{3.83}$$

Concluding the discussion on this method, the proposed algorithm allows to define an ancillary cost function and leads to an ideally decoupled primary ILC-system. The overall system itself is an over-actuated ILC system that is composed of a linear ILC update (3.9) and a lifted dynamics representation (3.8), if the controllers designed also lead to linear ILC updates. Doing so, the ILC-updates' matrices are lower block triangular. Therefore, this composition may not need special analysis tools, as it can be represented as a standard system. Also, it is shown that in the case of an unconstrained uniquely actuated plant, the composition of a primal point-to-point problem and an ancillary tracking cost without a Q-filters leads to another learn rule to solve the tracking problem. Therefore, it only differs in the convergence behavior but not in the solution. The main benefit occurs in exploiting the constraints, as the primary controller has access to the full range of values while the ancillary needs to use the rest. Contrary, there are some properties lost, i. e.the lifted dynamics representation is not lower triangular block Toeplitz if the model is assumed to be imperfect as it is for simple causal systems.

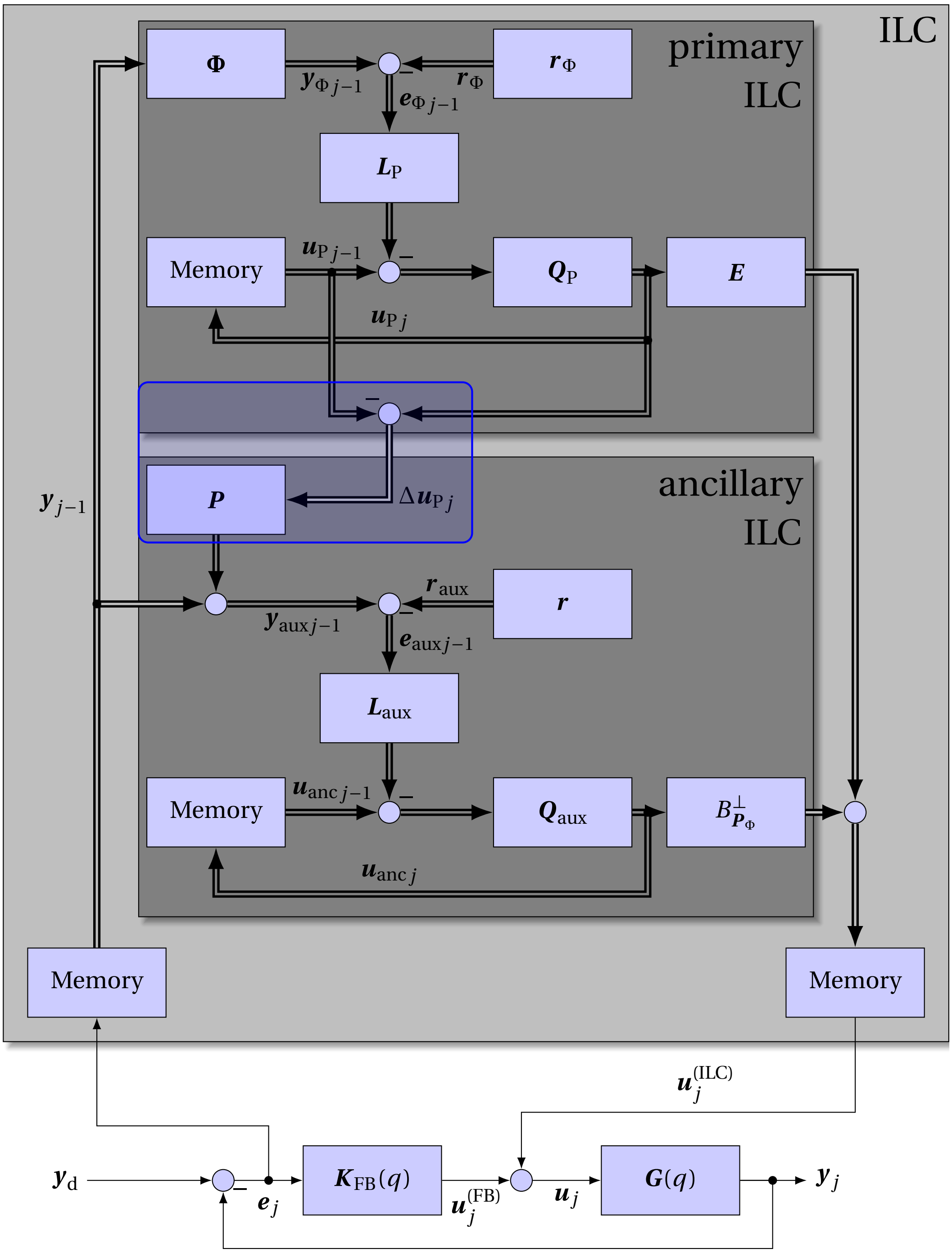
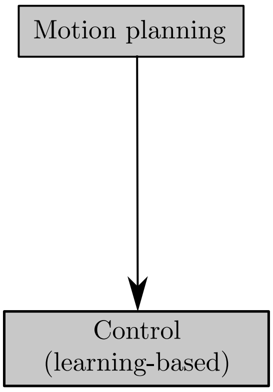


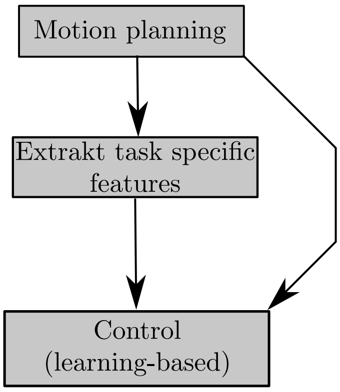
Figure 3-6. Concept of the ancillary optimization as it is introduced by Freeman [3]. The author decoupled the ILC systems using the nullspace projection in the ancillary problem. If the model was exact this assumption would hold. The prediction of the influence of the ILC update of the primary controller to the observation of the ancillary controller, highlighted in blue, is novel in this work.

3.6 Hybridization of planned Motions

This section presents the methods selected to generate references to apply the ILC algorithms to perform the planned motion. The approaches are split up into two branches illustrated in Figure 3-7. The first applies the algorithms to the trajectory naively 3.7(a). The second optimizes the trajectory online by extracting task-specific features 3.7(b). The hybridization strategies 3.1 and 3.2 are of the first category while the following are the second type.



(a) Straight forward.



(b) Extract features.

Figure 3-7. Overview over the embedding strategy.

There are two task specific feature in the case. The first is the obstacle (Figure 1-2). To extract the information the clearence of the trajectory is used that is know from the planning using the flexible collision library [32]. The second is the arrival at the goal configuration.

Hybridization 3.1: Hybridization of geometric paths in metric spaces

Given a *path* (Def. 4) and a metric $||\cdot||$ on the configuration space and a number of samples. Assign to each path segment (z_{i-1}, z_i) an integer amount of samples in ratio of the segment's length compared to the overall trajectory (using a linear interpolation). Sample each path segment on a linear interpolation equidistantly except for the initial configuration of the segment.

For a fully actuated robot apply standard Single Input Single Output (SISO) Iterative Learning Control (ILC), point to point ILC for the path vertices or apply point to point ILC with ancillary optimization assigning a primary cost to the waypoints and the auxiliary to the intermediate sections.

This method is the baseline case that has been successfully applied to fully actuated systems in simulation. It requires that the interpolation is a feasible trajectory that is in general not the case for underactuated robots.

Hybridization 3.2: Hybridization of trajectories by equidistant sampling.

Given a trajectory (Def. 3), a number of samples n_b and a desired waypoint clearance n_c. Sample the trajectory equidistantly with the number of samples n_b and assign each n_c sample to be a waypoint.

For the underactuated robot, apply SISO, Single Input Multiple Output (SIMO) or Point-to-Point Iterative Learning Control (P2P-ILC).

Hybridization 3.3: Hybridization of trajectories with clearance weighting.

Given a trajectory (Def. 3) and clearance information for the configuration space $c(\boldsymbol{q})$, a number of samples n_b and a weighting function $f(c)$. Sample the trajectory equidistantly and assign a weight $f(\boldsymbol{q}_k)$ to each sample.

Apply SIMO ILC to the under-actuated robot considering the clearance weighting in the $\boldsymbol{Q}_\mathrm{W}$.

Hybridization 3.4: Hybridization of trajectories as narrow passage points.

Given a trajectory (Def. 3) and clearance information for the configuration space $c(\boldsymbol{q})$, a number of samples n_b and a number of waypoints n_Φ. Sample the trajectory equidistantly in time and assign the most proximal sample as well as the samples n_off apart as waypoints.

Apply P2P-ILC to the under-actuated robot track the three points.

Hybridization 3.5: Hybridization of trajectories with clearance weighting and a narrow passage point.

Given a trajectory (Def. 3) and clearance information for the configuration space $c(\boldsymbol{q})$, a number of samples n_b and a weighting function $f(c)$. Sample the trajectory equidistantly and assign a weight $f(\boldsymbol{q}_k)$ to each sample. Furthermore, sample the trajectory equidistantly in time and assign the most proximal sample as way point.

Apply SIMO ILC to the underactuated robot and track the number of outputs that is equal to the number of actuators and track the other variables and their derivatives only at the waypoint, namely the most proximal.

Adopting ILC

These techniques allow the application of ILC algorithms to a planned trajectory.

The first Hybridization strategy is chosen for fully actuated robots as in this case, a metric can be constructed easily. It is tested in a simulation study (Section 3.8).

The second Hybridization strategy is the naive realization. They correspond to a strait application of single output multiple outputs or a point-to-point controller. An experiment is done. The tuning is presented in the corresponding part in the experimental validation section.

The latter ones (Hybridization 3.4 to 3.5) have in common that they generate a feasible control problem, i. e.the control problem becomes over-actuated or stays uniquely actuated for the first example. The third strategy (Hybridization 3.4) does not generate a feasible control problem, as the trajectory is bound for a time span and not only some dedicated waypoints. This issue is discussed in detail in Section 3.6.4. The ILC design strategy chosen is Norm Optimal Iterative Learning Control (NO-ILC), which requires a linear system model to compute the Markov parameters and weight matrices. The tuning is done in the following steps.

- selection of the weighting function
- selection of the cut off weight. (Method 3.3 and 3.5)
- design of a selector. (Method 3.4)
- design of the weights on the refrence $\boldsymbol{Q}_{\mathrm{W}}$
- design of the input update weights $\boldsymbol{S}_{\mathrm{W}}$
- deisgn of the input weight $\boldsymbol{R}_{\mathrm{W}}$
- formulation of the constraints

The input weighting $\boldsymbol{R}_{\mathrm{W}}$ is chosen to be the zero matrix in all cases. And the constraint are chosen $5°$ loose as it has been done in during planning, namely $-55°$ and $75°$. The reasoning behind this decision is discussed in the Experimental validation section. The selection of the other parameters is derived in the following.

3.6.1 Tuning of the clearance weighting and the cutoff weight

The clearance weighting function $f(c)$ used in the Hybridization methods 3.3 and 3.5 is chosen problem specific for the case that the terminal configuration and is one meter apart from the initial one and the obstacle is located halfway to at the height of 15 cm. So the initial and the goal configuration have a clearance of 4.5 cm. The weighting should exclude these configurations because otherwise, the problem becomes a standard problem from the beginning. This is not desirable since it discards the task-specific characteristic of the algorithm.

The fundamental function chosen is sigmoidal. It is positive and bound to one, which assures that the weights neither explode nor become negative. Furthermore, values below a threshold are set to zero to ensure that the widow selected is finite. The parameter α, called steepness, is chosen to be $\alpha = 50$, the clearance threshold $c_{\mathrm{th}} = 12\,\mathrm{cm}$ and the weight threshold $w_{\mathrm{th}} = 0.02$. This results in vanishing weights for clearances above 21 cm. Also, every trajectory passing the obstacle will contain weights larger than zero as the contour line of 10 cm needs to be intersected. Those values are chosen by experience throughout development. The number of parameters is 3. The decision is taken as it seems to robustly identify the passage in trajectories planned by the kinodynamic Rapidliy expanding Random Tree (RRT).

$$f(c) = \begin{cases} \frac{1}{1+e^{-\alpha(c-c_{\mathrm{th}})}} & \frac{1}{1+e^{-\alpha(c-c_{\mathrm{th}})}} > w_{\mathrm{th}} \\ 0 & \text{else} \end{cases} \tag{3.84}$$

The resulting clearance cost is depicted in Figure 3-8. It shows the discussed properties.

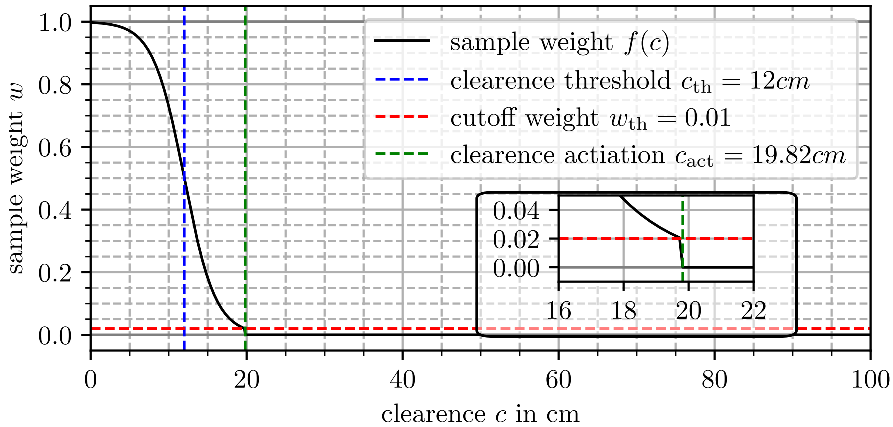


Figure 3-8. Chosen weighing function.

3.6.2 Selection of the sample offset for Hybridization 3.4

The offset of the points n_{off} is chosen by experience to the value of $n_{\mathrm{off}} = 10$. The Selector is designed according to Definition 20. The selected waypoints are the most proximal point, the points n_{off} apart, and the final waypoint to ensure the goal location. The individual selection matrices $\boldsymbol{\Phi}_i$ are chosen to be identity, and the reference conforms to the reference trajectory.

The decision is done by the observation in simulation that the three waypoints represent the waypoints are in close to the obstacle while still being apart from each other so that they mimic the movement to pass the obstacle..

3.6.3 Tuning of the Weight matrices

The realization of a motion using NO-ILC requires the tuning of the weight matrices $\boldsymbol{Q}_{\mathrm{W}}$, $\boldsymbol{S}_{\mathrm{W}}$, and $\boldsymbol{R}_{\mathrm{W}}$. The output weight matrix $\boldsymbol{Q}_{\mathrm{W}}$ encodes how well the reference is tracked, it is, therefore, the most important parameter to be tuned and characteristic for the method.

The weight selection is made on the basis of the full state to be the controlled output $\boldsymbol{y}_{\mathrm{c}} = \boldsymbol{x}$. The state of the system is $\boldsymbol{x} = \begin{bmatrix} \varphi & \dot{\varphi} & s_{\mathrm{W}} & \dot{s_{\mathrm{W}}} \end{bmatrix}^{\mathrm{T}}$.

The Hybridization strategies 3.3 and 3.5 are implemented using a multiple output controllers for the whole sample, i. e. take all samples of the trajectory into account. The weights are therefore designed as block diagonal matrix, where each block corresponds to a weight on a specific sample.

$$\boldsymbol{Q}_{\mathrm{W}} = \mathrm{Blockdiag}\begin{bmatrix}\boldsymbol{Q}_1 & \boldsymbol{Q}_2 & \dots & \boldsymbol{Q}_{n_\mathrm{b}}\end{bmatrix}$$

The Strategy 3.4 is implemented using a point-to-point ILC controller and the weight matrix is also composed as block diagonal. The entries on the diagonal blocks serve to correspond to the weights on the selected points.

$$\boldsymbol{Q}_{\mathrm{W}} = \mathrm{Blockdiag}\begin{bmatrix}\boldsymbol{Q}_{k_{\mathrm{prox}}-n_{\mathrm{off}}} & \boldsymbol{Q}_{k_{\mathrm{prox}}} & \boldsymbol{Q}_{k_{\mathrm{prox}}+n_{\mathrm{off}}} & \boldsymbol{Q}_{n_\mathrm{b}}\end{bmatrix}$$

Hybridization 3.3 Narrow passage weighting.

This Hybridization requires the selection of weight on the pitch and the position. In the present example, the system weights are chosen to be

$$\boldsymbol{Q}_k = \begin{cases} f(c_k)\begin{bmatrix}0 & 0 & 0 & 0\\ 0 & 40 & 0 & 0\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 800\,\mathrm{m}^{-2}\end{bmatrix} & k < n_\mathrm{b}\\ \begin{bmatrix}100 & 0 & 0 & 0\\ 0 & 100 & 0 & 0\\ 0 & 0 & 100\,\mathrm{s}^2\,\mathrm{m}^{-2} & 0\\ 0 & 0 & 0 & 100\,\mathrm{m}^{-2}\end{bmatrix} & k = n_\mathrm{b}\end{cases} \tag{3.85}$$

Hybridization 3.4 Narrow Passage Waypoints.

As the method only introduces cost on three intermediate waypoints, the weight is given for those. For the waypoints, only the configuration is included in the cost but not the velocity. Furthermore, on the final state, the cost to achieve arrival in rest.

$$\boldsymbol{Q}_k = \begin{cases} \begin{bmatrix}0 & 0 & 0 & 0\\ 0 & 0.4 & 0 & 0\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 8\,\mathrm{m}^{-2}\end{bmatrix} & k < n_\mathrm{b}\\ \begin{bmatrix}100 & 0 & 0 & 0\\ 0 & 100 & 0 & 0\\ 0 & 0 & 100\,\mathrm{s}^2\,\mathrm{m}^{-2} & 0\\ 0 & 0 & 0 & 100\,\mathrm{m}^{-2}\end{bmatrix} & k = n_\mathrm{b}\end{cases}$$

Hybridization 3.5 narrow passage with most proximal point.

In this strategy, the tracking in the window is only done on the pitch. The waypoint is used to control the position and the velocity. This fixes the trajectory from an intuitive point of view. To do so, the weights are defined as follows.

$$
\boldsymbol{Q}_k = \begin{cases}
f(c_k)\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 400 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} & k \neq k_{\text{prox}} \wedge k < n_{\text{b}} \\
\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 400 f(c_k) & 0 & 0 \\ 0 & 0 & 200\,\text{s}^2\,\text{m}^{-2} & 0 \\ 0 & 0 & 0 & 4000\,\text{m}^{-2} \end{bmatrix} & k = k_{\text{prox}} \\
\begin{bmatrix} 100 & 0 & 0 & 0 \\ 0 & 100 & 0 & 0 \\ 0 & 0 & 100\,\text{s}^2\,\text{m}^{-2} & 0 \\ 0 & 0 & 0 & 100\,\text{m}^{-2} \end{bmatrix} & k = n_{\text{b}}
\end{cases}
$$

The following section discusses an analytical argument that shows the actuation state.

3.6.4 Analysis of the state of actuation

In this Section, it is discussed how the state of actuation, as it is discussed in Section 3.2, is affected by the proposed methods Hybridization 3.3 and 3.5. The state of actuation is characterized by the dimension and rank of the lifted dynamics representation of the system. As the methods are proposed, they form a multiple-output iterative leaning controller on the full state of the robot. The corresponding weight matrix is diagonal but also has zero entries on the diagonal. This is equivalent to ignoring the output in the ILC control loop. Thus, the system can be reformulated to a point-to-point problem as it is discussed in Section 3.4. The state of actuation is then evaluated on the selected lifted dynamics.

To do the analysis, the proposed weights are decomposed in a selection matrix (Def. 20) and a full rank and diagonal weight matrix $\boldsymbol{\Phi}^{\text{T}}\tilde{\boldsymbol{Q}}_{\text{W}}\boldsymbol{\Phi} = \boldsymbol{Q}_{\text{W}}$.

Hybridization 3.3 Narrow passage weighting.

The sample weight can be decomposed in a sample selector and a diagonal matrix.

$$
\boldsymbol{Q}_k = \begin{cases} \underbrace{\begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}}_{\boldsymbol{\Phi}_k^{\mathrm{T}}} \underbrace{f(c_k) \begin{bmatrix} 40 & 0 \\ 0 & 800\,\mathrm{m}^{-2} \end{bmatrix}}_{\tilde{\boldsymbol{Q}}_k} \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\boldsymbol{\Phi}_k} & f(c_k) > 0 \\ \underbrace{\boldsymbol{E}}_{\boldsymbol{\Phi}_k^{\mathrm{T}}} \underbrace{\begin{bmatrix} 100 & 0 & 0 & 0 \\ 0 & 100 & 0 & 0 \\ 0 & 0 & 100\,\mathrm{s}^2\,\mathrm{m}^{-2} & 0 \\ 0 & 0 & 0 & 100\,\mathrm{m}^{-2} \end{bmatrix}}_{\tilde{\boldsymbol{Q}}_k} \underbrace{\boldsymbol{E}}_{\boldsymbol{\Phi}_k} & k = n_{\mathrm{b}} \end{cases} \tag{3.86}
$$

Therefore, the lifted selector is composed by the sequence of tuples $(k, \boldsymbol{r}_k, \boldsymbol{\Phi}_k)$ for all k so that $f(c_k) \neq 0$ or $k = n_{\mathrm{b}}$. In doing so, the state of actuation is analyzed for the system. Therefore, the lifted dynamics are composed of a sequence of consecutive selected samples i to j. The values of i and j are assumed to be the limit of the window with weights that are non-zero. Specifically, i is the first value so that $f(c_i) > 0$ and j the last value so that $f(c_j) > 0$ holds.

$$
\boldsymbol{P}_{\Phi} = \begin{bmatrix} \boldsymbol{\Phi}_i \boldsymbol{A}^{i-1}\boldsymbol{B} & \dots & \boldsymbol{\Phi}_i \boldsymbol{A}\boldsymbol{B} & \boldsymbol{\Phi}_i \boldsymbol{B} & \boldsymbol{0} & \boldsymbol{0} & \dots & \boldsymbol{0} \\ \boldsymbol{\Phi}_{i+1} \boldsymbol{A}^{i}\boldsymbol{B} & \dots & \boldsymbol{\Phi}_{i+1} \boldsymbol{A}^2\boldsymbol{B} & \boldsymbol{\Phi}_{i+1} \boldsymbol{A}\boldsymbol{B} & \boldsymbol{\Phi}_{i+1}\boldsymbol{B} & \boldsymbol{0} & \dots & \boldsymbol{0} \\ \boldsymbol{\Phi}_{i+2} \boldsymbol{A}^{i+1}\boldsymbol{B} & \dots & \boldsymbol{\Phi}_{i+2} \boldsymbol{A}^3\boldsymbol{B} & \boldsymbol{\Phi}_{i+2} \boldsymbol{A}^2\boldsymbol{B} & \boldsymbol{\Phi}_{i+2}\boldsymbol{A}\boldsymbol{B} & \boldsymbol{\Phi}_{i+2}\boldsymbol{B} & \dots & \boldsymbol{0} \\ \vdots & & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{\Phi}_j \boldsymbol{A}^{j-1}\boldsymbol{B} & \dots & \boldsymbol{\Phi}_j \boldsymbol{A}^{j-i+2}\boldsymbol{B} & \boldsymbol{\Phi}_j \boldsymbol{A}^{j-i+1}\boldsymbol{B} & \boldsymbol{\Phi}_j \boldsymbol{A}^{j-i}\boldsymbol{B} & \boldsymbol{\Phi}_j \boldsymbol{A}^{j-i-1}\boldsymbol{B} & \dots & \boldsymbol{\Phi}_j\boldsymbol{B} \end{bmatrix} \tag{3.87}
$$

The lifted dynamics $\boldsymbol{P}_{\Phi}$ can be decomposed into three matrices to analyze the rank.

$$
\begin{aligned}
\boldsymbol{P}_{\Phi} = & \underbrace{\begin{bmatrix} \boldsymbol{\Phi}_i & \boldsymbol{0} & \boldsymbol{0} & \dots & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{\Phi}_{i+1} & \boldsymbol{0} & \dots & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{\Phi}_{i+2} & \dots & \boldsymbol{0} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0} & \dots & \boldsymbol{\Phi}_j \end{bmatrix}}_{\text{full row rank}} \cdot \underbrace{\begin{bmatrix} \boldsymbol{E} & \boldsymbol{0} & \boldsymbol{0} & \dots & \boldsymbol{0} \\ \boldsymbol{A} & \boldsymbol{E} & \boldsymbol{0} & \dots & \boldsymbol{0} \\ \boldsymbol{A}^2 & \boldsymbol{A} & \boldsymbol{E} & \dots & \boldsymbol{0} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{A}^{j-i+1} & \boldsymbol{A}^{j-i} & \boldsymbol{A}^{j-i-1} & \dots & \boldsymbol{E} \end{bmatrix}}_{\text{full rank}} \\
& \cdot \underbrace{\begin{bmatrix} \boldsymbol{A}^{i-1}\boldsymbol{B} & \dots & \boldsymbol{A}^2\boldsymbol{B} & \boldsymbol{A}\boldsymbol{B} & \boldsymbol{B} & \boldsymbol{0} & \boldsymbol{0} & \dots & \boldsymbol{0} \\ \boldsymbol{0} & \dots & \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{B} & \boldsymbol{0} & \dots & \boldsymbol{0} \\ \boldsymbol{0} & \dots & \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{B} & \dots & \boldsymbol{0} \\ \vdots & & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{0} & \dots & \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0} & \dots & \boldsymbol{B} \end{bmatrix}}_{\text{rank} = n + m(j-i)}
\end{aligned} \tag{3.88}
$$

The first matrix is full rank by Definition 20, and the second as it is lower triangular. Therefore, the rank of the selected lifted dynamics is limited to $n + m(j - i)$, the rank of the third matrix, due to Sylvester's Inequality [40, p. 62, (546)]. For the proposed system with selectors $\boldsymbol{\Phi}_i \in \mathbb{R}^{2\times 4}$ and an input matrix $\boldsymbol{B} \in \mathbb{R}^{4\times 1}$, the selected lifted dynamics $\boldsymbol{P}_\Phi$ are of the dimension $2(j-i)\times j$ and the rank is limited to $n + m(j-i) = 4 + (j-i)$. Thus, the selected dynamics $\boldsymbol{P}_\Phi$ is not full row rank if the window has more than two samples. This means that it is both over and under-actuated.

Remark 5. *The analysis does not take the goal configuration into account. Doing so, the lifted dynamics $\boldsymbol{P}_\Phi$ is augmented by additional rows, that are full rank. Thus, the rank can not be complete.*

Proposition 1. *Hybridization 3.3 is under and over actuated.*

Hybridization 3.4 Narrow Passage Waypoints.

This method is over-actuated and not under-actuated for the reason that is formulated in Section 3.4, namely, the system becomes not over-actuated as the clearance of two consecutive waypoints is at least equal to the system order. This holds true as the system order n is limited to the waypoint clearance n_c.

Proposition 2. *Hybridization 3.4 is over actuated.*

Hybridization 3.5 Narrow passage weighting with waypoint.

The decomposed lifted weights are in this case:

$$
\boldsymbol{Q}_k = \begin{cases}
\underbrace{\begin{bmatrix}0\\1\\0\\0\end{bmatrix}}_{\boldsymbol{\Phi}_k^\mathrm{T}} \underbrace{f(c_k)\begin{bmatrix}40\end{bmatrix}}_{\tilde{\boldsymbol{Q}}_k} \underbrace{\begin{bmatrix}0 & 1 & 0 & 0\end{bmatrix}}_{\boldsymbol{\Phi}_k} & f(c_k) > 0 \land k \neq k_\text{prox} \\
\underbrace{\begin{bmatrix}0&0&0\\1&0&0\\0&1&0\\0&0&1\end{bmatrix}}_{\boldsymbol{\Phi}_k^\mathrm{T}} \underbrace{\begin{bmatrix}40 f(c_k) & 0 & 0\\ 0 & 200\,\mathrm{s}^2\,\mathrm{m}^{-2} & 0\\ 0 & 0 & 4000\,\mathrm{m}^{-2}\end{bmatrix}}_{\tilde{\boldsymbol{Q}}_k} \underbrace{\begin{bmatrix}0&1&0&0\\0&0&1&0\\0&0&0&1\end{bmatrix}}_{\boldsymbol{\Phi}_k} & k = k_\text{prox} \\
\underbrace{\boldsymbol{E}}_{\boldsymbol{\Phi}_k^\mathrm{T}} \underbrace{\begin{bmatrix}100 & 0 & 0 & 0\\ 0 & 100 & 0 & 0\\ 0 & 0 & 100\,\mathrm{s}^2\,\mathrm{m}^{-2} & 0\\ 0 & 0 & 0 & 100\,\mathrm{m}^{-2}\end{bmatrix}}_{\tilde{\boldsymbol{Q}}_k} \underbrace{\boldsymbol{E}}_{\boldsymbol{\Phi}_k} & k = n_\text{b}
\end{cases}
\tag{3.89}
$$

In this case, the system is not under-actuated. In contrast to the Hybridization 3.3 the selectors are mostly of the dimension $\boldsymbol{\Phi}_k \in \mathbb{R}^{1\times 4}$ except the most proximal point $\boldsymbol{\Phi}_{k_{\text{prox}}} \in \mathbb{R}^{3\times 4}$, therefore the selected lifted dynamics $\boldsymbol{P}_\Phi$ may have full row rank 3.88. Specifically, the selected lifted dynamics has $\underbrace{3}_{k_{\text{prox}}} + \underbrace{(j-i-1)}_{\text{window}}$ rows, less than the upper limit of the rank. The indices i is the first index so that $f(c_i) > 0$ holds and j the last so that $f(c_j) > 0$ holds.

Proposition 3. *Hybridization 3.5 is over actuated and potentially under actuated. To assure that Hybridization 3.5 is not under actuated, the rank of the lifted dynamics must be checked.*

In the present design, it is checked using matlab that the selected lifted dynamics are full row rank and therefore, the method is not under-actuated.

3.6.5 Cost on the input trajectory based on system knowledge

The Hybridization leads to over-actuated systems. Therefore, the selection of the input weighting $\boldsymbol{S}_\text{W}$ influences the appearance of the final trajectory. As the controller is chosen stable (see Section 4.2.1), the input trajectory should not be chosen trivially. This is why the input to the controlled system does not correspond to a physical quantity of interest. The stabilizing controller of the Two Wheeled Inverted Pendulum Robot (TWIPR) is designed so that a static input corresponds to a stable equilibrium position $(\varphi, \dot{\varphi}, s_\text{W}, \dot{s_\text{W}}) = (0, 0, {s_\text{W}}^{\text{eq}}(u), 0)$. Therefore, interactively it is reasonable to allow constant outputs, as the lead to upright standing. To achieve this behavior, the update weights shall be assigned on the inputs derivative $\dot{u}$. This is achieved with a modified update weight built by a discrete-time derivation filter, where h is the sampling period.

$$D(q) = \frac{1-q^{-1}}{h} \tag{3.90}$$

Furthermore, due to physical insight, it is assumed that the signal is zero before the input is applied and constant afterward, i. e. $\boldsymbol{u}(k) = \boldsymbol{0}$ for all $k < 0$ and $\boldsymbol{u}(k) = \boldsymbol{u}(n_\text{b}-1)$ for all $k > n_\text{b}$. The cost matrix is composed using the lifted dynamics representation.

$$\boldsymbol{S}_{\text{W}D} = \frac{1}{h^2}\begin{bmatrix} 2 & -1 & 0 & \dots & 0 & 0 \\ -1 & 2 & -1 & \dots & 0 & 0 \\ 0 & -1 & 2 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 2 & -1 \\ 0 & 0 & 0 & \dots & -1 & 1 \end{bmatrix} = \frac{1}{h^2}\begin{bmatrix} 1 & -1 & \dots & 0 & 0 \\ 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & -1 \\ 0 & 0 & \dots & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & \dots & 0 & 0 \\ -1 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \\ 0 & 0 & \dots & 1 & 0 \\ 0 & 0 & \dots & -1 & 1 \end{bmatrix} \tag{3.91}$$

This method is similar to the auxiliary cost in the algorithm introduced by Owens [23, pp. 323], who introduces an auxiliary cost in the learn rule.

Beside filtering, it is necessary to balance input update weights and output weights to control the convergence speed. The weights are chosen by experience to be:

$$\boldsymbol{S}_\text{W} = \frac{1}{2}\boldsymbol{S}_{\text{W}D} \tag{3.92}$$

3.6.6 Metrics of the resulting controllers

As the controllers are tuned, the metrics can be obtained. In doing so the author observed a spectral radius of one. They indicate that the controllers are not asymptotically stable. Also the error norm evolution is larger than one, namely, it is in the order of 30. This is not considered to be a problem, as the controller converged in simulation. But, in an application, this issue must be faced.

3.7 Benchmark Trajectories

To test the hybridization methods, a planned trajectory needs to be chosen. Since the trajectories produced by an Rapidliy expanding Random Tree (RRT) are stochastic by design, many solutions exist. The author has made the selection so that trajectories contain the characteristic features. Experimentally two trajectories have been reviewed. The first is the *immediate dive with proximal obstacle*, which does a dive movement from the initial state. The second is the *delayed dive* that moves to get into proximity of the obstacle first and then dives. The two trajectories solve different tasks as the obstacle has a different position in space. In simulation, a third trajectory is investigated: an *immediate dive with distant obstacle*. This remains in simulation as the experiments' effort exceeds this thesis's scope.

Immediate Dive with proximal obstacle

This results are drawn in earlier stage of the project. The solutions solve a motion problem with the obstacle located at 30 cm apart the initial configuration this required an immediate dive. This immediate dive is characteristic for all the solution of this planning problem. All of them perform an immediate dive and perform a compensation movement to reach the goal location. The trajectory is visualized in Figure 3-9. It can be observed that the trajectory immediately passes the obstacle and does a breaking maneuver as the obstacle is passed. The number of samples that are outside the activation clearance c_{act} in the beginning is small (in the order of 10 samples or 0.2s receptively).

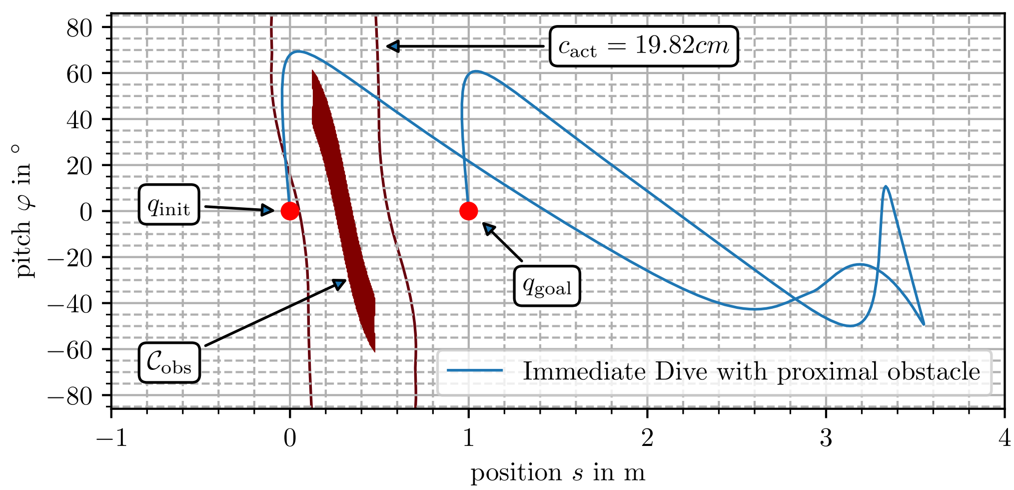


Figure 3-9. Benchmark trajectories with proximal obstacle. The red contour line indicates the activation clearance, so that weights greater or than zero are only assigned within the this contour.

Trajectory	File Reference
Immediate Dive with distant obstacle	`trj_immediate_dive_dsit.csv`
Delayed Dive	`trj_delayed_dive.csv`
Immediate Dibe with proximal obstacle	`trj_immediate_dive_prox.csv`

Table 3.2. Benchmark trajectory data. The files can be found in the git repro in `9_Python_Motion_Planning/results`.

Delayed Dive

These results are drawn in an earlier stage of the project. The solutions solve a motion problem with the obstacle located at 30 cm apart from the initial configuration. This required an immediate dive. This immediate dive is characteristic of all the solutions of this planning problem. They all perform an immediate dive and compensation movement to reach the goal location. The trajectory is visualized in Figure 3-9. It can be observed that the trajectory immediately passes the obstacle and does a breaking maneuver as the obstacle is passed. The number of samples that are outside the activation clearance c_{act}, in the beginning, is small (in the order of 10 samples or 0.2s receptively).

Immediate Dive with distant obstacle

The trajectory solves the same task as the one mentioned earlier. In contrast, the movement to get into proximity of the obstacle is more aggressive, forming a characteristic difference from the earlier one. It is also presented in Figure 3-10. In contrast to the former movement, the movement to get proximal to the obstacle is shorter. It takes 50 samples/1s to arrive in the activation clearance.

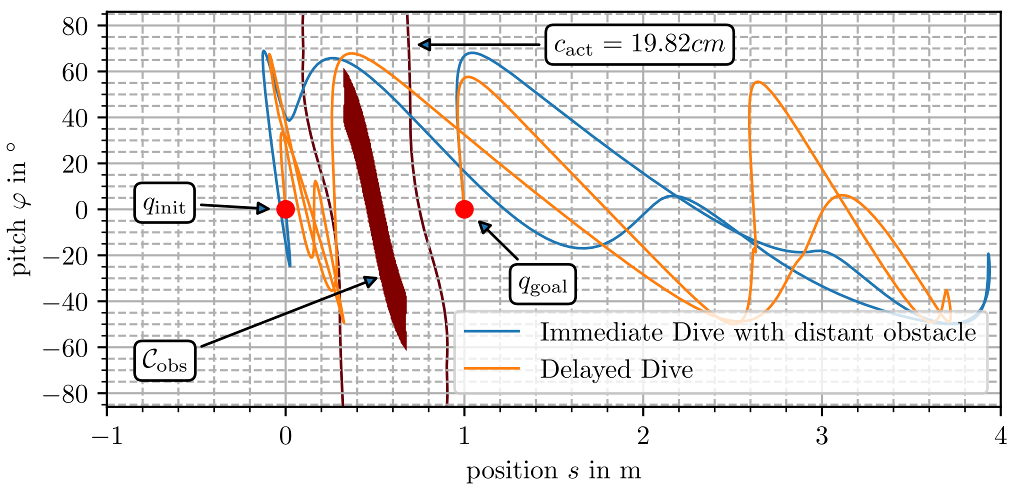


Figure 3-10. Benchmark trajectories with distant obstacle. The red contour line indicates the activation clearance, so that weights greater or than zero are only assigned within the this contour.

3.8 Simulation study on the Iterative Learning Control (ILC) algorithms

As the algorithms are wrapped up theoretically, they will be applied to systems in simulation to evaluate their performance. Furthermore, it is presented how the algorithms are combined with motion planning.

The study is structured as follows. First, the Single Input Single Output (SISO) problems are considered using the double integrator system (3.93). As the system is MIMO but with completely decoupled dynamics in its coordinates, this is suitable. I. e. one finds a SISO system along each axis in free space. In this setup, the previously discussed algorithms are applied, and the differences are pointed out. The procedure is done for a constrained and an unconstrained plant. The constraints are imposed on the input as follows. In the second step, the same algorithms are applied to the dynamic model of the TWIPR being an under-actuated SIMO system.

The model chosen for the SISO analysis is a planar robot with double integrator dynamics. This could also be an industrial robot with compensated non-linearities using exact linearization, which is possible due to its flatness [43, p. 375 Theorem 66, pp. 212].

$$\begin{aligned} \boldsymbol{\chi} = \boldsymbol{q} &= \begin{bmatrix} x & y \end{bmatrix}^{\mathrm{T}} \\ \ddot{\boldsymbol{q}} &= \boldsymbol{\tau} \\ ||\boldsymbol{\tau}||_{\infty} &\leq \tau_{\max} \end{aligned} \tag{3.93}$$

.

The performance is presented in an environment constructed by a simple Rapidliy expanding Random Tree (RRT) planning algorithm so that it provides the characteristics of geometrically planned trajectories. The setup is shown in Figure 3-11. A planar space is presented that consists of a free space and an obstacle region. The path to be tracked is generated using a RRT whose sample points are given by the black crosses, and the interconnections are given in red. To construct a reference, the red trajectory is sampled equidistantly at 400 points. Thus a traversal at constant speed is desired. Furthermore, the robot starts at rest. In the final waypoint, the robot is also desired to arrive at rest. This is achieved by repetition of the final waypoint.

The final waypoint is added and repeated in the end so that the robot will also stop in rest at the end of the trajectory. The change of the direction at the waypoints requires an instantaneous change ot the velocity's direction and, thus, a singular impact of the control input.

$$|\tau_i| \leq \tau_{\max} \tag{3.94}$$

To verify if the controller also works in the light of model uncertainty, the controller of the nominal plant is applied to a uncertain plant. That is chosen to be a integrator and a PT1.

$$\ddot{\boldsymbol{q}} = -T_{\mathrm{D}} \cdot \dot{\boldsymbol{q}} + \boldsymbol{\tau} \tag{3.95}$$

The time constant is chosen to be $T_{\mathrm{D}} = 20\,\mathrm{s}$.

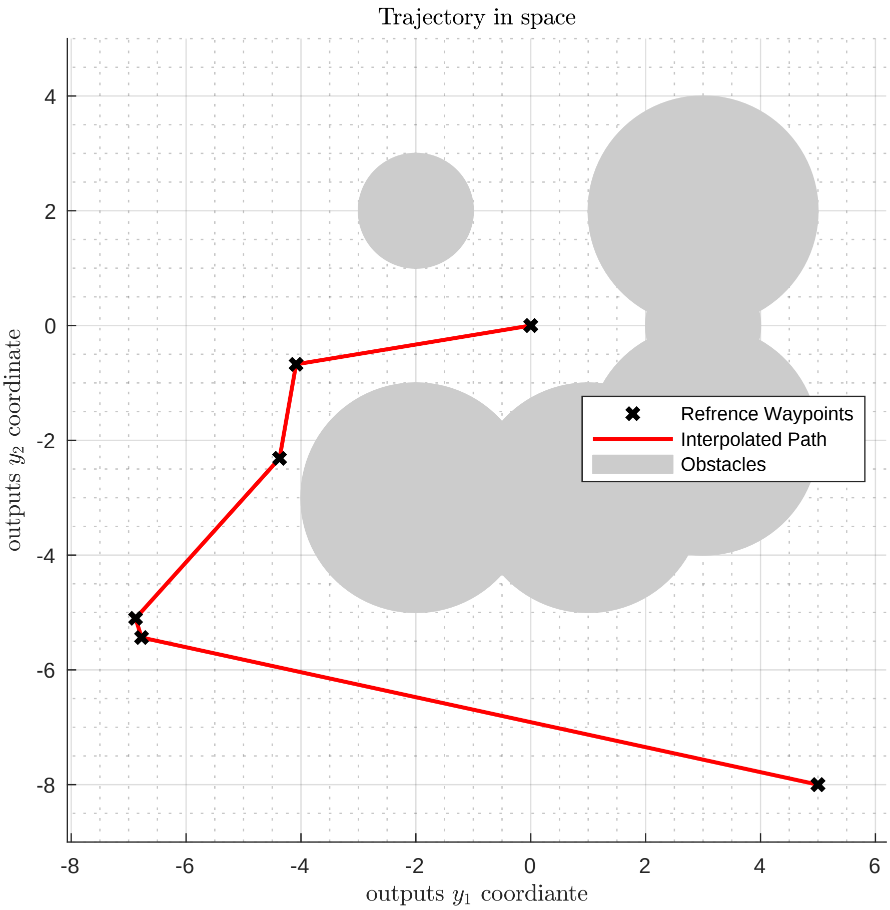


Figure 3-11. Problem setup for evaluations in the C-Space. The double integrator has to traverse a path generated by a RRT that does not take the dynamics into account. The time is for each segment is allocated proportional to its length.

In the latter a Single Input Multiple Output (SIMO) system is investigated to review the underacutated case. To do so the TWIPR model will be used. The investigation will start with the application of standard ILC to one chosen output, going to applying ILC to the under actuated system. Then converging P2P-ILC with and without ancillary cost term. The TWIPR dynamics are derived in Chapter 4. They take the format of a general under actuated robot.

$$\ddot{\boldsymbol{q}} = \boldsymbol{M}(\boldsymbol{q})^{-1}\left(-\boldsymbol{c}(\dot{\boldsymbol{q}},\boldsymbol{q})\dot{\boldsymbol{q}} - \boldsymbol{g}(\boldsymbol{q}) + \begin{bmatrix} \boldsymbol{\tau} \\ 0_n \end{bmatrix}\right) \qquad \boldsymbol{\tau} = \boldsymbol{u} \tag{3.96}$$

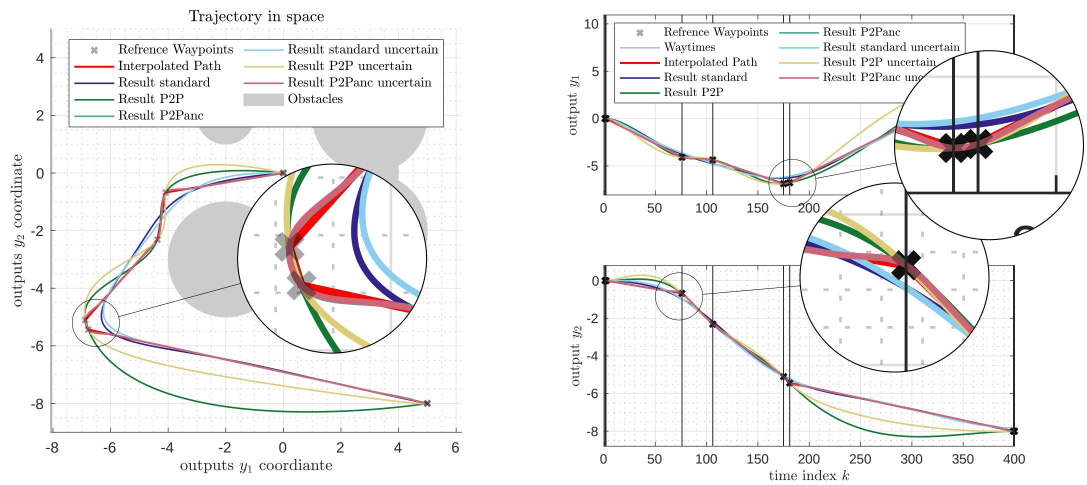
The representation is derived in detail in Chapter 4. The configuration in this case is given by $\boldsymbol{q} = \begin{bmatrix}\varphi & s_\mathrm{w}\end{bmatrix}^\mathrm{T}$.

The simulation study is done using `matlab` and the code can be found on `github`. As preliminary, the implementation of the sample systems is presented in the following.

The double integrator model (3.93) is discretized using the zero-order-hold method. The sampling period h is normalized to one w.o.l.g..

3.8.1 Unconstrained system

It is now investigated how the controller works in the absence of input constraints. How the controller is tuned is given in Table 3.3. The final trajectories are given in Figure 3.12(a) the Configuration space, in Figure 3.12(b) the evolution over time and in 3-13.



(a) Configuration Space.

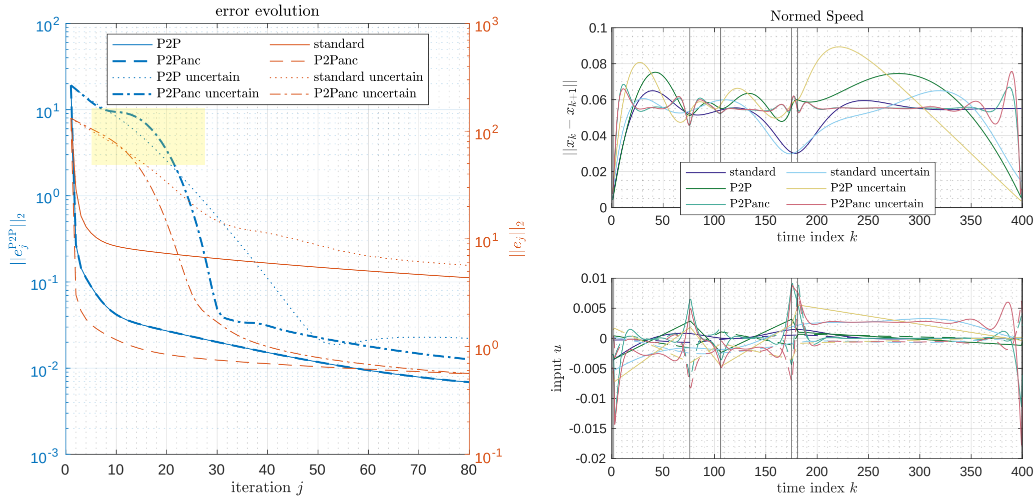
(b) Over time.

Figure 3-12. The ILC algorithms applied to a geometric plan. The unconstrained case.

Standard ILC	Point to Point ILC	Point to Point with anchillary ILC	
$\boldsymbol{Q}_{\mathrm{W}} = \boldsymbol{E}$	$\boldsymbol{Q}_{\mathrm{W}} = \frac{1}{{n_\Phi}^2}\boldsymbol{E}$	$\boldsymbol{Q}_{\mathrm{WP2P}} = \frac{1}{{n_\Phi}^2}\boldsymbol{E}$	$\boldsymbol{Q}_{\mathrm{Wanc}} = \boldsymbol{E}$
$\boldsymbol{S}_{\mathrm{W}} = 10^{-2}\,\|\|\boldsymbol{P}\|\|^2\,\boldsymbol{E}$	$\boldsymbol{S}_{\mathrm{W}} = 10^{-3}\frac{\|\|\boldsymbol{P}\|\|^2}{{n_\mathrm{b}}^2}\boldsymbol{E}$	$\boldsymbol{S}_{\mathrm{WP2P}} = 10^{-3}\frac{\|\|\boldsymbol{P}\|\|^2}{{n_\mathrm{b}}^2}\boldsymbol{E}$	$\boldsymbol{S}_{\mathrm{Wanc}} = 10^{-2}\left\|\left\|\boldsymbol{P}\boldsymbol{B}^{\perp}_{\boldsymbol{P}_\Phi}\right\|\right\|^2\boldsymbol{E}$
$\boldsymbol{Q} = \boldsymbol{E}$	$\boldsymbol{Q} = \boldsymbol{E}$	$\boldsymbol{Q}_{\mathrm{P2P}} = \boldsymbol{E}$	$\boldsymbol{Q}_{\mathrm{anc}} = \boldsymbol{E}$

Table 3.3. Controller tuning for the first simulation study.

All algorithms are learning over 80 trials. First, consider the results of the experiment on the nominal system. In configuration space, only the P2P-ILC achieved perfect tracking, while the trajectories of the other method do not converge to perfect tracking within 80 iterations. Both point-to-point methods track the path points to a satisfactory level. The best performance regarding the tracking behavior of the whole trajectory is achieved using the point-to-point controller with ancillary cost. In comparison, in the time evolution, the deviation of the standard ILC algorithm appears to be smaller, as it might be expected from the configuration space. Considering the norm evolution, it is observed that in the nominal case, all algorithms converge quickly in the early phase and



(a) Error convergence.

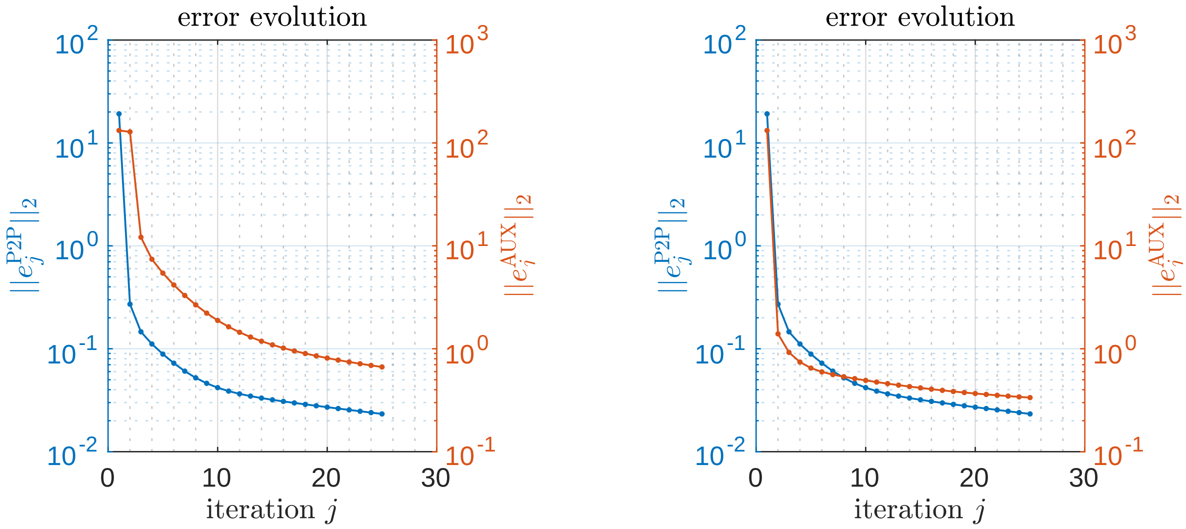
(b) Input trajectories.

Figure 3-13. Error convergence of the study. In the nominal case the point to point error is the same regardless of the consideration of an auxiliary cost. In the uncertain case in contrast the convergence differs. In the beginning, highlighted in yellow, the point to point error of the controller with ancillary cost performs worse than the pure point to point controller, but in the latter it outperformed the pure point to point method till the 80th trial.

eventually saddle to a slower convergence rate. Perfect tracking can not be expected to occur in a reasonable amount of iterations, as the improvement is one order of magnitude in approximately 300 trials. The point-to-point error is independent of the optimization of the ancillary cost, as expected in theory. Taking the uncertain system into account, the results in configuration space differ for the point-to-point and the standard algorithm. But, regarding the tracking behavior, the results coincide with those observed on the nominal system. The same holds true for evolution over time. The solution trajectories of the point-to-point method with ancillary cost function are almost indistinguishable for certain and uncertain plants. The error convergence differs significantly. The previously fast convergence in the early phase is not present anymore. But, all algorithms converge faster in the earlier phase and saddle eventually. In this case, also the error convergence of the point-to-point error trajectories do not coincide with and without ancillary optimization. In the highlighted phase, the point-to-point error suffers, introducing the ancillary optimization, while in the latter, it is supported. As it is discussed in theory, in the unconstrained case, the and the point to point algorithms with the ancillary cost are equivalent to linear ILC update. Therefore, this observation is not surprising. In this particular case, the ancillary cost also supports the point-to-point tracking rendering a conflict that occurs if the ancillary cost problem optimizes quantities like the energy consumption of the actuator.

Concluding, this study serves as a proof of concept that the point-to-point algorithm is beneficial for planned trajectory. The introduction of an ancillary cost results in desirable tracking behavior, especially in the waypoints. This is helpful for robotic tasks in cluttered environments where precise tracking is mandatory for the whole trajectory.

To validate the compensation introduced in Section 3.5 (see also Figure 3-6) the algorithm is performed once with and once without the compensation, and the corresponding error norm



(a) Uncompensated.

(b) Compensated.

Figure 3-14. Comparison of the error evolution if the feedforward compensation is introduced. One can observe that the error norm of the whole trajectory converges faster if the compensation is implemented.

evolution is depicted in Figure 3-14. It can be observed that the ancillary error converges faster in the compensated case.

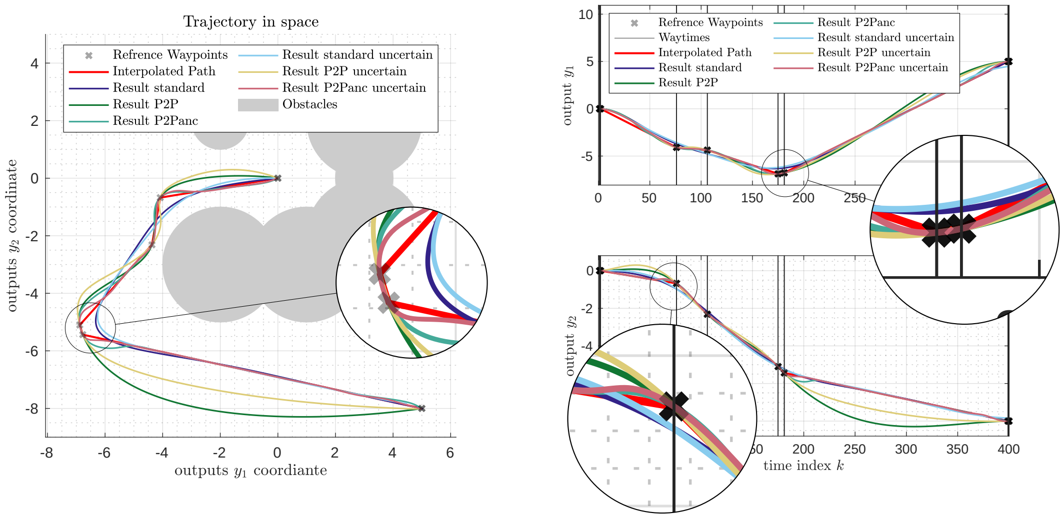
3.8.2 Constrained system

The same study is repeated for constrained inputs. The results are presented in Figure 3.15(b) to 3.16(a). The results turn out to be similar. The constraints are set to $\boldsymbol{\tau}_{\max} = 2.5 \times 10^{-3}$ for prefect plant knowledge and to $\boldsymbol{\tau}_{\max}$ for the uncertain (damped) system respectively.

First, consider the case with exact plant knowledge. The point-to-point error evolution coincides with the case with and without ancillary optimization, as the systems are perfectly decoupled. But contrarily to the previous case, the error on the whole trajectory no longer decays monotonically. This is due to conflicting optimizations. As the point-to-point error is prioritized, it is further optimized at the expense of the ancillary cost. This occurs in constrained and in unconstrained cases. That the constraints are active is obvious in Figure 3.16(b). Furthermore, they become active if the trajectory passes a waypoint. This leads to a different shape than the previous study, as observed in Figure 3.15(a).

Takeaways

This simulation study shows how trajectory generation can be omitted for fully actuated robots using learning control methods such as point-to-point ILC. Furthermore, it turned out that, even in the unconstrained case, the tracking of the waypoints does not suffer. That is surprising, considering the results found in theory. This observation is likely due to the compatibility of the point-to-point cost and the ancillary cost.



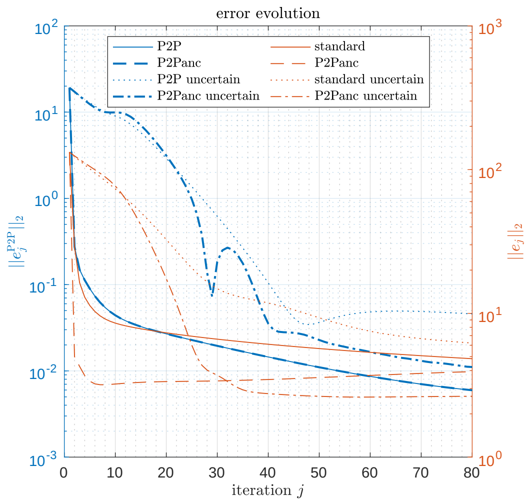
(a) Configuration space.

(b) Over time.

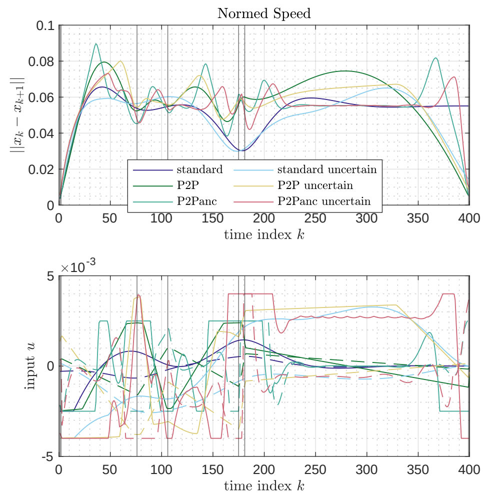
Figure 3-15. The proposed ILC algorithms applied to the proposed problem presented in the configuration space (left) and over time (right). The trajectory is tracked well in the straight parts between the way points in the corners it is not. In the limit the tracking will be perfect but convergence has not completed yet.

Case	Source File
Standard ILC unconstrained	`exp_1_standard.m`
Point to Point ILC unconstrained	`exp_1_p2p.m`
Point to Point with ancillary ILC unconstrained	`exp_1_aux_opt_freeman.m`
Standard ILC constrained	`exp_2_standard.m`
Point to Point ILC constrained	`exp_2_p2p.m`
Point to Point with ancillary ILC constrained	`exp_2_aux_opt_freeman.m`
Point to Point with ancillary ILC uncompensated	`exp_1_aux_opt_freeman_uncompensated.m`
Point to Point with ancillary ILC compensated	`exp_1_aux_opt_freeman.m`

Table 3.4. Source of the Simulation studies. They are located in the git in `20_Comp_Double_Integrator`. The comparisons with and without friction are conducted using the script `exp_2_comparison.m`.



(a) Error convergence.



(b) Input trajectories.

Figure 3-16. Error convergence (left) of the study. In the nominal case, the point-to-point error is the same regardless of the consideration of an ancillary cost. In the uncertain case in contrast, the convergence differs. Compared to the unconstrained case, the ancillary cost does not decay monotonically since the point-to-point cost is prioritized in exploiting constraints. The input trajectories (right) reflect that the constraints are active.

3.9 Case Study Two Wheeled Inverted Pendulum Robot (TWIPR) with low amplitude

This study is drawn to evaluate the performance of the ILC algorithms on an under-actuated system. In contrast to the former study, the robot model that is reviewed is that of the TWIPR, which is under-actuated. The reference trajectory is generated using a given pitch, i. e. a sinusoidal trajectory and the corresponding position trajectory. The position trajectory is obtained using the linearized model. The study observed that single input single output controllers achieve good tracking behavior but lead to bad results in position tracking. The underactuated multiple output control improves the position at the expense of a deviation in the pitch. The point-to-point algorithms improve the tracking in the waypoints in both variables, while greater clearance leads to better tracking. Incorporating an ancillary cost function improves the pitch error in the cases with a clearance larger or equal to 20. In the case with a period of only 10 samples, the ancillary cost almost did not influence the solution in the configuration space. The corresponding simulation scripts can be found in the files listed in Table 3.5. Metrics are also calculated and presented in Table 3.6 and given in the barplot Figure 3-17. In this section, results are first discussed qualitatively based on results in the configuration space. Then the link to the metrics is done.

The methods that have been tested are the following (The term in brackets is the intuitive formulation inspiring the abbreviation.):

SISO (Single Input Single Output) Reference tracking only on the pitch trajectory.

SIMO (Single Input Multiple Output) Reference tracking on both pitch and position.

P2P10 (Point to Point equidistant with a period of 10) Point to Point iterative learning control with a period of 10 samples.

P2P20 (Point to Point equidistant with a period of 20) Point to Point iterative learning control with a period of 20 samples.

P2Panc10 (Point to Point equidistant with a period of 10 and ancillary optimization) Point to Point iterative learning control with ancillary optimization with a period of 10 and an ancillary cost that optimizes reference tracking only on the pitch trajectory.

P2Panc20 (Point to Point equidistant with a period of 20 and ancillary optimization) Point to Point iterative learning control with ancillary optimization with a period of 20 and an ancillary cost that optimizes reference tracking only on the pitch trajectory.

P2Pstrat (Point to Point strategically selected) Point to point iterative learning control with samples manually chosen in the beginning as the pitch takes its maximum values to rise the importance of the points that are in the narrow area of the configuration space. The samples are chosen with a periodicity of 5 samples. Furthermore, the goal, i.e. the last point of the trajectory, is included.

P2PancStrat (Point to Point strategically selected with ancillary optimization) The same approach as the former but with ancillary optimization of the pitch apart the waypoints.

P2PstratBW (Point to Point strategically selected for backward dive) The same as P2Pstrat but with the way point strategically chosen in latter phase of the dive so that they mimic dive of the TWIPR leaning backwards.

P2PoneWP (Point to Point with one way point) Only one waypoint is selected in early phase and one in the end.

P2PancOndeWP (Point to Point with one way point and ancillary optimization) Same as the former but with ancillary optimization of the pitch.

Name	Source File
SISO	`EXP_std_SO_ref45.m`
SIMO	`EXP_std_MO_ref45.m`
P2P10	`EXP_std_MO_P2P10_ref45.m`
P2P20	`EXP_std_MO_P2P20_ref45.m`
P2Panc10	`EXP_std_MO_P2Panc10_ref45.m`
P2Panc20	`EXP_std_MO_P2Panc20_ref45.m`
P2Pstrat	`EXP_std_MO_P2Pstrat_ref45.m`
P2PancStrat	`EXP_std_MO_P2PancStrat_ref45.m`
P2PstratBW	`EXP_std_MO_P2PstratBW_ref45.m`
P2PoneWP	`EXP_std_MO_P2PoneWP_ref45.m`
P2PancOneWP	`EXP_std_MO_P2PancOneWP_ref45.m`

Table 3.5. Experiment scripts in the software developed. Git path `7_Exp45degVgl`.

Name	Pitch Err in °	Pos Err in cm	Goal Pitch Err in °	Goal Pos Err in cm	WP Pitch Err in °	WP Pos Err in cm
SISO	1.28×10^{-1}	9.87	7.91×10^{-1}	2.29×10^{1}		
SIMO	2.99	1.34	1.14×10^{1}	2.20		
P2P10	2.97	1.01	1.30	2.14	2.19	1.19
P2Panc10	2.99	1.03	1.30	2.14	2.19	1.19
P2P20	4.90	5.58×10^{-1}	3.09×10^{-1}	8.52×10^{-1}	8.07×10^{-1}	6.68×10^{-1}
P2Panc20	4.24	4.50×10^{-1}	3.10×10^{-1}	8.48×10^{-1}	8.16×10^{-1}	6.66×10^{-1}
P2Pstrat	5.99	2.39	1.82	1.46	9.69×10^{-1}	8.53×10^{-1}
P2PancStrat	5.31	1.84	2.17×10^{-1}	1.92×10^{-1}	1.86×10^{-1}	3.79×10^{-1}
P2PstratBW	4.10	1.21	1.81	3.31	1.41	1.65
P2PoneWP	8.01	3.70	1.03×10^{-1}	8.81×10^{-2}	1.34×10^{-1}	2.19×10^{-1}
P2PancOneWP	3.98	2.58×10^{-1}	1.06×10^{-1}	8.32×10^{-2}	1.50×10^{-1}	2.31×10^{-1}

Table 3.6. Metrics observed in the first simulation study.

The tuning of the applied controllers is presented in Table 3.7. The controllers are tuned so that the convergence is reasonably fast to converge within 30 iterations. The weight selection is made on the author's experience. As the convergence is of minor interest in the following, only the final trajectories are discussed. The controller acts on pitch and position as output, and the selector are also chosen to track both quantities. Thus, they are not repeated. In the table 3.7 are given the matching size identity matrices.

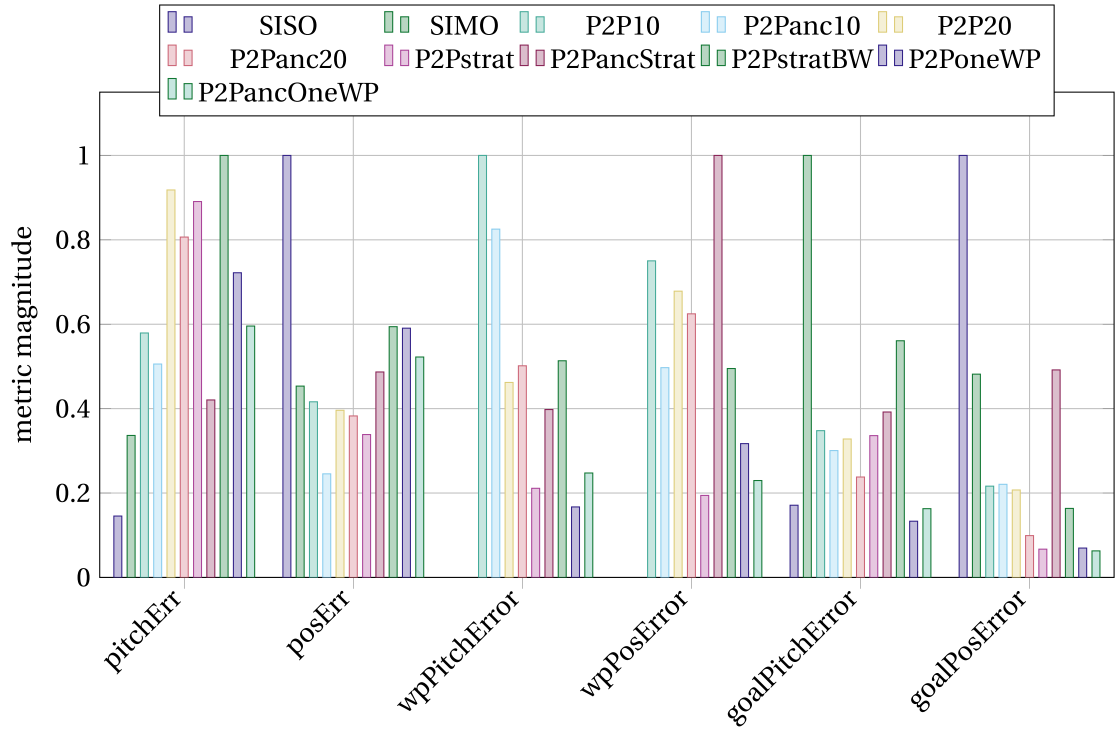


Figure 3-17. Metrics observed in the first case study. The maximal value is normed to one for each metric.

Name	$\boldsymbol{Q}_\mathrm{W}$	$\boldsymbol{S}_\mathrm{W}$	$\boldsymbol{R}_\mathrm{W}$	$\boldsymbol{Q}_\mathrm{Wanc}$	$\boldsymbol{S}_\mathrm{Wanc}$	$\boldsymbol{Q}_\mathrm{Wanc}$
SISO	$\boldsymbol{E}$	$0.25\boldsymbol{E}$	$10^{-3}\boldsymbol{E}$			
SIMO	$\boldsymbol{E}\otimes\mathrm{Diag}\,[[0.4 \quad 2]]$	$0.25\boldsymbol{E}$	$10^{-3}\boldsymbol{E}$			
P2P10	$\boldsymbol{E}\otimes\mathrm{Diag}\,[[4 \quad 20]]$	$0.25\boldsymbol{E}$	$\boldsymbol{0}$			
P2Panc10	$\boldsymbol{E}\otimes\mathrm{Diag}\,[[4 \quad 20]]$	$0.25\boldsymbol{E}$	$\boldsymbol{0}$	$\boldsymbol{E}\otimes\mathrm{Diag}\,[[1 \quad 0]]$	$0.25\boldsymbol{E}$	$10^{-3}\boldsymbol{E}$
P2P20	$\boldsymbol{E}\otimes\mathrm{Diag}\,[[8 \quad 40]]$	$0.25\boldsymbol{E}$	$\boldsymbol{0}$			
P2Panc20	$\boldsymbol{E}\otimes\mathrm{Diag}\,[[8 \quad 40]]$	$0.25\boldsymbol{E}$	$\boldsymbol{0}$	$\boldsymbol{E}\otimes\mathrm{Diag}\,[[1 \quad 0]]$	$0.25\boldsymbol{E}$	$10^{-3}\boldsymbol{E}$
P2Pstrat	$\boldsymbol{E}\otimes\mathrm{Diag}\,[[4 \quad 20]]$	$0.25\boldsymbol{E}$	$\boldsymbol{0}$			
P2PancStrat	$\boldsymbol{E}\otimes\mathrm{Diag}\,[[4 \quad 20]]$	$0.25\boldsymbol{E}$	$\boldsymbol{0}$	$\boldsymbol{E}\otimes\mathrm{Diag}\,[[1 \quad 0]]$	$0.25\boldsymbol{E}$	$10^{-3}\boldsymbol{E}$
P2PstratBW	$\boldsymbol{E}\otimes\mathrm{Diag}\,[[4 \quad 20]]$	$0.25\boldsymbol{E}$	$\boldsymbol{0}$			
P2PoneWP	$\boldsymbol{E}\otimes\mathrm{Diag}\,[[8 \quad 40]]$	$0.25\boldsymbol{E}$	$\boldsymbol{0}$			
P2PancOneWP	$\boldsymbol{E}\otimes\mathrm{Diag}\,[[8 \quad 40]]$	$0.25\boldsymbol{E}$	$\boldsymbol{0}$	$\boldsymbol{E}\otimes\mathrm{Diag}\,[[1 \quad 0]]$	$0.25\boldsymbol{E}$	$10^{-3}\boldsymbol{E}$

Table 3.7. Controller tuning for the first simulation study.

The reference trajectory is generated using a sinusoidal on the pitch and computing the corresponding position trajectory using the linearized model without friction. Practically this is done using the lifted dynamics representation. Therefore, two lifted dynamics are computed. One $\left(\boldsymbol{P}_\varphi, \boldsymbol{d}_\varphi\right)$ only considers the pitch as output and, therefore, regular. A second $(\boldsymbol{P}, \boldsymbol{d})$ considers both quantities pitch and position as outputs.

$$r_\varphi(k) = \hat{r}_\varphi \sin\left(2\pi \frac{k}{n_\mathrm{b}}\right) \tag{3.97}$$

$$\boldsymbol{r} = \boldsymbol{P}\boldsymbol{P}_\varphi^{-1}\left(\boldsymbol{r}_\varphi - \boldsymbol{d}_\varphi\right) + \boldsymbol{d} \tag{3.98}$$

Metrics of the reuslts are given in Table 3.6. All of them are the Root Mean Square Error (RMSE) (3.99) of all trajectory points of intrest.

$$\mathrm{RMSE}_y = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - r_i\right)^2} \quad [44] \tag{3.99}$$

Pitch Err and **Pos Err** are the RMSE of the whole pitch and position trajectory regardless of the fact if it was of interest for the control algorithm. **Goal Pitch** and **Goal Pos** only consider the final point. **WP Pitch** and **WP Pos** consider all points that are assigned as a waypoint for the Point to Point controller (and, therefore, does not exist for the SISO and SIMO). A bar chart is given in Figure 3-17 to get a visual impression of the data. To enhance compatibility, the values are indexed so that the experiment with the largest RMSE takes the value of one, i. e. the maximum of each metric is one, while the ratio is maintained with regard to the original values.

To evaluate the trajectories that are realized, they are represented in the configuration space in Figures 3-18 to 3-22. Each plot shows a selection of special experiments discussed in the following.

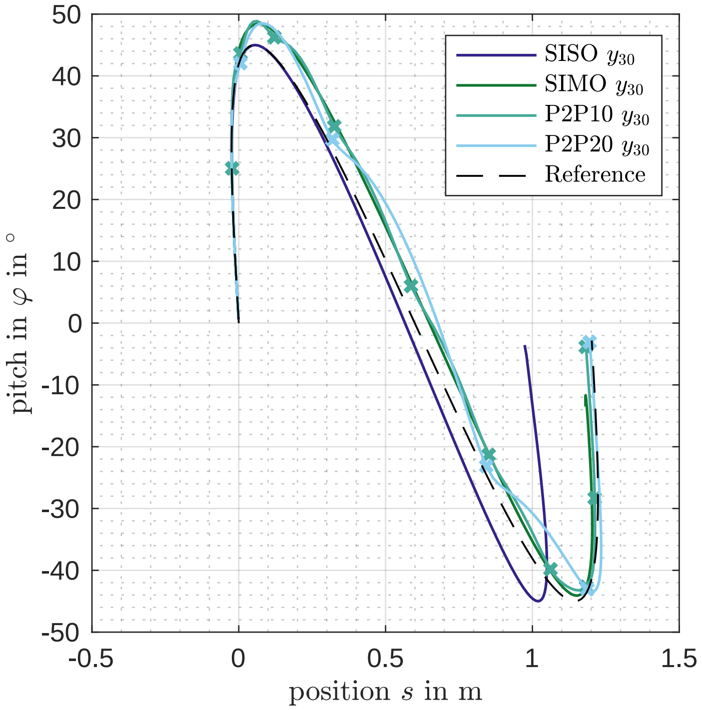


Figure 3-18. Comparison of the trajectories obtained in the case study on the reference movement. Given are the Single and Multiple Output cases as well as the Point to Point methods with periods of 10 and 20 samples.

The qualitative discussion of the results starts with the Single Input Single Output (SISO) case, which forms the basis for the state of the project [4]. It is compared to the Single Input Multiple Output (SIMO), and Point-to-Point Iterative Learning Control (P2P-ILC) approaches in Figure 3-18. It is observed that the tracking behaiviour of the pitch trajectory is almost perfect but in contrast, is not perfect in the position. In the figure, the trajectory is given in the configuration space, which is the basis for planning problems. Changing the cost so that both pitch and position are taken into account, the tracking fails for both variables. More precisely, the tracking in the position improves at the expense of the pitch. This demonstrates the major problem regarding planned trajectories. The solution deviates in a way that is not controlled and does not take specific features into account. That is e. g. the clearance at a given point or the fact that some specific points must be passed to fulfill the task. As discussed in theory, this problem can be overcome as the requirement is relaxed to track dedicated tracking points, namely the point-to-point method (Section 3.4). This changes the state of the control problem from under to over-actuated and therefore allows precise tracking. In the case of an equidistant sampling with a periodicity of 10 samples, no fundamental improvement is observed. The trajectory has changed a little but qualitatively did not lead to a major improvement regarding precise tracking of the waypoints. Also, the trajectory stays close to the SIMO solution apart from the waypoints. The solution of the point-to-point iterative learning control with a clearance of 20 samples significantly improved the tracking of the waypoints. This comes at the expense of more deviation apart from the waypoints. The author supposes that this fact is due to the higher damping of the linearized model for the frequency of 2.5 Hz in contrast to 5 Hz, see also Figure 4-6, where it is observed that the damping differs by about 10 dB corresponding to a factor of 3. As observed for the case with a period of 20 samples, the tracking of the waypoints requires a periodic compensation movement in between the waypoints. As this compensation needs to be twice the frequency of the case with a sample period of 10 samples, it converges slower using the Norm Optimal Iterative Learning Control (NO-ILC) design strategy. Inputs of a higher frequency need higher amplitude in the input to pass the system dynamics. Thus the optimal update balancing input update and output cost is slower. I.e., the convergence is slower. Furthermore, the demand for tracking the points may exceed the system's capabilities.

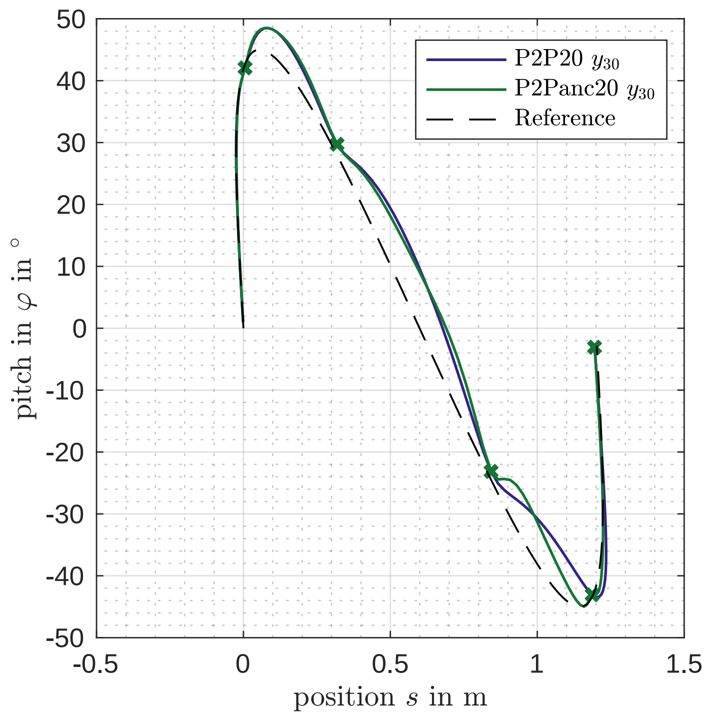


Figure 3-19. Comparison of the trajectories obtained in the case study on the reference movement. Presented are the cases point to point cases with a period of 20 samples once with ancillary cost and one without.

In the next step, the influence of the ancillary cost is reviewed. Therefore, observe the trajectories of the point-to-point with a clearance of 20 samples or 2.5 Hz in Figure 3-19. One observes no very obvious changes in the trajectories. The only difference is in the phase form 0.8 m to 1.2 m, where the reference is closer to the reference trajectory given in dashed black. The influence is more significant in the strategic selection case presented in Figure 3-20, which compares the strategic selection once with and once without ancillary cost on the pitch. It is observed that the trajectory differs significantly apart from the waypoints. The shape mimics the reference trajectory but is shifted by approximately 8°. This is due to the requirement to arrive at the goal location.

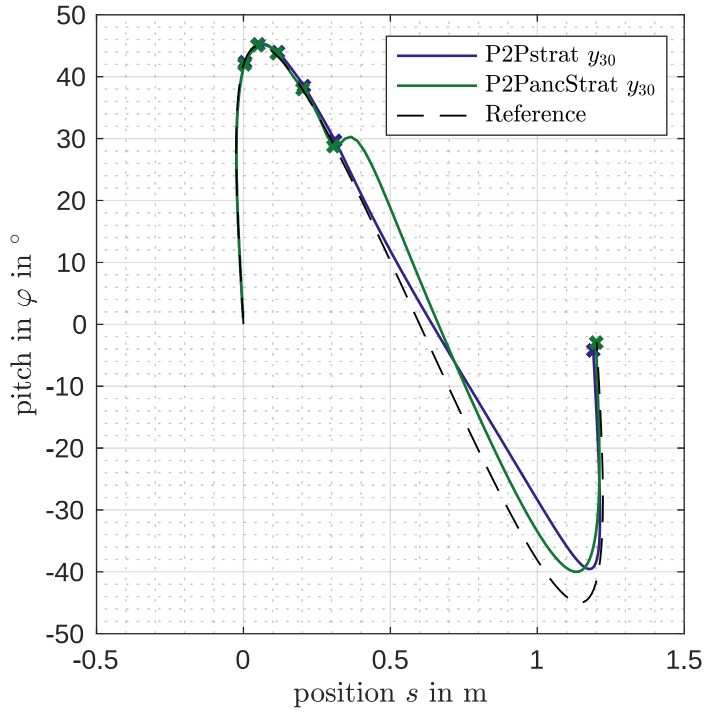


Figure 3-20. Comparison of the trajectories obtained in the case study on the reference movement. Presented are the cases point to point cases with strategic sample selection once with ancillary cost and one without.

To get a clearer understanding of the result the trajectories are also presented in the time domain in Figure 3-21. That the shape the diffrence in the trajectories shapes is also clearly visible. It also shows that the pitch trajectory mimics the reference apart from the way points, while it tracks the reference well within the way points (i. e. 0.3 s to 0.8 s). Recover, the ancillary cost is only proposed on the pitch trajectory but not on the position. This shows that the ancillary optimization works and also the decoupling is working. In the time domain it is also visible that the pitch trajectory has a break a the time instant 0.75 s. At this instant, the impact of the primary controller, the point to point controller, vaniches so that the ancillary shapes the output. This behaviour occur not very natural in a movement.

Finally, the experiments with strategic selection are compared in Figure 3-22. The three experiments, P2Pstrat, P2PoneWP, and P2PstratBW, are presented. It can be observed that trajectories differ in shape. While the strategic selection mimics the trajectory in the earlier phase (from 0 m to 0.3 m), the method that only takes care of one waypoint, namely P2PoneWP, steers to this in a different shape, more directly. Furthermore, it is shown that the selection for the backward dive also realized

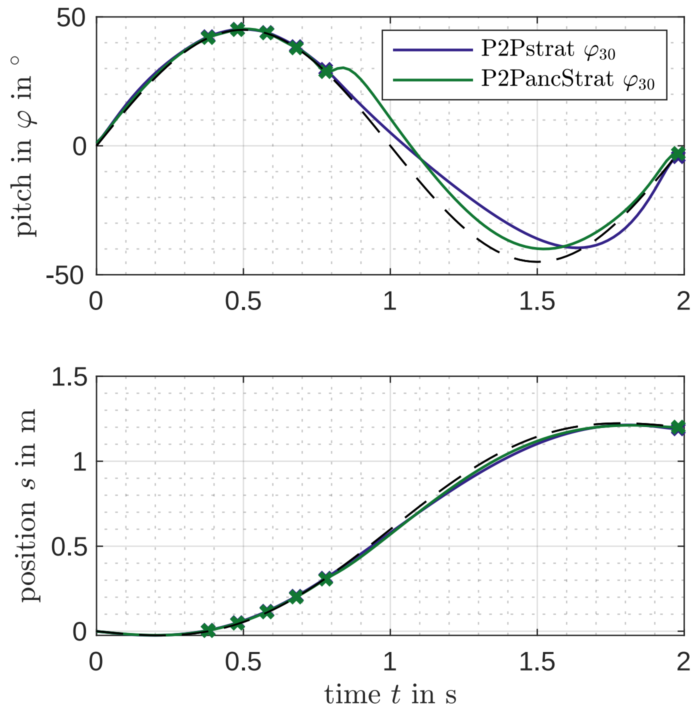


Figure 3-21. Comparison of the trajectories obtained in the case study on the reference movement. Presented are the cases point to point cases with strategic sample selection once with ancillary cost and one without in the time domain.

desired behavior. The tracking is the best in the phase from 0.8 m to 1.2 m, while it deviates in the earlier phase. But in contrast to the other strategic methods, the error in the waypoints and in the goal location is higher. This is also observed in the metrics (Table 3.6) as the waypoint error in pitch and position in about 40 % (pitch) to 70 % (position) larger. In the goal configuration, the error in the position is almost twice as big as that of the P2Pstrat experiment. In the pitch variable, the error is almost equal. The author supposes this is due to a specific conflict for the breaking maneuver.

Concluding the simulation study the following conclusions are drawn.

- Tracking only the pitch leads to undesired and hard-to-predict deviation in the position. Thus, it is not useful to fulfill task-specific requirements.
- Introducing a cost on the pitch as well leads to a compromise that does not prioritize task-specific features.
- Equidistant sampling of the trajectory with a period of 10 samples or a frequency of 5 Hz does not change the result in comparison to the multiple output case.
- In contrast, equidistant sampling with a clearance of 20 samples or a frequency of 2.5 Hz leads to better tracking at the waypoints. The expense is a more significant deviation in between the waypoints.

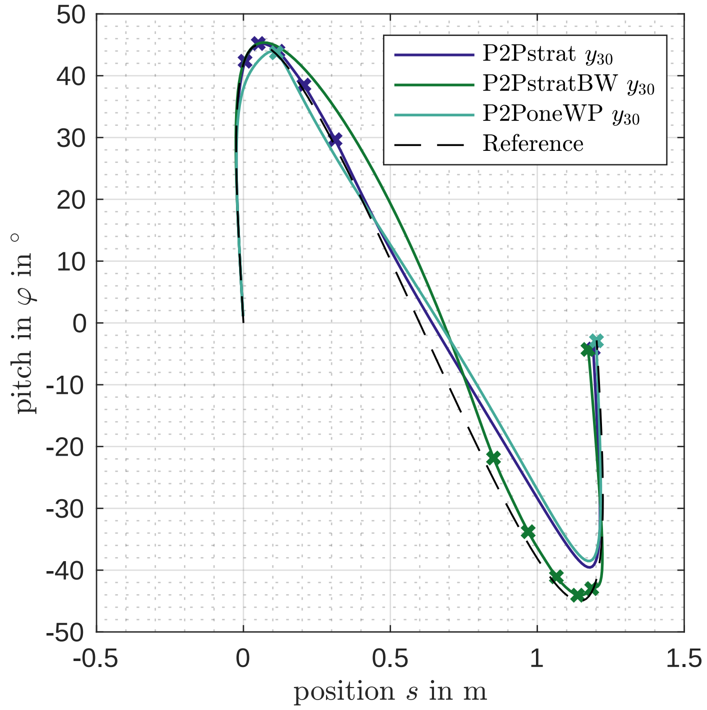


Figure 3-22. Comparison of the trajectories carried out by the point to point controller comparing different sampling strategies.

- Strategic sampling of a portion of the trajectory with a period of 5 samples or 10 Hz leads to precise tracking in the selected phase. It spreads the compensation to the rest of the trajectory. This holds for both the forward and the backward dive, while the backward dive appears to be more challenging.

- The ancillary cost leads to a result that mimics the reference pitch to the extent that the primary problem is solved.

3.9.1 Fulfill a task specific requirement using point to point ILC

This simulation demonstrates how P2P-ILC is used to solve a simple motion task. The task is to traverse from one spot to another. An additional demand is to arrive in rest, meaning that the robot's velocity and the pitch's angular velocity is zero. This leads to a mini example as the outputs to be tracked are only of dimension four, that is, the full state information at one instance in time. Furthermore, the requirement to arrive at a specified location at rest is a major requirement in motion tasks.

In this example, the outputs of the controller are the robots' configuration only $q = \begin{bmatrix} \varphi & s_{\mathrm{w}} \end{bmatrix}$ with a duration of 100 samples or 2 seconds. Therefore, the velocity needs to be calculated within the cost function. The algorithm chosen is a point-to-point iterative learning controller, synthesized using NO-ILC. Constraints are not considered. To tune this controller, the following steps are performed:

- Design the selector
- Choose the weights on the output
- Choose the weights on the input update
- Choose the weights on the input

The selector is designed according to Definition 20.

$$\mathcal{P} = \left(\left(k_1 = 99, \boldsymbol{\Phi}_1 = \boldsymbol{E}, \boldsymbol{r}_1 = \begin{bmatrix} 0° \\ 150\,\mathrm{cm} \end{bmatrix} \right), \left(k_2 = 100, \boldsymbol{\Phi}_2 = \boldsymbol{E}, \boldsymbol{r}_2 = \begin{bmatrix} 0° \\ 150\,\mathrm{cm} \end{bmatrix} \right) \right) \tag{3.100}$$

This means that the last two samples of the trajectory are selected. This enables the calculation of the velocity in the goal location.

The following weights on the output and its derivative are chosen.

$$\boldsymbol{Q}_{\mathrm{W}}^{(\boldsymbol{q}_{\mathrm{fin}})} = \begin{bmatrix} 8 & 0 \\ 0 & 40\,\mathrm{m}^{-2} \end{bmatrix} \tag{3.101}$$

$$\boldsymbol{Q}_{\mathrm{W}}^{(\dot{\boldsymbol{q}}_{\mathrm{fin}})} = \begin{bmatrix} 8\,\mathrm{s}^2 & 0 \\ 0 & 40\,\mathrm{s}^2\,\mathrm{m}^{-2} \end{bmatrix} \tag{3.102}$$

$$\boldsymbol{S}_{\mathrm{W}} = 0.05\,\mathrm{s}^2 \boldsymbol{S}_{\mathrm{W}D} \tag{3.103}$$

$$\boldsymbol{R}_{\mathrm{W}} = \boldsymbol{0} \tag{3.104}$$

The overall cost function is composed as follows:

$$\begin{aligned} J = {} & \boldsymbol{y}_{\Phi j}^{\mathrm{T}} \left(\begin{bmatrix} -\boldsymbol{E} \\ \boldsymbol{E} \end{bmatrix} \boldsymbol{Q}_{\mathrm{W}}^{(\dot{\boldsymbol{q}}_{\mathrm{fin}})} \begin{bmatrix} -\boldsymbol{E} & \boldsymbol{E} \end{bmatrix} \frac{1}{h^2} + \begin{bmatrix} \boldsymbol{0} \\ \boldsymbol{E} \end{bmatrix} \boldsymbol{Q}_{\mathrm{W}}^{(\boldsymbol{q}_{\mathrm{fin}})} \begin{bmatrix} \boldsymbol{0} & \boldsymbol{E} \end{bmatrix} \right) \boldsymbol{y}_{\Phi j} \\ & + \left(\boldsymbol{u}_j - \boldsymbol{u}_{j-1} \right)^{\mathrm{T}} 0.05\,\mathrm{s}^2 \boldsymbol{S}_{\mathrm{W}D} \left(\boldsymbol{u}_j - \boldsymbol{u}_{j-1} \right) \end{aligned} \tag{3.105}$$

Therein, the first term serves the computation of the derivative of the output using Finite Differences (FD). The second term is the static cost of the goal. The third is the input derivative cost, where $\boldsymbol{S}_{\mathrm{W}D}$ is designed as it is explained in Section 3.6.5 if it is designed so that the input derivative is weighted only.

The result in the configuration space is given in Figure 3-23. The presented trajectory is obtained with the simulation of the controlled model (4.52) labeled sim. In top plot presents the trajectory in the configuration space, and the bottom plot the robot in the workspace. The lines correspond to the top edges of the robot. An obstacle is included in the workspace and configuration space

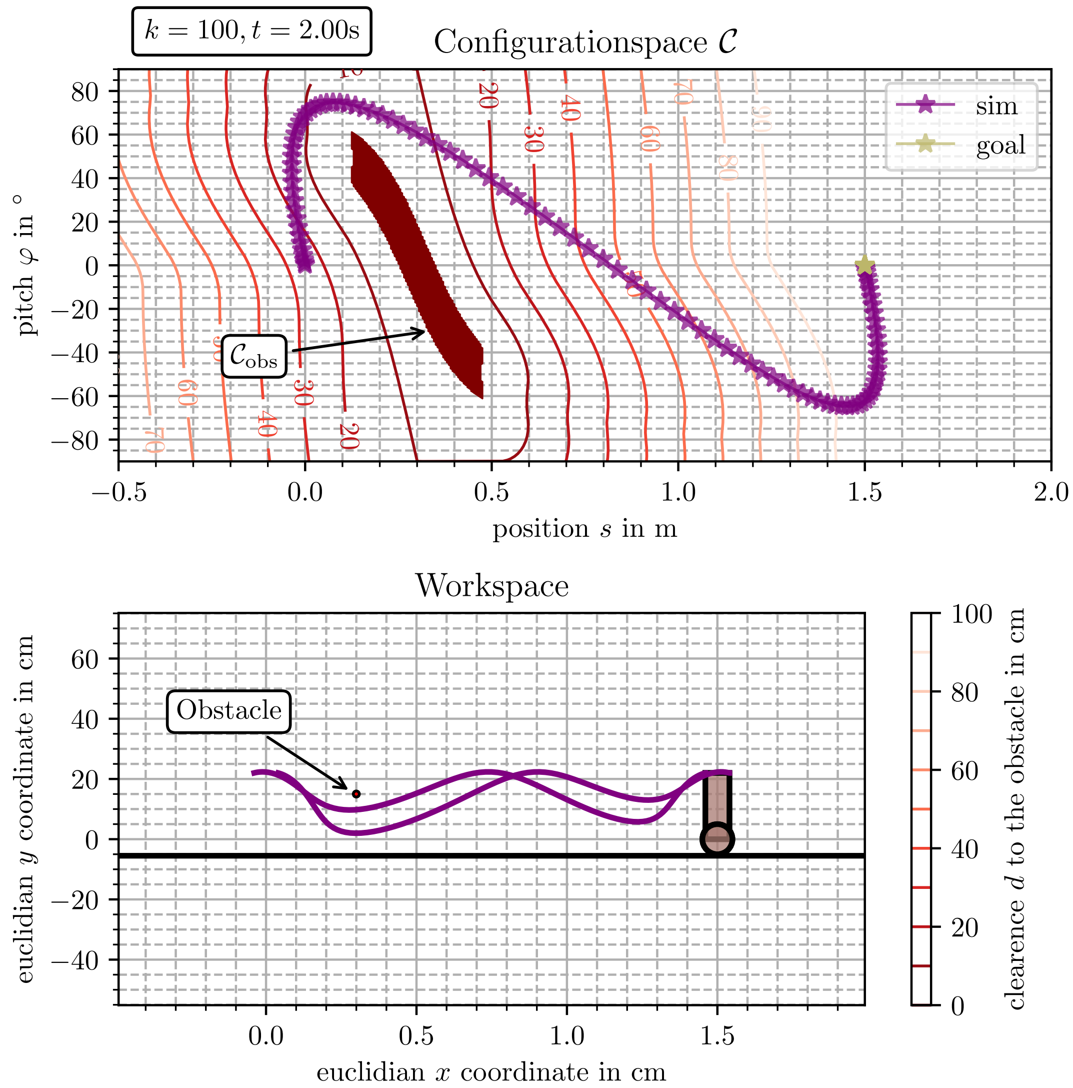


Figure 3-23. Result of the simulation of the simple point to point task using point to pint ILC.

to illustrate that the movement also solves a dive maneuver even though it is not taken care of in planning or control.

The resulting controller metrics are. The values are calculated using the results form Table 3.1 :

- $\rho \approx 0.74$.
- $\epsilon \approx 35$.

The metrics show that the controller is stable on the nominal model. The value of the spectral radius is surprisingly small, as in the preliminary project, a controller without input weights $\boldsymbol{R}_{\mathrm{W}} = \boldsymbol{0}$ the spectral radius is always one [4, p. 50 Figure 6.7]. Furthermore, input trajectories evolve in the image of the learn rule $\boldsymbol{L}$ that is of dimension 4. In contrast to standard ILC that forms a discrete system in the trail domain with a state that is the full input vector. So a usually high dimensional vector. Thus, the dynamics are characterized only by four eigenvalues, which are 0.7413, 0.2052, 0.0747, and 0.0233.

3.9.2 Simulation on the reference cases

In this simulation study the proposed Hybridizations 3.3 to 3.5 are tested.

Name	**Source File**
Window MO	`EXP_std_MO_constraint_MO_cost_small_portion.m`
Window SO + one WP	`EXP_std_MO_constraint_one_WP_small_portion.m`
3 Way points	`EXP_std_MO_constraint_P2P_NPP.m`

Table 3.8. Scripts that realize the methods in Simulation and experiment. Git path `8_Exps_Hybridization`.

The first evaluation is done in for the reference trajectory *Delayed Dive* they are presented in the configuration space in Figure 3-24. The trajectories displayed are the converged 30th iteration. The Hybridization 3.3 is given in red, labeled with MO importance. The Hybridization 3.4 in blue labeled with three waypoints. The selected waypoints are marked with crosses. The method Hybridization 3.5 is given in green. For the red and green trajectories, the values samples with non-zero weights are highlighted.

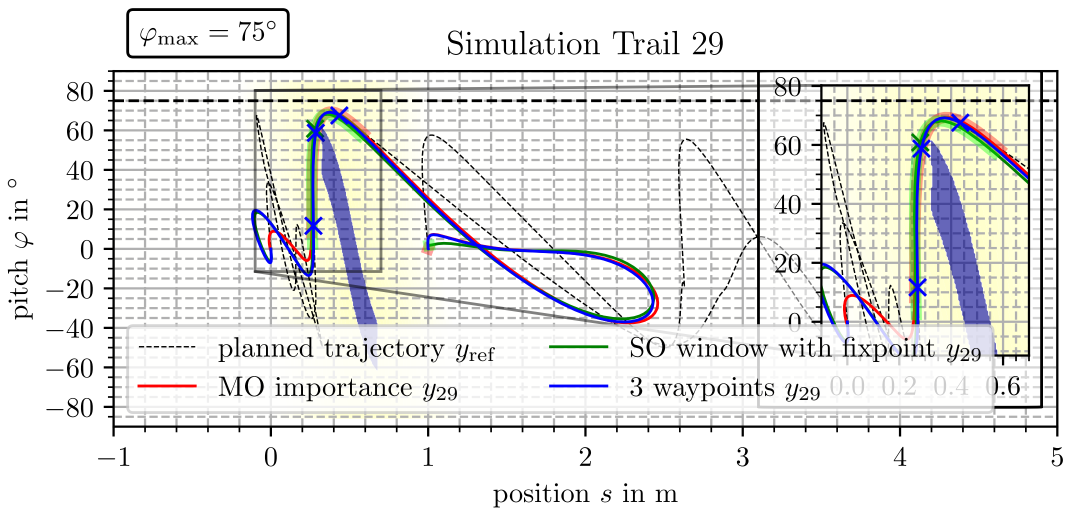


Figure 3-24. Final trajectory produced by the iterative learning control in the delayed dive reference case.

The first observation is that the three trajectories are very similar, and all of them solve the motion task, as the trajectories do not intersect the obstacle region $\mathcal{C}_{\text{obs}}$ highlighted in blue. The green and the blue one are almost indistinguishable when the actual passing movement is performed (i. e. 0.3 m to 0.75 m).

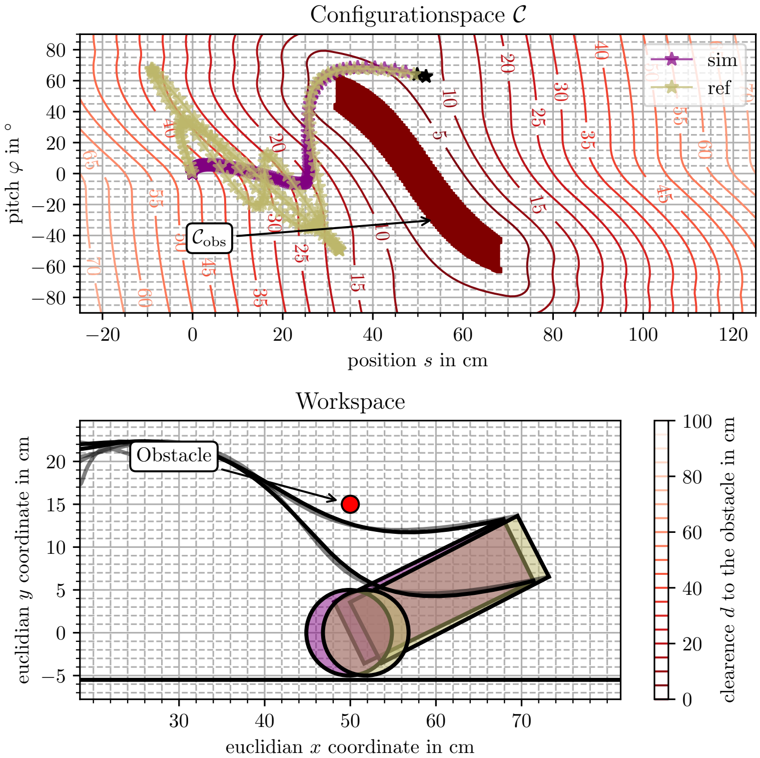


Figure 3-25. Illustration of the movement on the case *Delayed Dive* with Hybridization Method 3.5. In the top part one the trajectory is given in the configuration space, while it is given in the workspace in the bottom.

In the early phase before the obstacle is passed (0 m to 0.3 m) the blue and green trajectory are almost the same, while the red one slightly differs. The same holds true for the breaking maneuver in the phase from (0.75 m to 3 m).

The three observed phases serve different purposes. The first phase before the dive is meant to learn how to get proximal to the obstacle so that the robot is in a suitable state to pass the obstacle. The trajectory is to be tracked precisely to pass the obstacle in the actual pass. In the third phase, tracking is not required anymore, and the input is adopted to arrive at the goal configuration. The reference trajectory, in this case, does complicated movements in the first (early) phase and in the third. This only requires a very slow movement. That the movement is not very agile becomes clear as one observes that the pitch does not exceed an absolute value of 20°. Respectively in the breaking maneuver, the maximum absolute value is 40°.

In Figure 3-26 the reference case *Immediate Dive with distant obstacle* is reviewed. The fundamental difference of this case is that the selected part of the movement chosen from the trajectory starts in an agile state. That means the pitch needs to be at a value of 60° before a reference value is given to the controller.

The first observation to be done is that all of three trajectories solve the motion task. Again the early phase is meant to reach a state that suits the demand to pass the obstacle. In contrast to the previous case, the trajectories differ. The best tracking in of the selected points regarding the

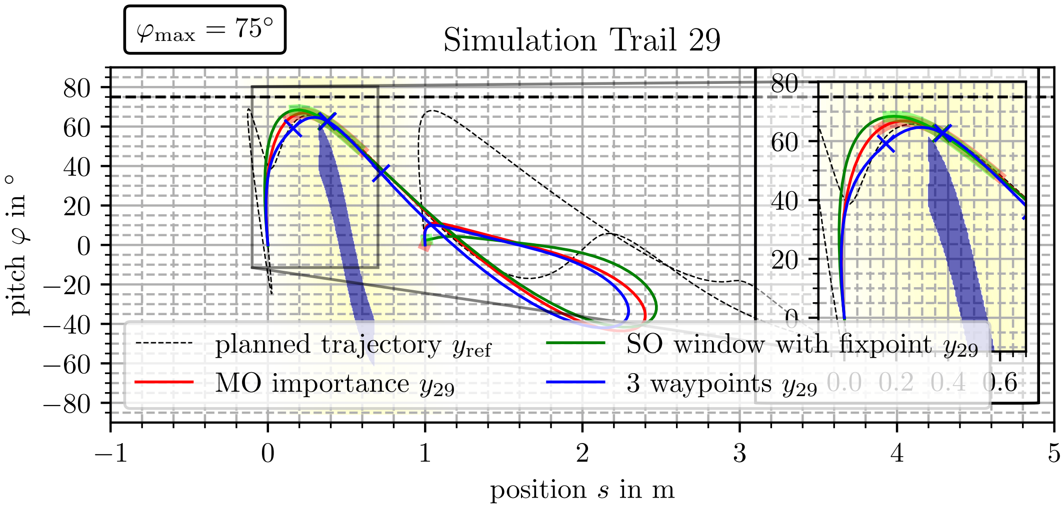


Figure 3-26. Final trajectory produced by the iterative learning control in the *immediate dive with distant obstacle* reference case.

reference is observed in the Hybridization 3.4 that attempts to track three-way points. The others do not track the reference very accurately, even in the highlighted phase.

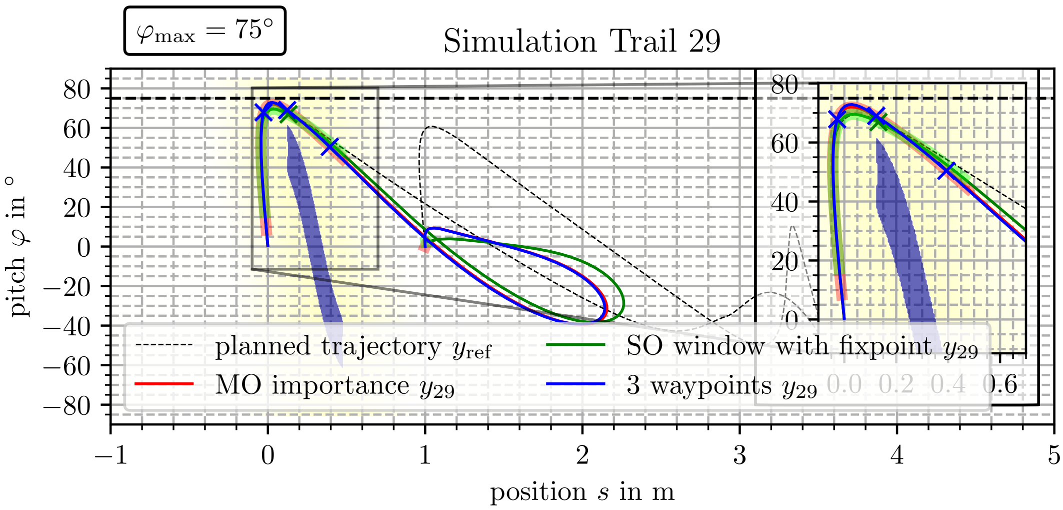


Figure 3-27. Final trajectory produced by the iterative learning control in the *immediate dive with proximal obstacle* reference case.

The last simulation shows the results of the *Immediate Dive with proximal obstacles*. The results are given in Figure 3-27. In this case, the results differ more than in the first case, while all three solve the motion task to pass the obstacle collision-free. In contrast to the previous cases, the movement requires an agile movement from the initial state, meaning there is no part at the beginning where the trajectory can evolve freely to reach an initial state that suits the demands of the following part of the trajectory to pass the obstacle.

Concluding, the simulation study has shown that the proposed combination of iterative learning control and sampling base motion planning solve the motion task. For the three benchmark trajectories the trajectory is optimized in the trail domain, to pass the obstacle precisely in the narrow passage. This is not possible for the whole trajectory due to the robots under-actuation. Therefore, the methods are validated to be a solution for the task in simulation. In comparison to other experiment, drawn the reference has been generated using sampling based motion planning.

Chapter 4

Software Components and Device under test (DuT)

This chapter presents the robot that is used for the experimental validation. Therefore, first, in Section 4.1 the system is analyzed using methods from analytical mechanics leading to insights into the system behavior. In Section 4.1.1, the model is derived using the Lagrange equation of motion. In the overall modeling and analysis, it turns out that the suitable representation of degrees of freedom exposes the constraint needed for motion planning. The advances in hardware and software in contrast to the preliminary project [4] are presented, which consist of some minor changes to the robot and a test library in `matlab` to test the controllers.

Figure 4-1. Two Wheeled Inverted Pendulum Robot (TWIPR) CAD rendering [4, p. 16].

Figure 4-2. Experimental setup.

In Figure 4-1 a rendering of the robot is depicted as it has been developed in the project [4]. The Figure 4-2 shows the setup where experiments have been performed in.

4.1 Device under test (DuT)

In the experimental validation, the Two Wheeled Inverted Pendulum Robot (TWIPR) is used. This system is one of the standard test systems in control research as it has some properties. Namely, it is unstable, non-minimum-phase, and non-holonomic. The project relies on the design of Michael Meindl, who presented the essential details in his master thesis [4]. In this section, an overview of the system is given. Furthermore, the system is augmented by a safety loop and a test system which is described as well.

4.1.1 Modeling

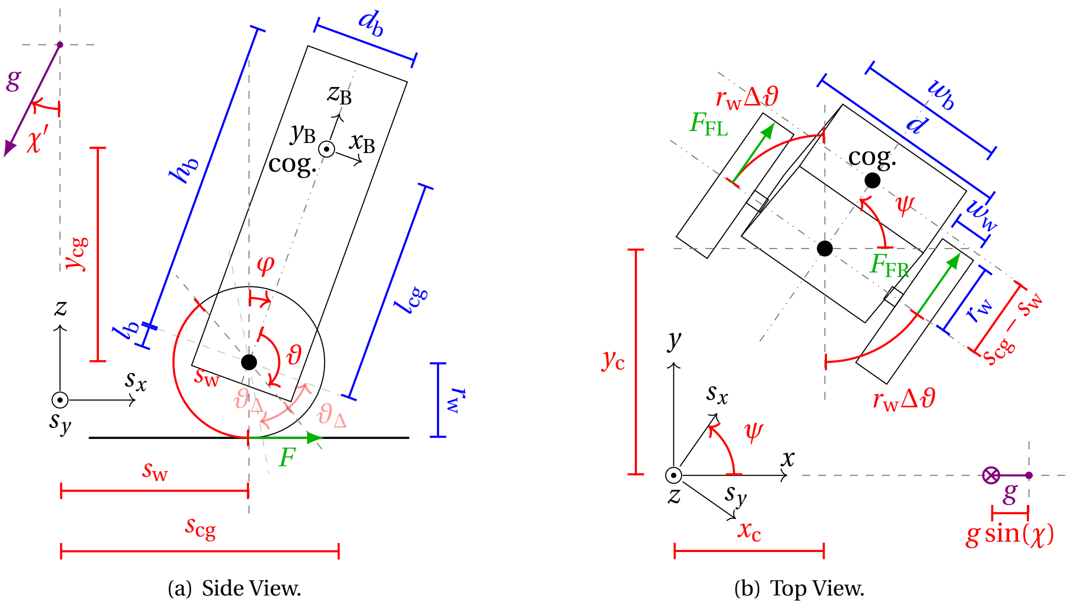


Figure 4-3. Two Wheeled Inverted Pendulum Robot (TWIPR) Kinetics. Blue quantities are physical dimensions and red are dynamic coordinates. The non conservative friction between ground and wheel is given in green.

In this section, a dynamic model is derived. Therefore, consider the kinetic model in Figure 4.3(a) and 4.3(b). To derive the model the Lagrange-Equations of motion are used [45, E44] [46, pp. 37]. To perform the procedure, the following steps are done. Firstly, the energy is quantified. Secondly, the kinematic constraints are stated and a reduced set of generalized coordinates describes the energy. Next, according to the set of generalized coordinates, the generalized non-conservative forces are calculated [45, E22]. Finally, the equations of motion are derived using `sympy` [47]. In contrast to the model from literature [48] used in the preliminary project, a different set of generalized coordinates is chosen. The selection is inspired by Ramasamy [49], leading to a representation that reflects the actuated joints and reduces the set of non-holonomic constraints by one to two. This representation forms a basis for showing the PDF property in Appendix A.8 granting insights for planning and control.

dimension		value	dimension		value
body height	h_b	22 cm	lever center of gravity	l_cg	2.6 cm
body depth	d_b	8 cm	body mass	m_b	2.5 kg
body width	w_b	17 cm	bodys moment of inertia	$J_{\mathrm{b}y}$	$1.65 \times 10^{-2}\,\mathrm{kgm}^2$
body height below axel	l_b	3 cm		$J_{\mathrm{b}x}$	-
wheel distance	d	26 cm		$J_{\mathrm{b}y}$	$1.65 \times 10^{-2}\,\mathrm{kgm}^2$
wheel width	w_w	4.8 cm	wheels mass	m_w	0.636 kg
wheel radius	r_w	5.25 cm	wheels moment of inertia	$J_{\mathrm{w}z}$	-
				$J_{\mathrm{w}z}$	-

Table 4.1. Physical dimensions and physical quantities. The left values are measured on the DuT, the right are taken form [4, p. 21]

.

The generalized coordinates that describe all the systems degrees of freedom are chosen as $q = \left(x_\mathrm{c}, y_\mathrm{c}, \varphi, \vartheta, \vartheta_\Delta\right)^\mathrm{T}$. Furthermore, slipping is assumed absent and both wheels are driven by electric motors. The model is obtained using the Lagrange function.

$$L = \underbrace{T}_{\text{Kinetic energy.}} - \underbrace{V}_{\text{Potential energy.}} \tag{4.1}$$

The system consists of three rigid bodies. For each body the rotational and translational velocities of the center of gravity (cog.) are calculated. Given those the energy of a rigid body is given by

$$T = \frac{1}{2} m \,||\boldsymbol{v}||^2 + \frac{1}{2} \boldsymbol{\omega}^\mathrm{T} \boldsymbol{J} \boldsymbol{\omega}, \tag{4.2}$$

where m is the bodies mass and J is the bodies rotational momentum of inertia tensor.

In case the axes are chosen carefully, the representing matrix tensor of the rotational moment of intertia becomes diagonal. The energies of the bodies are the following:

$$T = T_B + T_{WR} + T_{WL} \tag{4.3}$$

$$T_{WR} = \frac{1}{2} m_b \left(\left(\dot{x}_c + r_w \cos(\psi) \dot{\vartheta_\Delta} \right)^2 + \left(\dot{y}_c + r_w \sin(\psi) \dot{\vartheta_\Delta} \right)^2 \right) + \frac{1}{2} J_{yW} \left(\dot{\varphi} + \dot{\vartheta} + \dot{\vartheta_\Delta} \right)^2 + \frac{1}{2} J_{zW} \dot{\psi}^2 \tag{4.4}$$

$$T_{WL} = \frac{1}{2} m_b \left(\left(\dot{x}_c - r_w \cos(\psi) \dot{\vartheta_\Delta} \right)^2 + \left(\dot{y}_c - r_w \sin(\psi) \dot{\vartheta_\Delta} \right)^2 \right) + \frac{1}{2} J_{yW} \left(\dot{\varphi} + \dot{\vartheta} - \dot{\vartheta_\Delta} \right)^2 + \frac{1}{2} J_{zW} \dot{\psi}^2 \tag{4.5}$$

$$T_B = \frac{1}{2} m_B \left(\dot{x}_c + \cos(\varphi) \cos(\psi) l_{cg} \dot{\varphi} + \sin(\varphi) \sin(\psi) l_{cg} \dot{\psi} \right)^2 + \frac{1}{2} m_B \left(\left(\dot{y}_c + \cos(\varphi) \sin(\psi) l_{cg} \dot{\varphi} + \sin(\varphi) \cos(\psi) l_{cg} \dot{\psi} \right)^2 + \sin^2(\varphi) \sin^2(\varphi) {l_{cg}}^2 \dot{\varphi}^2 \right) \tag{4.6}$$

$$V = V_B \tag{4.7}$$

$$V_B = m_B l_{cg} g \cos\varphi \tag{4.8}$$

The energy of the transnational motion is described with respect to an inertial frame to assure that the centrifugal and Coriolis forces are captured correctly (x, y, z) is an inertial frame, while (x_c, y_c, z) is not.

The chosen coordinates are not jet independent. Thus, the following kinematic constraints are introduced within which (4.9) and (4.10) are non-holonomic.

$$\cos(\psi) \dot{x}_c + \sin(\psi) \dot{y}_c = \dot{s_w} \tag{4.9}$$

$$\sin(\psi) \dot{x}_c - \cos(\psi) \dot{y}_c = 0 \tag{4.10}$$

$$\frac{d}{2} \dot{\psi} = r_w \dot{\vartheta_\Delta} \tag{4.11}$$

$$\dot{s_w} = r_w \left(\dot{\vartheta} + \dot{\varphi} \right) \tag{4.12}$$

$$\dot{s_{cg}} - \dot{s_w} = \cos(\varphi) l_{cg} \dot{\varphi} \tag{4.13}$$

In the latter also the time derivative of the non hohlonomic constrains (4.9) and (4.10) are necessary. Therefore, observe:

$$\cos(\psi) \ddot{x}_c + \sin(\psi) \ddot{y}_c \underbrace{- \sin(\psi) \dot{\psi} \dot{x}_c + \cos(\psi) \dot{\psi} \dot{y}_c}_{=0 \ (4.10)} = \ddot{s_w} \qquad \cos(\psi) \ddot{x}_c + \sin(\psi) \ddot{y}_c = \ddot{s_w} \tag{4.14}$$

$$\sin(\psi) \ddot{x}_c - \cos(\psi) \ddot{y}_c + \underbrace{\cos(\psi) \dot{\psi} \dot{x}_c + \sin(\psi) \dot{\psi} \dot{y}_c}_{= \dot{s_w} \dot{\psi} \ (4.9)} = 0 \qquad \sin(\psi) \ddot{x}_c - \cos(\psi) \ddot{y}_c = -\dot{s_w} \dot{\psi} \tag{4.15}$$

To use the non-holonomic constraints later, they are represented in the generalized coordinates only and transformed to match the representation given in [46, p. 38 (1.95)].

$$\cos\left(\frac{2r_\text{w}}{d}\vartheta_\Delta\right)\dot{x}_\text{c} + \sin\left(\frac{2r_\text{w}}{d}\vartheta_\Delta\right)\dot{y}_\text{c} - r_\text{w}\left(\dot{\vartheta} + \dot{\varphi}\right) = 0 \qquad \sin\left(\frac{2r_\text{w}}{d}\right)\dot{x}_\text{c} - \cos\left(\frac{2r_\text{w}}{d}\right)\dot{y}_\text{c} = 0$$

$$\underbrace{\cos\left(\frac{2r_\text{w}}{d}\vartheta_\Delta\right)}_{a_{11}}\mathrm{d}q_1 + \underbrace{\sin\left(\frac{2r_\text{w}}{d}\vartheta_\Delta\right)}_{a_{12}}\mathrm{d}q_2 - \underbrace{r_\text{w}}_{a_{13}}\mathrm{d}q_3 - \underbrace{r_\text{w}}_{a_{14}}\mathrm{d}q_4 = 0 \quad \underbrace{\sin\left(\frac{2r_\text{w}}{d}\vartheta_\Delta\right)}_{a_{21}}\mathrm{d}q_1 - \underbrace{\cos\left(\frac{2r_\text{w}}{d}\vartheta_\Delta\right)}_{a_{22}}\mathrm{d}q_2 = 0$$

The generalized non-conservative forces are obtained using the method of virtual displacement [45, E22].

$$\boldsymbol{\tau}_i = \boldsymbol{f}\frac{\partial \boldsymbol{r}}{\partial q_i} \tag{4.16}$$

In the first step the only non-conservative forces taken into account are the motor torques. Therefore, define:

$$\begin{aligned}
\delta W &= \tau_\text{R}\left(\delta\vartheta + \delta\vartheta_\Delta\right) + \tau_\text{L}\left(\delta\vartheta - \delta\vartheta_\Delta\right)\\
\tau_{x_\text{c}} &= 0\\
\tau_{y_\text{c}} &= 0\\
\tau_{\varphi} &= 0\\
\tau_{\vartheta_\Delta} &= \tau_\text{R} - \tau_\text{L}\\
\tau_{\vartheta} &= \tau_\text{R} + \tau_\text{L}
\end{aligned}$$

Remark 6. *In this step it becomes clear whether the selection of generalized coordinates has been proper. If all generalized forces are either zero or con be chosen arbitrarily by the present controllers. The vanishing generalized forces result in the dynamic constraints later.*

The system is prone to many imperfections, e. g. the wheel friction [4]. A model that does not omit friction could be more worthy. Therefore, the friction of rolling is introduced. In contrast, the viscose damping of the actuators is not introduced directly as they can be compensated completely by modification of the input torque. As the input will be learned by the controller, a precise description of the input is not needed (refer to Chapter 3). This marks the model out to the observations found in the literature [48], and [49], of whom Kim [48] only took the damping imposed by the motors into account and Ramasamy [49] omitted friction at all.

$$\delta W = \tau_\text{R}\left(\delta\vartheta + \delta\vartheta_\Delta\right) + \tau_\text{L}\left(\delta\vartheta - \delta\vartheta_\Delta\right) - F_\text{FR}\left(\delta\vartheta + \delta\vartheta_\Delta + \delta\varphi\right)r_\text{w} - F_\text{FL}\left(\delta\vartheta - \delta\vartheta_\Delta + \delta\varphi\right)r_\text{w} \tag{4.17}$$

$$\tau_{x_\text{c}} = 0 \tag{4.18}$$

$$\tau_{y_\text{c}} = 0 \tag{4.19}$$

$$\tau_{\varphi} = -F_\text{FR}r_\text{w} - F_\text{FL}r_\text{w} \tag{4.20}$$

$$\tau_{\vartheta_\Delta} = \tau_\text{R} - \tau_\text{L} - F_\text{FR}r_\text{w} + F_\text{FL}r_\text{w} \tag{4.21}$$

$$\tau_{\vartheta} = \tau_\text{R} + \tau_\text{L} - F_\text{FR}r_\text{w} - F_\text{FL}r_\text{w} \tag{4.22}$$

Remark 7. *In this case the non-conservative forces acting on the not actuated coordinates do not depend on the input forces.*

$$\boldsymbol{\tau} = \begin{bmatrix}\boldsymbol{\tau}_\text{u}(\boldsymbol{q},\dot{\boldsymbol{q}})\\ \boldsymbol{\tau}_\text{a}(\boldsymbol{q},\dot{\boldsymbol{q}},\boldsymbol{u})\end{bmatrix} \tag{4.23}$$

Finally, the Lagrangian equation of motion are evaluated [45, p. E44].

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = \tau_i + \lambda_1 a_{1i} + \lambda_2 a_{2i} \tag{4.24}$$

Evaluating (4.24) for the local coordinates $(x_\mathrm{c}, y_\mathrm{c})$ leads to

$$\begin{aligned}
x_\mathrm{c}: \quad & m_\mathrm{b}\left(\ddot{x}_\mathrm{c} + \cos(\varphi)\cos(\psi)\, l_\mathrm{cg}\ddot{\varphi} - \sin(\varphi)\sin(\psi)\, l_\mathrm{cg}\ddot{\psi} - 2\cos(\varphi)\sin(\psi)\, l_\mathrm{cg}\dot{\varphi}\dot{\psi} - 2\sin(\varphi)\cos(\psi)\, l_\mathrm{cg}\left(\dot{\varphi}^2 + \dot{\psi}^2\right)\right) \\
& + 2m_\mathrm{w}\ddot{x}_\mathrm{c} \\
&= \lambda_1 \underbrace{\cos(\psi)}_{a_{11}} + \lambda_2 \underbrace{\sin(\psi)}_{a_{21}} && (4.25) \\
y_\mathrm{c}: \quad & m_\mathrm{b}\left(\ddot{y}_\mathrm{c} + \cos(\varphi)\sin(\psi)\, l_\mathrm{cg}\ddot{\varphi} + \sin(\varphi)\cos(\psi)\, l_\mathrm{cg}\ddot{\psi} + 2\cos(\varphi)\cos(\psi)\, l_\mathrm{cg}\dot{\varphi}\dot{\psi} - 2\sin(\varphi)\sin^2(\psi)\, l_\mathrm{cg}\left(\dot{\varphi}^2 + \dot{\psi}^2\right)\right) \\
& + 2m_\mathrm{w}\ddot{y}_\mathrm{c} \\
&= \lambda_1 \underbrace{\sin(\psi)}_{a_{12}} - \lambda_2 \underbrace{\cos(\psi)}_{a_{22}} && (4.26)
\end{aligned}$$

The second Lagrange-Multiplier λ_2 can be eliminated. Therefore, multiply the (4.25) with $\cos(\psi)$ and the second with $\sin(\psi)$ and add them.

$$\begin{aligned}
x_\mathrm{c}: \quad & m_\mathrm{b}\left(\ddot{x}_\mathrm{c}\cos(\psi) + \cos(\varphi)\cos^2(\psi)\, l_\mathrm{cg}\ddot{\varphi} - \sin(\varphi)\cos^2(\psi)\, l_\mathrm{cg}\left(\dot{\varphi}^2 + \dot{\psi}^2\right)\right) + 2m_\mathrm{w}\cos(\psi)\ddot{x}_\mathrm{c} \\
y_\mathrm{c}: \quad & + m_\mathrm{b}\left(\ddot{y}_\mathrm{c}\sin(\psi) + \cos(\varphi)\sin^2(\psi)\, l_\mathrm{cg}\ddot{\varphi} - \sin(\varphi)\sin^2(\psi)\, l_\mathrm{cg}\left(\dot{\varphi}^2 + \dot{\psi}^2\right)\right) + 2m_\mathrm{w}\sin(\psi)\ddot{y}_\mathrm{c} \\
&= \lambda_1\left(\cos^2(\psi) + \sin^2(\psi)\right) && (4.27)
\end{aligned}$$

Simplification yields the solution for λ_1.

$$m_\mathrm{b}\left(\ddot{x}_\mathrm{c}\cos(\psi) + \ddot{y}_\mathrm{c}\sin(\psi) + \cos(\varphi)\, l_\mathrm{cg}\ddot{\varphi} - \sin(\varphi)\, l_\mathrm{cg}\left(\dot{\varphi}^2 + \dot{\psi}^2\right)\right) + 2m_\mathrm{w}\underbrace{\left(\cos(\psi)\ddot{x}_\mathrm{c} + \sin(\psi)\ddot{y}_\mathrm{c}\right)}_{= r_\mathrm{w}(\ddot{\varphi}+\ddot{\vartheta})\ (4.14)} = \lambda_1 \tag{4.28}$$

As λ_2 does not occur in the other equations of motion, it does not need to be computed explicitly.

$$\left(m_\mathrm{b} r_\mathrm{w} + 2m_\mathrm{w} r_\mathrm{w}\right)\left(\ddot{\vartheta} + \ddot{\varphi}\right) + m_\mathrm{b}\cos(\varphi)\, l_\mathrm{cg}\ddot{\varphi} - m_\mathrm{b}\sin(\varphi)\, l_\mathrm{cg}\left(\dot{\varphi}^2 + \dot{\psi}^2\right) = \lambda_1 \tag{4.29}$$

This yields the following equations of motion obtained using `sympy`:

$$
\begin{aligned}
\varphi: \quad -\lambda_1 r_\mathrm{w} = {} & \left(J_{y\mathrm{B}} + m_\mathrm{b} l_\mathrm{cg}{}^2 + 2J_{y\mathrm{W}}\right)\ddot{\varphi} + 2J_{y\mathrm{W}}\ddot{\vartheta} \\
& + m_\mathrm{b} l_\mathrm{cg} \cos(\varphi) \underbrace{\left(\ddot{x}_\mathrm{c}\cos(\psi) + \ddot{y}_\mathrm{c}\sin(\psi)\right)}_{= r_\mathrm{w}\left(\ddot{\vartheta}+\ddot{\varphi}\right)\ (4.14)} \\
& - 4\frac{r_\mathrm{w}{}^2}{d^2}\left(J_{x\mathrm{B}} - J_{z\mathrm{B}} + m_\mathrm{B} l_\mathrm{cg}{}^2\right)\cos(\varphi)\sin(\varphi)\dot{\vartheta}_\Delta{}^2 \\
& + g m_\mathrm{B} l_\mathrm{cg} \sin(\varphi) \\
\varphi: \quad 0 = {} & \left(J_{y\mathrm{B}} + m_\mathrm{b} l_\mathrm{cg}{}^2 + 2J_{y\mathrm{W}} + 2m_\mathrm{W} r_\mathrm{w}{}^2 + m_\mathrm{b} r_\mathrm{w}{}^2 + 2m_\mathrm{b} l_\mathrm{cg} r_\mathrm{w}\cos(\varphi)\right)\ddot{\varphi} \\
& + \left(2J_{y\mathrm{W}} + 2m_\mathrm{W} r_\mathrm{w}{}^2 + m_\mathrm{b} r_\mathrm{w}{}^2 + m_\mathrm{b} l_\mathrm{cg} r_\mathrm{w}\cos(\varphi)\right)\ddot{\vartheta} \\
& - m_\mathrm{b}\sin(\varphi) l_\mathrm{cg} r_\mathrm{w}\left(\dot{\varphi}^2 + \dot{\psi}^2\right) \\
& - 4\frac{r_\mathrm{w}{}^2}{d^2}\left(J_{x\mathrm{B}} - J_{z\mathrm{B}} + m_\mathrm{B} l_\mathrm{cg}{}^2\right)\cos(\varphi)\sin(\varphi)\dot{\vartheta}_\Delta{}^2 \\
& + g m_\mathrm{B} l_\mathrm{cg} \sin(\varphi)
\end{aligned}
\tag{4.30}
$$

$$
\begin{aligned}
\vartheta: \quad \tau_\mathrm{R} + \tau_\mathrm{L} - \lambda_1 r_\mathrm{w} = {} & 2J_{y\mathrm{W}}\ddot{\varphi} + 2J_{y\mathrm{W}}\ddot{\vartheta} \\
\vartheta: \quad \tau_\mathrm{R} + \tau_\mathrm{L} = {} & \left(2J_{y\mathrm{W}} + 2m_\mathrm{W} r_\mathrm{w}{}^2 + m_\mathrm{b} r_\mathrm{w}{}^2 + \cos(\varphi) l_\mathrm{cg} r_\mathrm{w}\right)\ddot{\varphi} + \left(2J_{y\mathrm{W}} + 2m_\mathrm{W} r_\mathrm{w}{}^2 + m_\mathrm{b} r_\mathrm{w}{}^2\right)\ddot{\vartheta} \\
& - \sin(\varphi) m_\mathrm{b} l_\mathrm{cg} r_\mathrm{w}\left(\dot{\varphi}^2 + \dot{\psi}^2\right)
\end{aligned}
\tag{4.31}
$$

$$
\begin{aligned}
\vartheta_\Delta: \quad \tau_\mathrm{R} - \tau_\mathrm{L} = {} & \left(2J_{y\mathrm{W}} + 2m_\mathrm{W} r_\mathrm{w}{}^2 + 2\frac{4r_\mathrm{w}{}^2}{d^2}J_{z\mathrm{W}} + \frac{4r_\mathrm{w}{}^2}{d^2}\left(J_{x\mathrm{B}} + m_\mathrm{B} l_\mathrm{cg}{}^2\right)\sin^2(\varphi) + \frac{4r_\mathrm{w}{}^2}{d^2}J_{z\mathrm{B}}\cos^2(\varphi)\right)\ddot{\vartheta}_\Delta \\
& + \frac{8r_\mathrm{w}{}^2}{d^2}\left(J_{x\mathrm{B}} + m_\mathrm{B} l_\mathrm{cg}{}^2 - J_{z\mathrm{B}}\right)\cos(\varphi)\sin(\varphi)\dot{\varphi}\dot{\vartheta}_\Delta \\
& + \frac{2r_\mathrm{w}}{d} m_\mathrm{b} l_\mathrm{cg}\sin(\varphi)\underbrace{\left(\ddot{y}_\mathrm{c}\cos(\psi) - \ddot{x}_\mathrm{c}\sin(\psi)\right)}_{= r_\mathrm{w}\left(\dot{\vartheta}+\dot{\varphi}\right)\dot{\psi}\ (4.15)} \\
\vartheta_\Delta: \quad \tau_\mathrm{R} - \tau_\mathrm{L} = {} & \left(2J_{y\mathrm{W}} + 2m_\mathrm{W} r_\mathrm{w}{}^2 + 2\frac{4r_\mathrm{w}{}^2}{d^2}J_{z\mathrm{W}} + \frac{4r_\mathrm{w}{}^2}{d^2}\left(J_{x\mathrm{B}} + m_\mathrm{B} l_\mathrm{cg}{}^2\right)\sin^2(\varphi) + \frac{4r_\mathrm{w}{}^2}{d^2}J_{z\mathrm{B}}\cos^2(\varphi)\right)\ddot{\vartheta}_\Delta \\
& + \frac{8r_\mathrm{w}{}^2}{d^2}\left(J_{x\mathrm{B}} + m_\mathrm{B} l_\mathrm{cg}{}^2 - J_{z\mathrm{B}}\right)\cos(\varphi)\sin(\varphi)\dot{\varphi}\dot{\vartheta}_\Delta \\
& + \frac{4r_\mathrm{w}{}^2}{d^2} m_\mathrm{b} l_\mathrm{cg} r_\mathrm{w}\sin(\varphi)\left(\dot{\vartheta} + \dot{\varphi}\right)\dot{\vartheta}_\Delta
\end{aligned}
\tag{4.32}
$$

Now define for simplification purposes:

$$
\begin{aligned}
J'_{y\mathrm{B}} &= J_{y\mathrm{B}} + m_\mathrm{B} l_\mathrm{cg}{}^2 & \qquad J'_{y\mathrm{W}} &= J_{y\mathrm{W}} + m_\mathrm{W} r_\mathrm{w}{}^2 \\
J'_{z\mathrm{W}} &= \frac{4r_\mathrm{w}{}^2}{d^2}J_{z\mathrm{W}} & \qquad J'_{x\mathrm{B}} &= \frac{4r_\mathrm{w}{}^2}{d^2}\left(J_{x\mathrm{B}} + m_\mathrm{B} l_\mathrm{cg}{}^2\right) \\
J'_{z\mathrm{B}} &= \frac{4r_\mathrm{w}{}^2}{d^2}J_{z\mathrm{B}}
\end{aligned}
$$

The equations of motion (4.30) to (4.32) take the general form with regard to the vector of generalized coordinates $\boldsymbol{q} = \begin{bmatrix}\vartheta & \varphi & \vartheta_\Delta\end{bmatrix}^\mathrm{T}$:

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{c}(\boldsymbol{q},\dot{\boldsymbol{q}}) + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{\tau} \tag{4.33}$$

$$\boldsymbol{M}\big(\boldsymbol{q}\big) = \begin{bmatrix} \mu_{11}(\varphi) & \mu_{12}(\varphi) & 0 \\ \mu_{12}(\varphi) & \mu_{22}(\varphi) & 0 \\ 0 & 0 & \mu_{33}(\varphi) \end{bmatrix} \quad \boldsymbol{\tau} = \begin{bmatrix} \tau_{\mathrm{R}} + \tau_{\mathrm{L}} - F_{\mathrm{FR}} r_{\mathrm{w}} - F_{\mathrm{FL}} r_{\mathrm{w}} \\ -F_{\mathrm{FR}} r_{\mathrm{w}} - F_{\mathrm{FL}} r_{\mathrm{w}} \\ \tau_{\mathrm{R}} - \tau_{\mathrm{L}} - F_{\mathrm{FR}} r_{\mathrm{w}} + F_{\mathrm{FL}} r_{\mathrm{w}} \end{bmatrix} \quad \boldsymbol{g}(\boldsymbol{q}) = \begin{bmatrix} 0 \\ \underbrace{-g m_{\mathrm{B}} l_{\mathrm{cg}}}_{\nu} \sin(\varphi) \\ 0 \end{bmatrix}$$

$$\boldsymbol{c}(\boldsymbol{q},\dot{\boldsymbol{q}}) = \begin{bmatrix} -\underbrace{m_{\mathrm{B}} r_{\mathrm{w}} l_{\mathrm{cg}}}_{\beta} \sin(\varphi)\left(\dot{\varphi}^2 + \frac{4 r_{\mathrm{w}}{}^2}{d^2}\dot{\vartheta}_\Delta{}^2\right) \\ -\underbrace{m_{\mathrm{B}} l_{\mathrm{cg}} r_{\mathrm{w}}}_{\beta} \sin(\varphi)\left(\dot{\varphi}^2 + \frac{4 r_{\mathrm{w}}{}^2}{d^2}\dot{\vartheta}_\Delta{}^2\right) - 4\underbrace{\frac{r_{\mathrm{w}}{}^2}{d^2}\left(J_{x\mathrm{B}} - J_{z\mathrm{B}} + m_{\mathrm{B}} l_{\mathrm{cg}}{}^2\right)}_{\epsilon} \cos(\varphi)\sin(\varphi)\dot{\vartheta}_\Delta{}^2 \\ \frac{4 r_{\mathrm{w}}{}^2}{d^2}\underbrace{m_{\mathrm{b}} l_{\mathrm{cg}} r_{\mathrm{w}}}_{\beta} \sin(\varphi)\left(\dot{\vartheta} + \dot{\varphi}\right)\dot{\vartheta}_\Delta + 8\underbrace{\frac{r_{\mathrm{w}}{}^2}{d^2}\left(J_{x\mathrm{B}} - J_{z\mathrm{B}} + m_{\mathrm{B}} l_{\mathrm{cg}}{}^2\right)}_{\epsilon} \cos(\varphi)\sin(\varphi)\dot{\varphi}\dot{\vartheta}_\Delta \end{bmatrix}$$

$$\mu_{11}(\varphi) = \left(\underbrace{2J'_{y\mathrm{W}} + m_{\mathrm{B}} r_{\mathrm{w}}{}^2}_{\alpha}\right) \qquad \mu_{22}(\varphi) = \left(\underbrace{2J'_{y\mathrm{W}} + m_{\mathrm{B}} r_{\mathrm{w}}{}^2}_{\alpha} + 2\underbrace{m_{\mathrm{B}} l_{\mathrm{cg}} r_{\mathrm{w}}}_{\beta}\cos(\varphi) + \underbrace{J'_{y\mathrm{B}}}_{\gamma}\right)$$

$$\mu_{12}(\varphi) = \left(\underbrace{2J'_{y\mathrm{W}} + m_{\mathrm{B}} r_{\mathrm{w}}{}^2}_{\alpha} + \underbrace{m_{\mathrm{B}} l_{\mathrm{cg}} r_{\mathrm{w}}}_{\beta}\cos(\varphi)\right) \qquad \mu_{33}(\varphi) = \left(\underbrace{2J'_{y\mathrm{W}} + 2J'_{z\mathrm{W}}}_{\delta} + \underbrace{J'_{x\mathrm{B}}}_{\delta_1}\sin^2(\varphi) + \underbrace{J'_{z\mathrm{B}}}_{\delta_2}\cos^2(\varphi)\right)$$

This concludes the derivation of the model. The result allows to modify the generalized force by an friction model for between wheel and ground. The second item forms the kinodynamic constraint for the planning problem, that is:

$$\mu_{12}(\varphi)\ddot{\varphi} + \mu_{22}(\varphi)\ddot{\vartheta} + \boldsymbol{c}_2(\boldsymbol{q},\dot{\boldsymbol{q}}) + \boldsymbol{g}_2(q) = -F_{\mathrm{FR}} r_{\mathrm{w}} - F_{\mathrm{FL}} r_{\mathrm{w}}$$

Remark 8. *Independence of the actuator friction. The kinodynamic constraint found does not depend on the actuator friction.*

A general discussion about this issue is found in [50].

4.1.2 Relation to Literature Results

This section discusses how the kinodynamic constraint is also derived from a literature result. The model derived in [48] is accepted in our research group [4]. The constraint is computed using a state transformation that allows to derive such constraints of present mechanical models.

As discussed earlier, the results of Kim in [48], rely on a different set of coordinates that relate to the one imposed in the previous section by a linear transform.

$$\underbrace{\begin{bmatrix} \dot{s}_\mathrm{w} \\ \dot{\varphi} \\ \dot{\psi} \end{bmatrix}}_{\boldsymbol{q}_\mathrm{K}} = \underbrace{\begin{bmatrix} r_\mathrm{w} & r_\mathrm{w} & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{2r_\mathrm{w}}{d} \end{bmatrix}}_{\boldsymbol{T}} \cdot \underbrace{\begin{bmatrix} \dot{\vartheta} \\ \dot{\varphi} \\ \dot{\vartheta}_\Delta \end{bmatrix}}_{\boldsymbol{q}} \quad \begin{bmatrix} \dot{\vartheta} \\ \dot{\varphi} \\ \dot{\vartheta}_\Delta \end{bmatrix} = \underbrace{\begin{bmatrix} \frac{1}{r_\mathrm{w}} & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{d}{2r_\mathrm{w}} \end{bmatrix}}_{\boldsymbol{T}^{-1}} \cdot \begin{bmatrix} \dot{s}_\mathrm{w} \\ \dot{\varphi} \\ \dot{\psi} \end{bmatrix} \tag{4.34}$$

The proposed equations of motion have the following form.

$$\boldsymbol{M}_\mathrm{K}(\boldsymbol{q}_\mathrm{K})\ddot{\boldsymbol{q}}_\mathrm{K} + \boldsymbol{C}_\mathrm{K}(\boldsymbol{q}_\mathrm{K}, \dot{\boldsymbol{q}}_\mathrm{K})\dot{\boldsymbol{q}}_\mathrm{K} + \boldsymbol{D}_\mathrm{K}\dot{\boldsymbol{q}}_\mathrm{K} + \boldsymbol{g}_\mathrm{K}(\boldsymbol{q}_\mathrm{K}) = \boldsymbol{B}_\mathrm{K}\boldsymbol{\tau} \tag{4.35}$$

To transfer the equations in a form similar to the one from the previous section, the equation is multiplied from the left with the regular matrix $\boldsymbol{T}^\mathrm{T}$. Furthermore, the representation of the generalized coordinates is changed coresponding to equation 4.36.

$$\boldsymbol{T}^\mathrm{T}\boldsymbol{M}_\mathrm{K}(\boldsymbol{T}\boldsymbol{q})\boldsymbol{T}\ddot{\boldsymbol{q}} + \boldsymbol{T}^\mathrm{T}\boldsymbol{C}_\mathrm{K}(\boldsymbol{T}\boldsymbol{q}, \boldsymbol{T}\dot{\boldsymbol{q}})\boldsymbol{T}\dot{\boldsymbol{q}} + \boldsymbol{T}^\mathrm{T}\boldsymbol{D}_\mathrm{K}\boldsymbol{T}\dot{\boldsymbol{q}} + \boldsymbol{T}^\mathrm{T}\boldsymbol{g}_\mathrm{K}(\boldsymbol{T}\boldsymbol{q}) = \boldsymbol{T}^\mathrm{T}\boldsymbol{B}_\mathrm{K}\boldsymbol{\tau} \tag{4.36}$$

$$\boldsymbol{T}^\mathrm{T}\boldsymbol{M}_\mathrm{K}(\boldsymbol{T}\boldsymbol{q})\boldsymbol{T} = \begin{bmatrix} a_{11}{r_\mathrm{w}}^2 & a_{11}{r_\mathrm{w}}^2 + a_{12}r_\mathrm{w} & 0 \\ a_{11}{r_\mathrm{w}}^2 + a_{12}r_\mathrm{w} & a_{11}{r_\mathrm{w}}^2 + 2a_{12}r_\mathrm{w} + a_{22} & 0 \\ 0 & 0 & a_{33}\frac{4{r_\mathrm{w}}^2}{d^2} \end{bmatrix} \tag{4.37}$$

$$\boldsymbol{T}^\mathrm{T}\boldsymbol{C}_\mathrm{K}(\boldsymbol{T}\boldsymbol{q}, \boldsymbol{T}\dot{\boldsymbol{q}})\boldsymbol{T} = \begin{bmatrix} 0 & c_{12}r_\mathrm{w} & c_{13}\frac{2{r_\mathrm{w}}^2}{d} \\ 0 & c_{12}r_\mathrm{w} & c_{13}\frac{2{r_\mathrm{w}}^2}{d} + c_{23}\frac{2r_\mathrm{w}}{d} \\ c_{31}\frac{4{r_\mathrm{w}}^2}{d^2} & c_{31}\frac{4{r_\mathrm{w}}^2}{d^2} + c_{32}\frac{4{r_\mathrm{w}}^2}{d^2} & c_{33}\frac{4{r_\mathrm{w}}^2}{d^2} \end{bmatrix} \tag{4.38}$$

$$\boldsymbol{T}^\mathrm{T}\boldsymbol{M}_\mathrm{K}(\boldsymbol{T}\boldsymbol{q})\boldsymbol{T} = \begin{bmatrix} d_{11}{r_\mathrm{w}}^2 & d_{11}{r_\mathrm{w}}^2 + d_{12}r_\mathrm{w} & 0 \\ d_{11}{r_\mathrm{w}}^2 + d_{12}r_\mathrm{w} & d_{11}{r_\mathrm{w}}^2 + 2d_{12}r_\mathrm{w} + d_{22} & 0 \\ 0 & 0 & d_{33}\frac{4{r_\mathrm{w}}^2}{d^2} \end{bmatrix} \tag{4.39}$$

$$\boldsymbol{T}^\mathrm{T}\boldsymbol{g}_\mathrm{K}(\boldsymbol{T}\boldsymbol{q}) = \begin{bmatrix} 0 \\ -m_\mathrm{b}l_\mathrm{cg}g\sin(\varphi) \\ 0 \end{bmatrix} \tag{4.40}$$

$$\boldsymbol{T}^\mathrm{T}\boldsymbol{B}_\mathrm{K} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ -1 & 1 \end{bmatrix} \tag{4.41}$$

In the following the equation are compared to those found in the previous section. Therefore, first validate the mass matrix.

$$
\begin{aligned}
a_{11} {r_\mathrm{w}}^2 &= m_\mathrm{b} {r_\mathrm{w}}^2 + 2 m_\mathrm{w} {r_\mathrm{w}}^2 + 2 J_{y\mathrm{W}} \\
a_{11} {r_\mathrm{w}}^2 + a_{12} r_\mathrm{w} &= m_\mathrm{b} {r_\mathrm{w}}^2 + 2 m_\mathrm{w} {r_\mathrm{w}}^2 + 2 J_{y\mathrm{W}} + m_\mathrm{b} l_\mathrm{cg} r_\mathrm{w} \cos(\varphi) \\
a_{11} {r_\mathrm{w}}^2 + 2 a_{12} r_\mathrm{w} + a_{22} &= m_\mathrm{b} {r_\mathrm{w}}^2 + 2 m_\mathrm{w} {r_\mathrm{w}}^2 + 2 J_{y\mathrm{W}} + m_\mathrm{b} l_\mathrm{cg} r_\mathrm{w} \cos(\varphi) + J_{y\mathrm{B}} + m_\mathrm{b} {l_\mathrm{cg}}^2 \\
a_{33} \frac{4 {r_\mathrm{w}}^2}{d^2} &= \left(J_{y\mathrm{B}} + 2 J_{z\mathrm{W}} + \left(m_\mathrm{w} + \frac{J_{y\mathrm{W}}}{{r_\mathrm{w}}^2} \right) \frac{d^2}{2} - \left(J_{z\mathrm{B}} - J_{x\mathrm{B}} - m_\mathrm{b} {l_\mathrm{cg}}^2 \right) \sin^2(\varphi) \right) \frac{4 {r_\mathrm{w}}^2}{d^2} \\
&= \left(J_{z\mathrm{B}} \frac{4 {r_\mathrm{w}}^2}{d^2} + 2 J_{z\mathrm{W}} \frac{4 {r_\mathrm{w}}^2}{d^2} + 2 \left(J_{y\mathrm{W}} + m_\mathrm{w} {r_\mathrm{w}}^2 \right) - \left(J_{z\mathrm{B}} - J_{x\mathrm{B}} - m_\mathrm{b} {l_\mathrm{cg}}^2 \right) \frac{4 {r_\mathrm{w}}^2}{d^2} \sin^2(\varphi) \right)
\end{aligned}
$$

It is observed that the mass matrix coincide.

The gerenralized coriolisforces:

$$
\begin{aligned}
c_{12} r_\mathrm{w} &= - m_\mathrm{b} l_\mathrm{cg} r_\mathrm{w} \dot{\varphi} \sin(\varphi) \\
c_{13} \frac{2 {r_\mathrm{w}}^2}{d} &= - m_\mathrm{b} l_\mathrm{cg} \frac{2 {r_\mathrm{w}}^2}{d} \dot{\psi} \sin(\varphi) \\
&= - m_\mathrm{b} l_\mathrm{cg} r_\mathrm{w} \frac{4 {r_\mathrm{w}}^2}{d^2} \dot{\vartheta}_\Delta \sin(\varphi) \\
c_{13} \frac{2 {r_\mathrm{w}}^2}{d} + c_{23} \frac{2 r_\mathrm{w}}{d} &= - m_\mathrm{b} l_\mathrm{cg} \frac{2 {r_\mathrm{w}}^2}{d} \dot{\psi} \sin(\varphi) + \left(J_{z\mathrm{B}} - J_{x\mathrm{B}} - m_\mathrm{b} {l_\mathrm{cg}}^2 \right) \frac{2 r_\mathrm{w}}{d} \dot{\psi} \sin(\varphi) \cos(\varphi) \\
&= - m_\mathrm{b} l_\mathrm{cg} r_\mathrm{w} \frac{4 {r_\mathrm{w}}^2}{d^2} \dot{\vartheta}_\Delta \sin(\varphi) + \left(J_{z\mathrm{B}} - J_{x\mathrm{B}} - m_\mathrm{b} {l_\mathrm{cg}}^2 \right) \frac{4 {r_\mathrm{w}}^2}{d^2} \dot{\vartheta}_\Delta \sin(\varphi) \cos(\varphi) \\
c_{31} \frac{2 {r_\mathrm{w}}^2}{d} &= m_\mathrm{b} l_\mathrm{cg} \frac{2 {r_\mathrm{w}}^2}{d} \dot{\psi} \sin(\varphi) \\
&= m_\mathrm{b} l_\mathrm{cg} r_\mathrm{w} \frac{4 {r_\mathrm{w}}^2}{d^2} \dot{\vartheta}_\Delta \sin(\varphi) \\
c_{31} \frac{2 {r_\mathrm{w}}^2}{d} + c_{32} \frac{2 r_\mathrm{w}}{d} &= m_\mathrm{b} l_\mathrm{cg} \frac{2 {r_\mathrm{w}}^2}{d} \dot{\psi} \sin(\varphi) - \left(J_{z\mathrm{B}} - J_{x\mathrm{B}} - m_\mathrm{b} {l_\mathrm{cg}}^2 \right) \frac{2 r_\mathrm{w}}{d} \dot{\psi} \sin(\varphi) \cos(\varphi) \\
&= m_\mathrm{b} l_\mathrm{cg} r_\mathrm{w} \frac{4 {r_\mathrm{w}}^2}{d^2} \dot{\vartheta}_\Delta \sin(\varphi) - \left(J_{z\mathrm{B}} - J_{x\mathrm{B}} - m_\mathrm{b} {l_\mathrm{cg}}^2 \right) \frac{4 {r_\mathrm{w}}^2}{d^2} \dot{\vartheta}_\Delta \sin(\varphi) \cos(\varphi) \\
c_{33} \frac{4 {r_\mathrm{w}}^2}{d^2} &= - \left(J_{z\mathrm{B}} - J_{x\mathrm{B}} - m_\mathrm{b} {l_\mathrm{cg}}^2 \right) \frac{4 {r_\mathrm{w}}^2}{d^2} \dot{\varphi} \sin(\varphi) \cos(\varphi)
\end{aligned}
$$

Now to adopt to the former representation compose the coriolis forces in a vector.

$$
\boldsymbol{T}^{\mathrm{T}}\boldsymbol{C}_{\mathrm{K}}(\boldsymbol{T}\boldsymbol{q},\boldsymbol{T}\dot{\boldsymbol{q}})\boldsymbol{T}\dot{\boldsymbol{q}} = \begin{bmatrix} -m_{\mathrm{b}}l_{\mathrm{cg}}r_{\mathrm{w}}\left(\dot{\varphi}^2 + \frac{4{r_{\mathrm{w}}}^2}{d^2}{\dot{\vartheta}_{\Delta}}^2\right)\sin(\varphi) \\ -m_{\mathrm{b}}l_{\mathrm{cg}}r_{\mathrm{w}}\left(\dot{\varphi}^2 + \frac{4{r_{\mathrm{w}}}^2}{d^2}{\dot{\vartheta}_{\Delta}}^2\right)\sin(\varphi) + \left(J_{z\mathrm{B}} - J_{x\mathrm{B}} - m_{\mathrm{b}}{l_{\mathrm{cg}}}^2\right)\frac{4{r_{\mathrm{w}}}^2}{d^2}{\dot{\vartheta}_{\Delta}}^2\sin(\varphi)\cos(\varphi) \\ m_{\mathrm{b}}l_{\mathrm{cg}}r_{\mathrm{w}}\frac{4{r_{\mathrm{w}}}^2}{d^2}\dot{\vartheta}_{\Delta}\left(\dot{\vartheta}+\dot{\varphi}\right)\sin(\varphi) - 2\left(J_{z\mathrm{B}} - J_{x\mathrm{B}} - m_{\mathrm{b}}{l_{\mathrm{cg}}}^2\right)\frac{4{r_{\mathrm{w}}}^2}{d^2}\dot{\vartheta}_{\Delta}\dot{\varphi}\sin(\varphi)\cos(\varphi) \end{bmatrix} \tag{4.42}
$$

It is observed that the Coriolis forces coincide.

The damping matrix:

$$
\begin{aligned}
d_{11}{r_{\mathrm{w}}}^2 &= 2c_{\alpha} \\
d_{11}{r_{\mathrm{w}}}^2 + d_{12}r_{\mathrm{w}} &= 2c_{\alpha} - 2c_{\alpha} = 0 \\
d_{11}{r_{\mathrm{w}}}^2 + 2d_{12}r_{\mathrm{w}} + d_{22} &= 2c_{\alpha} - 4c_{\alpha} + 2c_{\alpha} = 0 \\
d_{33}\frac{4{r_{\mathrm{w}}}^2}{d^2} &= 2c_{\alpha}
\end{aligned}
$$

Here one observes that the friction proposed by Kim does not effect the under-actuated degree of freedom. Furthermore, as the equation of motion fulfill is equivalent to the one found in the earlier section the kinodynamic constraint could have been conducted from literature result already.

Concluding this section, it is validated that the derived model coincides with the literature results and therefore is more reliable. This is necessary as an experiment has not validated the model. The main advantage of the work is that it enables modification. This is done in the next step introducing the friction between the wheel and ground to the original model.

Wheel Ground Friction

The author assumes that the friction between the wheel and the ground is the most significant for the experiments. As the literature model does not consider this, it is necessary to augment the model. To avoid errors, the simulation based on the preliminary project is extended rather than designing a new one based on the modeling results. The friction model will be kept as simple as possible. Therefore, the damping matrix $\boldsymbol{D}_{\mathrm{K}}$ is augmented. The friction forces are calculated in the current representation to reach this goal. Then they are transformed to the representation used in Kim's publication by applying the transformation (4.34).

The introduced friction model is a friction force proportional to the velocity $F = c_{\beta}v$, with v being the velocity of the wheel over the ground. The modeling of the generalized forces (4.20) - (4.21) is used to reformulate the damping matrix.

The velocity of the wheels over the ground are obtained from the kinematic sketch using the assumption of no slipping (Figure 4-3):

$$v_{\mathrm{WR}} = \left(\dot{\varphi} + \dot{\vartheta} + \dot{\vartheta}_{\Delta}\right)$$

$$v_{\mathrm{WL}} = \left(\dot{\varphi} + \dot{\vartheta} - \dot{\vartheta}_{\Delta}\right)$$

This leads to the following values of the friction forces:

$$F_{\mathrm{WR}} = c_{\beta} r_{\mathrm{w}} \left(\dot{\varphi} + \dot{\vartheta} + \dot{\vartheta}_{\Delta}\right)$$

$$F_{\mathrm{WL}} = c_{\beta} r_{\mathrm{w}} \left(\dot{\varphi} + \dot{\vartheta} - \dot{\vartheta}_{\Delta}\right)$$

The friciton of the wheels is intriduces as it is done by Kim [48].

$$\tau_{\mathrm{WR}} = c_{\alpha} \left(\dot{\vartheta} + \dot{\vartheta}_{\Delta}\right)$$

$$\tau_{\mathrm{WL}} = c_{\alpha} \left(\dot{\vartheta} - \dot{\vartheta}_{\Delta}\right)$$

So, this can be composed in a friction matrix $\boldsymbol{\tau}_{\mathrm{f}} = \boldsymbol{D}\dot{\boldsymbol{q}}$ using (4.20) - (4.22). Therein $\boldsymbol{\tau}_{\mathrm{f}}$ are the generalized non-conservative forces introduced by friction.

$$\boldsymbol{D} = \begin{bmatrix} 2c_{\alpha} + 2c_{\beta}{r_{\mathrm{w}}}^2 & 2c_{\beta}{r_{\mathrm{w}}}^2 & 0 \\ 2c_{\beta}{r_{\mathrm{w}}}^2 & 2c_{\beta}{r_{\mathrm{w}}}^2 & 0 \\ 0 & 0 & 2c_{\alpha} + 2c_{\beta}{r_{\mathrm{w}}}^2 \end{bmatrix} \tag{4.43}$$

Finally, the matrix is transformed to the set of coordinates used by Kim doing the following transformation.

$$\boldsymbol{D}_{\mathrm{K}} = \left[\boldsymbol{T}^{-1}\right]^{\mathrm{T}} \boldsymbol{D}\boldsymbol{T}^{-1} = \begin{bmatrix} 2c_{\beta} + \frac{2c_{\alpha}}{{r_{\mathrm{w}}}^2} & -\frac{2c_{\alpha}}{r_{\mathrm{w}}} & 0 \\ -\frac{2c_{\alpha}}{r_{\mathrm{w}}} & 2c_{\alpha} & 0 \\ 0 & 0 & -\frac{c_{\alpha}d^2}{2{r_{\mathrm{w}}}^2} - \frac{1}{2}c_{\beta}d^2 \end{bmatrix} \tag{4.44}$$

This leads to the suggestion that the coefficient d_{11} and d_{33} shall be modified to introduce the friction between wheel and ground.

$$d_{11} = 2c_{\beta} + \frac{2c_{\alpha}}{{r_{\mathrm{w}}}^2}$$

$$d_{33} = 2c_{\beta} + \frac{2c_{\alpha}}{{r_{\mathrm{w}}}^2}$$

4.1.3 Simulation Model

The simulation model is taken from [4, p. 20], respectively [48] as the original source, as it is validated in the experiment and reliable. In the process, it has been observed that the behavior does not reflect the physical desired properties, most notably the influence of the friction in the motor and gears that must not influence the trajectory in the configuration space. Therefore, the derivation of the explicit representation is repeated, and different solutions of the coefficients D_{11} and D_{12} are found. In the remainder, the model is validated using the energy balance. It is found that the model from [4] contradicts energy conservation and must therefore be wrong. The model as it is derived here satisfies the energy conservation and is therefore plausible. Furthermore, it yields results that are closer to observations.

The equation of motion (4.35) is solved explicitly for the second derivative of the generalized coordinates as the mass matrix $\boldsymbol{M}$ is regular.

$$\ddot{\boldsymbol{q}} = \boldsymbol{M}^{-1}(\boldsymbol{q})\left(-\boldsymbol{C}\left(\dot{\boldsymbol{q}},\boldsymbol{q}\right) - \boldsymbol{D}\dot{\boldsymbol{q}} - \boldsymbol{g}(\boldsymbol{q}) + \boldsymbol{B}\boldsymbol{u}\right) \tag{4.45}$$

Kim introduces the following representation of the matrices:

$$\boldsymbol{M} = \begin{bmatrix} a_{11} & a_{12} & 0 \\ a_{12} & a_{22} & 0 \\ 0 & 0 & a_{33} \end{bmatrix} \qquad \boldsymbol{C} = \begin{bmatrix} 0 & c_{12} & c_{13} \\ 0 & 0 & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix} \qquad \boldsymbol{D} = \begin{bmatrix} d_{11} & d_{12} & 0 \\ d_{12} & d_{22} & 0 \\ 0 & 0 & d_{33} \end{bmatrix} \tag{4.46}$$

$$\boldsymbol{M}^{-1} = \begin{bmatrix} \frac{a_{22}}{a_{11}a_{22}-a_{12}^2} & \frac{-a_{12}}{a_{11}a_{22}-a_{12}^2} & 0 \\ \frac{-a_{12}}{a_{11}a_{22}-a_{12}^2} & \frac{a_{11}}{a_{11}a_{22}-a_{12}^2} & 0 \\ 0 & 0 & \frac{1}{a_{33}} \end{bmatrix} \qquad \boldsymbol{g}(\boldsymbol{q}) = \begin{bmatrix} 0 \\ g_2 \\ 0 \end{bmatrix} \qquad \boldsymbol{B} = \begin{bmatrix} \frac{1}{r_w} & \frac{1}{r_w} \\ -1 & -1 \\ -\frac{d}{2r_w} & \frac{d}{2r_w} \end{bmatrix} \tag{4.47}$$

This leads to the explicit representations of the following matrix products:

$$\boldsymbol{M}^{-1}\boldsymbol{C} = \begin{bmatrix} 0 & \frac{a_{22}c_{12}}{a_{11}a_{22}-a_{12}^2} & \frac{a_{22}c_{13}-a_{12}c_{23}}{a_{11}a_{22}-a_{12}^2} \\ 0 & \frac{-a_{12}c_{12}}{a_{11}a_{22}-a_{12}^2} & \frac{-a_{12}c_{13}+a_{11}c_{23}}{a_{11}a_{22}-a_{12}^2} \\ \frac{c_{31}}{a_{33}} & \frac{c_{32}}{a_{33}} & \frac{c_{33}}{a_{33}} \end{bmatrix} \qquad \boldsymbol{M}^{-1}\boldsymbol{D} = \begin{bmatrix} \frac{a_{11}d_{11}-a_{12}d_{12}}{a_{11}a_{22}-a_{12}^2} & \frac{a_{11}d_{12}-a_{12}d_{22}}{a_{11}a_{22}-a_{12}^2} & 0 \\ \frac{-a_{12}d_{11}+a_{22}d_{12}}{a_{11}a_{22}-a_{12}^2} & \frac{-a_{12}d_{11}+a_{22}d_{12}}{a_{11}a_{22}-a_{12}^2} & 0 \\ 0 & 0 & \frac{d_{33}}{a_{33}} \end{bmatrix} \tag{4.48}$$

$$\boldsymbol{M}^{-1}\boldsymbol{g} = \begin{bmatrix} \frac{-a_{12}g_2}{a_{11}a_{22}-a_{12}^2} \\ \frac{a_{22}g_2}{a_{11}a_{22}-a_{12}^2} \\ 0 \end{bmatrix} \qquad \boldsymbol{M}^{-1}\boldsymbol{B} = \begin{bmatrix} \frac{a_{22}\frac{1}{r_w}+a_{12}}{a_{11}a_{22}-a_{12}^2} & \frac{a_{22}\frac{1}{r_w}+a_{12}}{a_{11}a_{22}-a_{12}^2} \\ \frac{-a_{12}\frac{1}{r_w}-a_{11}}{a_{11}a_{22}-a_{12}^2} & \frac{-a_{12}\frac{1}{r_w}-a_{11}}{a_{11}a_{22}-a_{12}^2} \\ -\frac{d_{33}d}{2a_{33}r_w} & \frac{d_{33}d}{2a_{33}r_w} \end{bmatrix} \tag{4.49}$$

The coefficients are taken from [48] with the proposed representation of the damping matrix.

$$
\begin{aligned}
a_{11} &= m_{\mathrm{b}} + 2m_{\mathrm{w}} + 2J_{y\mathrm{W}} \\
a_{12} &= m_{\mathrm{b}} l_{\mathrm{cg}} \cos(\varphi) \\
a_{22} &= J_{\mathrm{b}y} + m_{\mathrm{b}} l_{\mathrm{cg}}{}^2 \\
a_{33} &= J_{z\mathrm{B}} + 2J_{z\mathrm{W}} + \left(m_{\mathrm{w}} + \frac{J_{y\mathrm{W}}}{r_{\mathrm{w}}{}^2}\right)\frac{d}{2} - \left(J_{z\mathrm{B}} - J_{x\mathrm{B}} - m_{\mathrm{b}} l_{\mathrm{cg}}{}^2\right)\sin^2(\varphi) \\
c_{12} &= -m_{\mathrm{b}} l_{\mathrm{cg}} \dot{\varphi} \sin(\varphi) \\
c_{13} &= -m_{\mathrm{b}} l_{\mathrm{cg}} \dot{\psi} \sin(\varphi) \\
c_{23} &= \left(J_{z\mathrm{B}} - J_{x\mathrm{B}} - m_{\mathrm{b}} l_{\mathrm{cg}}{}^2\right)\dot{\psi}\sin(\varphi)\cos(\varphi) \\
c_{31} &= m_{\mathrm{b}} l_{\mathrm{cg}} \dot{\psi} \sin(\varphi) \\
c_{32} &= -\left(J_{z\mathrm{B}} - J_{x\mathrm{B}} - m_{\mathrm{b}} l_{\mathrm{cg}}{}^2\right)\dot{\psi}\sin(\varphi)\cos(\varphi) \\
c_{33} &= -\left(J_{z\mathrm{B}} - J_{x\mathrm{B}} - m_{\mathrm{b}} l_{\mathrm{cg}}{}^2\right)\dot{\varphi}\sin(\varphi)\cos(\varphi) \\
d_{11} &= 2\frac{c_\alpha}{r_{\mathrm{w}}{}^2} + 2c_\beta \\
d_{12} &= -2\frac{c_\alpha}{r_{\mathrm{w}}} \\
d_{22} &= 2c_\alpha \\
d_{33} &= \frac{c_\alpha d^2}{2r_{\mathrm{w}}{}^2} + \frac{c_\beta d^2}{2} \\
g_2 &= -m_{\mathrm{b}} l_{\mathrm{cg}} g \sin(\varphi)
\end{aligned}
$$

The explicit formulation of the dynamic equation of motion is now computed for the following state vector:

$$
\boldsymbol{x} = \begin{bmatrix} \varphi & \dot{\varphi} & s_{\mathrm{w}} & \dot{s_{\mathrm{w}}} & \psi & \dot{\psi} \end{bmatrix}^{\mathrm{T}} \tag{4.50}
$$

$$
\dot{\boldsymbol{x}} = \begin{bmatrix}
\dot{\varphi} \\
\frac{\sin(\varphi)}{V_1}\left(C_{21}g - C_{22}\dot{\varphi}^2 - C_{23}\dot{\psi}^2\right) + \frac{D_{21}}{V_1}\dot{s_{\mathrm{w}}} - \frac{D_{22}}{V_1}\dot{\varphi} - \frac{B_2}{V_1}(u_1 + u_2) \\
\dot{s_{\mathrm{w}}} \\
\frac{\sin(\varphi)}{V_1}\left(-C_{11}g + C_{12}\dot{\varphi}^2 + C_{13}\dot{\psi}^2\right) - \frac{D_{11}}{V_1}\dot{s_{\mathrm{w}}} + \frac{D_{12}}{V_1}\dot{\varphi} + \frac{B_1}{V_1}(u_1 + u_2) \\
\dot{\psi} \\
\frac{\sin(\varphi)}{V_2}\left(C_{31}\dot{\varphi}\dot{\psi} - m_{\mathrm{b}} l_{\mathrm{cg}}\dot{s_{\mathrm{w}}}\dot{\psi}\right) - \frac{D_{33}}{V_2}\dot{\psi} - \frac{B_3}{V_2}(u_1 - u_2)
\end{bmatrix} \tag{4.51}
$$

The coefficients are chosen to be positive. The ones that are also found in [48], C_{13}, C_{23} and C_{31} are confirmed, the ones that are given in [4] are as well, beside the D_{12} and D_{11}.

$$
\begin{aligned}
C_{11} &= m_{\mathrm{b}}^2 {l_{\mathrm{cg}}}^2 \cos(\varphi) \\
C_{12} &= \left(J_{y\mathrm{B}} + m_{\mathrm{b}} {l_{\mathrm{cg}}}^2\right) m_{\mathrm{b}} l_{\mathrm{cg}} \\
C_{13} &= m_{\mathrm{b}} l_{\mathrm{cg}} \left(J_{y\mathrm{B}} + m_{\mathrm{b}} {l_{\mathrm{cg}}}^2 + \left(J_{z\mathrm{B}} - J_{x\mathrm{B}} - m_{\mathrm{b}} {l_{\mathrm{cg}}}^2\right) \cos^2(\varphi)\right) \\
C_{21} &= \left(m_{\mathrm{b}} + 2 m_{\mathrm{w}} + 2 \frac{J_{y\mathrm{W}}}{{r_{\mathrm{w}}}^2}\right) m_{\mathrm{b}} l_{\mathrm{cg}} \\
C_{22} &= m_{\mathrm{b}}^2 {l_{\mathrm{cg}}}^2 \cos(\varphi) \\
C_{23} &= \left(m_{\mathrm{b}}^2 {l_{\mathrm{cg}}}^2 + \left(m_{\mathrm{b}} + 2 m_{\mathrm{w}} + 2 J_{y\mathrm{W}}\right) \left(J_{z\mathrm{B}} - J_{x\mathrm{B}} - m_{\mathrm{b}} {l_{\mathrm{cg}}}^2\right)\right) \cos(\varphi) \\
C_{31} &= 2 \left(J_{z\mathrm{B}} - J_{x\mathrm{B}} - m_{\mathrm{b}} {l_{\mathrm{cg}}}^2\right) \cos(\varphi) \\
V_1 &= \left(m_{\mathrm{b}} + 2 m_{\mathrm{w}} + \frac{2 J_{y\mathrm{W}}}{{r_{\mathrm{w}}}^2}\right) \left(J_{y\mathrm{B}} + m_{\mathrm{b}} {l_{\mathrm{cg}}}^2\right) - m_{\mathrm{b}}^2 {l_{\mathrm{cg}}}^2 \cos(\varphi)^2 \\
V_2 &= J_{z\mathrm{B}} + 2 J_{z\mathrm{W}} + \left(m_{\mathrm{w}} + \frac{J_{y\mathrm{W}}}{{r_{\mathrm{w}}}^2}\right) \frac{d}{2} - \left(J_{z\mathrm{B}} - J_{x\mathrm{B}} - m_{\mathrm{b}} {l_{\mathrm{cg}}}^2\right) \sin^2(\varphi) \\
D_{22} &= \left(m_{\mathrm{b}} + 2 m_{\mathrm{w}} + \frac{2 J_{y\mathrm{W}}}{{r_{\mathrm{w}}}^2}\right) 2 c_\alpha + m_{\mathrm{b}} l_{\mathrm{cg}} \cos(\varphi) \frac{2 c_\alpha}{r_{\mathrm{w}}} \\
D_{21} &= \left(m_{\mathrm{b}} + 2 m_{\mathrm{w}} + \frac{2 J_{y\mathrm{W}}}{{r_{\mathrm{w}}}^2}\right) \frac{2 c_\alpha}{r_{\mathrm{w}}} + m_{\mathrm{b}} l_{\mathrm{cg}} \cos(\varphi) \left(\frac{2 c_\alpha}{{r_{\mathrm{w}}}^2} + 2 c_\beta\right) \\
D_{12} &= \left(J_{y\mathrm{B}} + m_{\mathrm{b}} {l_{\mathrm{cg}}}^2\right) \frac{2 c_\alpha}{r_{\mathrm{w}}} + m_{\mathrm{b}} l_{\mathrm{cg}} \cos(\varphi) 2 c_\alpha \\
D_{11} &= \left(J_{y\mathrm{B}} + m_{\mathrm{b}} {l_{\mathrm{cg}}}^2\right) \left(\frac{2 c_\alpha}{{r_{\mathrm{w}}}^2} + 2 c_\beta\right) + m_{\mathrm{b}} l_{\mathrm{cg}} \cos(\varphi) \frac{2 c_\alpha}{r_{\mathrm{w}}} \\
D_{33} &= \frac{c_\alpha d^2}{2 {r_{\mathrm{w}}}^2} + \frac{c_\beta d^2}{2} \\
B_2 &= m_{\mathrm{b}} \frac{l_{\mathrm{cg}}}{r_w} \cos(\varphi) + m_{\mathrm{b}} + 2 m_{\mathrm{w}} + \frac{2 J_{y\mathrm{W}}}{{r_{\mathrm{w}}}^2} \\
B_1 &= \left(J_{y\mathrm{B}} + m_{\mathrm{b}} {l_{\mathrm{cg}}}^2\right) \frac{1}{r_{\mathrm{w}}} + m_{\mathrm{b}} l_{\mathrm{cg}} \cos(\varphi) \\
B_3 &= \left(\frac{c_\alpha d^2}{2 {r_{\mathrm{w}}}^2} + \frac{c_\beta d^2}{2}\right) \frac{d}{r_{\mathrm{w}}}
\end{aligned}
$$

Model of the planar TWIPR

The project focuses on the planar TWIPR, meaning that the heading ψ is constantly zero and, therefore, the wheel angle difference ϑ_Δ is set to zero. That means the model reduces to:

$$
\begin{aligned}
\boldsymbol{x} &= \begin{bmatrix} \varphi & \dot{\varphi} & s_{\mathrm{w}} & \dot{s_{\mathrm{w}}} \end{bmatrix}^{\mathrm{T}} \\
\dot{\boldsymbol{x}} &= \underbrace{\begin{bmatrix} \dot{\varphi} \\ \frac{\sin(\varphi)}{V_1} \left(C_{21} g - C_{22} \dot{\varphi}^2\right) + \frac{D_{21}}{V_1} \dot{s_{\mathrm{w}}} - \frac{D_{22}}{V_1} \dot{\varphi} - \frac{B_2}{V_1} u \\ \dot{s_{\mathrm{w}}} \\ \frac{\sin(\varphi)}{V_1} \left(-C_{11} g + C_{12} \dot{\varphi}^2\right) - \frac{D_{11}}{V_1} \dot{s_{\mathrm{w}}} + \frac{D_{12}}{V_1} \dot{\varphi} + \frac{B_1}{V_1} u \end{bmatrix}}_{\boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u})}
\end{aligned}
\tag{4.52}
$$

Testing the planning specific properties to validate the modeling

To validate the model in regard to the planning-specific properties, a standard ILC is applied to learn a given output trajectory to four models with different friction characteristics. In the model used in [4] the values of D_{11} and D_{12} differ by a sign flip. To compare the results and show the physical misbehavior, it is also simulated and introduced as "ref" case that also introduces the friction in the engine. Furthermore, a result that is also used in the experimental validation (in the low amplitude reference case Sec. 5.1) is used to show that the introduced friction leads to a superior representation of the real behavior.

The method to control the pitch is a standard Iterative Learning Control (ILC) as it is also used in Section 3.9 in the case **SISO**.

Model	Engine friction c_α	Tire friction c_β
I	$0\,\mathrm{Nms}$	$0\,\mathrm{Nsm}^{-1}$
II (ref)	$4\times10^{-3}\,\mathrm{Nms}$	$0\,\mathrm{Nsm}^{-1}$
III (sim)	$4\times10^{-3}\,\mathrm{Nms}$	$6\times10^{-1}\,\mathrm{Nsm}^{-1}$
IV	$0\,\mathrm{Nms}$	$6\times10^{-1}\,\mathrm{Nsm}^{-1}$

Table 4.2. Values of the damping coefficients, used for identification and validation. Case III is the selection for the simulation study.

The selection of the friction parameters is given in Table 4.2 and the resulting trajectories in Figure 4-4. In Figure 4.4(a), it can be observed that the trajectories of models I and II coincide, and so do the trajectories of models III and IV. Figure 4.4(b) show that the learned input changes for each model. The trajectory in the ref case differs from cases I and II, which conflicts with the result from the theory that the kinodynamic constraint is not influenced by the engine's friction. Furthermore, it is observed that the trajectories of cases III and IV mimic the behavior of the experiment more accurately than the case without that friction modeled, namely I and II.

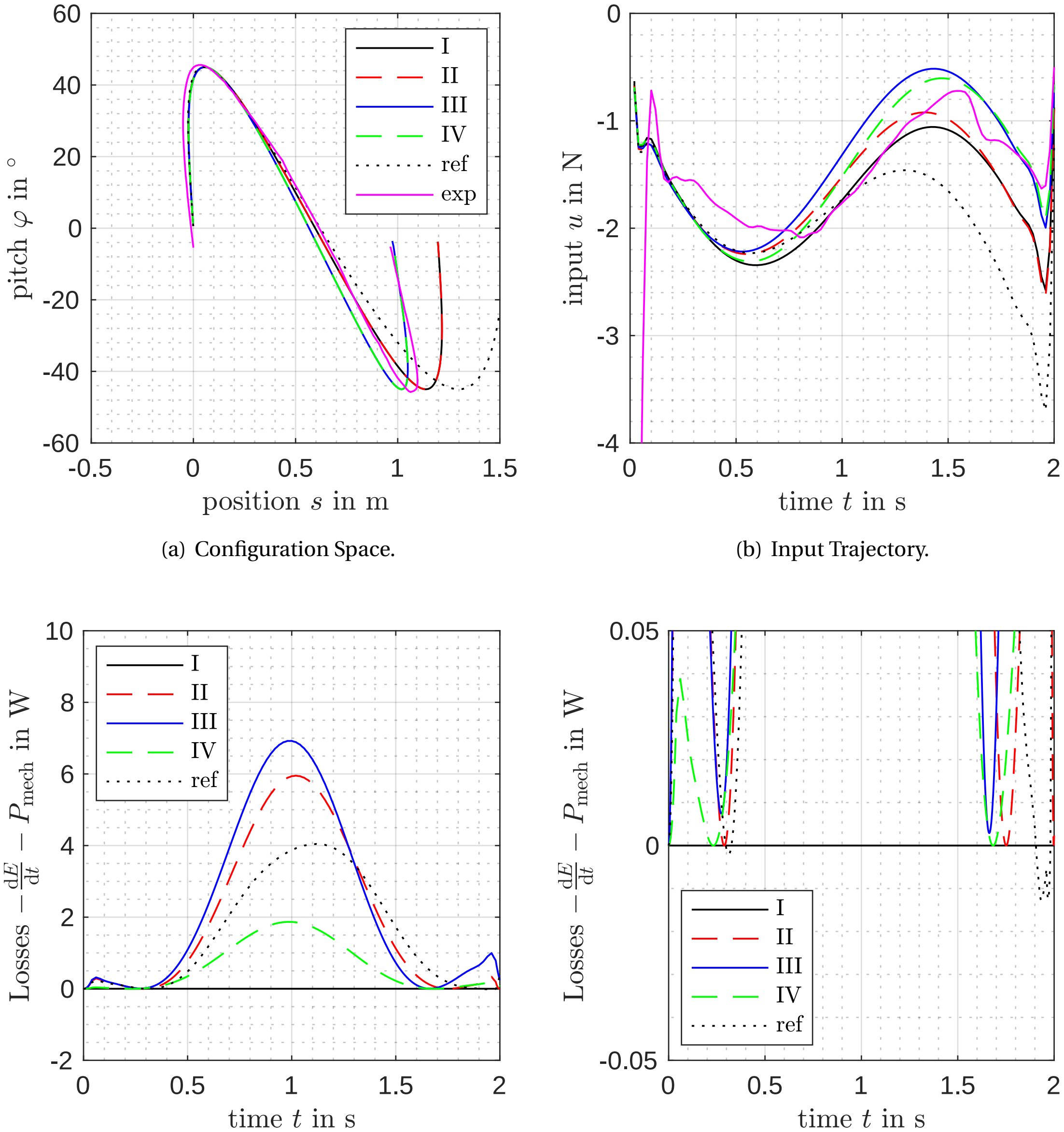
To ensure that the simulation implementation is correct, the energy balance is computed. The only power inflow is given by the mechanical power introduced by the engine. Therefore, in the presence of friction, the balance

$$\frac{\mathrm{d}(T+V)}{\mathrm{d}t} - \underbrace{\dot{\boldsymbol{q}}^{\mathrm{T}}\boldsymbol{\tau}}_{P_{\mathrm{mech}}} = \frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{1}{2}\dot{\boldsymbol{q}}^{\mathrm{T}}\boldsymbol{M}(\boldsymbol{q})\dot{\boldsymbol{q}} + m_{\mathrm{b}} l_{\mathrm{cg}}\cos(\varphi)\right] - \dot{\boldsymbol{q}}\boldsymbol{B}\boldsymbol{u}$$

is checked to be strictly non-negative and zero in the absence of friction. That these properties are satisfied can be validated in Figure 4.4(c). In case I, the system is conservative, and the balance is zero constantly. Furthermore, it can be seen that the requirement is violated for the ref model, showing that it is physically not reasonable. This strategy is also motivated by the passivity theory of mechanical systems and the Hamiltonian formulation as it is explained in [43, p. 154].

Linearization

The model is stabilized in the upper equilibrium (i. e. $\boldsymbol{x} = \mathbf{0}$ and $\boldsymbol{u} = \mathbf{0}$), therefore a linearized model regarding this equilibrium is used. Furhtermore, only the position and the pitch are taken into account as planning is later done for this coordinates exclusively. The resulting equations are given in the following.



(a) Configuration Space.

(b) Input Trajectory.

(c) Energy Balance.

(d) Energy Balance zoomed in.

Figure 4-4. Comparison of friction modeling.

$$\boldsymbol{x} = \begin{bmatrix} \varphi & \dot{\varphi} & s_{\mathrm{W}} & \dot{s_{\mathrm{W}}} \end{bmatrix}^{\mathrm{T}} \tag{4.53}$$

$$\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{u} \tag{4.54}$$

$$\boldsymbol{A} = \left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right|_{\boldsymbol{x}=\boldsymbol{0},\boldsymbol{u}=\boldsymbol{0}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ \frac{C_{21}g}{V_1} & -\frac{D_{22}}{V_1} & 0 & \frac{D_{21}}{V_1} \\ 0 & 0 & 0 & 1 \\ -\frac{C_{11}g}{V_1} & \frac{D_{12}}{V_1} & 0 & -\frac{D_{11}}{V_1} \end{bmatrix} \qquad \boldsymbol{B} = \left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{u}}\right|_{\boldsymbol{x}=\boldsymbol{0},\boldsymbol{u}=\boldsymbol{0}} = \begin{bmatrix} 0 \\ -\frac{B_2}{V_1} \\ 0 \\ \frac{B_2}{V_1} \end{bmatrix} \tag{4.55}$$

In the following sections often the case without friction underlies the planning methods. Thus, in case friction is omitted the model reduces to:

$$\boldsymbol{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ \frac{C_{21}g}{V_1} & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ -\frac{C_{11}g}{V_1} & 0 & 0 & 0 \end{bmatrix} \qquad \boldsymbol{B} = \begin{bmatrix} 0 \\ -\frac{B_2}{V_1} \\ 0 \\ \frac{B_2}{V_1} \end{bmatrix} \tag{4.56}$$

Remark 9. *Linear model. Observe that in contrast to the complex structure of the non-linear model the linear model without friciton has only two parameters to be estimated (4.56). This requires less effort in system identification.*

4.2 Technical design

In this section system design advances are presented that are done in this thesis project. As the work of Meindl [4] serves as a basis the details of the measurement system and the hardware control realizations are not repeated. In this thesis project the Iterative Learning Control (ILC) algorithms are not implemented on the embedded system but in a computer program processed in matlab. To maintain the ability to reproduce a matlab program is written that reports the experiments' results in an unified format. Commands are given via a Transmission Control Protocol (TCP) socket interface while the data is transfered using File Transfer Protocol (FTP). The description is not exhaustive but shall make the structure and functionality clear.

4.2.1 Control system

In this section, the robot's control system is described, so the interconnections become clear. As the development is done in [4], it focuses on the essentials. The points discussed are the feedback control algorithm, inputs, and outputs, as well as the sensor systems and the actuator system design.

Figure 4-5 shows a schematic of the control system. Measurement is realized using a wheel encoder and mems-based IMUs. The motor driver takes torque as the control input. The measurement system and the drivers' zero-order hold operate on a sampling frequency of 50 Hz. The sampling period h ist therefore 20 ms. The feedback controller is implemented on a beagle-bone black. The ILC algorithm is calculated on an operator PC. Therefore, the input and output trajectories are exchanged using `ftp`. The remote application, triggering a trail, is controlled using a `tcp` connection to a Matlab program.

The linear state feedback controller $\boldsymbol{K}$ is chosen using a Linear Quadratic Regulator (LQR) [51], which is sufficient for a swing-up maneuver. From a design point of view, it appears to be natural to

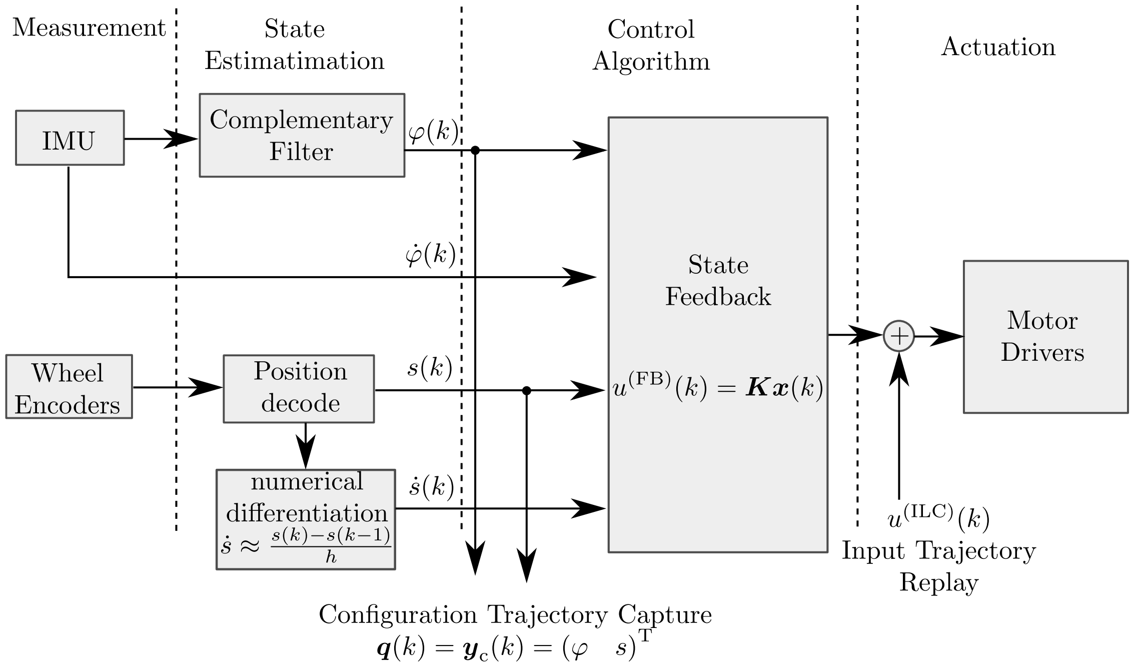


Figure 4-5. Structure of the implemented controller.

stabilize only $\dot{s}$ or even φ and leave the position in an open loop as it associates acceleration and speed as the controlled input. This is not desirable in the ILC system as it leaves integral behavior on the plant and is therefore not asymptotically stable. The problem that arises is that the assumption of a trail-invariant disturbance sequence is not realistic, as trail-invariant disturbances might occur. In the case of a stable controller, those trail invariant disturbances are damped in the control loop over time while using "intuitive" controllers leads to integration, and therefore high amplitude state variant disturbances will occur of whom the ILC algorithms suffer. This leaves the operator with a counter-intuitive plant behavior. This issue can be overcome using a feed-forward controller that introduces integrator or double integrator behavior and compensates known plant roots. As the system is autonomous in the present setup, an intuitive input is not required directly, but as discussed in the ILC analysis section, the input signals norm is an optimized quantity and, therefore, significantly influences the resulting trajectory.

In the present setup, the design strategy is the LQR optimizing the cost function given in (4.57). The design parameters (4.58) and the resulting plant roots (4.59) are given in the following:

$$J(u) = \int_0^\infty (\boldsymbol{x}^{\mathrm{T}}\boldsymbol{Q}\boldsymbol{x} + r u^2)\mathrm{d}t \tag{4.57}$$

$$\boldsymbol{Q} = \mathrm{Diag}\begin{bmatrix}0.1 & 0.5\,\mathrm{s}^2 & 10\,\mathrm{m}^{-2} & 0\end{bmatrix} \qquad r = 5 \tag{4.58}$$

$$s_1 \approx -26.9\,\mathrm{s}^{-1} \qquad s_2 \approx -1.25\,\mathrm{s}^{-1} + \mathrm{j}1.64\,\mathrm{s}^{-1} \qquad s_3 \approx -1.25\,\mathrm{s}^{-1} - \mathrm{j}1.64\,\mathrm{s}^{-1} \qquad s_4 \approx -2.34\,\mathrm{s}^{-1} \tag{4.59}$$

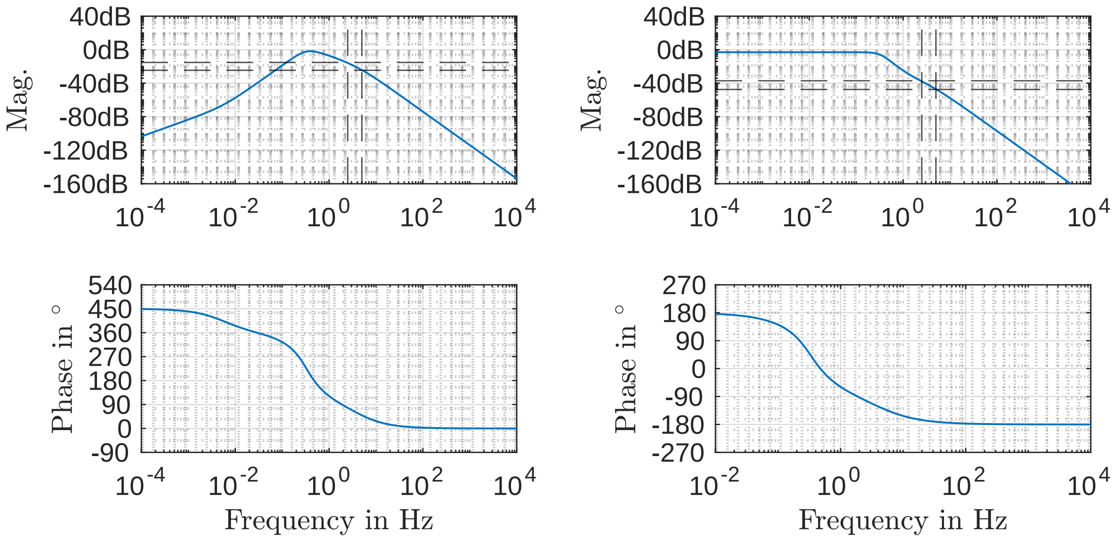
$$\boldsymbol{K} = \begin{bmatrix}-2.4471 & -0.5021 & -1.4142 & -1.4831\end{bmatrix} \tag{4.60}$$

4.2.2 Bandwidth of the controlled system and disctrization

Regarding the point-to-point method with equidistant reference points, it is important to characterize the system's bandwidth to select the clearance of consecutive samples. To characterize plants frequency behavior, the system's equations of motion are linearized around the upright equilibrium (4.56), and the feedback controller (4.60) is applied. The characteristic of the closed loop is obtained:

$$\dot{\boldsymbol{x}} = (\boldsymbol{A} - \boldsymbol{B}\boldsymbol{K})\,\boldsymbol{x} + \boldsymbol{B}\boldsymbol{u}_{\mathrm{c}} \tag{4.61}$$

$$\boldsymbol{y}_{\mathrm{c}} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \boldsymbol{x} \tag{4.62}$$



(a) Input to the pitch.

(b) Input to the position.

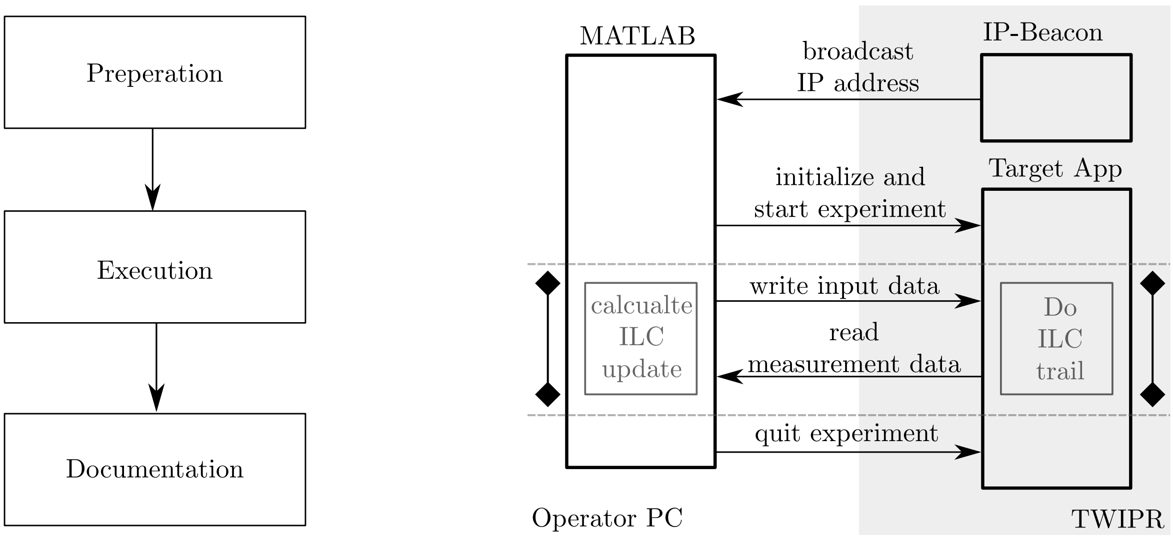
Figure 4-6. Bode diagram of the controlled robot in contentious-time. The frequencies that correspond to the equidistant sampling (2.5 Hz≙20samples and 5 Hz≙10samples) are indicated by the dashed lines. As the magnitude characteristic decreases by 40 dB per decade the diffrence in the gain for the both frequencies is about 10 dB corresponding to a gain ratio of 3.

The characteristics are given in Figure 4-6. One observes that the system's cut-off frequency is in the range of 1 Hz. In the characteristics regarding the position (right), the amplification magnitude is constant on the left, i. e. low frequencies, an indication of neither integral nor differential behavior. On the right, i. e. damping of 40dB per decade is observed, indicating a second-order low pass behavior. The phase starts from 180° and ends at -180° indicating a non-minimum phase system as a minimum phase 2nd order low pass results in a phase shift of -180°. Accordingly, the behavior of the pitch is analyzed. In the lower frequencies, the gain rises with 20dB per decade, indicating differential behavior and decays at 40dB per decade in higher frequencies. The overall phase shift is 450° indicating non-minimum phase behavior as the required phase shift is only 270° [52] [9, pp. 33].

To obtain the lifted fundamental system for the system representation, as explained in (Definition 9) the system (4.61) and (4.62) is discretized using maltab c2d [36]. As the given robot only delivers measurement of the pitch and the position only those outputs can be selected or observed. How the problem is overcome is given in the following Section 4.3.

4.2.3 Experimental Setup

To perform the experiments, a program is developed on the basis of the work done in [4]. The parts added is the documentation as well as the implementation of the iterative learning controllers. The code is available under `https://git.tu-berlin.de/campe/mp_and_ilc`.



(a) Script setup. (b) Communication.

Figure 4-7. General design of the test system setup.

The code of the toolbox is in the matlab repository `4_ILC_lib`. It consistis of

Listing 4.1. Tree query of the `4_ILC_lib` directory

```
.
|---- adapter (wrappers to adopt to the given software components)
|---- docs (additional contents for the documentation)
|---- feedback (feedbackcontroler abstraction)
|---- input_signal_filter (additional feedforward controllers)
|---- metrics (utils to calculate the matrics on basis of ILC therory)
|---- q_filter (Q filters)
|---- selectors (Selector Abstraction to apply Point to Point Mehods)
|---- system_representation (System representaion as described in ILC System representation)
|---- trajectory_generator (Trajectory generators)
|---- plotting_utils (functions used for plotting)
|---- utils (usefull fuctions used in the other parts)

12 directories
```

For each experiment a script is designed, composed of the following parts. The design details are explained in the `Readme.`.

The lib is hold adoptable to future robot implementations. It can be adopted by modifying `4_ILC_lib/adapter/do_an_ilc_experiment.m`.

To design an autonomous connection of the TWIPR with the operators PC, a program is developed that broadcasts the IP that is designed as follows.

- the TWIPR shares its IP using a Python program autostarted on boot
- the program is stored in `/usr/local/ip_beacon` and depends on `/usr/python3`
- it is set as a system deamon and can be monitored in the shell using `systemctl status ip_beacon`
- it broadcast a hello message `I'am the TWIPR` to port 5005

4.2.4 Safety loop

To maintain the robots integrity even if the developed software contains bugs a safety loop was introduced. It on the hardware side it consists of a ESP32 D1-mini dev board that consist of a micro controller with a WLAN interface. It can be programmed in the arduino framework, that makes the development very easy. The robot is disabled by intercepting the enable line of the motor drivers. A client device controlling the system can be introduced by any other process that has access to the network. The overall design is depicted in Figure 4-8. The communication is done using User Datagram Protocol (UDP) to be robust against network issues.

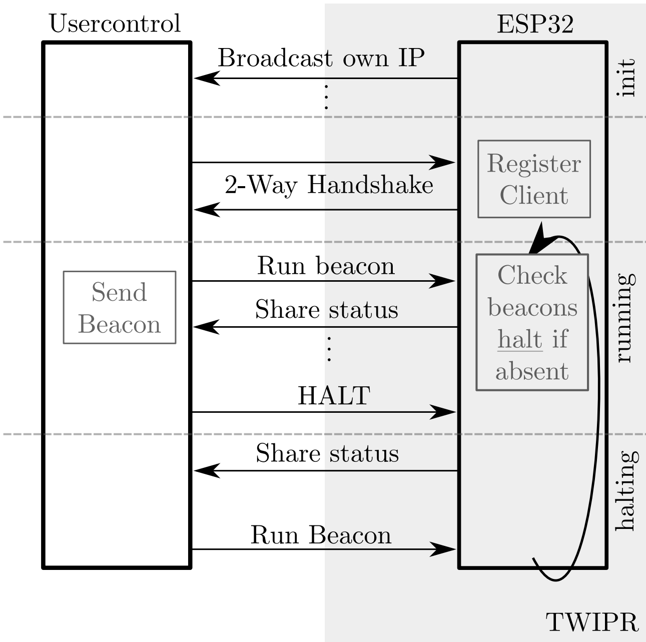


Figure 4-8. Architecture of the savetyloop.

The software on the ESP32 represents a state machine that has three states *init*, *run* and *halt*.

A user process can either send a *reset* signal that must contain a `R` in the first digits in its ascii encoded data packet and a *halt* signal that must contain a `H` instead. Any other signal is interpreted as a *run beacon*. Packages must be sent to port number 8888. The host device will return its status information by sharing a string containing to each registered client. The status will be sent to each client that has sent a `run beacon` or a `halt` signal once. In case the host receives a *reset* signal it forgets all host information and switches to the *init* state again. The motor drivers are enabled in the running state only. The host can handle up to 5 clients that must all consent to operation via thier *run beacon*. If a host stays away more than 5 seconds the host changes to *halt*. In this thesis a android app is developed as a remote control.

- Host `https://github.com/BUE687/SavetyloopHost`.
- Android Remote `https://github.com/BUE687/SavetyloopAndroidRemote`

4.3 Cost on the output to learn derivatives

To introduce the outputs derivatives Finite Differences (FD) is used. This is due to the fact that the system does not expose the velocities explicitly (Figure 4-5). Furthermore, devices like wheel encoder expose a measurement of the angle but not measure the velocity. Throughout testing and development it became necessary to introduce costs on derivatives. To do this in a relatively low cost way, the differentiation is done using the ouptut weights $\boldsymbol{Q}_\mathrm{W}$ in Norm Optimal Iterative Learning Control (NO-ILC). This is done for two purposes one is the demand to assure to arrive in the goal configuration in rest, the second is to fix a velocity in Hybridization 3.5. That requires a good velocity measure within the trajectory. To accomplish this the veleocity is derived using numerical differentiation. In the final sample of the trajectroy this is done using pure FD while the intermediate sample is optimized using a Finite Impulse Response (FIR) filter.

The transfer function of at discrete-time filter estimating the derivative using backward Euler is given in (4.63).

$$D(q) = \frac{1 - q^{-1}}{h} \tag{4.63}$$

This transferfunction is translated in lifted dynamics representation. Therefore, observe the markov parameter to be

$$\boldsymbol{P}(k) = \begin{cases} \boldsymbol{E}\frac{1}{h} & k = 0 \\ -\boldsymbol{E}\frac{1}{h} & k = 1 \\ \boldsymbol{0} & \text{else} \end{cases}$$

considereing a Multiple Input Multiple Output (MIMO) system, that estimates the derivative of each input.

The calculation of the derivative of an output can be done in the $\boldsymbol{Q}_\mathrm{W}$ of a norm optimal control using the following update of the $\boldsymbol{Q}_\mathrm{W}$ matrix.

$$\left[\boldsymbol{Q}_\mathrm{W}\right]_{p\cdot(k-2)+1:p\cdot k,p\cdot(k-2)+1:p\cdot k} \leftarrow \left[\boldsymbol{Q}_\mathrm{W}\right]_{p\cdot(k-2)+1:p\cdot k,p\cdot(k-2)+1:p\cdot k} + \frac{1}{h^2}\begin{bmatrix} -\boldsymbol{E} & \boldsymbol{E} \end{bmatrix} \dot{\boldsymbol{Q}}_{\mathrm{W},k} \begin{bmatrix} -\boldsymbol{E} \\ \boldsymbol{E} \end{bmatrix} \tag{4.64}$$

This shall be understood as pseudo code updating a variable containing a vector $\boldsymbol{Q}_{\mathrm{W}}$. The value $\dot{\boldsymbol{Q}}_{\mathrm{W},k} \in \mathbb{R}^{p\times p}$ is the value of weight that is assigned to the derivative of the output. frequency.

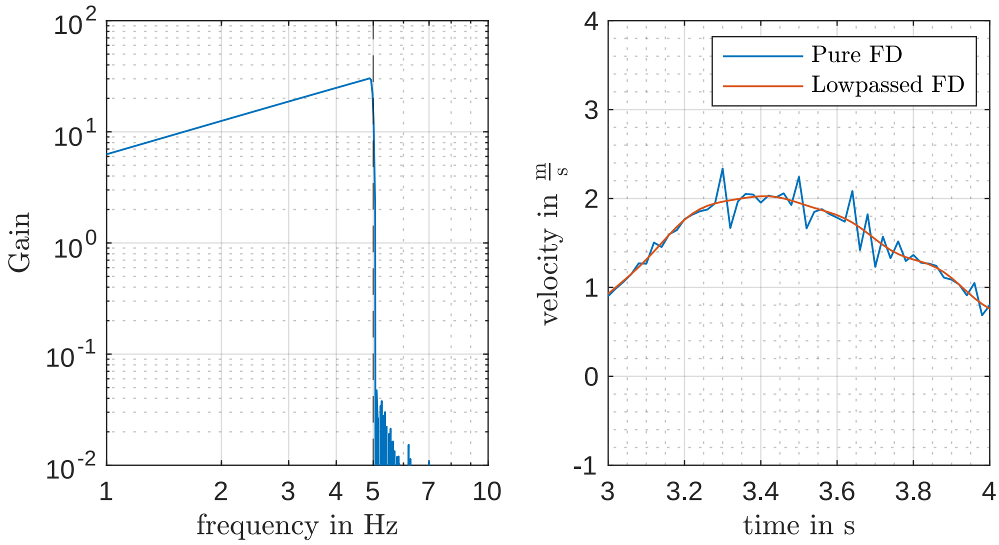


Figure 4-9. Characteristic of the applied low pass. Design in git `4_ILC_lib/utils/design_output_filter.m`.

To get the implementation of Hybridization 3.5 running it was necessary to get a more reliable estimate of the velocity. The application of a pure FD-filter lead to a noisy signal. A sufficient damping of the noise is achieved using an acausal low pass of order n_{b} and a cutoff frequency of 5 Hz using matlabs `fir1` [53] function.

The characteristic of the filter composed with the FD filter in frequency domain as well the perofor-mance in time domain on measured data is given in Figure 4-9. It can be observed that the filter damps the transients while maintaining the trajectories shape.

As FIR filter introduce windowing effects esspecially at the limits of a window the method has not been applied for the final configuration. Furhtermore, there was appernetly no need do so. To neglect windowing effects one could provide a measurement that is beyond the limits of the trail, to omit windowing effects.

In future developments it appears to be beneficial to calculate an estimate of the velocity using sensor fusion.

Chapter 5

Experimental Validation

In this chapter the methods discussed are applied to the Two Wheeled Inverted Pendulum Robot (TWIPR) as testbed in a planar setup. The case study is designed as follows. In the first case study a trajectory is generated on the basis of the linearized model, that has a low amplitude in the pitch φ allowing the application of controllers that do not deal with constraints. In the next step the movements are planned using the Local Motion Planning (LMP) as it is described in Section 2.2.1 that reflects the behaviour of a motion planner. Two traversals are reviewed, the first leads to an amplitude similar to the one of the first test. The second led to a amplitude of 70 deg that ist the most that could savely be learned throughout the experiments. In the latter the algorithms are applied to planned trajectories.

5.1 Case study on low amplitude

In this study, the algorithms' behavior is evaluated on the physical system with a low amplitude sinusoidal pitch trajectory with a duration of 2 s or 100 samples respectively and an amplitude of 45 deg. The corresponding position is calculated using the linearized model omitting friction. The TWIPR operates safely apart from constraints so that constraints do not impact the controller performance. To evaluate the results, the following metrics are chosen. The RMSE of the pitch and position trajectories as standard metrics. Also, the error of the last sample, called goal, is considered task-specific metrics. For the point-to-point methods the errors in the waypoints are also considered as metrics. The experiment data can be found in data directory in `30_comparisons/45Grad_exp`.

The control algorithms are chosen as follows: The baseline is the standard SISO ILC with equal weighting of the trajectory. As naive extensions, the Single Input Multiple Output (SIMO) controller is considered. In the next step, the point-to-point methods with and without ancillary optimization are reviewed. Also, the samples are first chosen naively, equidistantly with a period of 10 and 20 samples. In the latter, the waypoints are chosen strategically to mimic a dive maneuver. The waypoints are selected to be at the indices 20, 25, 30, 35, and 40. To mimic a backward dive, waypoints have been selected at 60, 65, 70, 75, 80 and the final point in another experiment. As last example, one way point was chosen at sample 30. In both cases also the final point is added. This is the task-specific goal configuration. In all cases the ancillary cost is chosen to be the offset from the pitch trajectory. The controllers are tuned empirically throughout simulations and experiments to obtain a desirable convergence rate. Furthermore, as the number of waypoints reduces the weight is

Name	$\boldsymbol{Q}_{\mathrm{W}}$	$\boldsymbol{S}_{\mathrm{W}}$	$\boldsymbol{R}_{\mathrm{W}}$	$\boldsymbol{Q}_{\mathrm{Wanc}}$	$\boldsymbol{S}_{\mathrm{Wanc}}$	$\boldsymbol{Q}_{\mathrm{Wanc}}$
SISO	$\boldsymbol{E}$	$0.25\boldsymbol{E}$	$10^{-3}\boldsymbol{E}$			
SIMO	$\boldsymbol{E}\otimes\mathrm{Diag}[[0.4\quad 2]]$	$0.25\boldsymbol{E}$	$10^{-3}\boldsymbol{E}$			
P2P10	$\boldsymbol{E}\otimes\mathrm{Diag}[[4\quad 20]]$	$0.25\boldsymbol{E}$	$\boldsymbol{0}$			
P2Panc10	$\boldsymbol{E}\otimes\mathrm{Diag}[[4\quad 20]]$	$0.25\boldsymbol{E}$	$\boldsymbol{0}$	$\boldsymbol{E}\otimes\mathrm{Diag}[[1\quad 0]]$	$0.25\boldsymbol{E}$	$10^{-3}\boldsymbol{E}$
P2P20	$\boldsymbol{E}\otimes\mathrm{Diag}[[8\quad 40]]$	$0.25\boldsymbol{E}$	$\boldsymbol{0}$			
P2Panc20	$\boldsymbol{E}\otimes\mathrm{Diag}[[8\quad 40]]$	$0.25\boldsymbol{E}$	$\boldsymbol{0}$	$\boldsymbol{E}\otimes\mathrm{Diag}[[1\quad 0]]$	$0.25\boldsymbol{E}$	$10^{-3}\boldsymbol{E}$
P2Pstrat	$\boldsymbol{E}\otimes\mathrm{Diag}[[4\quad 20]]$	$0.25\boldsymbol{E}$	$\boldsymbol{0}$			
P2PancStrat	$\boldsymbol{E}\otimes\mathrm{Diag}[[4\quad 20]]$	$0.25\boldsymbol{E}$	$\boldsymbol{0}$	$\boldsymbol{E}\otimes\mathrm{Diag}[[1\quad 0]]$	$0.25\boldsymbol{E}$	$10^{-3}\boldsymbol{E}$
P2PstratBW	$\boldsymbol{E}\otimes\mathrm{Diag}[[8\quad 40]]$	$0.25\boldsymbol{E}$	$\boldsymbol{0}$			
P2PoneWP	$\boldsymbol{E}\otimes\mathrm{Diag}[[8\quad 40]]$	$0.25\boldsymbol{E}$	$\boldsymbol{0}$			
P2PancOneWP	$\boldsymbol{E}\otimes\mathrm{Diag}[[8\quad 40]]$	$0.25\boldsymbol{E}$	$\boldsymbol{0}$	$\boldsymbol{E}\otimes\mathrm{Diag}[[1\quad 0]]$	$0.25\boldsymbol{E}$	$10^{-3}\boldsymbol{E}$

Table 5.1. Controller tuning for the first simulation study.

increased so that the proportion of the costs is maintained. The precise tuning is given in Table 5.1.

The metrics are presented in Table 5.2 and Figure 5-1. In the latter the maximum value is normed to one to enhance comparability. Selected trajectories are represented in Figures 5-2 to 5-9.

Name	**Pitch Err** in °	**Pos Err** in cm	**Goal Pitch Err** in °	**Goal Pos Err** in cm	**WP Pitch Err** in °	**WP Pos Err** in cm
SISO	1.56	1.04×10^{1}	9.40×10^{-1}	2.98×10^{1}		
SIMO	3.62	4.70	5.49	1.44×10^{1}		
P2P10	6.23	4.32	1.91	6.45	4.78	4.71
P2Panc10	5.44	2.55	1.65	6.58	3.95	3.12
P2P20	9.87	4.11	1.80	6.18	2.21	4.25
P2Panc20	8.67	3.97	1.31	2.96	2.40	3.92
P2Pstrat	9.58	3.51	1.85	2.00	1.01	1.22
P2PancStrat	4.52	5.05	2.15	1.47×10^{1}	1.90	6.27
P2PstratBW	1.07×10^{1}	6.16	3.08	4.88	2.46	3.10
P2PoneWP	7.76	6.13	7.32×10^{-1}	2.08	7.99×10^{-1}	1.99
P2PancOneWP	6.40	5.42	8.95×10^{-1}	1.88	1.18	1.44

Table 5.2. Metrics observed in the first case study.

First, consider the results in Figure 5-2 and 5-3. The trajectories presented are the results without ancillary cost. It is observed that in the single input, single output case, the tracking performs best in the pitch, while it is the poorest in the position. The multiple output design in contrast, balances between both errors but as it does not prioritize any point of the trajectory, the compromise is not helpful for the task goal. The point-to-point methods are performing better. Both solution trajectories mimic the desired in the waypoints. But also bending is observed between the way points. They are due to the fact that the unmodeled wheel friction force must be compensated. This can only be done if the TWIPR leans forward. The effect becomes larger as the clearance between waypoints rises. The most promising results are taken out for the strategic sampling. In this case, the trajectory tracks the reference at the waypoints almost indistinguishable. The same holds true for the selection of a single-way point. After passing the waypoints, the reference trajectory is left to gain enough energy to reach the goal configuration. Leading to the most precise tracking of the

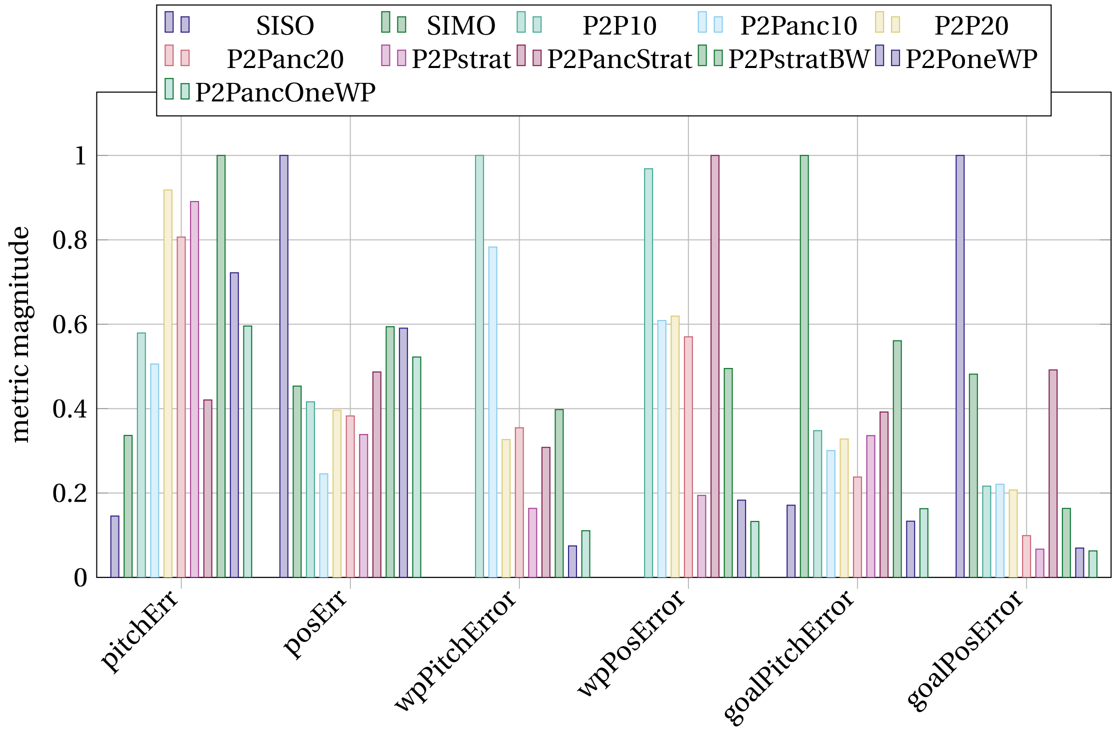


Figure 5-1. Metrics observed in the first case study. The maximal value is normed to one for each metric.

goal configuration. The clearance of only four samples between the strategically chosen waypoints is likely to gain any extra energy. Therefore, concluding this case, it is shown that the tracking of predefined waypoints is improved using point-to-point methods but the method also introduces artifacts on the trajectory. Theoretically, they could be predicetd if it is known where unmodeled friction is introduced.

Now the results are reviewed that incorporate the ancillary cost. Therefore, the same selection is presented in 5-4 and 5-5. The results are very similar. Investigating the metrics given in Figure 5-1 it can be observed that the RMSE on the pitch trajectory is improved as desired for all methods. But contrarily, the errors on the waypoints increase as it was expected from theory. Recover, the decoupling of the costs rely on the model. Furthermore, as the bandwidth of the system is in the order of 1 Hz, the ancillary cost update is of low importance, especially in the case of the waypoint period of 10 samples. For the promising method of strategic selection, the goal position error increased significantly. As a single experiment can not serve as a prove for bad decoupling.

5.1.1 Relation to the simulation study

As this experiment corresponds to the simulation study (Section 3.9) the results are related in the following. Almost all observations made in the simulation are not that easy to observe in the experiment. The influence of ancillary optimization is less obvious. Furthermore, the tracking of the goal configuration is worse than in the simulation.

Figure 5-6 shows the results of the Single Output, Multiple Output, and equidistant sampling with a period of 10 samples, receptively 20 samples. The corresponding results in the simulation is given

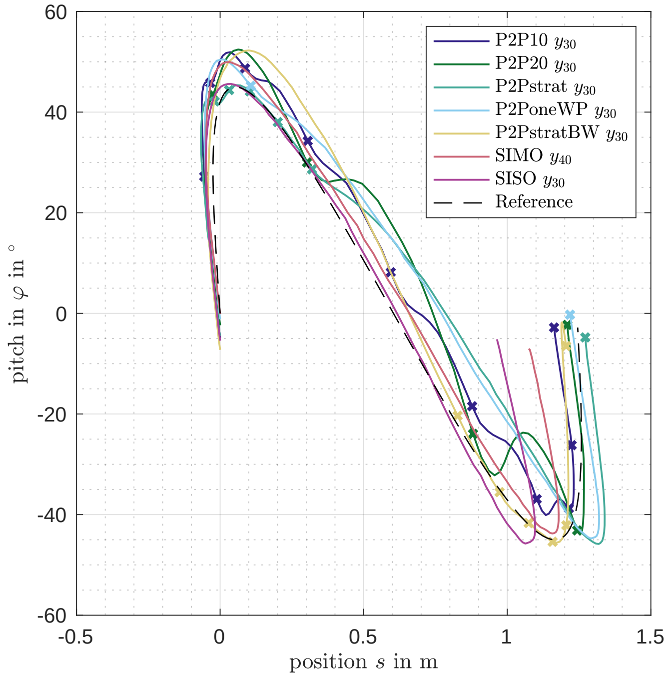


Figure 5-2. Comparison of the trajectories obtained by point to point methods.

in Figure 3-19. The results appear to be similar but are not that obvious to interpret. Possibly this is due to external reasons such as different inertial configurations.

Firstly, it is observed that the Single Input Single Output (SISO) and SIMO cases are similar to the simulation. Both are not perfect, but in the Single output case, the pitch is tracked well, while the position is bad. Also, the SIMO case is closer to the reference but still of in the goal configuration. This is not conflicting with the simulation results. But the in the case of the central sampling with a period of 10 samples it is not close to the SIMO case. Nonetheless, it is observed that the trajectory is relatively smooth. A clear compensation as in the period of 20 samples can not be observed.

Next, observe the comparison of the ancillary optimization in Figure 5-8. Recover, in the simulation, a difference is observed clearly, even though it is not very large. In the experiment, the trajectories differ, but the changes are not very large compared to the differences observed due to different initial states. Nonetheless, it can be guessed that the impact is similar (Figure 3-20). In the strategic selection method, recover, in simulation the impact on the pitch is very clear (Figure 5-9). In the experiemnt, this behavior is is also present. One feature that is clearly reproduced is the break at the position 0.3 s. Also the trajectory mimics the reference. One major drawback of the experiment is that only 20 iterations have been performed. This meight be the reason why the tracking in the goal configuration is not perfect.

Finally, the comparison of the strategic selection methods is reviewed. The simulation results are given in Figure 3-22. The experiments are given in Figure 5-9. In the experiment, the desired behavior is observed, namely, the tracking of the reference is precise in the regions with waypoints.

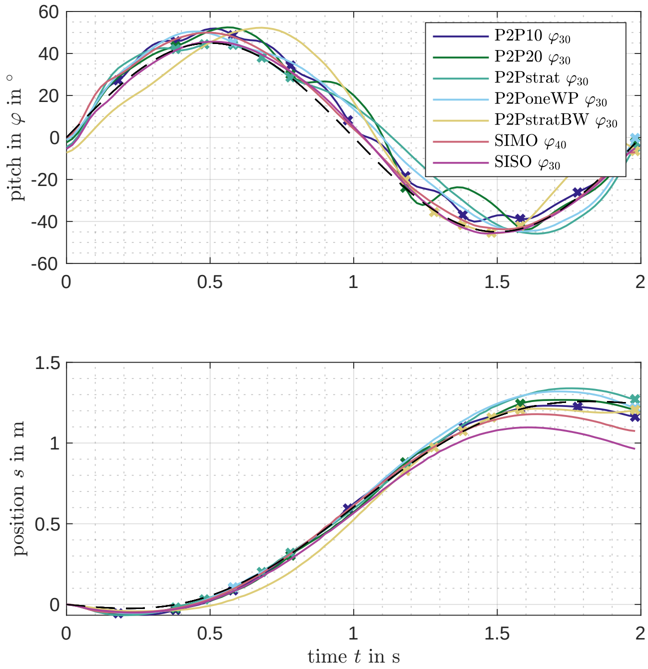


Figure 5-3. Comparison of the trajectories obtained by point to point methods.

In the regions appart from the waypoints, the observed trajectories differ. For example, in the P2PstratBW case, the positive pitch maximum value takes values around 55° in comparison to the simulation with 45°. The same holds true for the forward dive movement. In the simulation, the breaking maneuver took a negative pitch of 40° and a position below 1.3 m. In the experiment, the breaking maneuver takes 1.35 m and a negative pitch of 45°. In a more intuitive term, this behavior can be referred to as more aggressive. The author assumes this is due to the wrong friction modeling. But, the fundamental features of the trajectories are observed in the experiment and in the simulation. Therefore, the author assumes that modeling the friction is beneficial as it is suggested in Section A.8 is a good starting point for improved models suitable for planning purposes.

To get an Impression of the convergence plots of the metrics in the trail domain are available in the Appendix A.3 - A.4.

Concluding the comparison of the results in the experiment and simulation of the reference case with low amplitude have shown that the behavior that is observed in simulation is mainly reproduced. But, different influences lead to different effects, e. g. changing initial conditions. Furthermore, the overall experiment shows that point-to-point Iterative Learning Control (ILC) allows the selection of a dedicated portion of a reference that shall be tracked precisely.

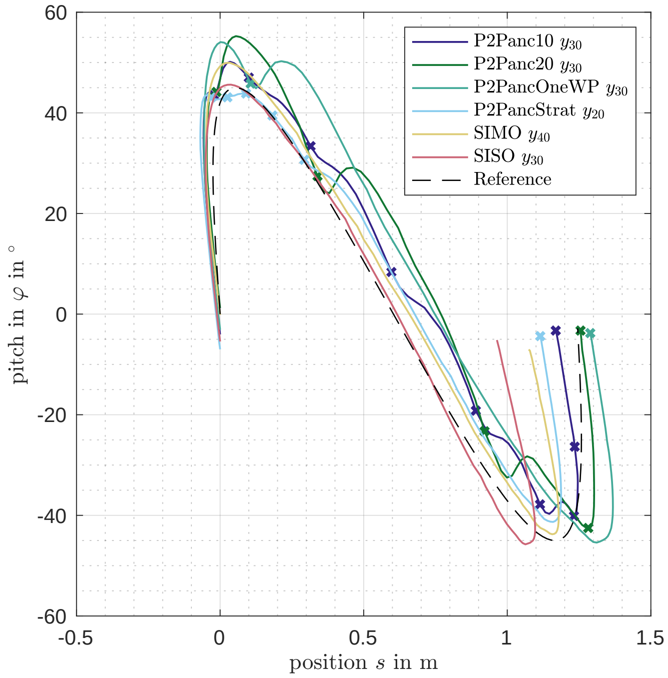


Figure 5-4. Comparison of the trajectories carried out by the point to point methods with ancillary controller.

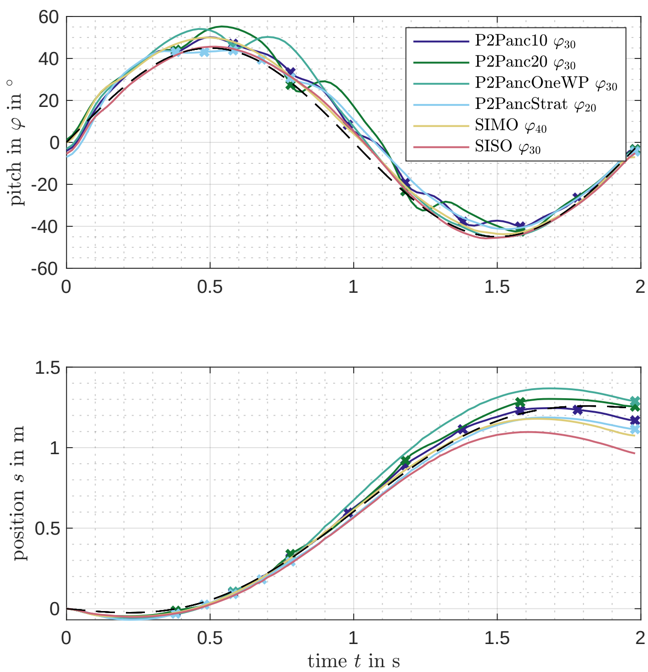


Figure 5-5. Comparison of the trajectories carried out by the point to point methods with ancillary controller.

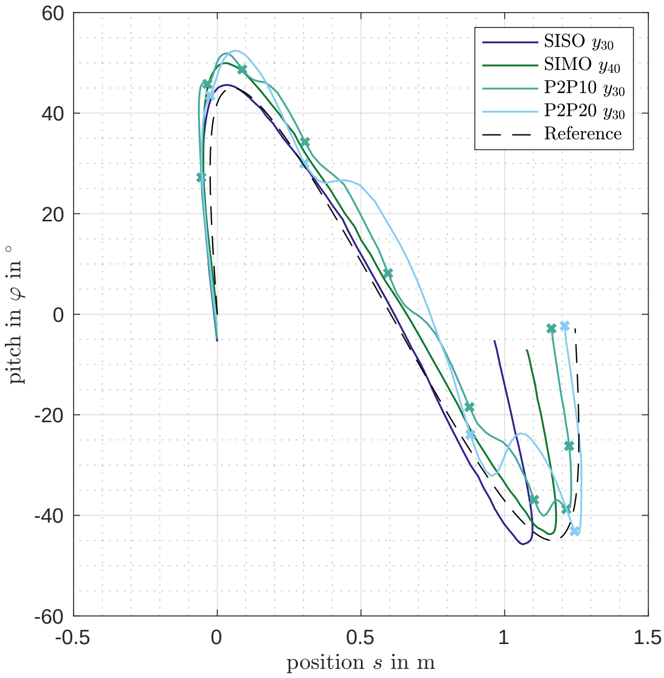


Figure 5-6. Comparison of the trajectories obtained in the case study on the reference movement. Given are the Single and Multiple Output cases as well as the Point to Point methods with periods of 10 and 20 samples.

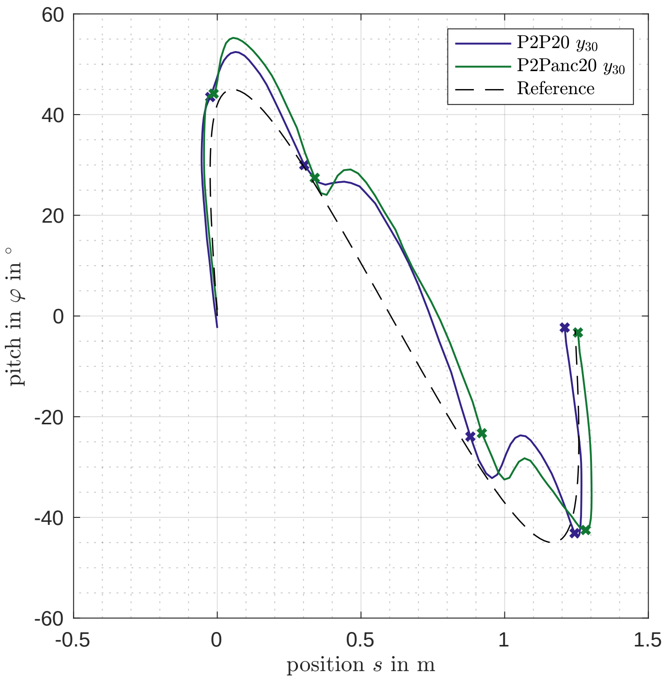


Figure 5-7. Comparison of the trajectories obtained in the case study on the reference movement. Presented are the cases point to point cases with a period of 20 samples once with ancillary cost and one without.

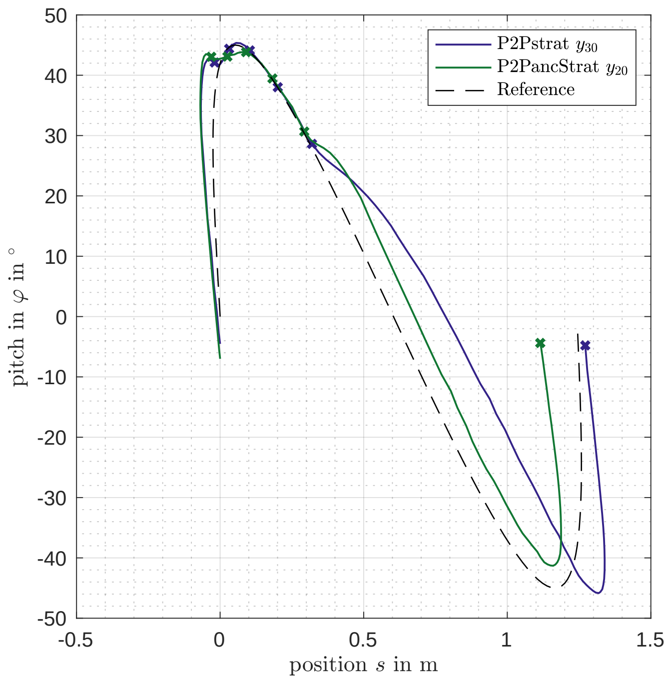


Figure 5-8. Comparison of the trajectories obtained in the case study on the reference movement. Presented are the cases point to point cases with strategic sample selection once with ancillary cost and one without.

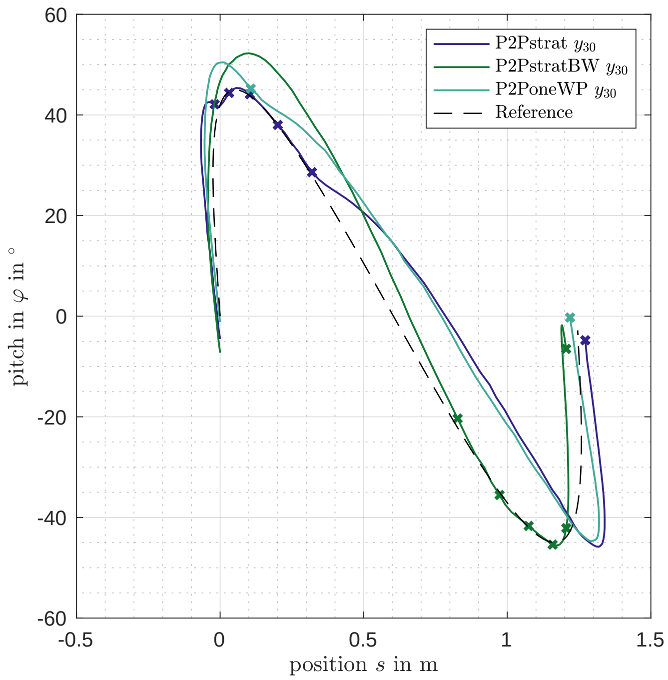


Figure 5-9. Comparison of the trajectories carried out by the point to point methods with ancillary controller. 4

5.2 Experiment: Arrive at a goal location

In the Hybridization experiments the robot approaches the goal configuration very slow. To validate that the method is also capeable to arrive stable in an agile maneuver, the controller was designed to traverse to a goal location in the given amount of time. To achieve this the controller is tuned as in the simulation described in Section 3.9.1.

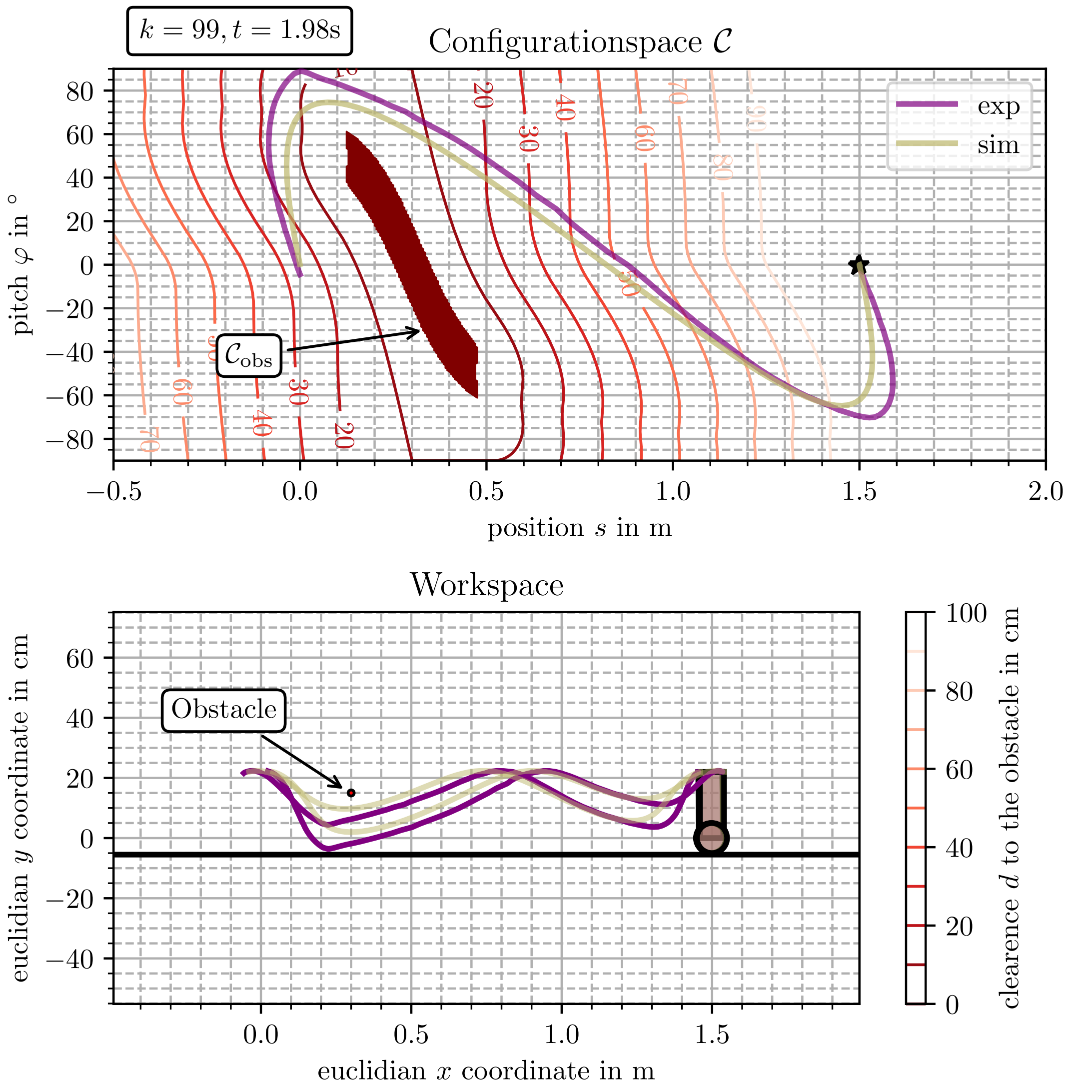


Figure 5-10. Experiment to arrive in a goal location.

The results are given in Figure 5-10 in the top. It is given in the configuration space in the bottom in the workspace. The both top vertices trajectories are highlighted and tracked by the given lines. The presented trajectory is that found as the Iterative Learning Control (ILC) converged. In the configuration and work space the obstacle is included as well. The obstacle is at 0.3 m and of the height of 15 cm, as it is also reviewed in the reference case *Immediate Dive with proximal obstacle*. This is just to illustrate that the motion problem has been solved even if obstacle has been present. It is not part of the method. In the Figure, the experiment is compared to the simulation form Section

3.9.1.

The trajectories in the configuration space have the same shape but deviate significantly. In the experiment, the trajectory takes higher pitch values than in the simulation (Simulation: Figure 3-23). Furthermore, the initial condition is different. A characteristic feature of the movement is that re robot hits the ground. This can be observed in configuration space as the trajectory touches the 90° line. In this instant, the trajectory is not smooth, as touching the ground leads to a change of the dynamics. This changes the plats characteristic, as the plant becomes a hybrid system, i. e. a system that consists of continuous states and discrete ones. Lastly, it is observed that the resulting trajectory solves the motion task, even though the obstacle has not been considered in the design.

Takeaways

- It is demonstrated that the Point to Point ILC enables reaching the goal configuration without any reference trajectory in experiment.
- The ILC converged, even though the Two Wheeled Inverted Pendulum Robot (TWIPR) body hits the ground.

5.3 Experiments using the naive approach

This Experiment serves as a comparison of how the results behave in naive using naive approaches. That means approaches that do not take task-specific information into account.

The controllers are tuned as follows:

SISO: $\boldsymbol{Q}_\mathrm{W} = \boldsymbol{E}$, $\boldsymbol{S}_\mathrm{W} = 0.25\boldsymbol{E}$, $\boldsymbol{R}_\mathrm{W} = \boldsymbol{0}$

SIMO: $\boldsymbol{Q}_\mathrm{W} = \boldsymbol{E} \otimes \mathrm{Diag}\left[\begin{bmatrix}4 & 20\end{bmatrix}\right]$, $\boldsymbol{S}_\mathrm{W} = 0.5\boldsymbol{S}_{\mathrm{W}D}$, $\boldsymbol{R}_\mathrm{W} = \boldsymbol{0}$

P2P10: $\boldsymbol{Q}_\mathrm{W} = \boldsymbol{E} \otimes \mathrm{Diag}\left[\begin{bmatrix}8 & 40\end{bmatrix}\right]$, $\boldsymbol{S}_\mathrm{W} = 0.5\boldsymbol{S}_{\mathrm{W}D}$, $\boldsymbol{R}_\mathrm{W} = \boldsymbol{0}$

P2P20: $\boldsymbol{Q}_\mathrm{W} = \boldsymbol{E} \otimes \mathrm{Diag}\left[\begin{bmatrix}8 & 40\end{bmatrix}\right]$, $\boldsymbol{S}_\mathrm{W} = 0.5\boldsymbol{S}_{\mathrm{W}D}$, $\boldsymbol{R}_\mathrm{W} = \boldsymbol{0}$

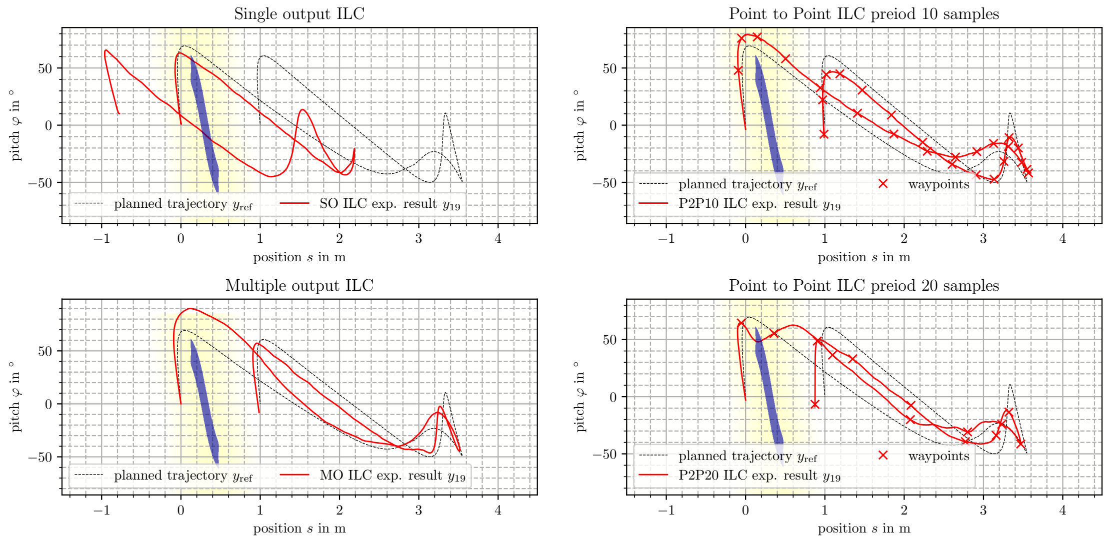
All controllers are constrained to $-55°$ and $70°$.

The update weight matrix is build using $\boldsymbol{S}_{\mathrm{W}D}$ derived in

The data is found in the data drive under `rrt_cases`.

In this Experiment, the naive approaches are examined, meaning a Single Input Single Output (SISO) controller is applied to the pitch only. A Single Input Multiple Output (SIMO) controller is used, and two point-to-point controllers with a period of 10 respectively 20 samples.

The results are given in the configuration space in Figure 5-11.



(a) Singe and Multiple Output.

(b) Point to Point methods.

Figure 5-11. Experimental results of the naive approaches

It is observed that all methods except the single output controller solves the motion task. In the single output case, the trajectory diverges over the evolution of the trajectory. The other trajectories solve the motion task as they do not intersect the obstacle region. The trajectory is shifted for the case of the Multiple Input Multiple Output (MIMO) controller. For movements from left to right, the trajectory is above and the other way below the reference trajectory. The same phenomenon is observed for the point-to-point methods. Furthermore, the method that samples the trajectory with a period of 10 samples appears to be almost smooth. This coincides with the observations made in the reference experiment. Even though the trajectories solve the task, the imperfection in the tracking may lead to problems. In this case, the deviation pushes the trajectory away from the obstacle region. This can be the case in some situations. If the trajectory is on the other side of the trajectory, the result might intersect the obstacle region and therefore collide with the obstacle.

5.4 Experiments on the Reference cases with Hybridization Methods

The experiments are done for the reference trajectory *Delayed Dive* and the Hybridization Methods 3.5 to 3.3. The experiments are performed using the same algorithms as in the simulation study (Section 3.9.2). The simulation serves as a preliminary to this experiment.

The tuning coincides with the methods applied in the simulation. The results obtained applying the first Hybridization 3.5 assigning a multiple output weight to the trajectory. The trajectory is given as a red line, while the window with weights larger than zero is highlighted with a red background. One observes that the trajectory do not track the reference perfectly in both cases in the highlighted part. In the experiment, the resulting trajectory in the 30th tail even intersects the obstacle region so that the motion task is not solved. Furthermore, one observes that the trajectory in the movement from 2 m to the goal configuration at 1 m is not as smooth in the simulation. In the experiment, the initial

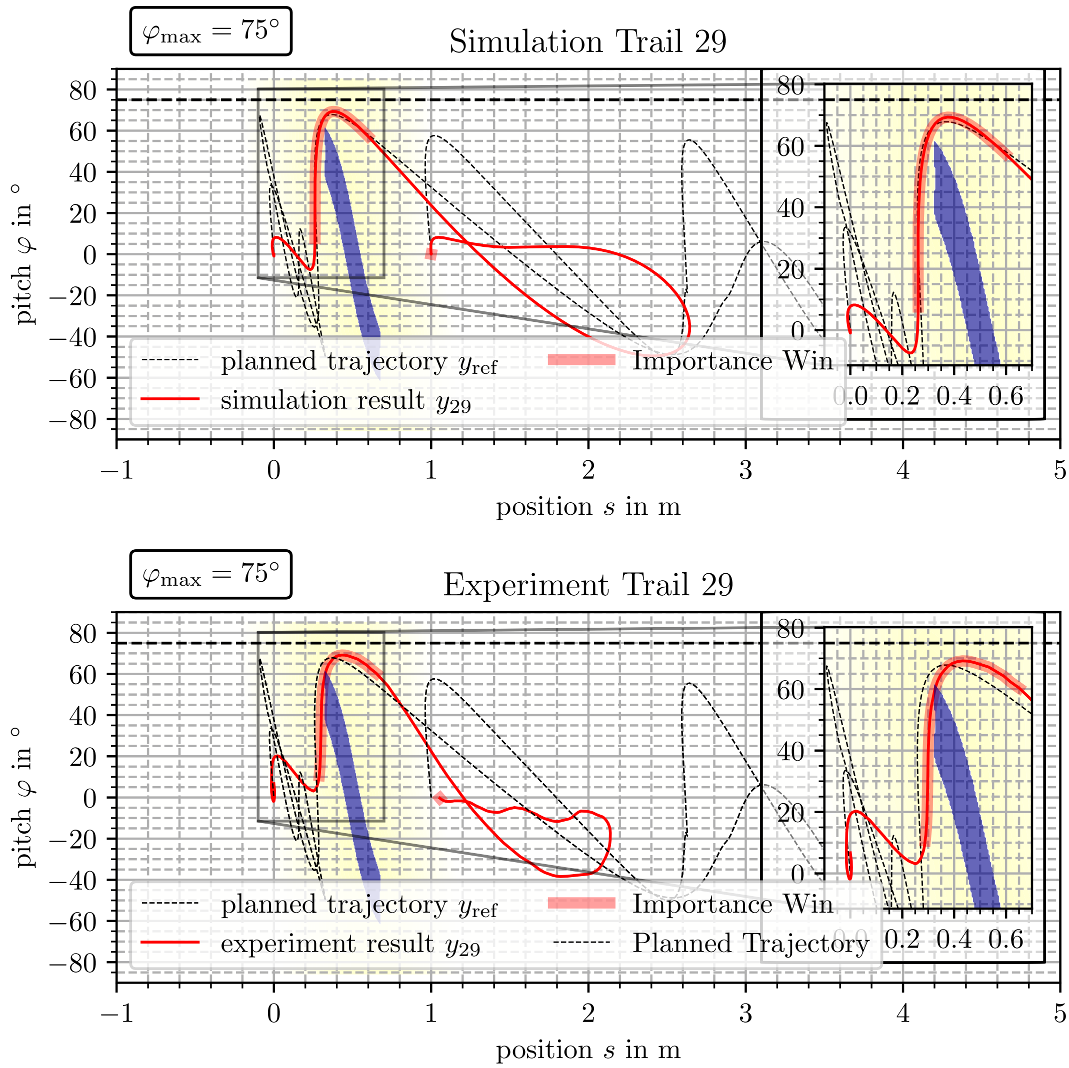


Figure 5-12. Experimental results of the Hybridization method 3.5 given in configuration space. The trajectories are compared to the simulation in the top.

state has been assured to be in the range of 2° by the operator to minimize the influence of an initial state difference.

The trajectories are also given in the time domain in Figure 5-13 to interpret the results further. At the top, the position trajectory is provided at the bottom of the pitch. This shows that the window with weights larger than zero is shorter in the time domain that it appears in the configuration space. This is due to the fast velocity the robot has as the obstacle is passed. Furthermore, it is observed that the trajectory's deviation appears differently. The incline of the trajectory, in the beginning, is different than the one of the reference. The difference in the earlier phase is also observed in the configuration space. Also, the difference appears to be more balanced than in the configuration space. One observation that is also made is that the movement before and after the actual dive is very slow.

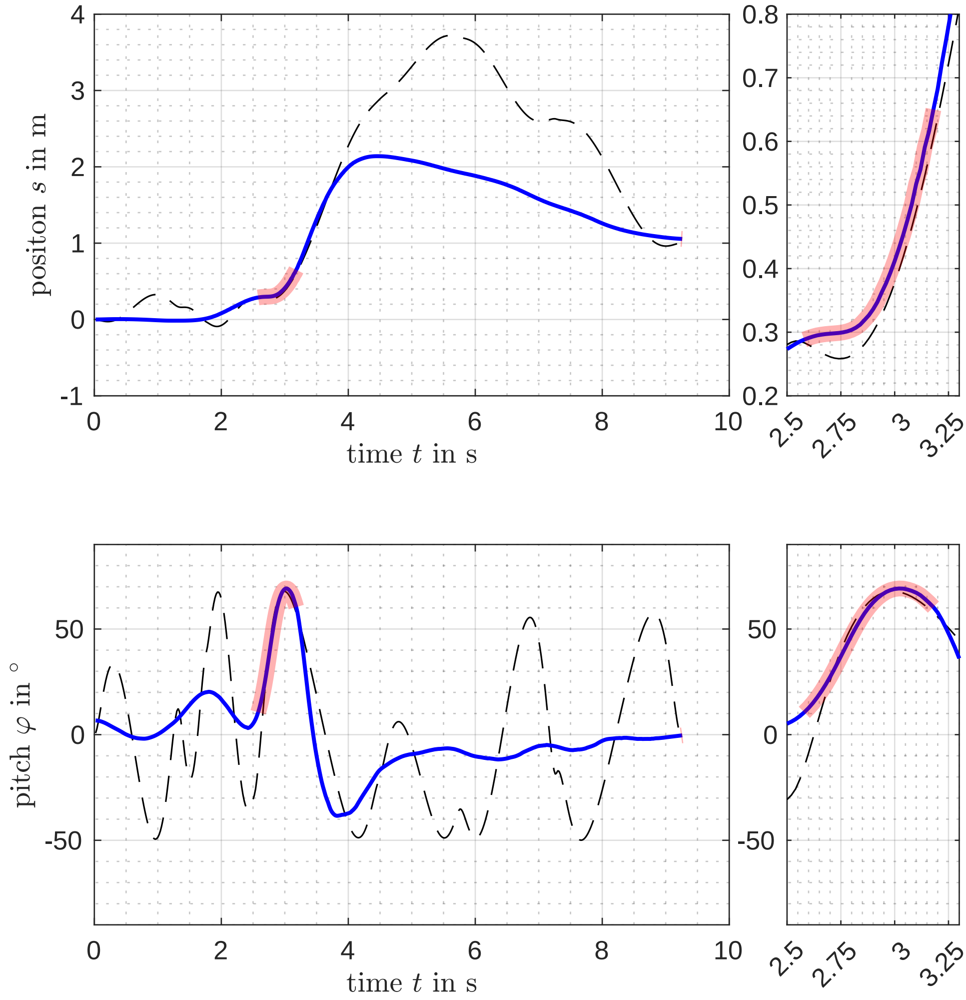


Figure 5-13. Experimental results of the Hybridization method 3.5 given over time. On the right a zoom in of the tracked window is given.

The author assumes that this is due to the under-actuation of the solution, as discussed in Section 3.6.4. Furthermore, in all cases, the trajectories differ from trail to trail in contrast to the simulation. The Figures are implemented as animations, and the reader can inspect the other trails. In the under-actuated case, the trajectory can not be tracked perfectly. The resulting trajectory will differ. In this case, the difference observed in the simulation does not lead to a collision, while the one in the real system might.

The results of the Hybridization strategy 3.5 are presented in Figure 5-14. Recover, the methods control the position and speed only at the waypoint. The pitch is controlled to mimic the reference trajectory in the window within the activation clearance c_{act}. It is observed that the trajectory does not intersect the obstacle region and therefore solves the motion task. Furthermore, the trajectory mimics the behavior of the reference well within the highlighted window. The movement to get in proximity of the obstacle differs from that one observed in Figure 5-15, the case with three waypoints. The maneuver to get in the proximity reaches a pitch of 20° and does not take negative values.

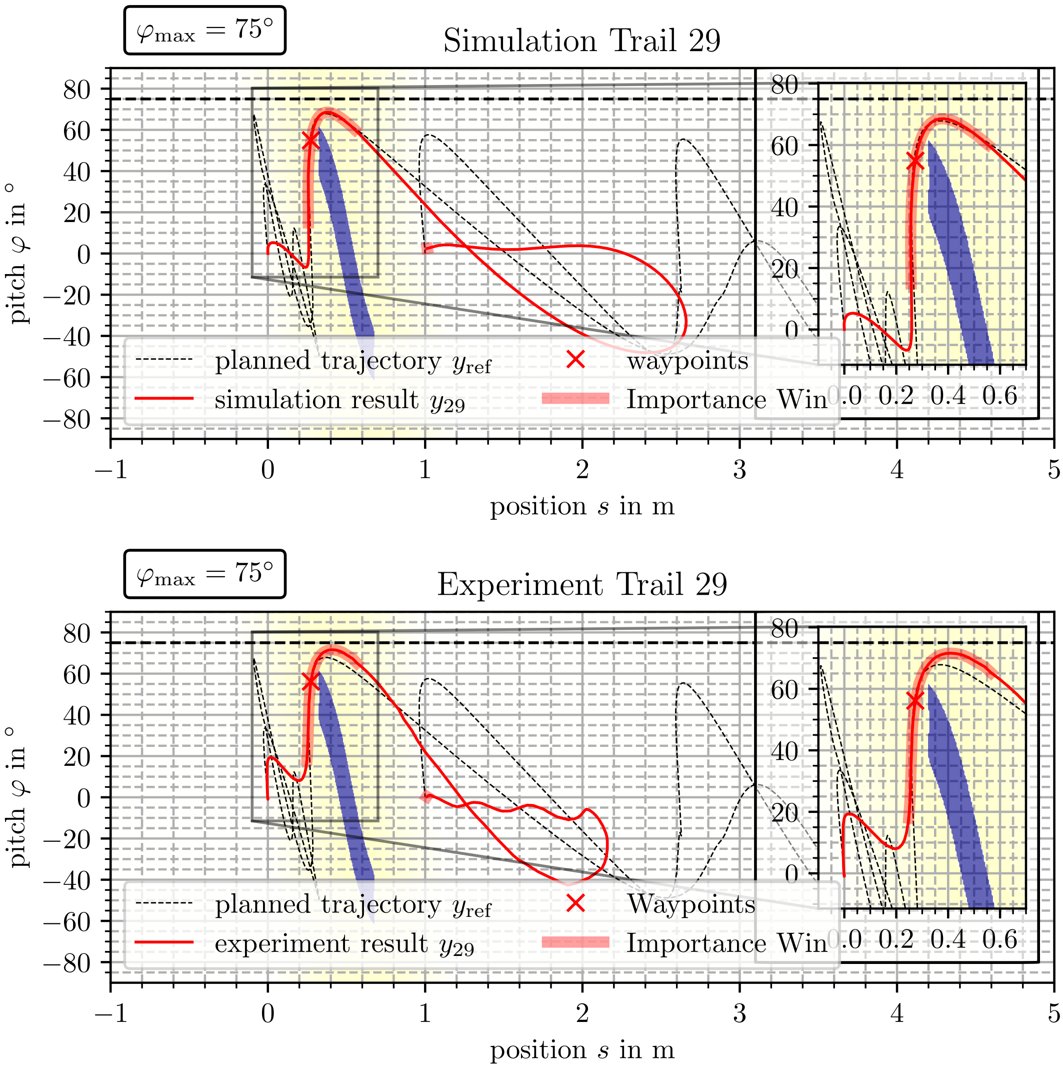


Figure 5-14. Experimental results of the Hybridization method 3.5 given in the configuration space. The trajectories are compared to the simulation in the top.

The results for the Hybridization 3.4 are given in Figure 5-14 in the configuration space. Recover, the method aims to track the three waypoints that are indicated by the red crosses. In this case, the trajectory does not intersect the obstacle region and therefore solves the motion task. Nevertheless, the points are slightly off the reference. The shape of the trajectory appears different than the one found in the simulation. The movement before the robot passes, seen on the left in the range from −0.5 m to 0.3 m differs form the simulation result. The amplitude in pitch and position is higher. The breaking maneuver, right of the obstacle region, is much shorter in the experiment than in the simulation. Especially the third waypoint appears to be more off the reference than the others.

To get an impression of the dynamics observed in the experiment, a video of one experiment is on

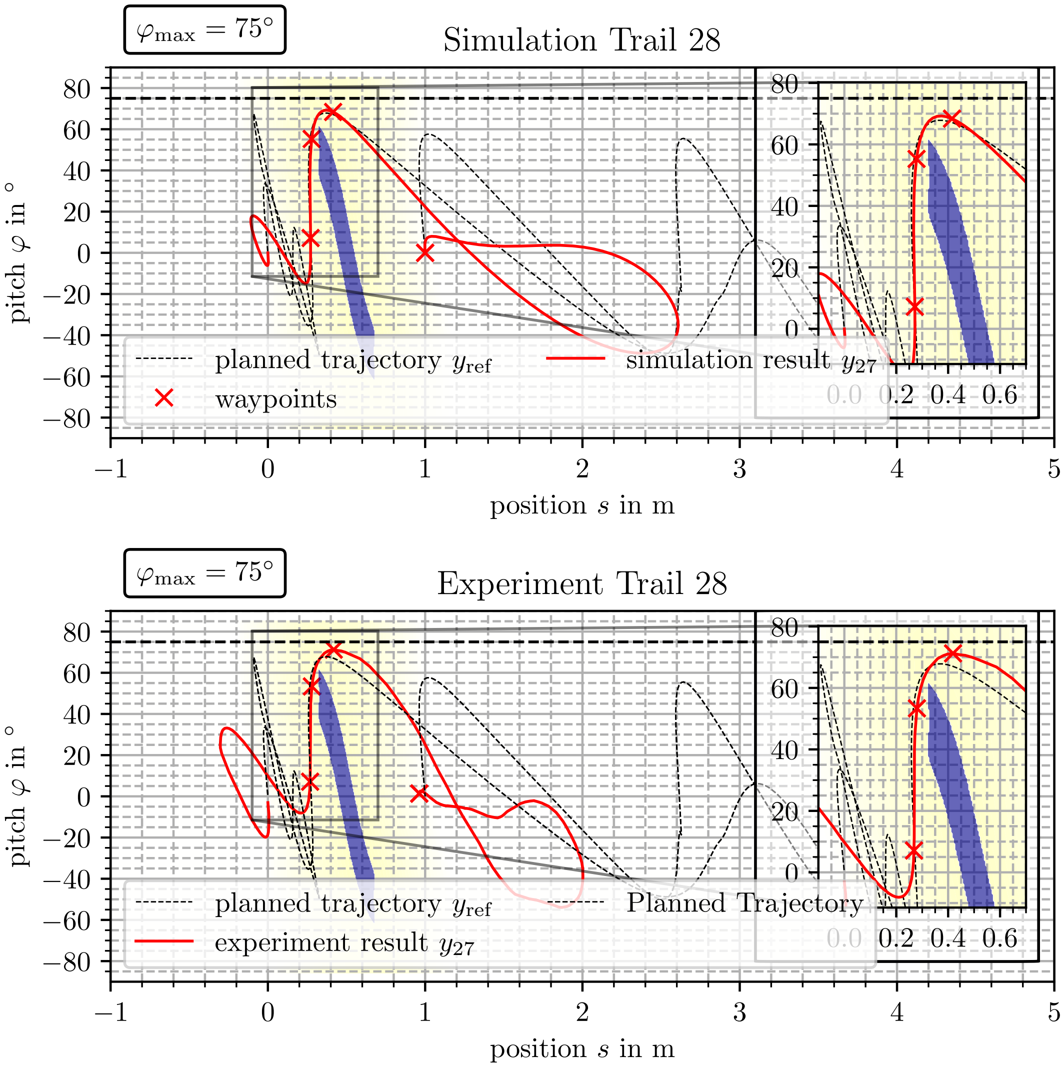


Figure 5-15. Experimental results of the Hybridization method 3.4 given in the configuration space. The trajectories are compared to the simulation in the top.

Youtube `https://youtu.be/Wgl8O614za8`. This video shows the movement presented in Figure 5-14.

Comparison discussion

In this part, some remarks are made that are also due to the author's observation in the experiment.

In the experiments, all proposed methods solved the task in some trials but in others collided. This shows that the methods can solve the problem. But, they not very robust in the current setup. This means that the trajectories fluctuate even though the algorithm has converged in the simulation. So, the results can only serve as a starting point. For example, the theory developed needs to include more in the characterization of uncertainty. This has not been possible in this work, along with the other tasks that had to be completed. Furthermore, the changes can also be due to disturbances from the outside, of whom none have been identified as a reason. Examples of these influences are the tires' friction behavior, the ground surface, imperfections of the axle, and much more. In the project of D. Lehmann, another Two Wheeled Inverted Pendulum Robot (TWIPR) is designed that appears to have a more reliable design. Repeating the experiments on that device may lead to improved behavior.

5.5 Discussion of the selection of constraints

In this section, an experiment shows that the constraints for the Hybridization 3.3, that is, the application of a weighted multiple output cost, requires loose constraints for the Iterative Learning Control (ILC) algorithm. In the experiment, it is observed that the ILC fails if the constraints used for planning are chosen as constraints for the controller. In contrast, the controller succeeded as the constraint has been chosen 5° loose.

The reference trajectory is the *Immediate dive with proximal obstacle*. Figure 5-16 shows the results in the configuration space. To present the evolution of the trajectories, trials 25 to 30 are given. In the top plot, trajectories 26, 28 and 30 fail to solve the motion task while the others do. But, the trajectories that solve the task (trail 25, 27 and 29) violate the constraint. Therefore, no solution is valid. This issue is because the problem is underactuated, as discussed in Section 3.6.4. Therefore, feasible trajectories are likely of the reference trajectory. Furthermore, it is observed that the 25th and the 29th, as well as the 26th and the 30th trajectory, almost coincide. This indicates that the controller operates on a limit circle in the trail-domain. A limit circle is a phenomenon that is observed in non-linear systems. The constrained ILC is non-linear. A general introduction on limit circles is found in [43, pp. 26].

The constraint is relaxed to 75° to mitigate this problem. The results are given in the lower plot. In this experiment, it is observed that all trajectories are feasible. Furthermore, all trajectories take maximum pitch values larger than 70°. Therefore, in this case, it is beneficial to relax the constraints. This comes for the price that the motion planner is not able to exploit the available range in the pitch. I. e. one has to introduce a margin to allow the realization of the trajectory, additionally to the margin necessary to assure collision-free operation of the ILC controller.

This is a drawback of the hybridization method 3.3 in contrast to 3.5. The problem of under-actuation is not present in the second design. If beside the waypoint, not both coordinates of the trajectory but only the pitch is tracked, a perfectly converged controller will not exceed the limit in the pitch. Thus, the constraint could be exploited totally by the planner.

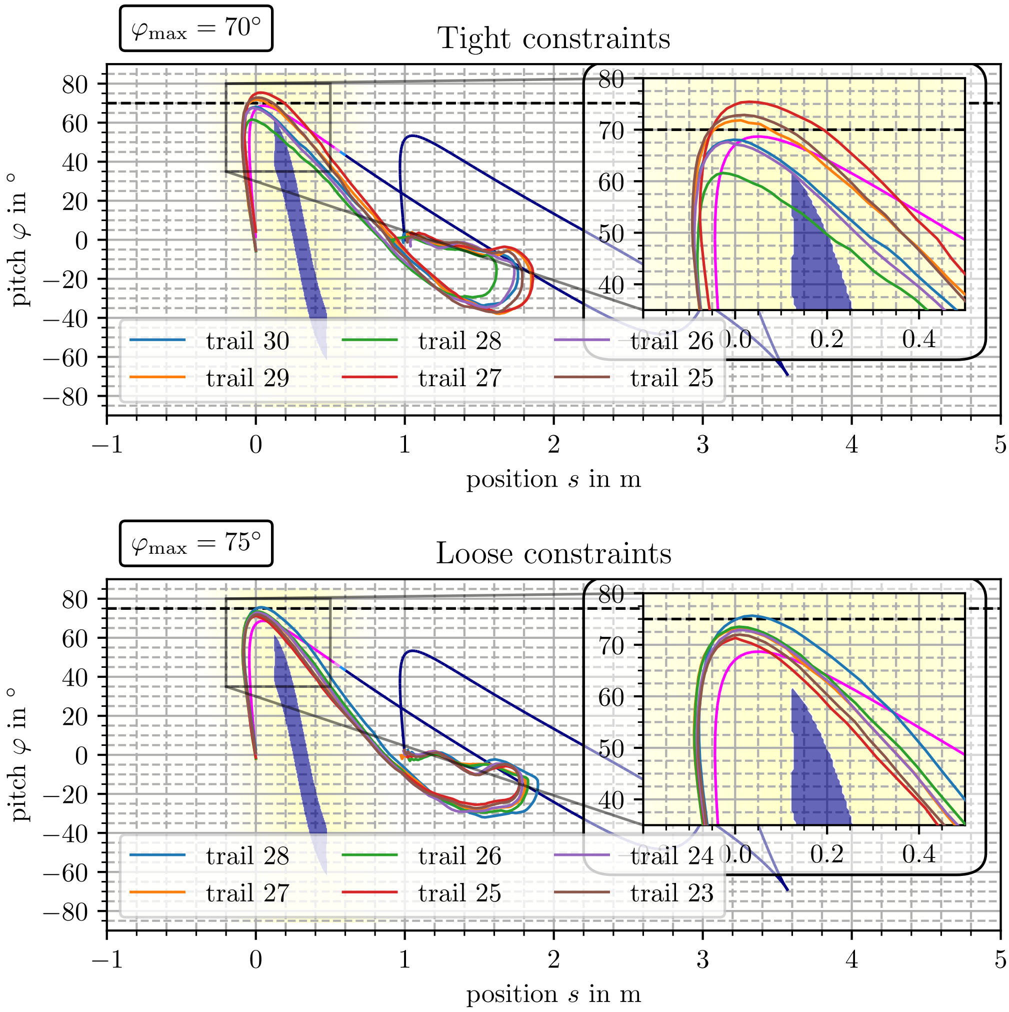


Figure 5-16. Norm optimal multiple output ILC applied to a trajectory planned by a Rapidliy expanding Random Tree (RRT). The weights are selected using proximity weighting. The weighing is indicated on the reference trajectory. It is given by the line whose color drifts from blue to purple. Purple indicates high weights and blue low. The two experiments show are differently constrained. In the first, the constraint is tight and conformal to the planning, while in the second, the results are chosen softer, i. e.70° compared to 75°.

5.6 Forward dive by hand designed reference

This experiment is done in the earlier phase of the project. It is performed to validate the constraint satisfaction and shows if a movement can be done with a very sparse selection of waypoints. The applied method is a Point to Point controller. The reference is generated using the stable model and an integral feed-forward control. The waypoint is chosen as the 30th sample on the reference trajectory.

The controller used is a point to point iterative learning controller designed by NO-ILC with constraints. The design requires the following steps:

- Design of a the selector.
- selection of the weights.
- selection of the constraints.

The selector is designed according to Definition 20

$$\mathcal{P} = \left(\left(k_1 = 30, \boldsymbol{\Phi}_1 = \boldsymbol{E}, \boldsymbol{r}_1 = \begin{bmatrix} 67.61° \\ 17\,\mathrm{cm} \end{bmatrix}\right), \left(k_2 = 100, \boldsymbol{\Phi}_2 = \boldsymbol{E}, \boldsymbol{r}_2 = \begin{bmatrix} -4.39° \\ 194\,\mathrm{cm} \end{bmatrix}\right)\right) \tag{5.1}$$

Meaning that the 30th and the 100th samples are selected.

The final way point is not located at 0° as a trajectory generator generates it, but in the accuracy of the system this can be considered negligible. The weight matrices have been chosen so that both points are weighted equally. The weight matrices are:

$$\boldsymbol{Q}_\mathrm{W} = \mathrm{Diag}\begin{bmatrix} 8 & 40\,\mathrm{m}^{-2} & 8 & 40\,\mathrm{m}^{-2} \end{bmatrix} \tag{5.2}$$
$$\boldsymbol{S}_\mathrm{W} = 0.25\boldsymbol{E} \tag{5.3}$$
$$\boldsymbol{R}_\mathrm{W} = \boldsymbol{0} \tag{5.4}$$

The constraints are set to −55° and 70° for the pitch.

To achieve an integral behavior of the plant, an integral input is introduced. Meaning that an integrator filters an the controlled input before it is fed into actual input. This is equivalent to assigning an the cost on the inputs derivative. This is because the experiment has been drawn in an earlier state of the project. The results are eqivalent to omitting the Filter and updating the update cost to $\boldsymbol{S}_\mathrm{W} = 0.25\boldsymbol{S}_{\mathrm{W}D}$

The resulting trajectories are given in the configuration space in the workspace in Figure 5-17. The figure also includes the obstacle to show that the realized trajectory already solves the motion task for an obstacle 0.3 m apart from the Two Wheeled Inverted Pendulum Robot (TWIPR) initial configuration. The obstacle has not been considered in the design pipeline, as the trajectory has been designed without taking an obstacle into account.

Some selected trajectories in the configuration sapce are given in Figure 5-18. The trajectory given in green represents the trajectory an unconstrained controller would converge to on the nominal system. In the beginning, the shape is similar to the planned one. In the 15th trial, the constraints are violated, but contra intuitively, in the next trial, the constraints are violated more, almost leading to a collision. Eventually, in the 17th trail, the shape of the trajectory changes. In the analysis of the results, it turned out that the initial pitch angle significantly changed for both consecutive trails. Therefore, the controller inputs lead to a different output trajectory. Nonetheless, the conservatively chosen constraints ensured that the trajectory was still collision-free.

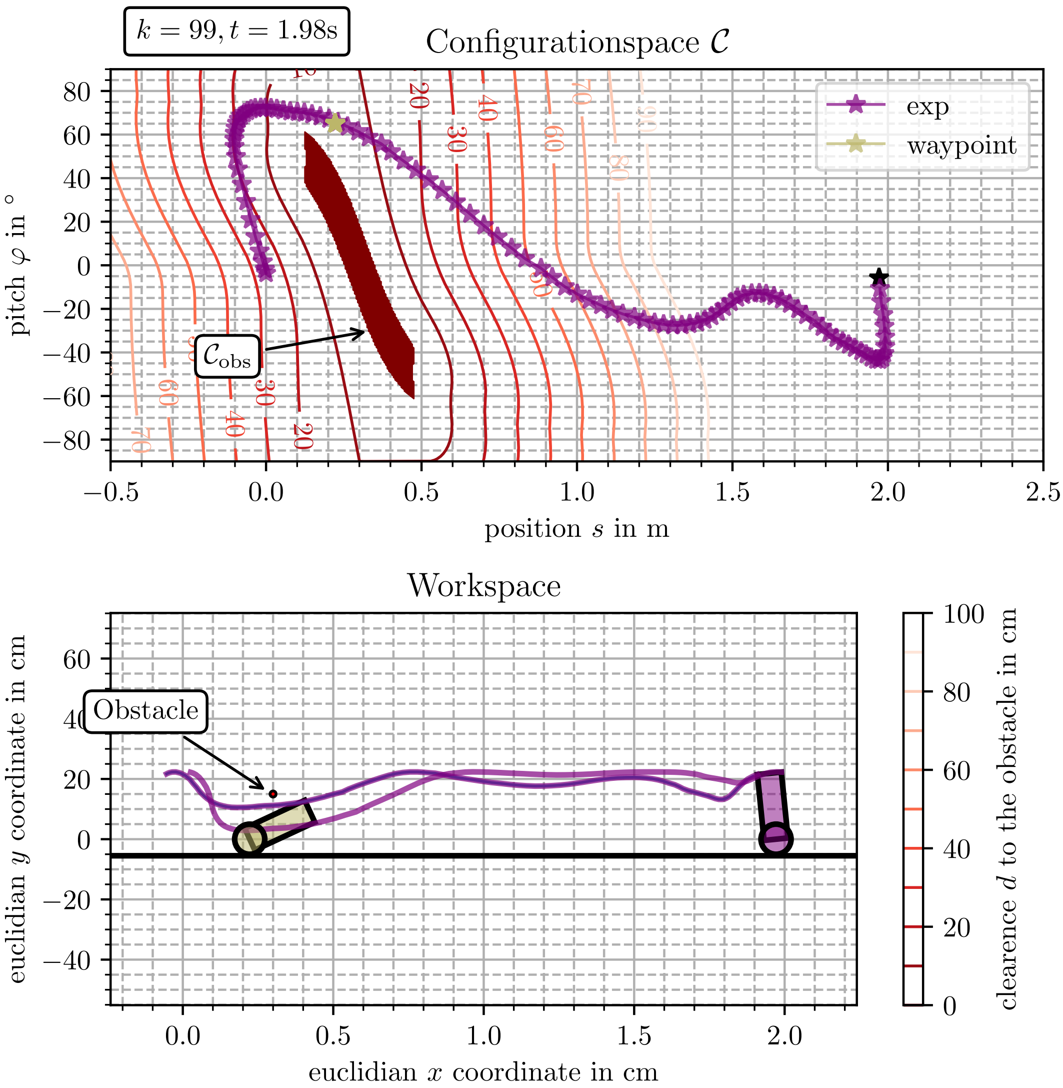


Figure 5-17. Results of the forward dive maneuver with one waypopint given. The lines that are drawn in the workspace are the positions of the both top vortexes of the robot. (When one opens this pdf in Abrobat reader or Debian ocular, the movement is also animated.)

This experiment also serves as validation for the constraint satisfaction functionality. Therefore, the computation result of the 16th trail is highlighted in Figure 5-19. In this figure, the past (15th) observation is given in blue, and the observed trajectory in black. The computed prediction done by the Norm Optimal Iterative Learning Control (NO-ILC) controller is shown in dashed black. It is observed that the prediction satisfies the constraints. But, the observed output still violates the constraints. The violation is even more severe than in the past trial. The reason is the deviation in the initial pitch, which is almost 10° of. Thus, the main requirement for applying the Iterative Learning Control (ILC) theory is violated. It is non the less remarkable that the experiment did not result in a collision. The experiment was repeated and worked three consecutive times. It has also been tested if the algorithms work with less conservative constraints. All those tests failed, regardless of which constraint is relaxed, the positive or the negative. The author supposes that

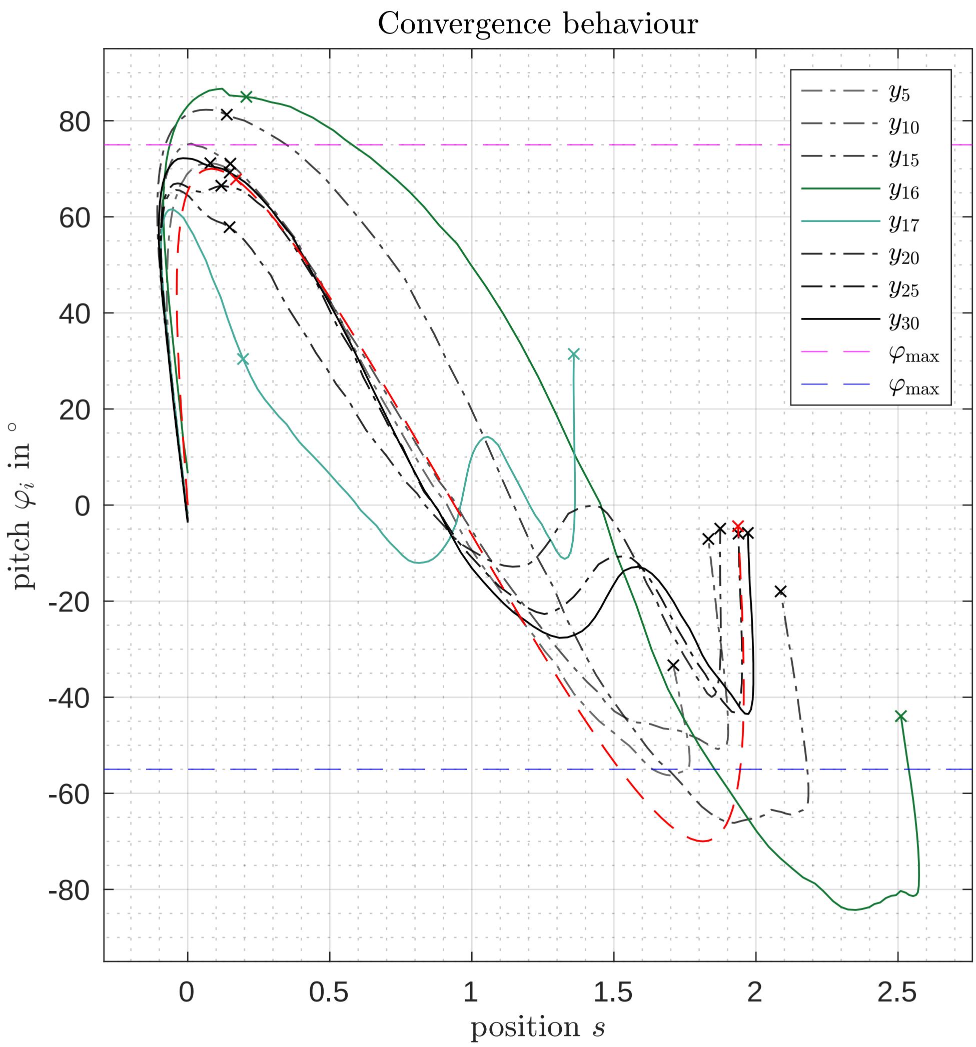


Figure 5-18. Results of the experiment given one way point and the goal. The dashed trajectories show each 5th trail from light gray to solid black. The colored trajectories are selected as they have shown an anomaly, i. e.the shape of the trajectory has changed leading to a movement that is able to reach the goal.

this is due to the significant error in the initial state. It is highly desirable to start a trail in the same initial state reliably for future developments. The experiment was also repeated manually, assuring that the initial configuration lies in a window of 2°. These did not lead to experiments with more relaxed constraints.

A video of the experiment is available on YouTube: `https://youtu.be/zw4BzynKbeA?t=308`

The data is found on the disc `30_Comparisons/70Grad_exp/OneWaypoint70`.

Concluding the discussion of this experiment the following insights can be taken.

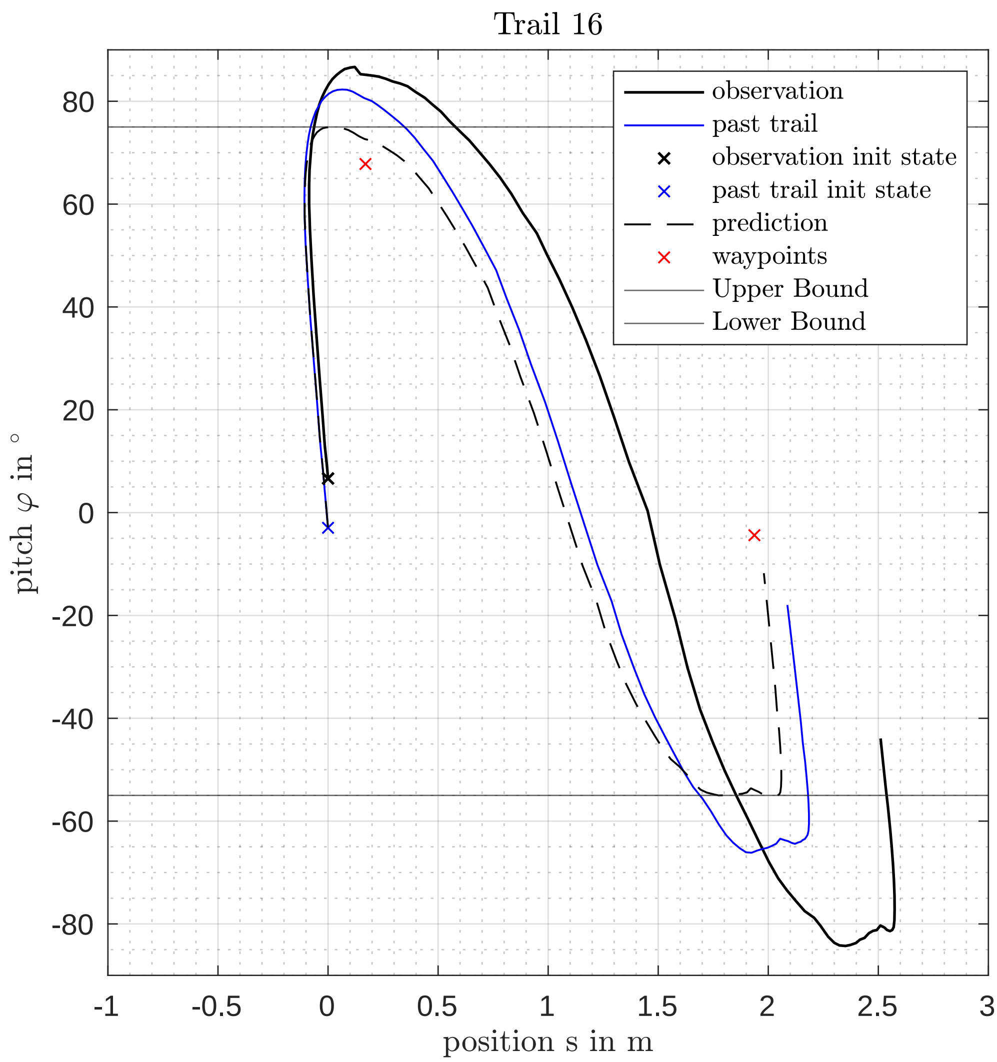


Figure 5-19. Observations in the 16th trail. One observes on the one hand that the past trail violated the constraints. The prediction satisfies the constraints. The observed trail violates the constraints even harder as the initial state is not the same. In the experiment the robot has initially been in rest in both trails.

- With carefully selected waypoints, point to point ILC solves the motion task with very sparse information.
- The functionality of the algorithm to assure that the constraints are satisfied for the one step ahead predictor is validated.
- The relatively conservative selected constraints chosen to be 75° and −55° assure that the algorithm works even in the presence of changes in the initial configuration.

Chapter 6

Conclusions and Outlook

The main result of this thesis project it that the proposed method combination of sampling base motion planning and iterative learning control solves the motion problem proposed in the beginning. The methods beats the naive application of single input single output control in the experiment for the tested device.

Another result of is the demonstration that methods from iterative learning control render the problem occurring with under-actuated robots out. It turned out standard methods solve the planning problem.

To close the sim to real gap ILC, methods have successfully solved the motion task with the under-actuated robot. It is shown by the prioritization of task-specific features, namely the goal configuration and the narrow passage.

The underlying methods in ILC are tested in a reference case to illustrate features and drawbacks. It clearly shows that ILC can balance the tracking performance regarding the task features. In contrast, naive approaches that do not take care of the features are not suitable for improving performance. In the reference experiment, a clearance of 10 samples is similar to the multiple output case. A period of 20 samples (0.4 s) improved the tracking.

The three Hybridization strategies for an under-actuated robot are presented, which have different properties. The most important is the state of actuation. The Hybridization 3.4 and 3.5 are not under-acutuated while, 3.3 is. This indicates that the Hybridization 3.3 might also behave like a multiple output solution. This drawback is not present for the Method 3.5 that only tracks a single waypoint in the position.

The proposed Hybridization strategies focus only on the part of a planned trajectory that has been important to pass the obstacle. This method enabled the traversal of an obstacle with a guess for a trajectory like those presented as reference cases (Section 3.7). All three have a complicated shape that operation shall not be performed. The methods result in trajectories that do not mimic the complicated shape while tracking the reference well in the important part, as only the latter is weighted in the controller.

The experiments verifying (ref to goal only) the goal configuration shows that the motion task is also solved without any reference trajectory. In this case, the algorithm realized a movement from one configuration to another. Furthermore, the experiment with one waypoint shows that the controller also does not need a reference trajectory to do the movement. But, as it includes a waypoint on the pass also, it introduces problem-specific information. The selected waypoint relies on a reference trajectory to ensure that the timing of the passage is correct.

Furthermore, besides this main observation, it shows how ILC acts on fully actuated robots. The shapes of different trajectories are compared. It is validated that the method introduced by Freeman [3] can be used to stage the cost functions for both fully actuated and under-actuated systems. Furthermore, it is observed that the theoretical property that the ILC controller converges to perfect tracking is of minor interest in practical application. It is seen that even in the unconstrained case, the reference is not tracked perfectly within a reasonable amount of iterations.

Besides this main observation, it is also shown how ILC acts on fully actuated robots. The shapes of different trajectories are compared. It is validated that the method introduced by Freeman [3] can be used to stage the cost functions for both fully actuated and under-actuated systems. Furthermore, it is observed that the theoretical property that the ILC controller converges to perfect tracking is of minor interest in practical application. It is seen that even in the unconstrained case, the reference is not tracked perfectly within a reasonable amount of iterations.

Beyond this central question, some details of the algorithm have been analyzed.

- It has been shown that equidistant sampling is only beneficial if the period reflects the system's bandwidth. For the special case of the TWIPR that is at least 20.
- The analysis of the mechanics led to insights regarding problem understanding. Especially the fact that actuator friction is irrelevant to the trajectory in the configurations space.
- A fiction model is introduced that reflects the problems in motion planning. This model enables more agile testing of the algorithms.

6.1 Outlook

In this section, the author shares his ideas on how to further improve the methodology.

The primary outcome of the experiment has been the successful application of the Hybridization strategies. The three methods solve the motion task successfully but introduce a significant drawback. Namely, the compensation movement is very slow. This is not desirable, as it is unnecessary. The main idea to improve this is to introduce the time in the control loop, leading to a time-optimal iterative learning controller. The challenge is that a time-optimal solution requires full exploitation of the actuator [43, pp. 258]. This leads to the requirement to introduce the constraint of the input to the Iterative Learning Control (ILC) controller.

Constraining the input real input to the robot's actuators is generally possible in the framework presented (Definition 9). Therefore, assign the observed output $\boldsymbol{y}_\mathrm{o}$ to be the actuator torque.

A starting point for this issue is to implement a controller that solves the goal-only problem while it optimizes the time as an auxiliary quantity.

Furthermore, the experiments suffered from the present design of the robot. Therefore, one might repeat the experiments with the improved robot that is currently designed.

Also, in light of the number of controllers that have been tested, the weights are not chosen systematically. Therefore, developing methods to select the values is potentially a starting point for another project.

As the system is a discrete-time system in the trail domain, application of analysis tool for this type of systems shall be used, to tune the parameters. In the autors point of view, the proposed weights for the controllers have a very aggressive tuning, and therefore, may lead to dead-beat behavior. That is know to be prone to system distrubanceses. [54] Furthermore, the sharp formulation of constraints in the constrained controller lead, intuitively, to dead-beat like behaviour. This is for the reason that the constraint is fully compenstated if it is violated. Thus, a soft formulation for the constraint, e. g. an augmented Lagrangian, shall be used in contrast to sharp constrained optimization.

Appendix A

Implementation

A.1 Matrix algebra

In the context of Iterative Learning Control (ILC) Operator algebra is discussed excessively. In the scope of this thesis focus was given to the time-discrete case. In this case one can restrict to fintidimensional linar algebra which is equivalent to matrix algebra fundamentally. To be clear in speech some concepts are recalled here.

A.1.1 Image, Kernel (Nullspace), Nullspaceprojection

A.1.2 More-Penrose-Pseudo Inverse

The pseudo inverse fullfills the following properties similar to the inverse [40, p. 22].

$$\boldsymbol{A}\boldsymbol{A}^{\ddagger}\boldsymbol{A} = \boldsymbol{A} \tag{A.1}$$

$$\boldsymbol{A}^{\ddagger}\boldsymbol{A}\boldsymbol{A}^{\ddagger} = \boldsymbol{A}^{\ddagger} \tag{A.2}$$

$$\left(\boldsymbol{A}\boldsymbol{A}^{\ddagger}\right)^{*} = \boldsymbol{A}\boldsymbol{A}^{\ddagger} \tag{A.3}$$

$$\left(\boldsymbol{A}^{\ddagger}\boldsymbol{A}\right)^{*} = \boldsymbol{A}^{\ddagger}\boldsymbol{A} \tag{A.4}$$

If the matrix has full row rank (A.1) implies that

$$\boldsymbol{A}\boldsymbol{A}^{\ddagger} = \boldsymbol{A}\boldsymbol{A}^{\rotatebox[origin=c]{180}{$\dagger$}} = \boldsymbol{E} \tag{A.5}$$

also holds, so does

$$\boldsymbol{A}^{\ddagger}\boldsymbol{A} = \boldsymbol{A}^{\dagger}\boldsymbol{A} = \boldsymbol{E} \tag{A.6}$$

if $\boldsymbol{A}$ has full column rank implied by (A.2). In this cases the matrix can be calculated using

$$\boldsymbol{A}^{\rotatebox[origin=c]{180}{$\dagger$}} = \boldsymbol{A}^{\mathrm{T}}\left(\boldsymbol{A}\boldsymbol{A}^{\mathrm{T}}\right)^{-1} \qquad \boldsymbol{A}^{\dagger} = \left(\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}\right)^{-1}\boldsymbol{A}^{\mathrm{T}}. \tag{A.7}$$

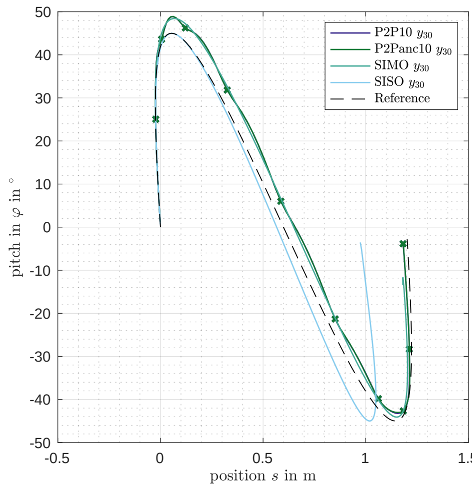
In general the pseudo inverse can be determined using the Singular Value Decomposition (SVD) as the pseudo inverse of a diagonal matrix is defined by an other diagonal matrix that inverts all non zero diagonal entries.

$$A = U_A \Sigma_A V_A^{\mathrm{T}} \tag{A.8}$$

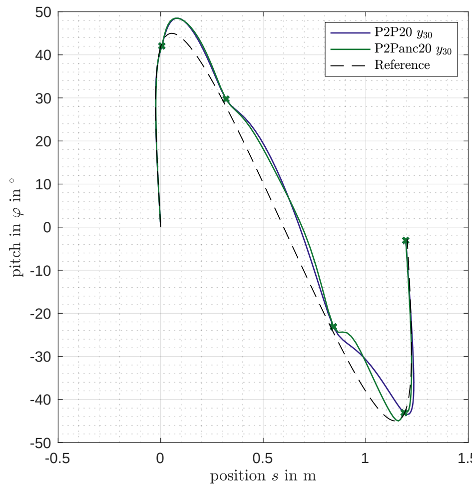
$$A^{\ddagger} = V_A \Sigma_A^{\ddagger} U_A^{\mathrm{T}} \tag{A.9}$$

$$\Sigma_{A\,ii}^{\ddagger} = \begin{cases} \frac{1}{\Sigma_{A\,ii}} & \Sigma_{A\,ii} \neq 0 \\ 0 & \Sigma_{A\,ii} = 0 \end{cases} \tag{A.10}$$

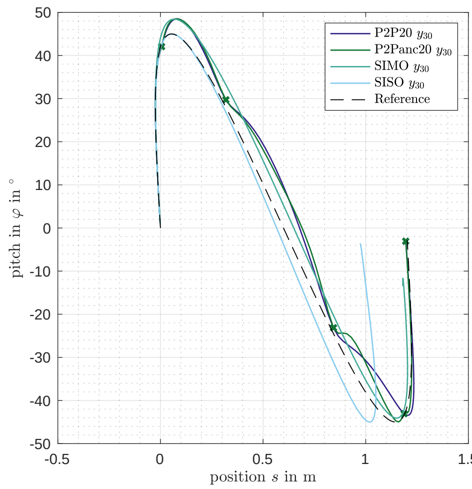
A.2 Reference Simulation case $45°$



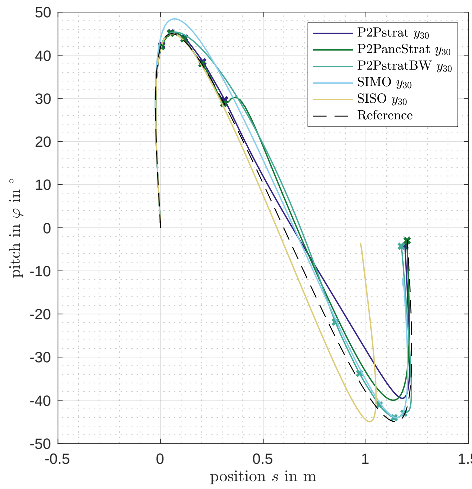
(a) P2P10. P2Panc10. SIMO. SISO.



(b) P2P20. P2Panc20.

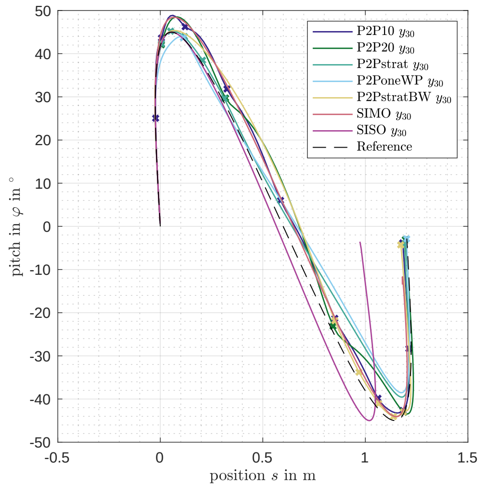


(c) P2P20. P2Panc20. SIMO. SISO.

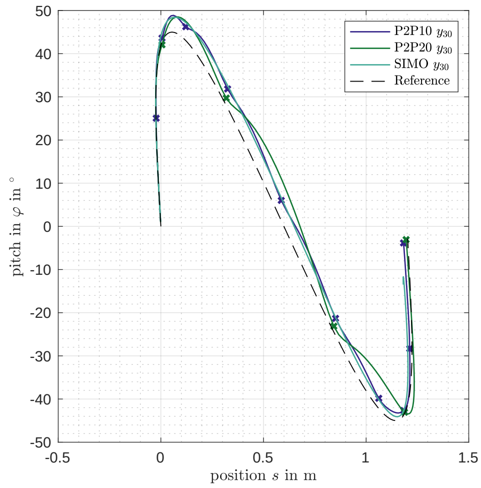


(d) P2Pstrat. P2PancStrat. P2PstratBW. SIMO. SISO.

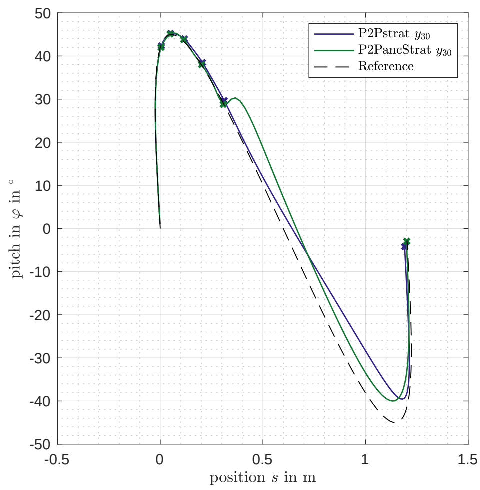
Figure A-1. Trajectory in the C-Space in the 45° reference case simulation 1.



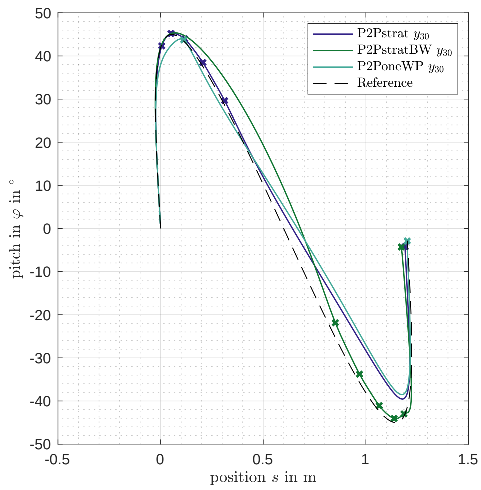
(a) Several P2P and SIMO SISO.



(b) P2P10 P2P20 and SIMO.

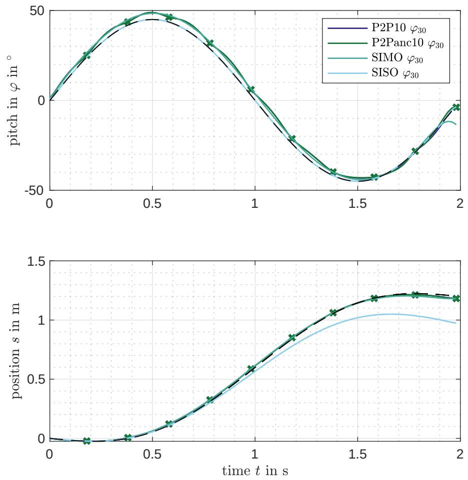


(c) P2Pstrat P2PancStrat.

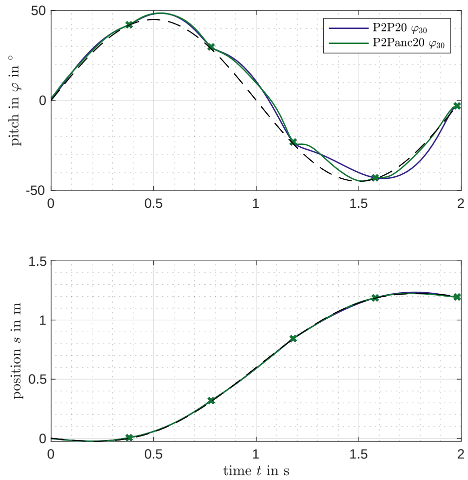


(d) P2Pstrat. P2PstratBW. P2PstratWP.

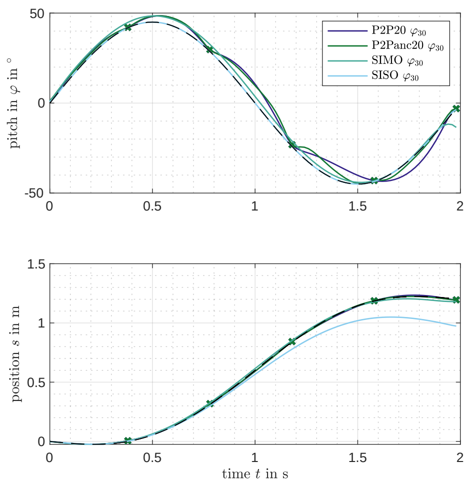
Figure A-2. Trajectory in the C-Space in the 45° reference case simulation 2.



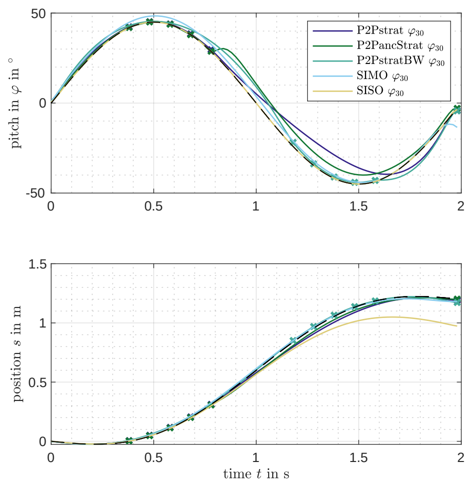
(a) P2P10. P2Panc10. SIMO. SISO.



(b) P2P20. P2Panc20.

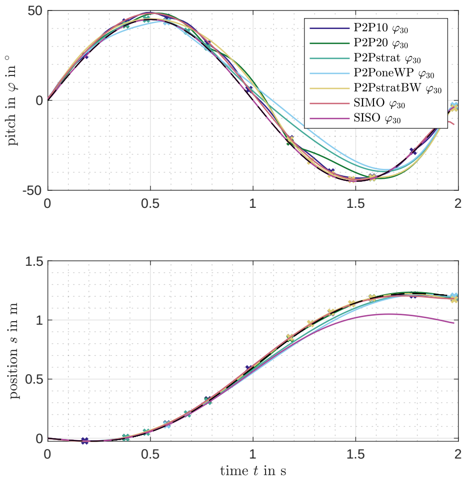


(c) P2P20. P2Panc20. SIMO. SISO.

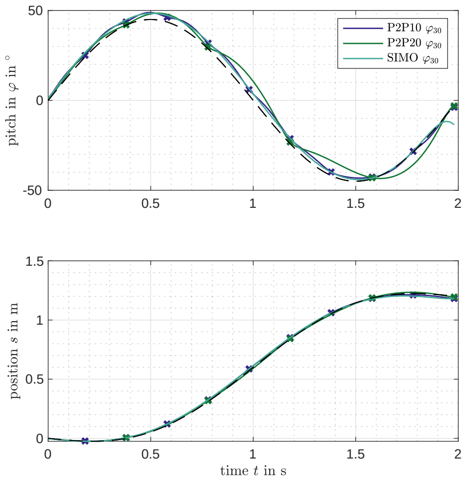


(d) P2Pstrat. P2PancStrat. P2PstratBW. SIMO. SISO.

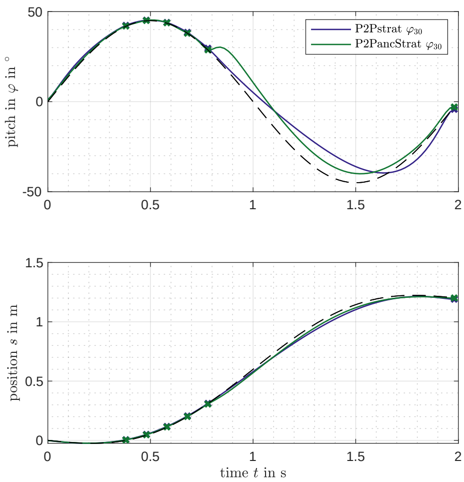
Figure A-3. Evolution over time in the 45° reference case simulation 1.



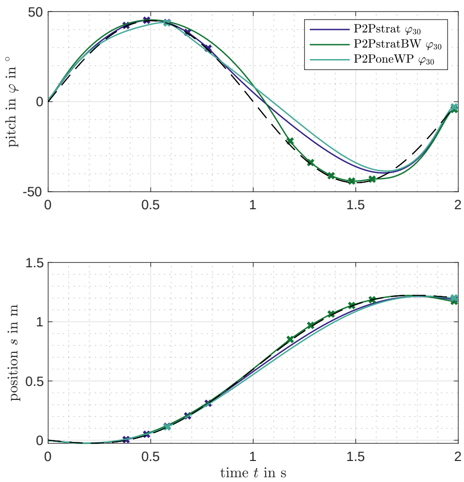
(a) Several P2P and SIMO SISO.



(b) P2P10 P2P20 and SIMO.

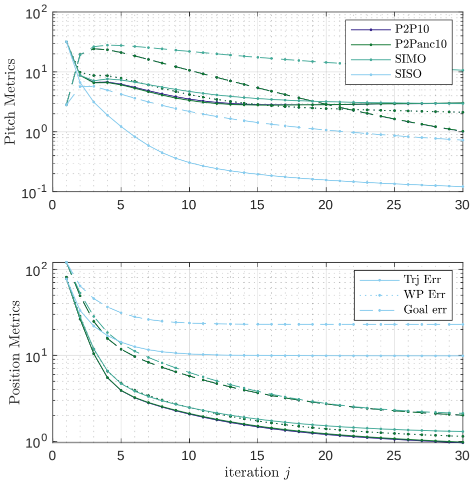


(c) P2Pstrat P2PancStrat.

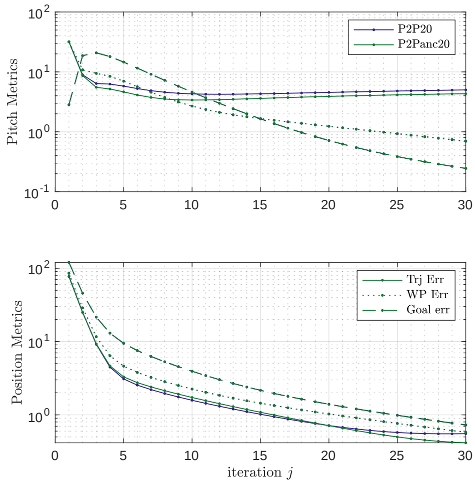


(d) P2Pstrat. P2PstratBW. P2PstratWP.

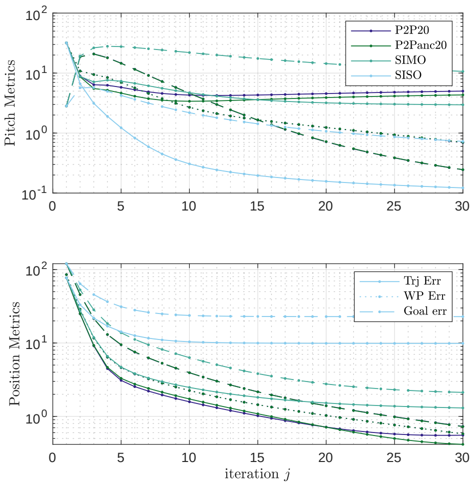
Figure A-4. Evolution over time in the 45° reference case simulation 2.



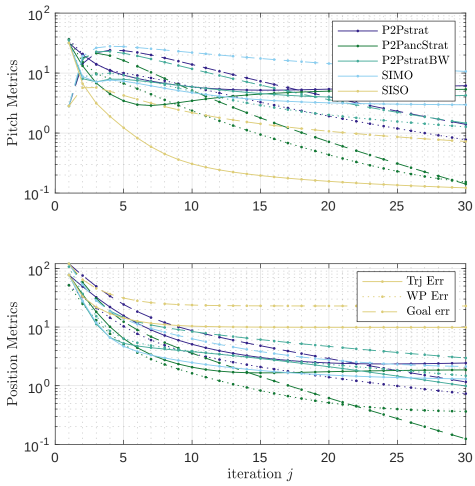
(a) P2P10. P2Panc10. SIMO. SISO.



(b) P2P20. P2Panc20.

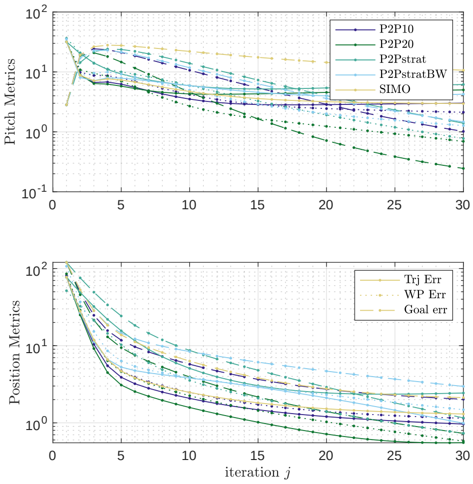


(c) P2P20. P2Panc20. SIMO. SISO.

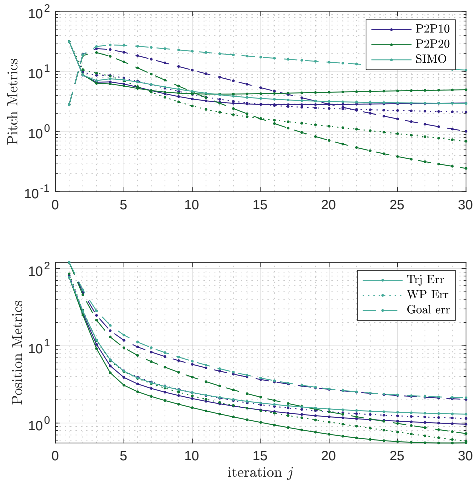


(d) P2Pstrat. P2PancStrat. P2PstratBW. SIMO. SISO.

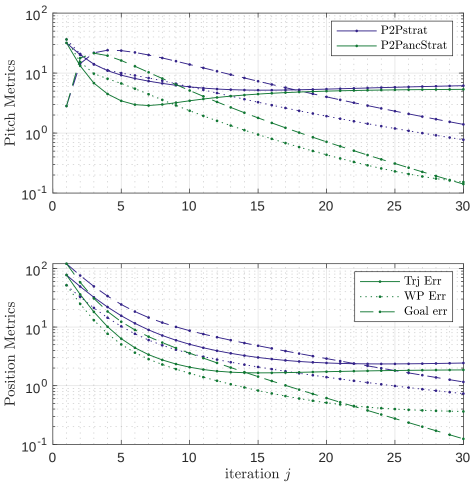
Figure A-5. Metrics of the 45° reference case simulation 1.



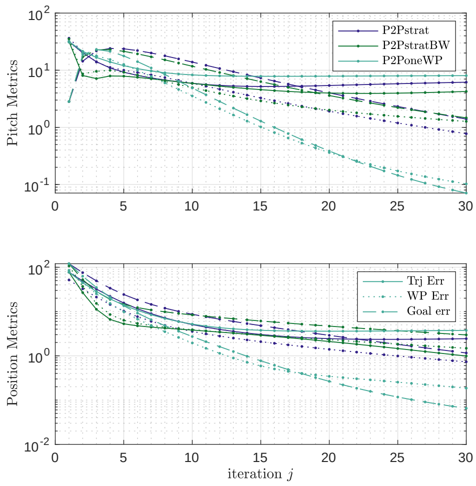
(a) Several P2P and SIMO SISO.



(b) P2P10 P2P20 and SIMO.



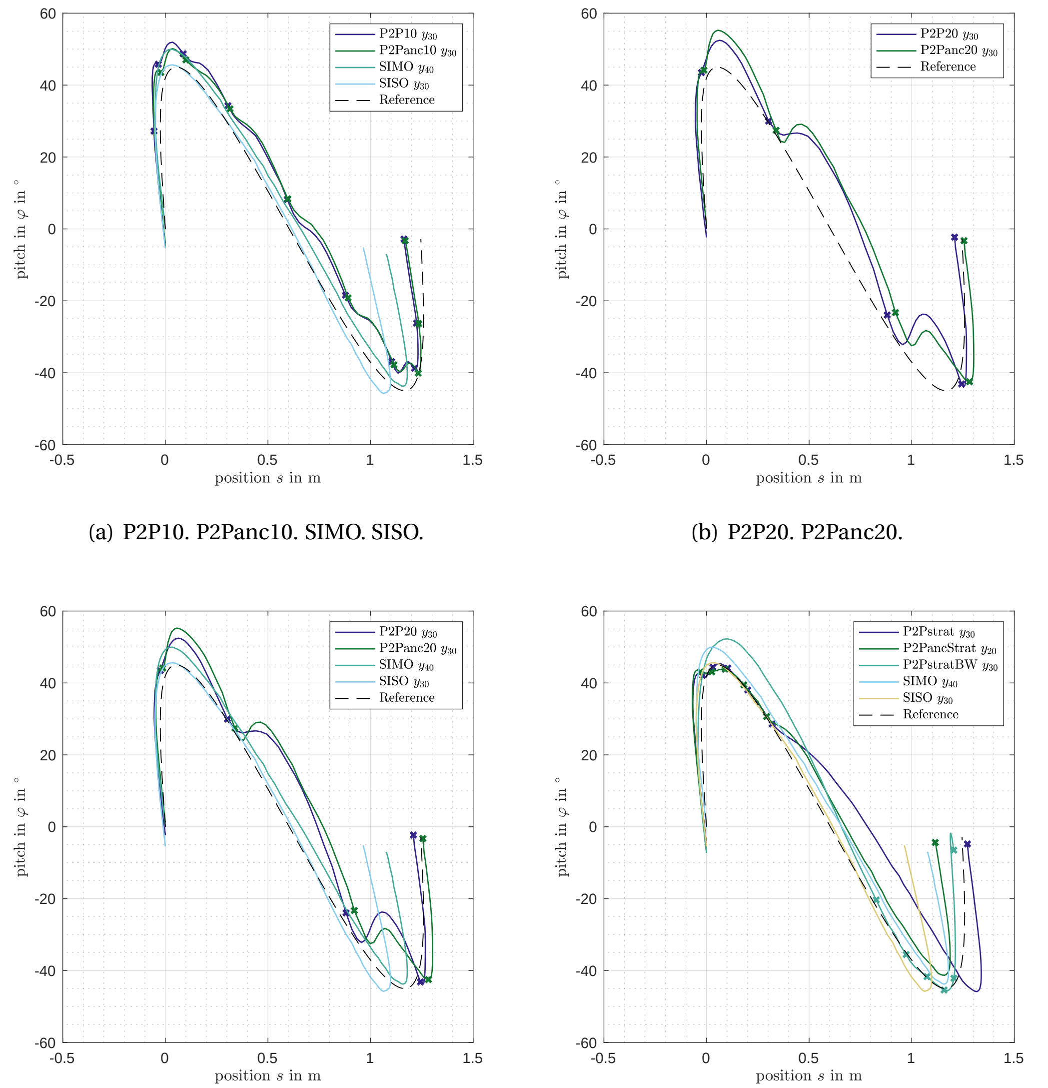
(c) P2Pstrat P2PancStrat.



(d) P2Pstrat. P2PstratBW. P2PstratWP.

Figure A-6. Metrics of the 45° reference case simulation 2.

A.3 Reference Experiment case $45°$



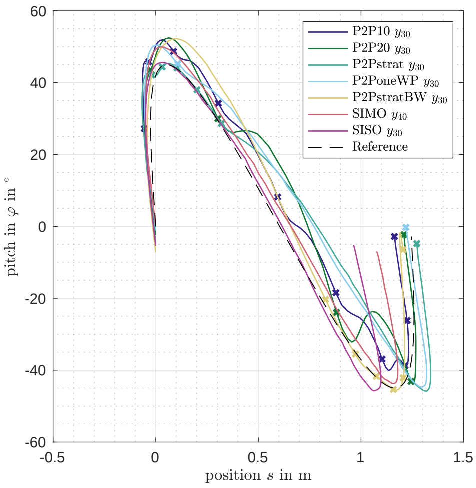
(a) P2P10. P2Panc10. SIMO. SISO.

(b) P2P20. P2Panc20.

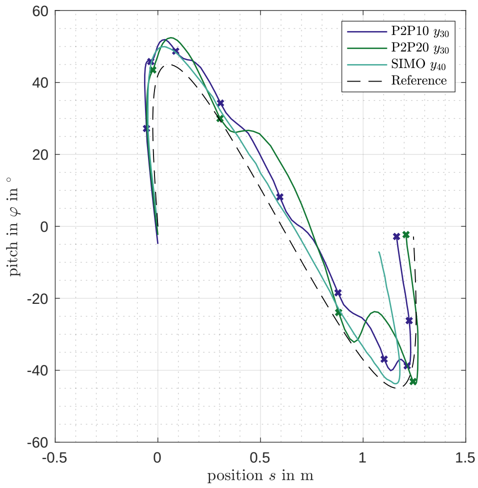
(c) P2P20. P2Panc20. SIMO. SISO.

(d) P2Pstrat. P2PancStrat. P2PstratBW. SIMO. SISO.

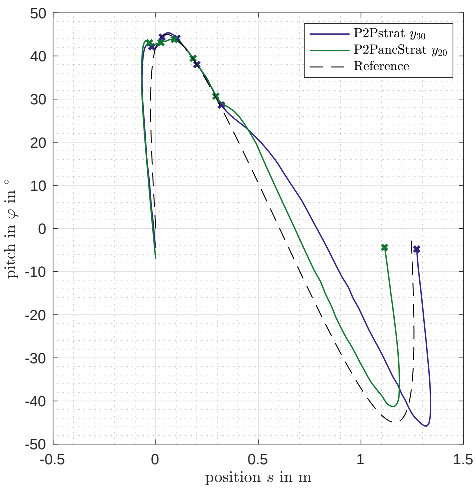
Figure A-7. Trajectory in the C-Space in the 45° reference case experiment 1.



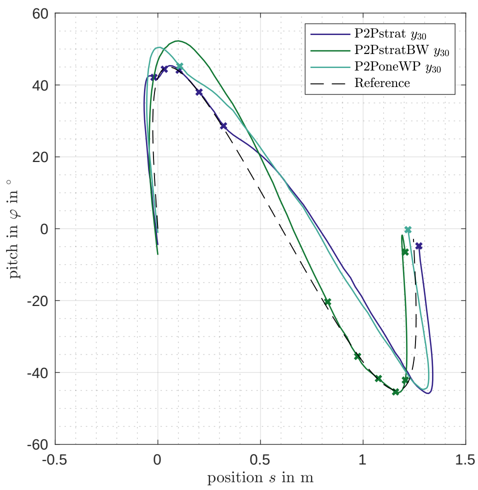
(a) Several P2P and SIMO SISO.



(b) P2P10 P2P20 and SIMO.

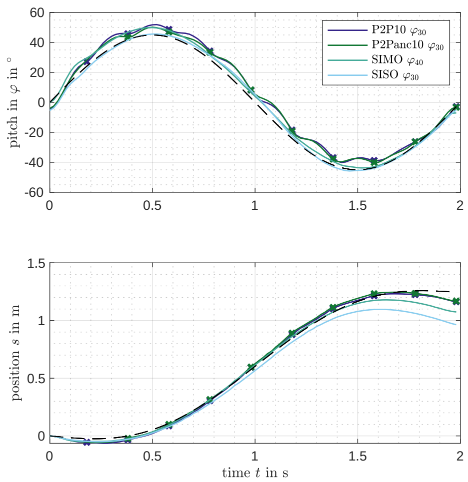


(c) P2Pstrat P2PancStrat.

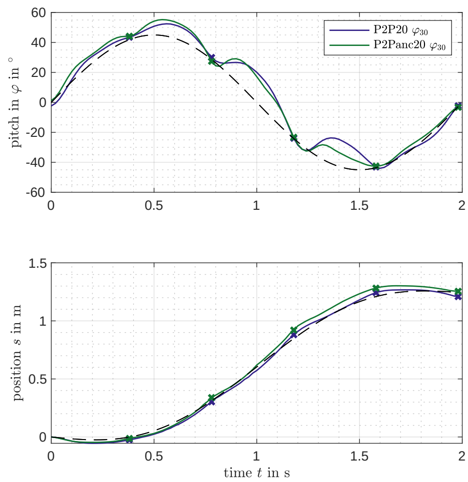


(d) P2Pstrat. P2PstratBW. P2PstratWP.

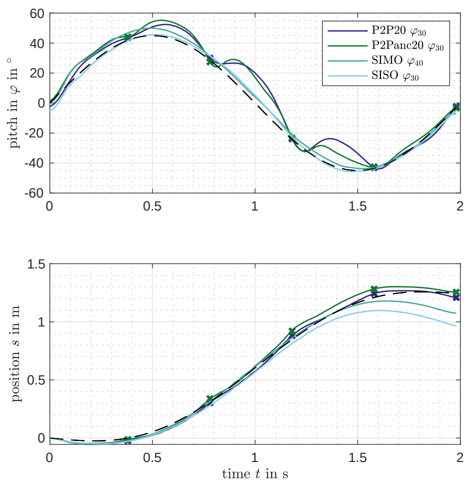
Figure A-8. Trajectory in the C-Space in the 45° reference case experiment 2.



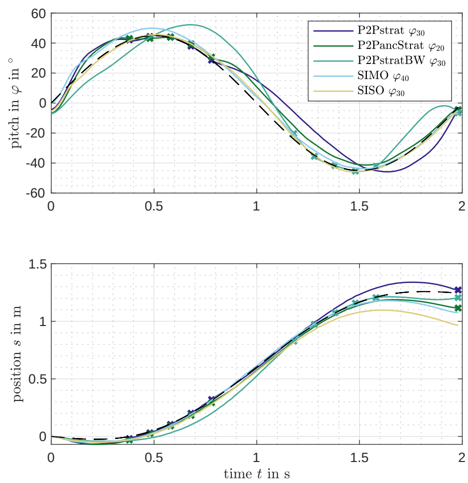
(a) P2P10. P2Panc10. SIMO. SISO.



(b) P2P20. P2Panc20.

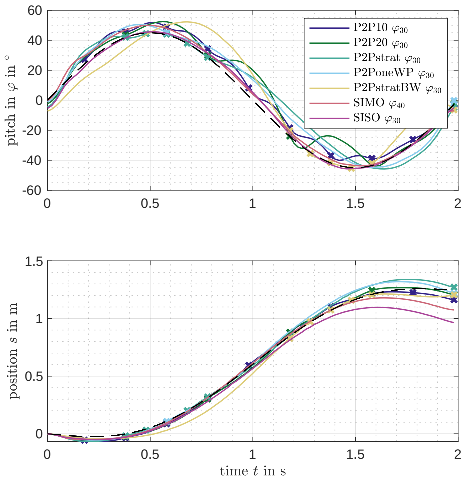


(c) P2P20. P2Panc20. SIMO. SISO.

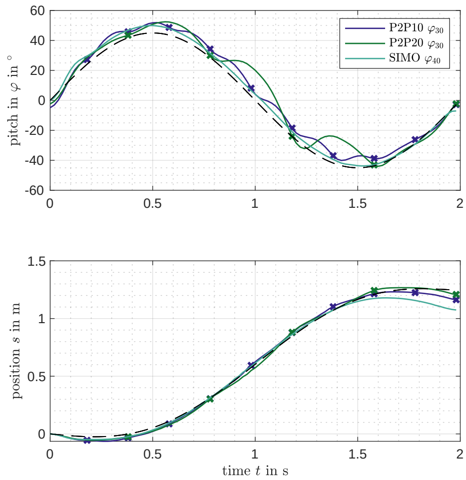


(d) P2Pstrat. P2PancStrat. P2PstratBW. SIMO. SISO.

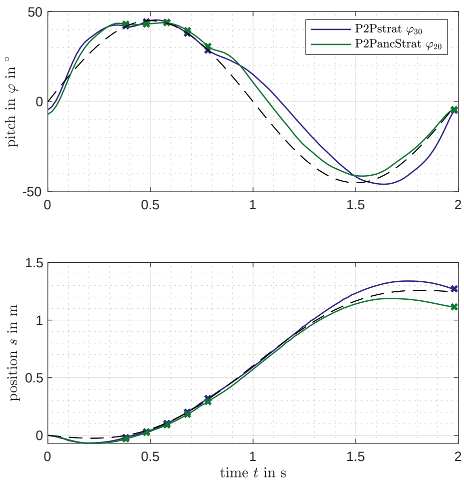
Figure A-9. Evolution over time in the 45° reference case experiment 1.



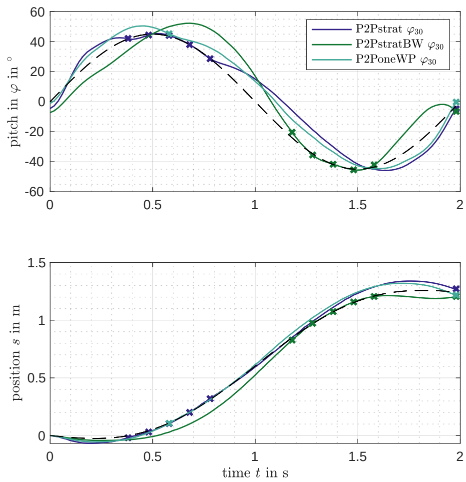
(a) Several P2P and SIMO SISO.



(b) P2P10 P2P20 and SIMO.

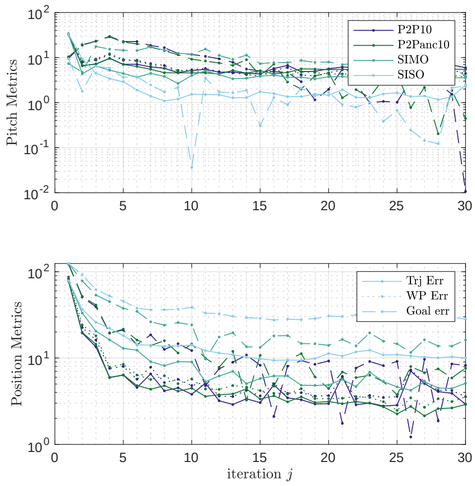


(c) P2Pstrat P2PancStrat.

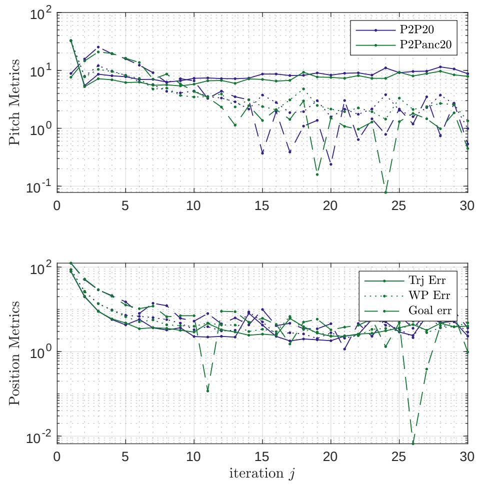


(d) P2Pstrat. P2PstratBW. P2PstratWP.

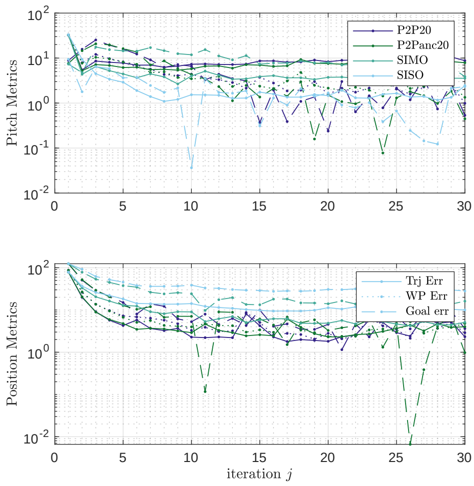
Figure A-10. Evolution over time in the 45° reference case experiment 2.



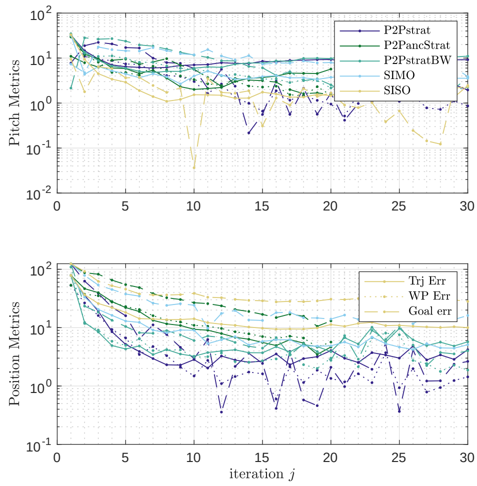
(a) P2P10. P2Panc10. SIMO. SISO.



(b) P2P20. P2Panc20.

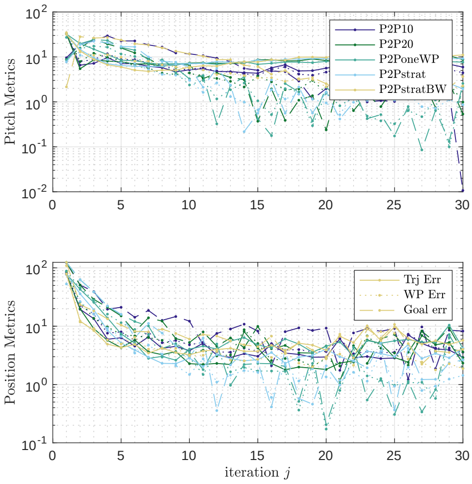


(c) P2P20. P2Panc20. SIMO. SISO.

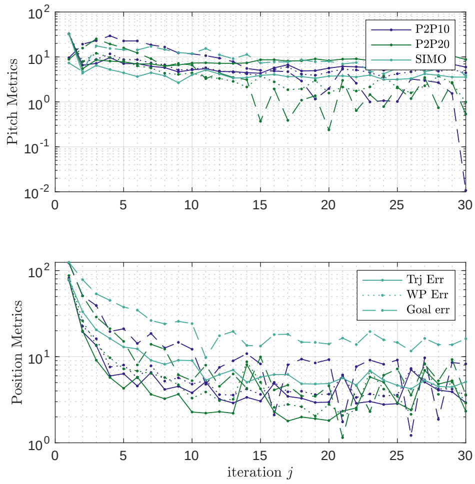


(d) P2Pstrat. P2PancStrat. P2PstratBW. SIMO. SISO.

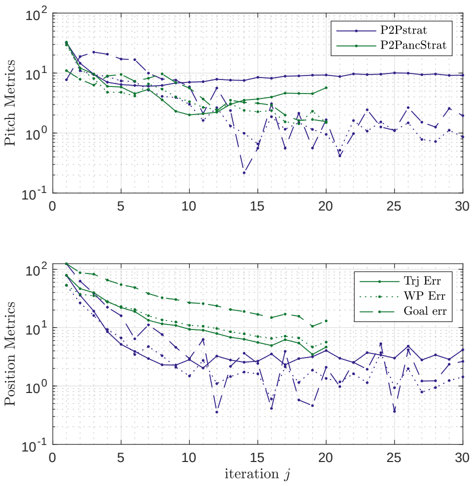
Figure A-11. Metrics of the 45° reference case experiment 1.



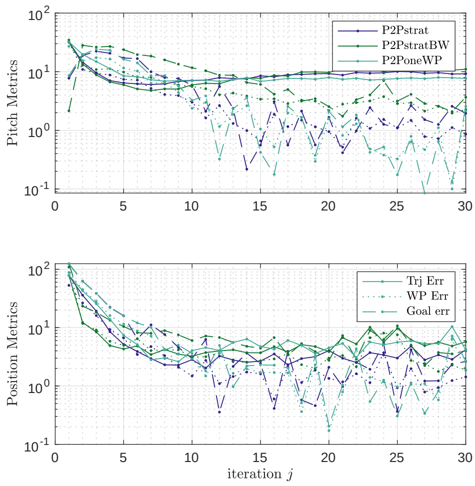
(a) Several P2P and SIMO SISO.



(b) P2P10 P2P20 and SIMO.



(c) P2Pstrat P2PancStrat.



(d) P2Pstrat. P2PstratBW. P2PstratWP.

Figure A-12. Metrics of the 45° reference case experiment 2.

A.4 Reference Experomiment case 70°

Case Study Local motion Planning.
In this case the trajectory is generated using a minimal input energy to the stabelized Two Wheeled Inverted Pendulum Robot (TWIPR) local motion planner. The reference is therefore of higher amplitude. The metrics are obtained similar to Section 5.1. The controllers an cases also coincide. Date is on the drive `30_Comparisons/LMP_exp`

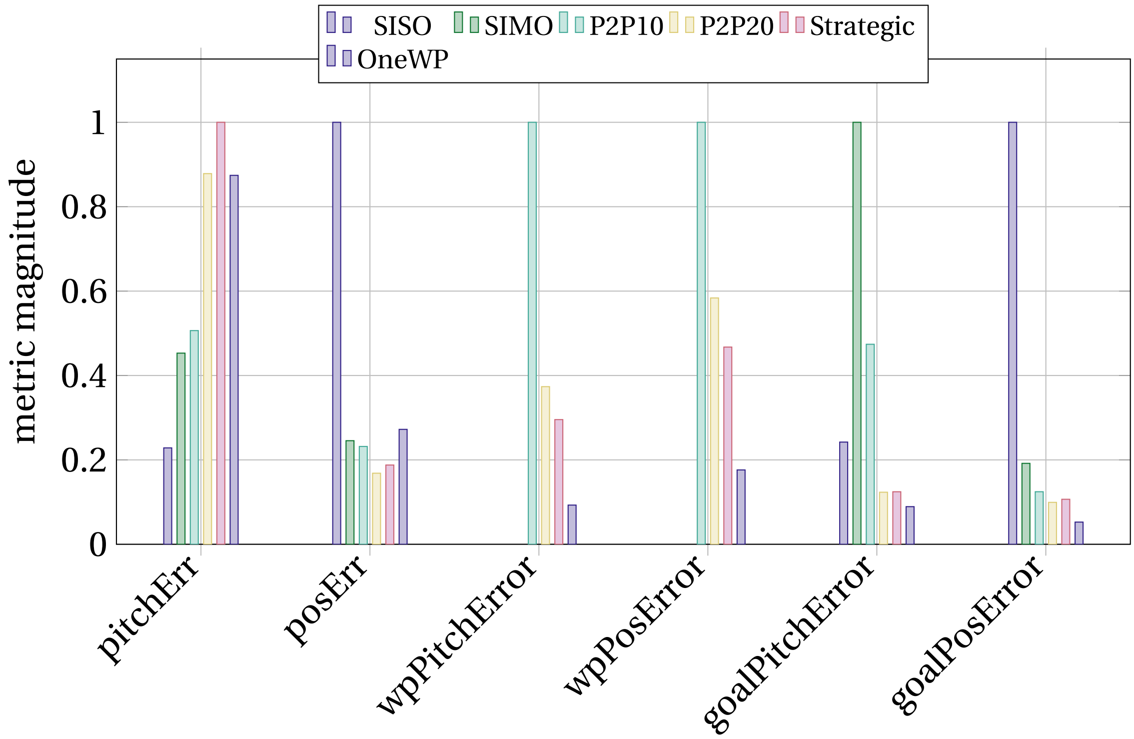
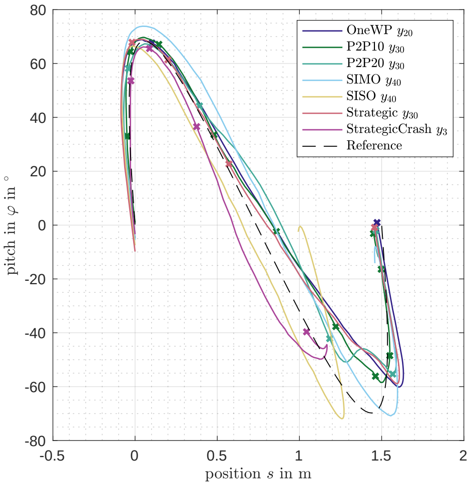


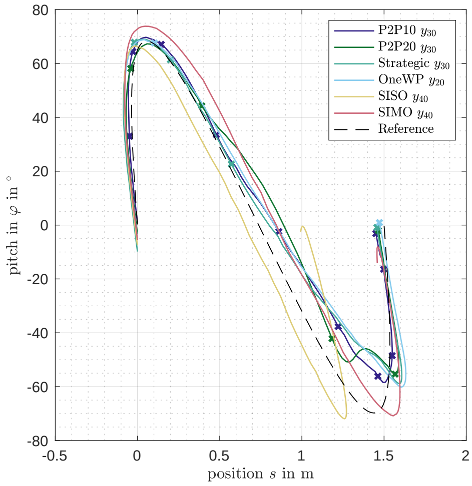
Figure A-13. Barplot of the metrics in the reference case 70° experiment.

Name	Pitch Err in °	Pos Err in cm	Goal Pitch Err in °	Goal Pos Err in cm	WP Pitch Err in °	WP Pos Err in cm
SISO	3.18	2.44×10^1	4.42	5.72×10^1		
SIMO	6.30	5.98	1.83×10^1	1.10×10^1		
P2P10	7.04	5.64	8.66	7.12	6.61	6.04
P2P20	1.22×10^1	4.10	2.25	5.68	3.49	4.98
Strategic	1.39×10^1	4.57	2.27	6.10	3.09	4.46
OneWP	1.22×10^1	6.63	1.63	3.01	1.37	2.38

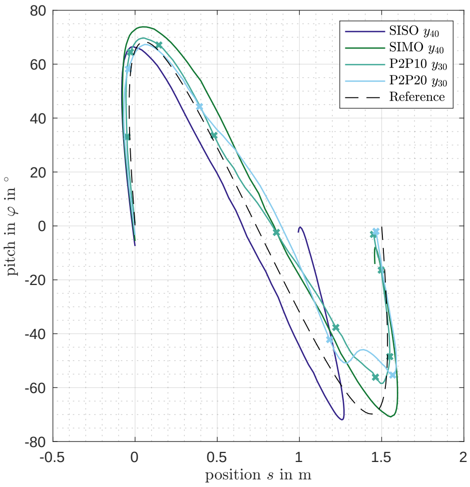
Table A.1. Metrics taken in the 70° experiment using the70° reference.



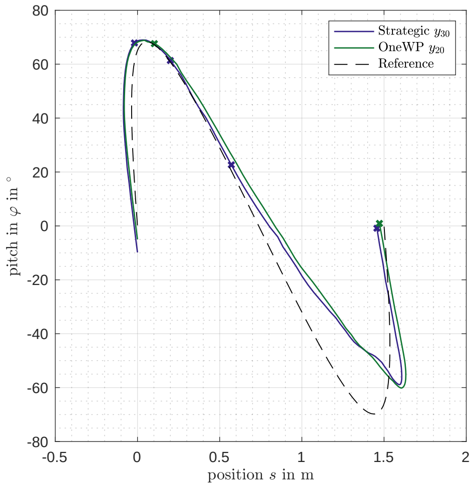
(a) All.



(b) P2P.

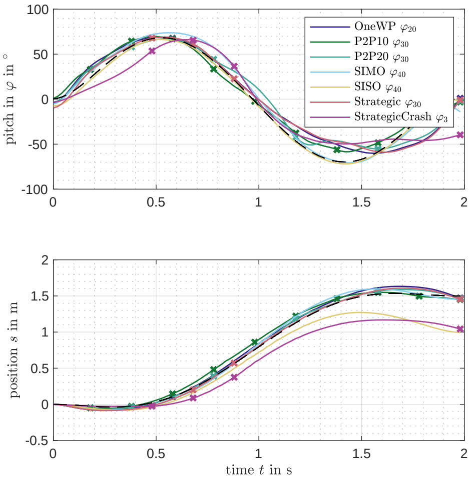


(c) P2P. SISO. SIMO.

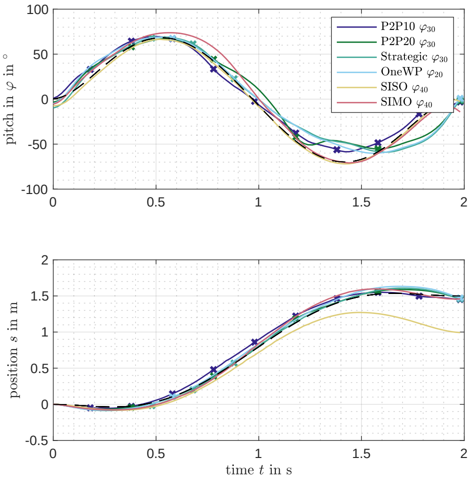


(d) strategic P2P.

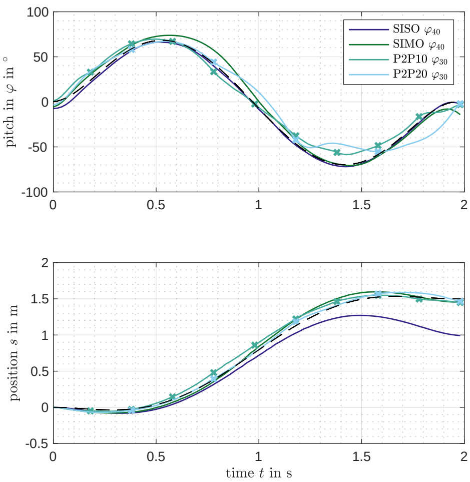
Figure A-14. Trajectory in C-space of the 70° reference case.



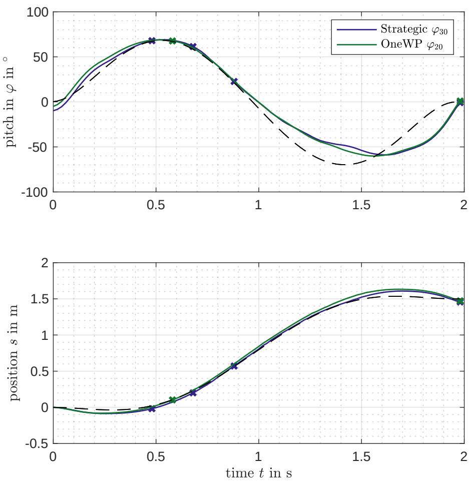
(a) All.



(b) P2P.

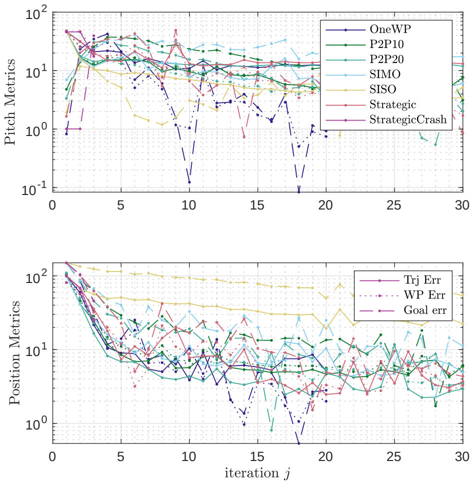


(c) P2P. SISO. SIMO.

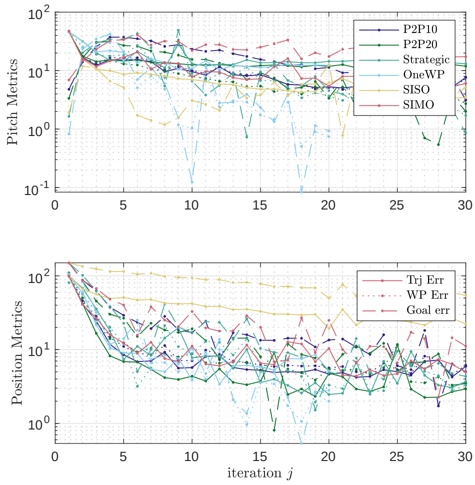


(d) strategic P2P.

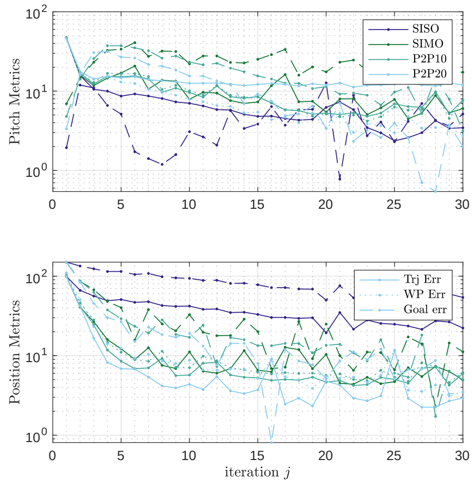
Figure A-15. Evolution over time in the 70° reference case.



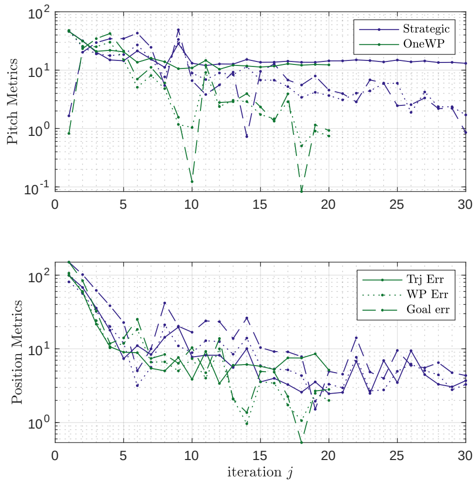
(a) All.



(b) P2P.



(c) P2P. SISO. SIMO.



(d) strategic P2P.

Figure A-16. Metrics of the 70° reference case.

A.5 Q-Filter design

In the scope of this thesis a custom Q-filter is used. As the filter does not need to be causal and may have a large order, a high order Finite Impulse Response (FIR) filter is seclected. For a given order an Infinite Impulse Response (IIR) filter outperforms a FIR filter in terms of damping and steepness. In contrast, for a higher order it is more esay to compute a FIR and it has a linear phase by design, so that the signal delay is constant for each frequency. The design is done in matlab without a window weighting. As the trail length is not to big, the FIR filter is chosen to have double the order of the signal length. To suppress windowing effects the resulting toeplitz matrix is designed as follows. The

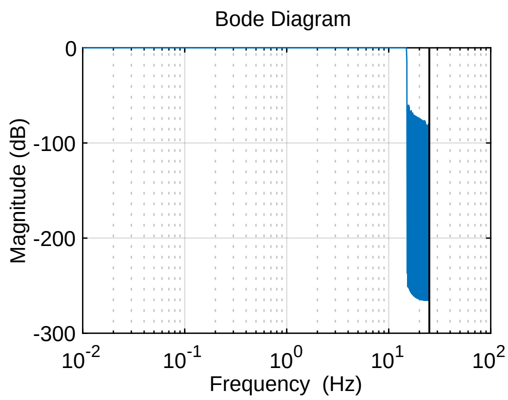


Figure A-17. Example of a frequency characteristic of a FIR Q-Filter with a cut off frequency of 15Hz an sample period of 20ms and a batchsize of 250.

first method $\boldsymbol{Q}_{\mathrm{FIR1}}$ assumes a cyclic repetition of the input signal, while the seceond assumes that the signal is constantly zero before the trail and stays on the final value in the end. An example of a frequency characteristic is given in Figure **??**. Due to numerical reasons some singular values are larger than one. To mitigate this problem, the SVD is performed and the singular values are cliped above one and the resulting representation is used in the process.

$$\boldsymbol{Q}_{\mathrm{FIR1}} = \begin{bmatrix} b_0 & b_1 + b_{1-n_\mathrm{b}} & b_2 + b_{2-n_\mathrm{b}} & \dots & b_{n_\mathrm{b}-1} + b_{-1} \\ b_{1-n_\mathrm{b}} + b_{-1} & b_0 & b_1 + b_{1-n_\mathrm{b}} & \dots & b_{n_\mathrm{b}-2} + b_{-2} \\ b_{2-n_\mathrm{b}} + b_{-2} & b_{1-n_\mathrm{b}} + b_{-1} & b_0 & \dots & b_{n_\mathrm{b}-3} + b_{-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b_{-1} + b_{n_\mathrm{b}-1} & b_{-2} + b_{n_\mathrm{b}-2} & b_{n_\mathrm{b}-3} + b_{-3} & \dots & b_0 \end{bmatrix} \qquad b_i = \frac{\sin(i \cdot \Omega_\mathrm{g})}{i \cdot \Omega_\mathrm{g}} \qquad \Omega_\mathrm{g} = 2\frac{f_\mathrm{g}}{T} \tag{A.11}$$

$$
\boldsymbol{Q}_{\text{FIR2}} = \begin{bmatrix} b_0 & b_1 & b_2 & \dots & b_{n_\text{b}-2} & b_{n_\text{b}-1} \\ b_{-1} & b_0 & b_1 & \dots & b_{n_\text{b}-3} & b_{n_\text{b}-2} + b_{n_\text{b}-1} \\ b_{-2} & b_{-1} & b_0 & \dots & b_{n_\text{b}-4} & \sum_{i=n_\text{b}-3}^{n_\text{b}-1} b_i \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ b_{2-n_\text{b}} & b_{3-n_\text{b}} & b_{4-n_\text{b}} & \dots & b_0 & \sum_{i=1}^{n_\text{b}-1} b_i \\ b_{1-n_\text{b}} & b_{2-n_\text{b}} & b_{3-n_\text{b}} & \dots & b_{-1} & \sum_0^{n_\text{b}-1} b_i \end{bmatrix} \qquad b_i = \frac{\sin(i \cdot \Omega_\text{g})}{i \cdot \Omega_\text{g}} \qquad \Omega_\text{g} = 2\frac{f_\text{g}}{T} \tag{A.12}
$$

A.5.1 Projection to the image and kernel and its complements

A linear projection is defined by the property that $\boldsymbol{A} = \boldsymbol{A} \circ \boldsymbol{A}$. Therefore,

$$
\boldsymbol{A}^\ddagger \boldsymbol{A} = \boldsymbol{A}^\ddagger \boldsymbol{A} \boldsymbol{A}^\ddagger \boldsymbol{A}
$$

and

$$
\boldsymbol{A}^\ddagger \boldsymbol{A} = \boldsymbol{A}^\ddagger \boldsymbol{A} \boldsymbol{A}^\ddagger \boldsymbol{A}
$$

are projections. (Obviously by (A.1))

The first acts in the domain of $\boldsymbol{A}$ and projects into the complment of the kernel. The second acts on the range of $\boldsymbol{A}$ and projects into the image of $\boldsymbol{A}$. So does $\left(\boldsymbol{E} - \boldsymbol{A}^\ddagger \boldsymbol{A}\right)$ in the range to the kernel and $\left(\boldsymbol{E} - \boldsymbol{A}\boldsymbol{A}^\ddagger\right)$ to the complement of the image. The complement of V being a subspace of X is defiened as $X \setminus Y$. Furthermore, the compementary projection is orthogonal as

$$
\begin{aligned}
\boldsymbol{A}^\text{T}\left(\boldsymbol{E} - \boldsymbol{A}\boldsymbol{A}^\ddagger\right) &= \boldsymbol{A}^\text{T}\left(\boldsymbol{E} - \left(\boldsymbol{A}\boldsymbol{A}^\ddagger\right)^\text{T}\right) && \text{(A.13)} \\
&= \boldsymbol{A}^\text{T}\left(\boldsymbol{E} - \left(\boldsymbol{A}^\ddagger\right)^\text{T} \boldsymbol{A}^\text{T}\right) && \text{(A.14)} \\
&= \boldsymbol{A}^\text{T} - \boldsymbol{A}^\text{T}\left(\boldsymbol{A}^\ddagger\right)^\text{T} \boldsymbol{A}^\text{T} && \text{(A.15)} \\
&= \boldsymbol{A}^\text{T} - \left(\boldsymbol{A}\boldsymbol{A}^\ddagger \boldsymbol{A}\right)^\text{T} && \text{(A.16)} \\
&= \boldsymbol{A}^\text{T} - \boldsymbol{A}^\text{T} && \text{(A.17)} \\
&= \boldsymbol{0} && \text{(A.18)}
\end{aligned}
$$

Contrariliy, in general $\boldsymbol{A}(\boldsymbol{B}\boldsymbol{A})^{-1}\boldsymbol{B}$ is not orthogonal. As it can be seen in a counter example:

$$
\boldsymbol{A} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \qquad \boldsymbol{B} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}
$$

$$
\boldsymbol{A}^\text{T}\left(\boldsymbol{E} - \boldsymbol{A}(\boldsymbol{B}\boldsymbol{A})^{-1}\boldsymbol{B}\right) = \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & 1 \\ 0 & 0 & 0 \end{bmatrix}
$$

A.5.2 Kronecker Product

The Kronecker product $\boldsymbol{A} \otimes \boldsymbol{B}$ of matrx $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ $\boldsymbol{B} \in \mathbb{R}^{r \times q}$ is defiend by

$$\boldsymbol{A} \otimes \boldsymbol{B} = \begin{bmatrix} a_{11}\boldsymbol{B} & a_{12}\boldsymbol{B} & \dots & a_{1n}\boldsymbol{B} \\ a_{21}\boldsymbol{B} & a_{22}\boldsymbol{B} & \dots & a_{2n}\boldsymbol{B} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}\boldsymbol{B} & a_{m2}\boldsymbol{B} & \dots & a_{mn}\boldsymbol{B} \end{bmatrix} \text{[40, p. 59].} \tag{A.19}$$

A.6 Derivation on Iterative Learning Control (ILC) dynamics

In this section the derivation of the input and error dynamics is shown step by step the results in the square case shown by Michael [4, pp. 42]. The derivations are done in atomic steps that are highlighted by colors. Only in rare cases where the whole term is reformulated the coloring is omitted. If in a step a color occurred, the part not highlighted will not be changed.

A.6.1 Input Dynamics

In this section the dynamics of the input is introduced. Also, the output and error trajectory dynamics are prepared as preliminary for the consecutive sections.

The input dynamics are independent on the state of actuation and are represented by

$$\begin{aligned} \boldsymbol{u}_{j+1} &= \boldsymbol{Q}\left(\boldsymbol{u}_j - \boldsymbol{L}\boldsymbol{e}_j\right) \\ &\quad \text{substitute } \boldsymbol{e}_j = \boldsymbol{y}_j - \boldsymbol{r} \\ &= \boldsymbol{Q}\left(\boldsymbol{u}_j - \boldsymbol{L}\left(\boldsymbol{y}_j - \boldsymbol{r}\right)\right) \\ &\quad \text{substitute } \boldsymbol{y}_j = \boldsymbol{P}\boldsymbol{u}_j - \boldsymbol{d} \\ &= \boldsymbol{Q}\left(\boldsymbol{u}_j - \boldsymbol{L}\left(\boldsymbol{P}\boldsymbol{u}_j + \boldsymbol{d} - \boldsymbol{r}\right)\right) \\ &\quad \text{reorganize} \\ &= \boldsymbol{Q}\left(\boldsymbol{E} - \boldsymbol{L}\boldsymbol{P}\right)\boldsymbol{u}_j - \boldsymbol{Q}\boldsymbol{L}\left(\boldsymbol{d} - \boldsymbol{r}\right) \end{aligned} \tag{A.20}$$

The input trajectory is the dynamic state of the system in the iteration domain and output and error depend on those algebraically. In the following derivation it is usefull to write them down explicitly.

$$\begin{aligned} \boldsymbol{y}_{j+1} &= \boldsymbol{P}\boldsymbol{u}_{j+1} + \boldsymbol{d} \\ &\quad \text{substitute (A.20) } \boldsymbol{u}_{j+1} = \boldsymbol{Q}\left(\boldsymbol{E} - \boldsymbol{L}\boldsymbol{P}\right)\boldsymbol{u}_j - \boldsymbol{Q}\boldsymbol{L}\left(\boldsymbol{d} - \boldsymbol{r}\right) \\ &= \boldsymbol{P}\left(\boldsymbol{Q}\left(\boldsymbol{E} - \boldsymbol{L}\boldsymbol{P}\right)\boldsymbol{u}_j - \boldsymbol{Q}\boldsymbol{L}\left(\boldsymbol{d} - \boldsymbol{r}\right)\right) + \boldsymbol{d} \\ &\quad \text{reorganize} \\ &= \boldsymbol{P}\boldsymbol{Q}\left(\boldsymbol{E} - \boldsymbol{L}\boldsymbol{P}\right)\boldsymbol{u}_j - \boldsymbol{P}\boldsymbol{Q}\boldsymbol{L}\left(\boldsymbol{d} - \boldsymbol{r}\right) + \boldsymbol{d} \end{aligned} \tag{A.21}$$

$$
\begin{aligned}
\boldsymbol{e}_{j+1} &= \boldsymbol{y}_{j+1} - \boldsymbol{r} \\
&\quad \text{substitute (A.21) } \boldsymbol{y}_{j+1} = \boldsymbol{Q}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{u}_j - \boldsymbol{QL}(\boldsymbol{d}-\boldsymbol{r}) \\
&\quad = \boldsymbol{P}\big(\boldsymbol{Q}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{u}_j - \boldsymbol{QL}(\boldsymbol{d}-\boldsymbol{r})\big) + \boldsymbol{d} - \boldsymbol{r} \\
&\quad \text{reorganize} \\
&\quad = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{u}_j + (\boldsymbol{E}-\boldsymbol{PQL})(\boldsymbol{d}-\boldsymbol{r})
\end{aligned} \tag{A.22}
$$

A.6.2 Unique Actuation

Output dynamics:

$$
\begin{aligned}
\boldsymbol{y}_{j+1} &= \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{u}_j - \boldsymbol{PQL}(\boldsymbol{d}-\boldsymbol{r}) + \boldsymbol{d} \quad \text{(A.21)} \\
&\quad \text{supplement (A.5) } \boldsymbol{E} = \boldsymbol{P}^{-1}\boldsymbol{P} \\
&\quad \text{supplement } \boldsymbol{0} = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{-1}\boldsymbol{d} - \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{-1}\boldsymbol{d} \\
&\quad = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{-1}\boldsymbol{P}\boldsymbol{u}_j - \boldsymbol{PQL}(\boldsymbol{d}-\boldsymbol{r}) + \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{-1}\boldsymbol{d} - \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{-1}\boldsymbol{d} + \boldsymbol{d} \\
&\quad = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{-1}\boldsymbol{P}\boldsymbol{u}_j - \boldsymbol{PQL}(\boldsymbol{d}-\boldsymbol{r}) + \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{-1}\boldsymbol{d} - \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{-1}\boldsymbol{d} + \boldsymbol{d} \\
&\quad \text{reorganize reorganize} \\
&\quad = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{-1}\big(\boldsymbol{P}\boldsymbol{u}_j + \boldsymbol{d}\big) - \boldsymbol{PQL}(\boldsymbol{d}-\boldsymbol{r}) - \boldsymbol{PQP}^{-1}\boldsymbol{d} + \boldsymbol{PQLd} + \boldsymbol{d} \\
&\quad = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{-1}\big(\boldsymbol{P}\boldsymbol{u}_j + \boldsymbol{d}\big) - \boldsymbol{PQL}(\boldsymbol{d}-\boldsymbol{r}) - \boldsymbol{PQP}^{-1}\boldsymbol{d} + \boldsymbol{PQLd} + \boldsymbol{d} \\
&\quad \text{cancel reorganize substitute } \boldsymbol{y}_j = \boldsymbol{P}\boldsymbol{u}_j + \boldsymbol{d} \\
&\quad = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{-1}\boldsymbol{y}_j + \boldsymbol{PQLr} + \big(\boldsymbol{E}-\boldsymbol{PQP}^{-1}\big)\boldsymbol{d} \\
\boldsymbol{y}_{j+1} &= \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{-1}\boldsymbol{y}_j + \boldsymbol{PQLr} + \big(\boldsymbol{E}-\boldsymbol{PQP}^{-1}\big)\boldsymbol{d}
\end{aligned} \tag{A.23}
$$

Error dynamics:

$$
\begin{aligned}
\boldsymbol{e}_{j+1} &= \boldsymbol{y}_{j+1} + \boldsymbol{r} \\
&\quad \text{sustitute (A.23)} \\
&\quad \text{supplement } \boldsymbol{0} = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{-1}\boldsymbol{r} - \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{-1}\boldsymbol{r} \\
&\quad = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{-1}\boldsymbol{y}_j + \boldsymbol{PQLr} + \big(\boldsymbol{E}-\boldsymbol{PQP}^{-1}\big)\boldsymbol{d} - \boldsymbol{r} + \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{-1}\boldsymbol{r} - \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{-1}\boldsymbol{r} \\
&\quad = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{-1}\boldsymbol{y}_j + \boldsymbol{PQLr} + \big(\boldsymbol{E}-\boldsymbol{PQP}^{-1}\big)\boldsymbol{d} - \boldsymbol{r} + \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{-1}\boldsymbol{r} - \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{-1}\boldsymbol{r} \\
&\quad \text{reorganize reorganize} \\
&\quad = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{-1}\big(\boldsymbol{y}_j - \boldsymbol{r}\big) + \boldsymbol{PQLr} + \big(\boldsymbol{E}-\boldsymbol{PQP}^{-1}\big)\boldsymbol{d} - \boldsymbol{r} + \boldsymbol{PQP}^{-1}\boldsymbol{r} - \boldsymbol{PQLr} \\
&\quad = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{-1}\big(\boldsymbol{y}_j - \boldsymbol{r}\big) + \boldsymbol{PQLr} + \big(\boldsymbol{E}-\boldsymbol{PQP}^{-1}\big)\boldsymbol{d} - \boldsymbol{r} + \boldsymbol{PQP}^{-1}\boldsymbol{r} - \boldsymbol{PQLr} \\
&\quad \text{substitute } \boldsymbol{e}_j = \boldsymbol{y}_j - \boldsymbol{r} \text{ cancel reorganize} \\
&\quad = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{-1}\boldsymbol{e}_j + \big(\boldsymbol{E}-\boldsymbol{PQP}^{-1}\big)\boldsymbol{d} - \big(\boldsymbol{E}-\boldsymbol{PQP}^{-1}\big)\boldsymbol{r} \\
&\quad = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{-1}\boldsymbol{e}_j + \big(\boldsymbol{E}-\boldsymbol{PQP}^{-1}\big)\boldsymbol{d} - \big(\boldsymbol{E}-\boldsymbol{PQP}^{-1}\big)\boldsymbol{r} \\
&\quad \text{reorganize} \\
&\quad = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{-1}\boldsymbol{e}_j + \big(\boldsymbol{E}-\boldsymbol{PQP}^{-1}\big)(\boldsymbol{d}-\boldsymbol{r}) \\
\boldsymbol{e}_{j+1} &= \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{-1}\boldsymbol{e}_j + \big(\boldsymbol{E}-\boldsymbol{PQP}^{-1}\big)(\boldsymbol{d}-\boldsymbol{r})
\end{aligned} \tag{A.24}
$$

A.6.3 Under Actuation

In the case of under actuation the matrix is "tall" [40, p.22] meaning that the range of $\boldsymbol{P}$ is higher dimensional than its domain. Furthermore, the lifteddynamics matrix is assumed to be full rank, so that the pseudoinverse The underlying sysmtem is the standard ILC system that is composed of the plant model (**??**) and the standard ILC update (3.9).

Output dynamics:

$$\boldsymbol{y}_{j+1} = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\,\boldsymbol{u}_j - \boldsymbol{PQL}(\boldsymbol{d}-\boldsymbol{r}) + \boldsymbol{d} \text{ (A.21)}$$

supplement (A.5) $\boldsymbol{E} = \boldsymbol{P}^{\dagger}\boldsymbol{P}$

supplement $\boldsymbol{0} = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\,\boldsymbol{P}^{\dagger}\boldsymbol{d} - \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\,\boldsymbol{P}^{\dagger}\boldsymbol{d}$

$$= \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\,\boldsymbol{P}^{\dagger}\boldsymbol{P}\boldsymbol{u}_j + \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\,\boldsymbol{P}^{\dagger}\boldsymbol{d} - \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\,\boldsymbol{P}^{\dagger}\boldsymbol{d} + \boldsymbol{PQL}(\boldsymbol{r}-\boldsymbol{d}) + \boldsymbol{d}$$

$$= \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\,\boldsymbol{P}^{\dagger}\boldsymbol{P}\boldsymbol{u}_j + \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\,\boldsymbol{P}^{\dagger}\boldsymbol{d} - \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\,\boldsymbol{P}^{\dagger}\boldsymbol{d} + \boldsymbol{PQL}(\boldsymbol{r}-\boldsymbol{d}) + \boldsymbol{d}$$

reorganize reorganize reorganize

$$= \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\,\boldsymbol{P}^{\dagger}\left(\boldsymbol{P}\boldsymbol{u}_j + \boldsymbol{d}\right) + \boldsymbol{PQLr} + \left(\boldsymbol{E} - \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\,\boldsymbol{P}^{\dagger} - \boldsymbol{PQL}\right)\boldsymbol{d}$$

substitute (3.8) $\boldsymbol{y}_j = \boldsymbol{P}\boldsymbol{u}_j + \boldsymbol{d}$

$$\boldsymbol{y}_{j+1} = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\,\boldsymbol{P}^{\dagger}\boldsymbol{y}_j + \boldsymbol{PQLr} + \left(\boldsymbol{E} - \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\,\boldsymbol{P}^{\dagger} - \boldsymbol{PQL}\right)\boldsymbol{d}$$

reorganize

$$\boldsymbol{y}_{j+1} = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\,\boldsymbol{P}^{\dagger}\boldsymbol{y}_j + \boldsymbol{PQLr} + \left(\boldsymbol{E} - \boldsymbol{PQ}\left(\boldsymbol{P}^{\dagger} - \boldsymbol{LPP}^{\dagger}\right) - \boldsymbol{PQL}\right)\boldsymbol{d}$$

supplement

$$\boldsymbol{y}_{j+1} = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\,\boldsymbol{P}^{\dagger}\boldsymbol{y}_j + \boldsymbol{PQLr} + \left(\boldsymbol{E} - \boldsymbol{PQ}\left(\boldsymbol{P}^{\dagger} - \boldsymbol{L}\left(\boldsymbol{E}-\boldsymbol{E}+\boldsymbol{PP}^{\dagger}\right)\right) - \boldsymbol{PQL}\right)\boldsymbol{d}$$

$$\boldsymbol{y}_{j+1} = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\,\boldsymbol{P}^{\dagger}\boldsymbol{y}_j + \boldsymbol{PQLr} + \left(\boldsymbol{E} - \boldsymbol{PQ}\left(\boldsymbol{P}^{\dagger} - \boldsymbol{L}\left(\boldsymbol{E}-\boldsymbol{E}+\boldsymbol{PP}^{\dagger}\right)\right) - \boldsymbol{PQL}\right)\boldsymbol{d}$$

reorganize

$$\boldsymbol{y}_{j+1} = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\,\boldsymbol{P}^{\dagger}\boldsymbol{y}_j + \boldsymbol{PQLr} + \left(\boldsymbol{E} - \boldsymbol{PQP}^{\dagger} + \boldsymbol{PQL} - \boldsymbol{PQL}\left(\boldsymbol{E}-\boldsymbol{PP}^{\dagger}\right) - \boldsymbol{PQL}\right)\boldsymbol{d}$$

$$\boldsymbol{y}_{j+1} = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\,\boldsymbol{P}^{\dagger}\boldsymbol{y}_j + \boldsymbol{PQLr} + \left(\boldsymbol{E} - \boldsymbol{PQP}^{\dagger} + \boldsymbol{PQL} - \boldsymbol{PQL}\left(\boldsymbol{E}-\boldsymbol{PP}^{\dagger}\right) - \boldsymbol{PQL}\right)\boldsymbol{d}$$

cancel

$$\boldsymbol{y}_{j+1} = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\,\boldsymbol{P}^{\dagger}\boldsymbol{y}_j + \boldsymbol{PQLr} + \left(\boldsymbol{E} - \boldsymbol{PQP}^{\dagger} + \boldsymbol{PQL}\left(\boldsymbol{E}-\boldsymbol{PP}^{\dagger}\right)\right)\boldsymbol{d}$$

reorganize

$$\boldsymbol{y}_{j+1} = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\,\boldsymbol{P}^{\dagger}\boldsymbol{y}_j + \boldsymbol{PQL}\left(\boldsymbol{r} - \left(\boldsymbol{E}-\boldsymbol{PP}^{\dagger}\right)\boldsymbol{d}\right) + \left(\boldsymbol{E}-\boldsymbol{PQP}^{\dagger}\right)\boldsymbol{d}$$

$$\boldsymbol{y}_{j+1} = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\,\boldsymbol{P}^{\dagger}\boldsymbol{y}_j + \boldsymbol{PQL}\left(\boldsymbol{r} - \left(\boldsymbol{E}-\boldsymbol{PP}^{\dagger}\right)\boldsymbol{d}\right) + \left(\boldsymbol{E}-\boldsymbol{PQP}^{\dagger}\right)\boldsymbol{d} \tag{A.25}$$

Error dynamics:

$$
\begin{aligned}
\boldsymbol{e}_{j+1} &= \boldsymbol{y}_{j+1} - \boldsymbol{r} \\
&= \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{\dagger}\boldsymbol{y}_j + \boldsymbol{PQL}\left(\boldsymbol{r} - \left(\boldsymbol{E}-\boldsymbol{PP}^{\dagger}\right)\boldsymbol{d}\right) + \left(\boldsymbol{E}-\boldsymbol{PQP}^{\dagger}\right)\boldsymbol{d} - \boldsymbol{r} \\
&\text{reorganize} \\
&= \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{\dagger}\boldsymbol{y}_j - (\boldsymbol{E}-\boldsymbol{PQL})\boldsymbol{r} - \boldsymbol{PQL}\left(\boldsymbol{E}-\boldsymbol{PP}^{\dagger}\right)\boldsymbol{d} + \left(\boldsymbol{E}-\boldsymbol{PQP}^{\dagger}\right)\boldsymbol{d} \\
&\text{supplement} \\
&= \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{\dagger}\boldsymbol{y}_j - (\boldsymbol{E}-\boldsymbol{PQL})\boldsymbol{r} - \boldsymbol{PQL}\left(\boldsymbol{E}-\boldsymbol{PP}^{\dagger}\right)\boldsymbol{d} + \left(\boldsymbol{E}-\boldsymbol{PQP}^{\dagger}\right)\boldsymbol{d} \\
&\quad - \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{\dagger}\boldsymbol{r} + \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{\dagger}\boldsymbol{r} \\
&= \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{\dagger}\boldsymbol{y}_j - (\boldsymbol{E}-\boldsymbol{PQL})\boldsymbol{r} - \boldsymbol{PQL}\left(\boldsymbol{E}-\boldsymbol{PP}^{\dagger}\right)\boldsymbol{d} + \left(\boldsymbol{E}-\boldsymbol{PQP}^{\dagger}\right)\boldsymbol{d} \\
&\quad - \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{\dagger}\boldsymbol{r} + \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{\dagger}\boldsymbol{r} \\
&\text{reorganize and substitute } \boldsymbol{e}_j = \boldsymbol{y}_j - \boldsymbol{r} \text{ reorganize} \\
&= \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{\dagger}\boldsymbol{e}_j - (\boldsymbol{E}-\boldsymbol{PQL})\boldsymbol{r} - \boldsymbol{PQL}\left(\boldsymbol{E}-\boldsymbol{PP}^{\dagger}\right)\boldsymbol{d} + \left(\boldsymbol{E}-\boldsymbol{PQP}^{\dagger}\right)\boldsymbol{d} \\
&\quad + \boldsymbol{PQ}\left(\boldsymbol{P}^{\dagger} - \boldsymbol{LPP}^{\dagger}\right)\boldsymbol{r} \\
&\text{supplement} \\
&= \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{\dagger}\boldsymbol{e}_j - (\boldsymbol{E}-\boldsymbol{PQL})\boldsymbol{r} - \boldsymbol{PQL}\left(\boldsymbol{E}-\boldsymbol{PP}^{\dagger}\right)\boldsymbol{d} + \left(\boldsymbol{E}-\boldsymbol{PQP}^{\dagger}\right)\boldsymbol{d} \\
&\quad + \boldsymbol{PQ}\left(\boldsymbol{P}^{\dagger} - \boldsymbol{L}\left(\boldsymbol{E}-\boldsymbol{E}+\boldsymbol{PP}^{\dagger}\right)\right)\boldsymbol{r} \\
&\text{reorganize} \\
&= \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{\dagger}\boldsymbol{e}_j - (\boldsymbol{E}-\boldsymbol{PQL})\boldsymbol{r} - \boldsymbol{PQL}\left(\boldsymbol{E}-\boldsymbol{PP}^{\dagger}\right)\boldsymbol{d} + \left(\boldsymbol{E}-\boldsymbol{PQP}^{\dagger}\right)\boldsymbol{d} \\
&\quad + \boldsymbol{PQP}^{\dagger}\boldsymbol{r} - \boldsymbol{PQLr} + \boldsymbol{PQL}\left(\boldsymbol{E}-\boldsymbol{PP}^{\dagger}\right)\boldsymbol{r} \\
&\text{reorganize} \quad \text{cancel} \\
&= \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{\dagger}\boldsymbol{e}_j - \boldsymbol{PQL}\left(\boldsymbol{E}-\boldsymbol{PP}^{\dagger}\right)\boldsymbol{d} + \left(\boldsymbol{E}-\boldsymbol{PQP}^{\dagger}\right)\boldsymbol{d} \\
&\quad + \boldsymbol{PQP}^{\dagger}\boldsymbol{r} + \boldsymbol{PQL}\left(\boldsymbol{E}-\boldsymbol{PP}^{\dagger}\right)\boldsymbol{r} - \boldsymbol{r} \\
&= \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{\dagger}\boldsymbol{e}_j - \boldsymbol{PQL}\left(\boldsymbol{E}-\boldsymbol{PP}^{\dagger}\right)\boldsymbol{d} + \left(\boldsymbol{E}-\boldsymbol{PQP}^{\dagger}\right)\boldsymbol{d} \\
&\quad + \boldsymbol{PQP}^{\dagger}\boldsymbol{r} + \boldsymbol{PQL}\left(\boldsymbol{E}-\boldsymbol{PP}^{\dagger}\right)\boldsymbol{r} - \boldsymbol{r} \\
&\text{reorganize} \\
&= \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{\dagger}\boldsymbol{e}_j - \boldsymbol{PQL}\left(\boldsymbol{E}-\boldsymbol{PP}^{\dagger}\right)\boldsymbol{d} + \left(\boldsymbol{E}-\boldsymbol{PQP}^{\dagger}\right)\boldsymbol{d} \\
&\quad - \left(\boldsymbol{E}-\boldsymbol{PQP}^{\dagger}\right)\boldsymbol{r} + \boldsymbol{PQL}\left(\boldsymbol{E}-\boldsymbol{PP}^{\dagger}\right)\boldsymbol{r} \\
&= \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{\dagger}\boldsymbol{e}_j - \boldsymbol{PQL}\left(\boldsymbol{E}-\boldsymbol{PP}^{\dagger}\right)\boldsymbol{d} + \left(\boldsymbol{E}-\boldsymbol{PQP}^{\dagger}\right)\boldsymbol{d} \\
&\quad - \left(\boldsymbol{E}-\boldsymbol{PQP}^{\dagger}\right)\boldsymbol{r} + \boldsymbol{PQL}\left(\boldsymbol{E}-\boldsymbol{PP}^{\dagger}\right)\boldsymbol{r} \\
&\text{reorganize} \\
&= \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{\dagger}\boldsymbol{e}_j - \boldsymbol{PQL}\left(\boldsymbol{E}-\boldsymbol{PP}^{\dagger}\right)(\boldsymbol{d}-\boldsymbol{r}) + \left(\boldsymbol{E}-\boldsymbol{PQP}^{\dagger}\right)(\boldsymbol{d}-\boldsymbol{r})
\end{aligned}
$$

$$\boldsymbol{e}_{j+1} = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{\dagger}\boldsymbol{e}_j - \boldsymbol{PQL}\left(\boldsymbol{E}-\boldsymbol{PP}^{\dagger}\right)(\boldsymbol{d}-\boldsymbol{r}) + \left(\boldsymbol{E}-\boldsymbol{PQP}^{\dagger}\right)(\boldsymbol{d}-\boldsymbol{r}) \tag{A.26}$$

Convergence sequence

$$\boldsymbol{e}_\infty = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^\dagger\boldsymbol{e}_\infty - \boldsymbol{PQL}\left(\boldsymbol{E}-\boldsymbol{PP}^\dagger\right)(\boldsymbol{d}-\boldsymbol{r}) + \left(\boldsymbol{E}-\boldsymbol{PQP}^\dagger\right)(\boldsymbol{d}-\boldsymbol{r}) \tag{A.27}$$

substract (A.26) and multipy by -1

$$\begin{aligned}
\boldsymbol{e}_{j+1}-\boldsymbol{e}_\infty &= \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^\dagger\boldsymbol{e}_j - \boldsymbol{PQL}\left(\boldsymbol{E}-\boldsymbol{PP}^\dagger\right)(\boldsymbol{d}-\boldsymbol{r}) + \left(\boldsymbol{E}-\boldsymbol{PQP}^\dagger\right)(\boldsymbol{d}-\boldsymbol{r})\\
&\quad - \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^\dagger\boldsymbol{e}_\infty + \boldsymbol{PQL}\left(\boldsymbol{E}-\boldsymbol{PP}^\dagger\right)(\boldsymbol{d}-\boldsymbol{r}) - \left(\boldsymbol{E}-\boldsymbol{PQP}^\dagger\right)(\boldsymbol{d}-\boldsymbol{r})\\
\boldsymbol{e}_{j+1}-\boldsymbol{e}_\infty &= \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^\dagger\left(\boldsymbol{e}_j-\boldsymbol{e}_\infty\right) \qquad (A.28)
\end{aligned}$$

A.6.4 Overactuation

Assume that the system is BROAD and full rank [40, p.22] meaning the system is under actuated $\boldsymbol{PP}^{+} = \boldsymbol{PP}^{\mathrm{T}}\left(\boldsymbol{PP}^{\mathrm{T}}\right)^{-1} = \boldsymbol{E}$.

$$\boldsymbol{y}_{j+1} = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{u}_j - \boldsymbol{PQL}(\boldsymbol{d}-\boldsymbol{r}) + \boldsymbol{d} \quad \text{taken from (A.21)}$$

supplement $\boldsymbol{E} = \boldsymbol{P}^{+}\boldsymbol{P} + \boldsymbol{E} - \boldsymbol{P}^{+}\boldsymbol{P}$

supplement $\boldsymbol{0} = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{+}\boldsymbol{d} - \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{+}\boldsymbol{d}$

$$\boldsymbol{y}_{j+1} = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\left(\boldsymbol{P}^{+}\boldsymbol{P}+\boldsymbol{E}-\boldsymbol{P}^{+}\boldsymbol{P}\right)\boldsymbol{u}_j - \boldsymbol{PQL}(\boldsymbol{d}-\boldsymbol{r}) + \boldsymbol{d} + \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{+}\boldsymbol{d} - \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{+}\boldsymbol{d}$$

$$\boldsymbol{y}_{j+1} = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\left(\boldsymbol{P}^{+}\boldsymbol{P}+\boldsymbol{E}-\boldsymbol{P}^{+}\boldsymbol{P}\right)\boldsymbol{u}_j - \boldsymbol{PQL}(\boldsymbol{d}-\boldsymbol{r}) + \boldsymbol{d} + \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{+}\boldsymbol{d} - \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{+}\boldsymbol{d}$$

reorganize reorganize

$$\boldsymbol{y}_{j+1} = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{+}\left(\boldsymbol{Pu}_j+\boldsymbol{d}\right) + \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\left(\boldsymbol{E}-\boldsymbol{P}^{+}\boldsymbol{P}\right)\boldsymbol{u}_j - \boldsymbol{PQL}(\boldsymbol{d}-\boldsymbol{r}) + \boldsymbol{d} - \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{+}\boldsymbol{d}$$

$$\boldsymbol{y}_{j+1} = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{+}\left(\boldsymbol{Pu}_j+\boldsymbol{d}\right) + \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\left(\boldsymbol{E}-\boldsymbol{P}^{+}\boldsymbol{P}\right)\boldsymbol{u}_j - \boldsymbol{PQL}(\boldsymbol{d}-\boldsymbol{r}) + \boldsymbol{d} - \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{+}\boldsymbol{d}$$

substitute (3.8) $\boldsymbol{y}_j = \boldsymbol{Pu}_j + \boldsymbol{d}$

$$\boldsymbol{y}_{j+1} = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{+}\boldsymbol{y}_j + \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\left(\boldsymbol{E}-\boldsymbol{P}^{+}\boldsymbol{P}\right)\boldsymbol{u}_j - \boldsymbol{PQL}(\boldsymbol{d}-\boldsymbol{r}) + \boldsymbol{d} - \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{+}\boldsymbol{d}$$

$$\boldsymbol{y}_{j+1} = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{+}\boldsymbol{y}_j + \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\left(\boldsymbol{E}-\boldsymbol{P}^{+}\boldsymbol{P}\right)\boldsymbol{u}_j - \boldsymbol{PQL}(\boldsymbol{d}-\boldsymbol{r}) + \boldsymbol{d} - \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{+}\boldsymbol{d}$$

reorganize reorganize

$$\begin{aligned}
\boldsymbol{y}_{j+1} &= \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{+}\boldsymbol{y}_j + \boldsymbol{PQ}\left(\boldsymbol{E}-\boldsymbol{P}^{+}\boldsymbol{P}\right)\boldsymbol{u}_j - \boldsymbol{PQL}\underbrace{\boldsymbol{P}\left(\boldsymbol{E}-\boldsymbol{P}^{+}\boldsymbol{P}\right)}_{\boldsymbol{0}}\boldsymbol{u}_j\\
&\quad - \boldsymbol{PQL}(\boldsymbol{d}-\boldsymbol{r}) + \boldsymbol{d} - \boldsymbol{PQP}^{+}\boldsymbol{d} + \boldsymbol{PQL}\underbrace{\boldsymbol{PP}^{+}}_{\boldsymbol{E}}\boldsymbol{d}
\end{aligned}$$

$$\boldsymbol{y}_{j+1} = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{+}\boldsymbol{y}_j + \boldsymbol{PQ}\left(\boldsymbol{E}-\boldsymbol{P}^{+}\boldsymbol{P}\right)\boldsymbol{u}_j - \boldsymbol{PQL}(\boldsymbol{d}-\boldsymbol{r}) + \boldsymbol{d} - \boldsymbol{PQP}^{+}\boldsymbol{d} + \boldsymbol{PQLd}$$

cancel reorganize

$$\boldsymbol{y}_{j+1} = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{+}\boldsymbol{y}_j + \boldsymbol{PQ}\left(\boldsymbol{E}-\boldsymbol{P}^{+}\boldsymbol{P}\right)\boldsymbol{u}_j + \boldsymbol{PQLr} + \left(\boldsymbol{E}-\boldsymbol{PQP}^{+}\right)\boldsymbol{d}$$

$$\boldsymbol{y}_{j+1} = \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\boldsymbol{P}^{+}\boldsymbol{y}_j + \boldsymbol{PQ}\left(\boldsymbol{E}-\boldsymbol{P}^{+}\boldsymbol{P}\right)\boldsymbol{u}_j + \boldsymbol{PQLr} + \left(\boldsymbol{E}-\boldsymbol{PQP}^{+}\right)\boldsymbol{d} \tag{A.29}$$

$$
\begin{aligned}
e_{j+1} &= y_{j+1} - r \\
&\quad \text{substitute (A.29)} \\
&\quad \text{supplement } 0 = PQ\,(E-LP)\,P^{\dagger}r - PQ\,(E-LP)\,P^{\dagger}r \\
&= PQ\,(E-LP)\,P^{\dagger}y_j + PQ\left(E-P^{\dagger}P\right)u_j + PQLr + \left(E-PQP^{\dagger}\right)d - r \\
&\quad + PQ\,(E-LP)\,P^{\dagger}r - PQ\,(E-LP)\,P^{\dagger}r \\
&= PQ\,(E-LP)\,P^{\dagger}y_j + PQ\left(E-P^{\dagger}P\right)u_j + PQLr + \left(E-PQP^{\dagger}\right)d - r \\
&\quad + PQ\,(E-LP)\,P^{\dagger}r - PQ\,(E-LP)\,P^{\dagger}r \\
&\quad \text{reoranize reorganize} \\
&= PQ\,(E-LP)\,P^{\dagger}\left(y_j - r\right) + PQ\left(E-P^{\dagger}P\right)u_j + PQLr + \left(E-PQP^{\dagger}\right)d - r \\
&\quad + PQP^{\dagger}r - PQL\underbrace{PP^{\dagger}}_{E}r \\
&= PQ\,(E-LP)\,P^{\dagger}\left(y_j - r\right) + PQ\left(E-P^{\dagger}P\right)u_j + PQLr + \left(E-PQP^{\dagger}\right)d - r \\
&\quad + PQP^{\dagger}r - PQLr \\
&\quad \text{cancel reorganize} \\
&= PQ\,(E-LP)\,P^{\dagger}\left(y_j - r\right) + PQ\left(E-P^{\dagger}P\right)u_j + \left(E-PQP^{\dagger}\right)(d-r) \\
&= PQ\,(E-LP)\,P^{\dagger}\left(y_j - r\right) + PQ\left(E-P^{\dagger}P\right)u_j + \left(E-PQP^{\dagger}\right)(d-r)
\end{aligned}
\tag{A.30}
$$

If $Q = E$

$$
\begin{aligned}
e_{j+1} &= P\,(E-LP)\,P^{\dagger}e_j + P\,(E-LP)\left(E-P^{\dagger}P\right)u_j + \left(E-PP^{\dagger}\right)(d-r) \\
e_{j+1} &= (E-PL)\,PP^{\dagger}e_j + (E-PL)\,P\left(E-P^{\dagger}P\right)u_j + \left(E-PP^{\dagger}\right)(d-r) \\
e_{j+1} &= (E-PL)\,PP^{\dagger}e_j + (E-PL)\left(E-PP^{\dagger}\right)Pu_j + \left(E-PP^{\dagger}\right)(d-r) \\
e_{j+1} &= (E-PL)\,e_j
\end{aligned}
\tag{A.31}
$$

Therefore, it is demonstrated that the error vaniches to zero as the Q-Filter is identity.

As it is not the error convergence also depends on the input convergence.

$$
\begin{aligned}
e_{j+1} - e_{\infty} &= PQ\,(E-LP)\,P^{\dagger}e_j + PQ\left(E-P^{\dagger}P\right)u_j + \left(E-PQP^{\dagger}\right)(d-r) \\
&\qquad - PQ\,(E-LP)\,P^{\dagger}e_{\infty} - PQ\left(E-P^{\dagger}P\right)u_{\infty} - \left(E-PQP^{\dagger}\right)(d-r) \\
e_{j+1} - e_{\infty} &= PQ\,(E-LP)\,P^{\dagger}\left(e_j - e_{\infty}\right) + PQ\left(E-P^{\dagger}P\right)\left(u_j - u_{\infty}\right)
\end{aligned}
\tag{A.32}
$$

Approach 1: try to upper bound the fraction:

$$\begin{aligned}
\boldsymbol{u}_{j+1} &= \boldsymbol{Q}(\boldsymbol{E}-\boldsymbol{LP})\,\boldsymbol{u}_j - \boldsymbol{QL}(\boldsymbol{d}-\boldsymbol{r}) \\
\boldsymbol{u}_{j+1}-\boldsymbol{u}_\infty &= \boldsymbol{Q}(\boldsymbol{E}-\boldsymbol{LP})\left(\boldsymbol{u}_j-\boldsymbol{u}_\infty\right) \\
\boldsymbol{e}_{j+1}-\boldsymbol{e}_\infty &= \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\,\boldsymbol{P}^{+}\left(\boldsymbol{e}_j-\boldsymbol{e}_\infty\right) + \boldsymbol{PQ}\left(\boldsymbol{E}-\boldsymbol{P}^{+}\boldsymbol{P}\right)\left(\boldsymbol{u}_j-\boldsymbol{u}_\infty\right) \\
\left|\left|\boldsymbol{e}_{j+1}-\boldsymbol{e}_\infty\right|\right| &= \left|\left|\boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\,\boldsymbol{P}^{+}\left(\boldsymbol{e}_j-\boldsymbol{e}_\infty\right) + \boldsymbol{PQ}\left(\boldsymbol{E}-\boldsymbol{P}^{+}\boldsymbol{P}\right)\left(\boldsymbol{u}_j-\boldsymbol{u}_\infty\right)\right|\right| \\
&\quad \text{tirangular inequality} \\
\left|\left|\boldsymbol{e}_{j+1}-\boldsymbol{e}_\infty\right|\right| &\leq \left|\left|\boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\,\boldsymbol{P}^{+}\left(\boldsymbol{e}_j-\boldsymbol{e}_\infty\right)\right|\right| + \left|\left|\boldsymbol{PQ}\left(\boldsymbol{E}-\boldsymbol{P}^{+}\boldsymbol{P}\right)\left(\boldsymbol{u}_j-\boldsymbol{u}_\infty\right)\right|\right| \\
&\quad \text{submltiplicativity} \\
\left|\left|\boldsymbol{e}_{j+1}-\boldsymbol{e}_\infty\right|\right| &\leq \left|\left|\boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\,\boldsymbol{P}^{+}\right|\right| \left|\left|\left(\boldsymbol{e}_j-\boldsymbol{e}_\infty\right)\right|\right| + \left|\left|\boldsymbol{PQ}\left(\boldsymbol{E}-\boldsymbol{P}^{+}\boldsymbol{P}\right)\right|\right| \left|\left|\left(\boldsymbol{u}_j-\boldsymbol{u}_\infty\right)\right|\right| \\
\left|\left|\boldsymbol{e}_{j+1}-\boldsymbol{e}_\infty\right|\right| &\leq \left(\left|\left|\boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\,\boldsymbol{P}^{+}\right|\right| + \left|\left|\boldsymbol{PQ}\left(\boldsymbol{E}-\boldsymbol{P}^{+}\boldsymbol{P}\right)\right|\right| \frac{\left|\left|\left(\boldsymbol{u}_j-\boldsymbol{u}_\infty\right)\right|\right|}{\left|\left|\left(\boldsymbol{e}_j-\boldsymbol{e}_\infty\right)\right|\right|}\right) \left|\left|\left(\boldsymbol{e}_j-\boldsymbol{e}_\infty\right)\right|\right| \\
\left|\left|\boldsymbol{e}_{j+1}-\boldsymbol{e}_\infty\right|\right| &\leq \left(\left|\left|\boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\,\boldsymbol{P}^{+}\right|\right| + \frac{\left|\left|\boldsymbol{PQ}\left(\boldsymbol{E}-\boldsymbol{P}^{+}\boldsymbol{P}\right)\left(\boldsymbol{u}_j-\boldsymbol{u}_\infty\right)\right|\right|}{\left|\left|\left(\boldsymbol{e}_j-\boldsymbol{e}_\infty\right)\right|\right|}\right) \left|\left|\left(\boldsymbol{e}_j-\boldsymbol{e}_\infty\right)\right|\right| \\
\left|\left|\boldsymbol{e}_{j+1}-\boldsymbol{e}_\infty\right|\right| &= \left(\left|\left|\boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{LP})\,\boldsymbol{P}^{+}\right|\right| + \left|\left|\boldsymbol{PQ}\left(\boldsymbol{E}-\boldsymbol{P}^{+}\boldsymbol{P}\right)\right|\right| \underbrace{\frac{\left|\left|\left(\boldsymbol{u}_j-\boldsymbol{u}_\infty\right)\right|\right|}{\left|\left|\left(\boldsymbol{e}_j-\boldsymbol{e}_\infty\right)\right|\right|}}_{\text{find a upper limit}}\right) \left|\left|\left(\boldsymbol{e}_j-\boldsymbol{e}_\infty\right)\right|\right|
\end{aligned} \tag{A.33}$$

Approach 2: look at the composition

$$\begin{aligned}
\begin{bmatrix} \boldsymbol{u}_{j+1}-\boldsymbol{u}_\infty \\ \boldsymbol{e}_{j+1}-\boldsymbol{e}_\infty \end{bmatrix} &= \begin{bmatrix} \boldsymbol{Q}(\boldsymbol{E}-\boldsymbol{LP}) & \boldsymbol{0} \\ \boldsymbol{PQ}\left(\boldsymbol{E}-\boldsymbol{P}^{+}\boldsymbol{P}\right) & \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{PL})\,\boldsymbol{P}^{+} \end{bmatrix} \begin{bmatrix} \boldsymbol{u}_j-\boldsymbol{u}_\infty \\ \boldsymbol{e}_j-\boldsymbol{e}_\infty \end{bmatrix} \\
\left|\left|\begin{matrix} \boldsymbol{u}_{j+1}-\boldsymbol{u}_\infty \\ \boldsymbol{e}_{j+1}-\boldsymbol{e}_\infty \end{matrix}\right|\right| &\leq \left|\left|\begin{matrix} \boldsymbol{Q}(\boldsymbol{E}-\boldsymbol{LP}) & \boldsymbol{0} \\ \boldsymbol{PQ}\left(\boldsymbol{E}-\boldsymbol{P}^{+}\boldsymbol{P}\right) & \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{PL})\,\boldsymbol{P}^{+} \end{matrix}\right|\right| \left|\left|\begin{matrix} \boldsymbol{u}_j-\boldsymbol{u}_\infty \\ \boldsymbol{e}_j-\boldsymbol{e}_\infty \end{matrix}\right|\right|
\end{aligned} \tag{A.34}$$

Idee:
Definiere eine Norm:

$$\left|\left|\begin{matrix} \boldsymbol{u}_j-\boldsymbol{u}_\infty \\ \boldsymbol{e}_j-\boldsymbol{e}_\infty \end{matrix}\right|\right| = \lim_{\epsilon\to 0}\left(\epsilon^2 \left|\left|\boldsymbol{u}_j-\boldsymbol{u}_\infty\right|\right| + \left|\left|\boldsymbol{e}_j-\boldsymbol{e}_\infty\right|\right|\right) \tag{A.35}$$

Das wäre je equivalent zu sowas hier:

$$\begin{aligned}
\begin{bmatrix} \epsilon\left(\boldsymbol{u}_{j+1}-\boldsymbol{u}_\infty\right) \\ \boldsymbol{e}_{j+1}-\boldsymbol{e}_\infty \end{bmatrix} &= \begin{bmatrix} \boldsymbol{Q}(\boldsymbol{E}-\boldsymbol{LP}) & \boldsymbol{0} \\ \epsilon^{-1}\boldsymbol{PQ}\left(\boldsymbol{E}-\boldsymbol{P}^{+}\boldsymbol{P}\right) & \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{PL})\,\boldsymbol{P}^{+} \end{bmatrix} \begin{bmatrix} \epsilon\left(\boldsymbol{u}_j-\boldsymbol{u}_\infty\right) \\ \boldsymbol{e}_j-\boldsymbol{e}_\infty \end{bmatrix} \\
\left|\left|\begin{matrix} \epsilon\left(\boldsymbol{u}_{j+1}-\boldsymbol{u}_\infty\right) \\ \boldsymbol{e}_{j+1}-\boldsymbol{e}_\infty \end{matrix}\right|\right| &\leq \left|\left|\begin{matrix} \boldsymbol{Q}(\boldsymbol{E}-\boldsymbol{LP}) & \boldsymbol{0} \\ \epsilon^{-1}\boldsymbol{PQ}\left(\boldsymbol{E}-\boldsymbol{P}^{+}\boldsymbol{P}\right) & \boldsymbol{PQ}(\boldsymbol{E}-\boldsymbol{PL})\,\boldsymbol{P}^{+} \end{matrix}\right|\right| \left|\left|\begin{matrix} \epsilon\left(\boldsymbol{u}_j-\boldsymbol{u}_\infty\right) \\ \boldsymbol{e}_j-\boldsymbol{e}_\infty \end{matrix}\right|\right|
\end{aligned} \tag{A.36}$$

Funktioniert nicht. Matrixnorm wird explodieren.

A.6.5 Overactuation input evolution

$$
\begin{aligned}
\boldsymbol{u}_{j+1} &= \boldsymbol{Q}\left(\boldsymbol{u}_j - \boldsymbol{L}\left(\boldsymbol{y}_j - \boldsymbol{r}\right)\right) \\
&= \boldsymbol{Q}\left(\boldsymbol{u}_j - \boldsymbol{L}\left(\boldsymbol{P}\boldsymbol{u}_j + \boldsymbol{d} - \boldsymbol{r}\right)\right) \\
&= \boldsymbol{Q}\left(\boldsymbol{u}_j - \boldsymbol{L}\boldsymbol{P}\boldsymbol{u} - \boldsymbol{L}(\boldsymbol{d} - \boldsymbol{r})\right) \\
\boldsymbol{u}_\infty &= \boldsymbol{Q}(\boldsymbol{E} - \boldsymbol{L}\boldsymbol{P})\,\boldsymbol{u}_\infty - \boldsymbol{Q}\boldsymbol{L}(\boldsymbol{d} - \boldsymbol{r}) \\
&= -(\boldsymbol{E} - \boldsymbol{Q}(\boldsymbol{E} - \boldsymbol{L}\boldsymbol{P}))^{-1}\boldsymbol{Q}\boldsymbol{L}(\boldsymbol{d} - \boldsymbol{r})
\end{aligned}
$$

If Q = E inversion not possible.

$$
\begin{aligned}
\boldsymbol{u}_{j+1} &= \boldsymbol{u}_j - \boldsymbol{L}\boldsymbol{P}\boldsymbol{u}_j - \boldsymbol{L}(\boldsymbol{d} - \boldsymbol{r}) \\
\boldsymbol{L}\boldsymbol{P}\boldsymbol{u}_\infty &= \boldsymbol{L}(\boldsymbol{d} - \boldsymbol{r}) \\
\boldsymbol{P}\boldsymbol{u}_\infty &= \boldsymbol{d} - \boldsymbol{r} \\
\boldsymbol{P}^{\dagger}\boldsymbol{P}\boldsymbol{u}_\infty &= \boldsymbol{P}^{\dagger}\boldsymbol{P}(\boldsymbol{d} - \boldsymbol{r}) \\
\boldsymbol{u}_\infty &= (\boldsymbol{E} - \boldsymbol{L}\boldsymbol{P})^\infty \boldsymbol{u}_0 - \sum_{i=0}^{\infty} (\boldsymbol{E} - \boldsymbol{L}\boldsymbol{P})^i \boldsymbol{L}(\boldsymbol{d} - \boldsymbol{r})
\end{aligned}
$$

Neumann Reihe (Wikipedia). Kovergiert nicht.

Aber, man kann einen Teil aus der Matrix rausnehmen, die nur die konvergiert.

Ehm$\boldsymbol{E} = B_{\boldsymbol{L}}^{\perp} + B_{\boldsymbol{L}}^{\top}$

$$
\boldsymbol{u}_\infty = (\boldsymbol{E} - \boldsymbol{L}\boldsymbol{P})^\infty \boldsymbol{u}_0 - \sum_{i=0}^{\infty} \left[\left(\boldsymbol{E} - \boldsymbol{L}\boldsymbol{L}^{\ddagger} + \boldsymbol{L}\boldsymbol{L}^{\ddagger}\right)(\boldsymbol{E} - \boldsymbol{L}\boldsymbol{P})\left(\boldsymbol{E} - \boldsymbol{L}\boldsymbol{L}^{\ddagger} + \boldsymbol{L}\boldsymbol{L}^{\ddagger}\right)\right]^i \boldsymbol{L}(\boldsymbol{d} - \boldsymbol{r})
$$

Das hier ist hoch plausiebel aber das ganze aber noch zu zeigen.

$$
\begin{aligned}
&= (\boldsymbol{E} - \boldsymbol{L}\boldsymbol{P})^\infty \boldsymbol{u}_0 - \sum_{i=1}^{\infty} \left[\left(\boldsymbol{E} - \boldsymbol{L}\boldsymbol{L}^{\ddagger}\right)(\boldsymbol{E} - \boldsymbol{L}\boldsymbol{P})\left(\boldsymbol{E} - \boldsymbol{L}\boldsymbol{L}^{\ddagger}\right)\right]^i \boldsymbol{L}(\boldsymbol{d} - \boldsymbol{r}) \\
&\quad - \sum_{i=0}^{\infty} \left[\boldsymbol{L}\boldsymbol{L}^{\ddagger}(\boldsymbol{E} - \boldsymbol{L}\boldsymbol{P})\boldsymbol{L}\boldsymbol{L}^{\ddagger}\right]^i \boldsymbol{L}(\boldsymbol{d} - \boldsymbol{r}) \\
&= (\boldsymbol{E} - \boldsymbol{L}\boldsymbol{P})^\infty \boldsymbol{u}_0 - \sum_{i=1}^{\infty} \left[\left(\boldsymbol{E} - \boldsymbol{L}\boldsymbol{L}^{\ddagger}\right)\left(\boldsymbol{E} - \boldsymbol{L}\boldsymbol{L}^{\ddagger}\right)\right]^i \boldsymbol{L}(\boldsymbol{d} - \boldsymbol{r}) \\
&\quad - \sum_{i=0}^{\infty} \left[\boldsymbol{L}\left(\boldsymbol{E} - \boldsymbol{L}^{\ddagger}\boldsymbol{L}\boldsymbol{P}\boldsymbol{L}\right)\boldsymbol{L}^{\ddagger}\right]^i \boldsymbol{L}(\boldsymbol{d} - \boldsymbol{r}) \\
&= (\boldsymbol{E} - \boldsymbol{L}\boldsymbol{P})^\infty \boldsymbol{u}_0 - \sum_{i=1}^{\infty} \left[\left(\boldsymbol{E} - \boldsymbol{L}\boldsymbol{L}^{\ddagger}\right)\left(\boldsymbol{E} - \boldsymbol{L}\boldsymbol{L}^{\ddagger}\right)\right]^i \boldsymbol{L}(\boldsymbol{d} - \boldsymbol{r}) \\
&\quad - \sum_{i=0}^{\infty} \boldsymbol{L}\left[\left(\boldsymbol{E} - \boldsymbol{L}^{\ddagger}\boldsymbol{L}\boldsymbol{P}\boldsymbol{L}\right)\right]^i \boldsymbol{L}^{\ddagger}\boldsymbol{L}(\boldsymbol{d} - \boldsymbol{r}) \\
&= (\boldsymbol{E} - \boldsymbol{L}\boldsymbol{P})^\infty \boldsymbol{u}_0 - \sum_{i=1}^{\infty} \left[\left(\boldsymbol{E} - \boldsymbol{L}\boldsymbol{L}^{\ddagger}\right)\left(\boldsymbol{E} - \boldsymbol{L}\boldsymbol{L}^{\ddagger}\right)\right]^i \boldsymbol{L}(\boldsymbol{d} - \boldsymbol{r}) \\
&\quad - \sum_{i=0}^{\infty} \boldsymbol{L}[(\boldsymbol{E} - \boldsymbol{P}\boldsymbol{L})]^i (\boldsymbol{d} - \boldsymbol{r})
\end{aligned}
$$

Nehmen wir erstmal an, dass diese Neumannreihe konvergiert.

$$
\begin{aligned}
&= (\boldsymbol{E} - \boldsymbol{L}\boldsymbol{P})^\infty \boldsymbol{u}_0 \\
&\quad - \boldsymbol{L}(\boldsymbol{E} - (\boldsymbol{E} - \boldsymbol{P}\boldsymbol{L}))^{-1}(\boldsymbol{d} - \boldsymbol{r}) \\
&= \left(\boldsymbol{E} - \boldsymbol{L}\boldsymbol{L}^{\ddagger}\right)\boldsymbol{u}_0 - \boldsymbol{L}(\boldsymbol{E} - (\boldsymbol{E} - \boldsymbol{P}\boldsymbol{L}))^{-1}(\boldsymbol{d} - \boldsymbol{r}) \\
&= \left(\boldsymbol{E} - \boldsymbol{L}\boldsymbol{L}^{\ddagger}\right)\boldsymbol{u}_0 - \boldsymbol{L}(\boldsymbol{P}\boldsymbol{L})^{-1}(\boldsymbol{d} - \boldsymbol{r})
\end{aligned}
$$

Based on this the final tracking error.

$$
\begin{aligned}
\boldsymbol{e}_\infty &= \boldsymbol{P}\left(\boldsymbol{E}-\boldsymbol{L}\boldsymbol{P}\right)^\infty \boldsymbol{u}_0 - \boldsymbol{P}\boldsymbol{L}(\boldsymbol{P}\boldsymbol{L})^{-1}\left(\boldsymbol{d}-\boldsymbol{r}\right)+\boldsymbol{d}-\boldsymbol{r}\\
&= \boldsymbol{P}\left(\boldsymbol{E}-\boldsymbol{L}\boldsymbol{P}\right)^\infty \boldsymbol{u}_0\\
&= \left(\boldsymbol{E}-\boldsymbol{P}\boldsymbol{L}\right)^\infty \boldsymbol{P}\boldsymbol{u}_0\\
&= \boldsymbol{0}
\end{aligned}
$$

Perfect tracking.

A.7 Iterative Learning Control with robust constraint satisfaction using a Markov Model

In this section of the appendix a method is presented that was meant to push the limits of ILC further in the field of robustness. The goal that is faced is to reach perfect tracking proximal to the constraints using learning. The methods as the have been developed by Meindl [55] rely on the nominal model. As well a the methods used in this text. In the ilc framework it is considered that a given input will lead to the same output. Only if the input is changed the output becomes uncertatain. Therefore, the profound was of choosing a safety margin is unsatisfactory as it leads to a margin as the algorithm converged. The following work piece attempts to overcome this limitation and reach perfect tracking while assuring a sufficient safety margin throughout learning.

The model selected is a Markov model. It has be chosen as it is a linear system and assures positive definiteness. This is worthy in two ways. The region that is considered unsafe is easy to compute, and the optimization problem only introduces a non linearity that relies on the sign of the input and therefore can be solved efficiently using sequential quadratic programming. A counter argument is the missing physical motivation. The development was pushed to a simulation that has shown the desired behavior, but it appeared that the uncertainty quantification is not suitable and requires a lot of hacking to set it up to work. Furthermore, the convergence is very slow. This will lead to no tight constraint satisfaction in real environments where noise occurs.

So the following section presents a promising idea but not a suitable solution and is not fully developed. Non the less the thoughts may help. This section starts with a description of the model. The characterizing the resulting linear system. And some further discussion about more proper uncertainty quantification methods. Finally, a simulation study is done that shows that in simulation without disturbance the method works fine.

A.7.1 Motivation

The main advantage of Iterative Learning control is that it enables exact tracking also in the light of model uncertainty. Non the less in the process of convergence limits can not be guaranteed. The approaches known always rely on the system model used for design and do not take model uncertainty into account [55]. In most cases the model uncertainty can be quantified.

In an Iterative Learning Control (ILC) update a new trajectory is generated that is predicted on the basis of a known old trajectory and an update that is predicted using a linear model that e. g.is derived by linearization around the equilibrium. Therefore, some formalism are introduced.

$$\boldsymbol{x}_{j+1}(k+1) = \boldsymbol{f}\left(\boldsymbol{x}_{j+1}(k), \boldsymbol{u}_{j+1}(k)\right)$$

The Norm Optimal Iterative Learning Control (NO-ILC) design pattern uses a state update prediction, that approximates the system to be linear.

$$\begin{aligned}
\boldsymbol{x}_{j+1}(k+1) &= && \text{(A.37)}\\
\Delta\boldsymbol{x}_{j+1}(k+1) + \boldsymbol{x}_j(k+1) &= \boldsymbol{f}\left(\boldsymbol{x}_{j+1}(k), \boldsymbol{u}_{j+1}(k)\right) && \text{(A.38)}\\
&= \boldsymbol{f}\left(\boldsymbol{x}_j(k) + \Delta\boldsymbol{x}_{j+1}(k), \boldsymbol{u}_j + \Delta\boldsymbol{u}_{j+1}\right) && \text{(A.39)}\\
&\approx \underbrace{\boldsymbol{f}\left(\boldsymbol{x}_j(k), \boldsymbol{u}_j(k)\right)}_{\boldsymbol{x}_j(k+1)} && \text{(A.40)}\\
&\quad + \underbrace{\left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right|_{\boldsymbol{x},\boldsymbol{u}=\boldsymbol{x}_j(k),\boldsymbol{u}_j(k)}}_{\approx \boldsymbol{A}} \Delta\boldsymbol{x}_{j+1}(k) + \underbrace{\left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{u}}\right|_{\boldsymbol{x},\boldsymbol{u}=\boldsymbol{x}_j(k),\boldsymbol{u}_j(k)}}_{\approx \boldsymbol{B}} \Delta\boldsymbol{u}_{j+1}(k) && \text{(A.41)}\\
\Delta\boldsymbol{x}_{j+1}(k+1) &\approx \boldsymbol{A}\Delta\boldsymbol{x}_{j+1}(k) + \boldsymbol{B}\Delta\boldsymbol{u}_{j+1}(k) && \text{(A.42)}
\end{aligned}$$

Therefore, the ILC update rule relies on the following one step ahead predictor in the trial domain.

$$\begin{aligned}
\hat{\boldsymbol{x}}_{j+1|j}(k) &= \Delta\hat{\boldsymbol{x}}_{j+1|j}(k) + \boldsymbol{x}_j(k) && \text{(A.43)}\\
\Delta\hat{\boldsymbol{x}}_{j+1|j}(k+1) &= \boldsymbol{A}\Delta\hat{\boldsymbol{x}}_{j+1|j}(k) + \boldsymbol{B}\Delta\boldsymbol{u}_{j+1}(k) && \text{(A.44)}\\
\Delta\hat{\boldsymbol{x}}_{j+1|j}(0) &= 0_n && \text{(A.45)}
\end{aligned}$$

A.7.2 Markov Model

The one step ahead predictor itself forms a Markov Model considering a state update noise.

$$\begin{aligned}
\Delta\boldsymbol{x}(k+1) &= \boldsymbol{A}\Delta\boldsymbol{x}(k) + \boldsymbol{B}\Delta\boldsymbol{u}(k) + \boldsymbol{v}(k)\\
\Delta\boldsymbol{y}(k) &= \boldsymbol{C}\Delta\boldsymbol{x}(k)
\end{aligned} \qquad \text{(A.46)}$$

The state noise is quantified using a heuristic $\boldsymbol{v}(k) \propto \mathcal{N}\left(0_n, \boldsymbol{B}\boldsymbol{P}_{\boldsymbol{u}}\boldsymbol{B}^{\mathrm{T}}\,|\Delta\boldsymbol{u}(k)|\right)$. This is motivated by the fact that change in the input variable only affects consecutive samples.

To predict the consecutive trial the one step ahead predictor is proposed.

$$\begin{aligned}
\Delta\boldsymbol{x}_{j+1|j}(k+1) &= \boldsymbol{A}\Delta\boldsymbol{x}_{j+1|j}(k) + \boldsymbol{B}\Delta\boldsymbol{u}_{j+1}(k) + \boldsymbol{v}(k)\\
\Delta\boldsymbol{y}_{j+1|j}(k) &= \boldsymbol{C}\Delta\boldsymbol{x}_{j+1|j}(k)
\end{aligned} \qquad \text{(A.47)}$$

Using that information the covariance can be computed.

$$
\begin{aligned}
\boldsymbol{P}_{\boldsymbol{x},j+1|j}(k+1) &= \boldsymbol{A}\boldsymbol{P}_{\boldsymbol{x},j+1|j}(k)\boldsymbol{A}^{\mathrm{T}} + \boldsymbol{B}\boldsymbol{P}_{\boldsymbol{u},j+1}(k)\boldsymbol{B}^{\mathrm{T}} \\
\boldsymbol{P}_{\boldsymbol{y},j+1|j}(k) &= \boldsymbol{C}\boldsymbol{P}_{\boldsymbol{x},j+1|j}(k)\boldsymbol{C}^{\mathrm{T}} + \boldsymbol{D}\boldsymbol{P}_{\boldsymbol{u},j+1}(k)\boldsymbol{D}^{\mathrm{T}}
\end{aligned} \tag{A.48}
$$

This simple model itself represents a Linear Time Invariant (LTI) system and can be represented by lifted dynamics with respect to the input $\boldsymbol{P}_{\boldsymbol{u},j+1}(k)$. The model represents a state space system of order $\frac{(n+1)\cdot n}{2}$ since the covariance matrix is symmetric.

$$
\boldsymbol{P}_{\Sigma} = \left[\boldsymbol{P}_{\Sigma}(j-i)\right]^{i,j} \tag{A.49}
$$

The observed next ILC-Trial must follow the distribution $\boldsymbol{y}_{j+1}(k) \propto \mathcal{N}\left(\boldsymbol{y}_{j+1|j}(k), \boldsymbol{P}_{\boldsymbol{y},j+1|j}(k)\right)$.

The system that maps the covariance of the input to the outputs covariance. Therefore, the input's covariance is flattens yielding the input vector.

$$
\mathtt{F}: \mathbb{R}^{m\times n} \to \mathbb{R}^{mn}; \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m} & \dots & a_{mn} \end{bmatrix} \mapsto \begin{bmatrix} a_{11} \\ \vdots \\ a_{m1} \\ a_{12} \\ \vdots \\ a_{nn} \end{bmatrix} \tag{A.50}
$$

The System itself is a LTI system with respect to the input covariance matrix flattened $\mathtt{F}(\boldsymbol{P}_{\boldsymbol{u}}(k))$. To define the markov parameter the indicaotr matrix is defined.

$$
\boldsymbol{I}_{i_1 j_1} = \left[\delta_{i-i_1}\delta_{j-j_1}\right]^{i,j} \in \mathbb{R}^{m\times n} \tag{A.51}
$$

Considering the index map $i_p(\bar{i}) = ((\bar{i}-1) \mod p) + 1$ and $j_p(\bar{i}) = ((\bar{i}-1) \div p) + 1$, the markov parameters are defined.

$$
\boldsymbol{P}_{\Sigma}(k) = \begin{cases} \boldsymbol{0}_{p^2\times m^2} & k<0 \\ \left[\left[\boldsymbol{D}\boldsymbol{I}_{i_m(j)j_m(j)}\boldsymbol{D}^{\mathrm{T}}\right]_{i_p(i),j_p(i)}\right]^{i,j} \in \mathbb{R}^{p^2\times m^2} & k=0 \\ \left[\left[\boldsymbol{C}\boldsymbol{A}^{k-1}\boldsymbol{B}\boldsymbol{I}_{i_m(j)j_m(j)}\left(\boldsymbol{C}\boldsymbol{A}^{k-1}\boldsymbol{B}\right)^{\mathrm{T}}\right]_{i_p(i),j_p(j)}\right]^{i,j} \in \mathbb{R}^{p^2\times m^2} & k>0 \end{cases} \tag{A.52}
$$

Using this we can form the lifted dynamics representation.

$$\boldsymbol{P}_\Sigma = \left[\boldsymbol{P}_\Sigma(i-j)\right]^{i,j} \in \mathbb{R}^{n_\mathrm{b}p^2 \times n_\mathrm{b}m^2} \tag{A.53}$$

The representation delivers the flattened covariance matrices that can be converted to the output covariance using the inverse flatten operator $\mathtt{F}^{-1}$.

Using this covariance estimation one can compute a confidence set.

$$\mathcal{D}(\Delta\boldsymbol{u}) = \left\{\Delta\boldsymbol{y} \in \mathcal{Y}^{n_\mathrm{b}} | \forall k \quad \left(\Delta\boldsymbol{y}(k) - \Delta\boldsymbol{y}_{j+1|j}(k)\right)^\mathrm{T} \boldsymbol{P}_{\boldsymbol{y},j+1|j}(k)\left(\Delta\boldsymbol{y}(k) - \Delta\boldsymbol{y}_{j+1|j}\right) \leq \delta\right\} \tag{A.54}$$

Dynamics of the uncertainty quantification

To analyse the solutions of the dynamics of the uncertainty quantification system (A.48) the eigenvectors and eigenvalues of the linear mapping $L: \mathbb{R}^{n\times n} \to \mathbb{R}^{n\times n}; \boldsymbol{P} \mapsto \boldsymbol{A}\boldsymbol{P}\boldsymbol{A}^\mathrm{T}$ need to be analysed. It can be shown that the eigenvalues of L are uniquely known as those of $\boldsymbol{A}$ are. The properties are shown in the following.

Lemma 1. *Linearity. The mapping L is linear.*

Proof. Let $\lambda \in \mathbb{R}$ and $\boldsymbol{V}_1, \boldsymbol{V}_2 \in \mathbb{R}^{n\times n}$ be arbitrary. Then $L(\boldsymbol{V}_1 + \lambda\boldsymbol{V}_2) = \boldsymbol{A}(v_1 + \lambda v_2)\boldsymbol{A}^\mathrm{T} = \boldsymbol{A}\boldsymbol{V}_1\boldsymbol{A}^\mathrm{T} + \lambda\boldsymbol{A}\boldsymbol{V}_2\boldsymbol{A}^\mathrm{T} = L(\boldsymbol{V}_1) + \lambda L(\boldsymbol{V}_2)$. Therefore L is linear. □

Lemma 2. *Eigenvalues and eigenvectors of L. If (λ_i, v_i) is a generalized eigenvalue eigenvector pair of order m_i of A and (λ_j, v_j) is a generalized eigenvalue eigenvector pair of order m_j of A, then $(\lambda_i\lambda_j, v_i v_j^\mathrm{T})$ is an generalized eigenvalue eigenvector pair of at most the order $m_i + m_j - 1$ of L.*

Proof. The statement is proven by induction. Therefore, first assume that $m_i = m_j = 1$. As (λ_i, v_i) and (λ_j, v_j) are ordinary eigenvalue eigenvector pairs of $\boldsymbol{A}$, also $\lambda_i v_i = \boldsymbol{A}v_i$ hold. Let $(J_{\boldsymbol{A}}, S_{\boldsymbol{A}})$ be the Jordan decomposition of $\boldsymbol{A}$ and the matrix is composed of the generalized eigenvectors of $\boldsymbol{A}$.

$$[v_1 \dots v_n] = S_{\boldsymbol{A}}$$

$$\begin{aligned}\left(L - \lambda_i\lambda_j Id\right)\left(v_i v_j^\mathrm{T}\right) &= \boldsymbol{A}v_i v_j^\mathrm{T}\boldsymbol{A}^\mathrm{T} - \lambda_i\lambda_j v_i v_j^\mathrm{T} \\ &= S_{\boldsymbol{A}}J_{\boldsymbol{A}}S_{\boldsymbol{A}}{}^{-1} v_i v_j^\mathrm{T}\left(S_{\boldsymbol{A}}J_{\boldsymbol{A}}S_{\boldsymbol{A}}{}^{-1}\right)^\mathrm{T} - \lambda_i\lambda_j v_i v_j^\mathrm{T} \\ &= S_{\boldsymbol{A}}J_{\boldsymbol{A}}e_i e_j^\mathrm{T} J_{\boldsymbol{A}}^\mathrm{T} S_{\boldsymbol{A}}^\mathrm{T} - \lambda_i\lambda_j v_i v_j^\mathrm{T} \qquad \text{t'} \\ &= \lambda_i\lambda_j S_{\boldsymbol{A}}e_i e_j^\mathrm{T} S_{\boldsymbol{A}}^\mathrm{T} - \lambda_i\lambda_j v_i v_j^\mathrm{T} \\ &= \lambda_i\lambda_j v_i v_j^\mathrm{T} - \lambda_i\lambda_j v_i v_j^\mathrm{T} = 0\end{aligned} \tag{A.55}$$

Therefore, $(\lambda_i^2, v_i v_i^\mathrm{T})$ is an ordinary eigenvalue eigenvector pair of L.

Assume that $m_i > 1$ and $m_j = 1$ and that $(\lambda_i\lambda_j^{\mathrm{T}}, v_{i-1}v_j^{\mathrm{T}})$ is a generalized eigenvalue eigenvector pair of order $m_i + m_j - 2$ of L.

$$
\begin{aligned}
\left(L-\lambda_i\lambda_j Id\right)^{m_i+m_j-1}\left(v_i v_j^{\mathrm{T}}\right) &= \left(L-\lambda_i\lambda_j Id\right)^{m_i+m_j-2}\left(\boldsymbol{A}v_i v_j^{\mathrm{T}}\boldsymbol{A}^{\mathrm{T}}-\lambda_i\lambda_j v_i v_j^{\mathrm{T}}\right)\\
&= \left(L-\lambda_i\lambda_j Id\right)^{m_i+m_j-2}\left(S_{\boldsymbol{A}}J_{\boldsymbol{A}}S_{\boldsymbol{A}}{}^{-1}v_i v_j^{\mathrm{T}}\left(S_{\boldsymbol{A}}J_{\boldsymbol{A}}S_{\boldsymbol{A}}{}^{-1}\right)^{\mathrm{T}}-\lambda_i\lambda_j v_i v_j^{\mathrm{T}}\right)\\
&= \left(L-\lambda_i\lambda_j Id\right)^{m_i+m_j-2}\left(S_{\boldsymbol{A}}J_{\boldsymbol{A}}e_i e_j^{\mathrm{T}}J_{\boldsymbol{A}}^{\mathrm{T}}S_{\boldsymbol{A}}^{\mathrm{T}}-\lambda_i\lambda_j v_i v_j^{\mathrm{T}}\right)\\
&= \left(L-\lambda_i\lambda_j Id\right)^{m_i+m_j-2}\left(S_{\boldsymbol{A}}\left(\lambda_i e_i+e_{i-1}\right)\lambda_j e_j^{\mathrm{T}}S_{\boldsymbol{A}}^{\mathrm{T}}-\lambda_i\lambda_j v_i v_j^{\mathrm{T}}\right)\\
&= \left(L-\lambda_i\lambda_j Id\right)^{m_i+m_j-2}\left(\lambda_i\lambda_j v_i v_j^{\mathrm{T}}+\lambda_j v_{i-1}v_j^{\mathrm{T}}-\lambda_i\lambda_j v_i v_j^{\mathrm{T}}\right)\\
&= \left(L-\lambda_i\lambda_j Id\right)^{m_i+m_j-2}\left(\lambda_j v_{i-1}v_j^{\mathrm{T}}\right)\\
&= 0
\end{aligned}
\tag{A.56}
$$

Obviously the same argument holds for $m_i = 1$ and $m_j > 1$ and under the assumption that $\left(\lambda_i\lambda_j^{\mathrm{T}}, v_i v_{j-1}^{\mathrm{T}}\right)$ is a generalized eigenvalue eigenvector pair of order $m_i + m_j - 2$ of L.

Finally, assume $m_i > 1$ and $m_j > 1$ and that $\left(\lambda_i\lambda_j^{\mathrm{T}}, v_{i-1}v_j^{\mathrm{T}}\right)$ and $\left(\lambda_i\lambda_j^{\mathrm{T}}, v_i v_{j-1}^{\mathrm{T}}\right)$ is a generalized eigenvalue eigenvector pair of order $m_i + m_j - 2$ of L. Furthermore, assume that $\left(\lambda_i\lambda_j^{\mathrm{T}}, v_{i-1}v_{j-1}^{\mathrm{T}}\right)$ is a generalized eigenvalue eigenvector pair of order $m_{i-1} + m_{j-1} - 3$ of L.

$$
\begin{aligned}
&(L-\lambda_i\lambda_j Id)^{m_i+m_j-1}\left(v_i v_j^{\mathrm{T}}\right)\\
&\quad= \left(L-\lambda_i\lambda_j Id\right)^{m_i+m_j-2}\left(\boldsymbol{A}v_i v_j^{\mathrm{T}}\boldsymbol{A}^{\mathrm{T}}-\lambda_i\lambda_j v_i v_j^{\mathrm{T}}\right)\\
&\quad= \left(L-\lambda_i\lambda_j Id\right)^{m_i+m_j-2}\left(S_{\boldsymbol{A}}J_{\boldsymbol{A}}S_{\boldsymbol{A}}{}^{-1}v_i v_j^{\mathrm{T}}\left(S_{\boldsymbol{A}}J_{\boldsymbol{A}}S_{\boldsymbol{A}}{}^{-1}\right)^{\mathrm{T}}-\lambda_i\lambda_j v_i v_j^{\mathrm{T}}\right)\\
&\quad= \left(L-\lambda_i\lambda_j Id\right)^{m_i+m_j-2}\left(S_{\boldsymbol{A}}J_{\boldsymbol{A}}e_i e_j^{\mathrm{T}}J_{\boldsymbol{A}}^{\mathrm{T}}S_{\boldsymbol{A}}^{\mathrm{T}}-\lambda_i\lambda_j v_i v_j^{\mathrm{T}}\right)\\
&\quad= \left(L-\lambda_i\lambda_j Id\right)^{m_i+m_j-2}\left(S_{\boldsymbol{A}}\left(\lambda_i e_i+e_{i-1}\right)\left(\lambda_j e_j+e_{j-1}\right)^{\mathrm{T}}S_{\boldsymbol{A}}^{\mathrm{T}}-\lambda_i\lambda_j v_i v_j^{\mathrm{T}}\right)\\
&\quad= \left(L-\lambda_i\lambda_j Id\right)^{m_i+m_j-2}\left(\lambda_i\lambda_j v_i v_j^{\mathrm{T}}+\lambda_i v_i v_{j-1}+\lambda_j v_{i-1}v_j^{\mathrm{T}}+v_{i-1}v_{j-1}^{\mathrm{T}}-\lambda_i\lambda_j v_i v_j^{\mathrm{T}}\right)\\
&\quad= \left(L-\lambda_i\lambda_j Id\right)^{m_i+m_j-2}\left(\lambda_i v_i v_{j-1}+\lambda_j v_{i-1}v_j^{\mathrm{T}}+v_{i-1}v_{j-1}^{\mathrm{T}}\right)\\
&\quad= 0
\end{aligned}
\tag{A.57}
$$

This concludes the induction. Therefore, $(\lambda_i\lambda_j, v_i v_j^{\mathrm{T}})$ is a generalized eigenvalue eigenvector pair of at least order $m_i + m_j - 1$ of L. □

Corollary 1. *Eigenvalues of L. The set of all eigenvalues of L is uniquely given by the set of all pairs* $\lambda_i\lambda_j$.

To give an intuition of the system's behavior lets consider the well known standard solutions [56, pp. A92]. The solutions are characterized in Table A.2. One observes different frequencies than in the original system. Furthermore, higher order generalized eigenvectors occur.

$\times$	$k^{m_1-1}e^{a_1k}e^{j\Omega_1jk}$	$k^{m_1-1}e^{a_1k}e^{-j\Omega_1k}$
$k^{m_1-1}e^{a_1k}e^{j\Omega_1jk}$	$k^{2m_1-2}e^{2a_1k}e^{j2\Omega_1k}$	$k^{2m_1-2}e^{2a_1k}$
$k^{m_1-1}e^{a_1k}e^{-j\Omega_1jk}$	$k^{2m_1-2}e^{2a_1k}$	$k^{2m_1-2}e^{2a_1k}e^{-j2\Omega_1k}$
$k^{m_2-1}e^{a_2k}e^{j\Omega_2jk}$	$k^{m_1+m_2-2}e^{(a_1+a_2)k}e^{j(\Omega_1+\Omega_2)k}$	$k^{m_1+m_2-2}e^{(a_1+a_2)k}e^{-j(\Omega_1-\Omega_2)k}$
$k^{m_2-1}e^{a_2k}e^{-j\Omega_2jk}$	$k^{m_1+m_2-2}e^{(a_1+a_2)k}e^{j(\Omega_1-\Omega_2)k}$	$k^{m_1+m_2-2}e^{(a_1+a_2)k}e^{-j(\Omega_1+\Omega_2)k}$

Table A.2. Behavior of characteristic solutions in the covariance evolution.

Remark 10. *For real valued systems the eigenvalues are either real or complex conjugated. For each complex conjugated root pair one observes one pair with twice the frequency and one pair in the aperiodic limit case both twice damped.*

As the stochastic model maintains the positive definiteness of the matrices positive input signals must lead to positive output signals for all times in the Single Input Single Output (SISO) case.

Discussion

- Pros
  - meets requirements
  - efficient implementation
- Cons
  - the model assumes strict knowledge in the quantification (see below)
  - the model can not be certain as it is used for any point of operation.

The uncertainty quantification chosen assumes the introduction of uncertainty if an input is applied. The evolution of the systems uncertainty is determined by the systems dynamics that are assumed to be known exactly. This is a not a proper assumption though, since the system is not linear. Non the less the description is simple and well suited for implementation.

Controlling the uncertainty dynamics

The algorithm requires a stable plant as otherwise the uncertainty grows to infinity for a finite time domain. Undesirably, the uncertainty also decays faster as the plant is stable. This is not very likely though, as trajectory is not stachastic but determistic. Therefore, one may modify the roots of uncertainty model. To do so one has two very straight forward methods. This can be interpreted as a

$$\begin{aligned} \boldsymbol{P}_{\boldsymbol{x},j+1|j}(k+1) &= \boldsymbol{A}\boldsymbol{P}_{\boldsymbol{x},j+1|j}(k)\boldsymbol{A}^{\mathrm{T}} + \boldsymbol{B}\boldsymbol{P}_{u}\boldsymbol{B}^{\mathrm{T}} + \boldsymbol{v} \quad \boldsymbol{v} = \mu\boldsymbol{P}_{\boldsymbol{x},j+1|j}(k) \\ \boldsymbol{P}_{\boldsymbol{y},j+1|j}(k) &= \boldsymbol{C}\boldsymbol{P}_{\boldsymbol{x},j+1|j}(k+1)\boldsymbol{C}^{\mathrm{T}} \end{aligned} \tag{A.58}$$

The real part of all eigenvalues is shifted this makes sense from an ocular point of view.

One could also selectively control the real part of dedicated eigenvalues. EG assume the eigenvector $v_i v_j^{\mathrm{T}}$.

$$\boldsymbol{P}_{\boldsymbol{x},j+1|j}(k+1) = A\boldsymbol{P}_{\boldsymbol{x},j+1|j}(k)A^{\mathrm{T}} + B\boldsymbol{P}_{\boldsymbol{u},j+1|j}(k)B^{T} + \boldsymbol{v} \quad \boldsymbol{v} = \sum_{j=1}^{n}\sum_{i=1}^{n}\mu_{ij}\frac{v_i v_i^{\mathrm{T}}}{v_i^{\mathrm{T}} v_i}\boldsymbol{P}\frac{v_j v_j^{\mathrm{T}}}{v_j^{\mathrm{T}} v_j} \quad \text{uncertainty if the deviation is e}$$
$$\boldsymbol{P}_{\boldsymbol{y},j+1|j}(k) = \boldsymbol{C}\boldsymbol{P}_{\boldsymbol{x},j+1|j}(k)\boldsymbol{C}^{\mathrm{T}} \tag{A.59}$$

This is equivalent to the modification of $\tilde{\boldsymbol{A}} = \boldsymbol{A} + \sum_{i=1}^{n}\sum_{j=1}^{n}\mu_{ij} v_i v_i^{\mathrm{T}}$ as one restricts $\mu_{ij} = \mu_{ji}$.

A reasonable modification is $\mu_{ij} = \mathcal{R}\left(\lambda_j^{-1}\right)$ as it asserts the decay speed to be equal to those of the eigenvalue of the original system. Unfortunately this does not fulfill the aforementioned restriction. This can be overcome using the modification

$$\mu_{ij} = \mathcal{R}\left(\frac{1}{2}\left(\frac{1}{\lambda_i} + \frac{1}{\lambda_j}\right)\right) = \mathcal{R}\left(\frac{\lambda_i + \lambda_j}{2\lambda_i\lambda_j}\right). \tag{A.60}$$

This has the same effect for all μ_{ij}.

```
modify_eigenvalues
```

Taking model uncertainty into account

Discuss how to the thing goes if we quantify the systems uncertainty. Unfortunately the system is represented by a matrix and we need to represent the uncertainty. This is a tensor of 4-th order. The second order statistics of a matrix are

$$\hat{a}_{ij} = \mathbb{E}\left[a_{ij}\right]$$

$$p_{A,ijkl} = \mathbb{E}\left[\left(a_{ij} - \hat{a}_{ij}\right)\left(a_{kl} - \hat{a}_{kl}\right)\right] = \mathbb{E}\left[a_{ij}a_{kl}\right] - \hat{a}_{ij}\hat{a}_{kl}$$

Further the second order statistics of the states are well known. $\hat{x}_i$ and $p_{x,ij}$. Under the assumption of second order statistics we the state update is analysed. Expectation:

$$\mathbb{E}[Ax] = \mathbb{E}[\mathbb{E}[A|x]\,x] = \mathbb{E}[\mathbb{E}[A]\,x] = \mathbb{E}[A]\,\mathbb{E}[x] = \hat{A}\hat{x} \tag{A.61}$$

Variance:

$$
\begin{aligned}
&\mathbb{E}\left[\left(Ax-\hat{A}\hat{x}\right)\left(Ax-\hat{A}\hat{x}\right)^{\mathrm{T}}\right] && \text{(A.62)}\\
=&\mathbb{E}\left[Axx^{\mathrm{T}}A^{\mathrm{T}}\right]-\mathbb{E}\left[Ax\hat{x}^{\mathrm{T}}\hat{A}^{\mathrm{T}}\right]-\mathbb{E}\left[\hat{A}\hat{x}x^{\mathrm{T}}A^{\mathrm{T}}\right]+\mathbb{E}\left[\hat{A}\hat{x}\hat{x}^{\mathrm{T}}\hat{A}^{\mathrm{T}}\right] && \text{(A.63)}\\
=&\mathbb{E}\left[Axx^{\mathrm{T}}A^{\mathrm{T}}\right]-\mathbb{E}\left[Ax\right]\hat{x}^{\mathrm{T}}\hat{A}^{\mathrm{T}}-\hat{A}\hat{x}\mathbb{E}\left[x^{\mathrm{T}}A^{\mathrm{T}}\right]+\hat{A}\hat{x}\hat{x}^{\mathrm{T}}\hat{A}^{\mathrm{T}} && \text{(A.64)}\\
=&\mathbb{E}\left[Axx^{\mathrm{T}}A^{\mathrm{T}}\right]-\hat{A}\hat{x}\hat{x}^{\mathrm{T}}\hat{A}^{\mathrm{T}} && \text{(A.65)}\\
=&\mathbb{E}\left[A\mathbb{E}\left[xx^{\mathrm{T}}|A\right]A^{\mathrm{T}}\right]-\hat{A}\hat{x}\hat{x}^{\mathrm{T}}\hat{A}^{\mathrm{T}} && \text{(A.66)}\\
=&\mathbb{E}\left[A\mathbb{E}\left[xx^{\mathrm{T}}\right]A^{\mathrm{T}}\right]-\hat{A}\hat{x}\hat{x}^{\mathrm{T}}\hat{A}^{\mathrm{T}} && \text{(A.67)}\\
=&\mathbb{E}\left[A\left(P_x+\hat{x}\hat{x}^{\mathrm{T}}\right)A^{\mathrm{T}}\right]-\hat{A}\hat{x}\hat{x}^{\mathrm{T}}\hat{A}^{\mathrm{T}} && \text{(A.68)}\\
&\text{switch to tensor notation} && \text{(A.69)}\\
=&\mathbb{E}\left[a_{ij}\left(p_{x,jk}+\hat{x}_j\hat{x}_k\right)a_{lk}\right]-\hat{a}_{ij}\hat{x}_j\hat{x}_k\hat{a}_{lk} && \text{(A.70)}\\
=&\mathbb{E}\left[a_{ij}a_{lk}\right]\left(p_{x,jk}+\hat{x}_j\hat{x}_k\right)-\hat{a}_{ij}\hat{x}_j\hat{x}_k\hat{a}_{lk} && \text{(A.71)}\\
=&\left(p_{a,ijlk}+\hat{a}_{ij}\hat{a}_{lk}\right)\left(p_{x,jk}+\hat{x}_j\hat{x}_k\right)-\hat{a}_{ij}\hat{x}_j\hat{x}_k\hat{a}_{lk} && \text{(A.72)}\\
=&p_{a,ijlk}\left(p_{x,jk}+\hat{x}_j\hat{x}_k\right)+\hat{a}_{ij}\left(p_{x,jk}+\hat{x}_j\hat{x}_k\right)\hat{a}_{lk}-\hat{a}_{ij}\hat{x}_j\hat{x}_k\hat{a}_{kl} && \text{(A.73)}\\
=&p_{a,ijlk}\left(p_{x,jk}+\hat{x}_j\hat{x}_k\right)+\hat{a}_{ij}p_{x,jk}\hat{a}_{lk} && \text{(A.74)}\\
&\text{assume}\quad p_{a,ijkl}=\bar{p}_{a,ij}\delta_{i-k}\delta_{j-l}\quad\text{diagonal} && \text{(A.75)}\\
=&\bar{p}_{a,ij}\delta_{i-l}\delta_{j-k}\left(p_{x,jk}+\hat{x}_j\hat{x}_k\right)+\hat{a}_{ij}p_{x,jk}\hat{a}_{lk} && \text{(A.76)}\\
=&\underbrace{\delta_{i-l}\bar{p}_{a,ij}\left(p_{x,jj}+\hat{x}_j\hat{x}_j\right)}_{\text{diagonal}}+\hat{a}_{ij}p_{x,jk}\hat{a}_{lk} && \text{(A.77)}\\
&\text{back to matrix} && \text{(A.78)}\\
=&\mathrm{Diag}\left[\bar{P}_a\mathrm{Diag}\left[P_x+\hat{x}\hat{x}^{\mathrm{T}}\right]\right]+\hat{A}P_x\hat{A}^{\mathrm{T}} && \text{(A.79)}
\end{aligned}
$$

Assume that the model is uncertain.

$$x(k+1)=A(k)x(k)+bu(k) \tag{A.80}$$
$$y(k)=c^{\mathrm{T}}x(k) \tag{A.81}$$

Assume that $A(k)\propto\mathcal{N}(\hat{A},\bar{P}_A)$. i.e. the the state matrices error of the state matrix is unform over the state update.

Then the uncertain model and the markov parameter change.

$$
\begin{aligned}
\boldsymbol{P}_{\boldsymbol{x},j+1|j}(k+1)&=\boldsymbol{A}\boldsymbol{P}_{\boldsymbol{x},j+1|j}(k)\boldsymbol{A}^{\mathrm{T}}+\boldsymbol{B}\boldsymbol{P}_u\boldsymbol{B}^{\mathrm{T}}+\mathrm{Diag}\left[\bar{\boldsymbol{P}}_{\boldsymbol{A}}\mathrm{Diag}\left[\boldsymbol{P}_{\boldsymbol{x},j+1|j}+\hat{x}_{j+1|j}\hat{x}_{j+1|j}^{\mathrm{T}}\right]\right]\\
\boldsymbol{P}_{\boldsymbol{y},j+1|j}(k)&=\boldsymbol{C}\boldsymbol{P}_{\boldsymbol{x},j+1|j}(k)\boldsymbol{C}^{\mathrm{T}}
\end{aligned}
\tag{A.82}
$$

One can observe that the problem became nonlinear to the predicted state sequence. The full system:

$$
\begin{aligned}
\Delta \boldsymbol{x}_{j+1|j}(k+1) &= \boldsymbol{A}\Delta \boldsymbol{x}_{j+1|j}(k) + \boldsymbol{B}\Delta \boldsymbol{u}_{j+1}(k) + \boldsymbol{v}(k)\\
\Delta \boldsymbol{y}_{j+1|j}(k) &= \boldsymbol{C}\Delta \boldsymbol{x}_{j+1|j}(k)\\
\boldsymbol{P}_{\boldsymbol{x},j+1|j}(k+1) &= \boldsymbol{A}\boldsymbol{P}_{\boldsymbol{x},j+1|j}(k)\boldsymbol{A}^{\mathrm{T}} + \boldsymbol{B}\boldsymbol{P}_{u}\boldsymbol{B}^{\mathrm{T}} + \mathrm{Diag}\left[\bar{\boldsymbol{P}}_{\boldsymbol{A}}\left(\mathrm{Diag}\left[\boldsymbol{P}_{\boldsymbol{x},j+1|j}\right] + \mathrm{Diag}\left[\Delta\hat{x}_{j+1|j}\right]\Delta\hat{x}_{j+1|j}\right)\right]\\
\boldsymbol{P}_{\boldsymbol{y},j+1|j}(k) &= \boldsymbol{C}\boldsymbol{P}_{\boldsymbol{x},j+1|j}(k)\boldsymbol{C}^{\mathrm{T}}
\end{aligned}
\tag{A.83}
$$

Linearize the sytem to enable markov parameter representation.

$$
\begin{aligned}
\Delta \boldsymbol{x}_{j+1|j}(k+1) &= \boldsymbol{A}\Delta \boldsymbol{x}_{j+1|j}(k) + \boldsymbol{B}\Delta \boldsymbol{u}_{j+1}(k) + \boldsymbol{v}(k)\\
\Delta \boldsymbol{y}_{j+1|j}(k) &= \boldsymbol{C}\Delta \boldsymbol{x}_{j+1|j}(k)\\
\boldsymbol{P}_{\boldsymbol{x},j+1|j}(k+1) &= \boldsymbol{A}\boldsymbol{P}_{\boldsymbol{x},j+1|j}(k)\boldsymbol{A}^{\mathrm{T}} + \boldsymbol{B}\boldsymbol{P}_{u}\boldsymbol{B}^{\mathrm{T}}\\
&\quad + \mathrm{Diag}\left[\bar{\boldsymbol{P}}_{\boldsymbol{A}}\left(\mathrm{Diag}\left[\boldsymbol{P}_{\boldsymbol{x},j+1|j}\right] + \mathrm{Diag}\left[\Delta\hat{x}^{0}_{j+1|j}\right]\left(\Delta\hat{x}_{j+1|j} - \Delta\hat{x}^{0}_{j+1|j}\right)\right)\right]\\
\boldsymbol{P}_{\boldsymbol{y},j+1|j}(k) &= \boldsymbol{C}\boldsymbol{P}_{\boldsymbol{x},j+1|j}(k)\boldsymbol{C}^{\mathrm{T}}
\end{aligned}
\tag{A.84}
$$

A.7.3 Simulation Study

To evaluate the performance the constraint satisfaction is applied to a standard ILC problem. The reference trajectory is given as.

$$
\boldsymbol{r}(k) = \hat{a}\sin\left(2\pi\frac{k}{n_{\mathrm{b}}}\right)\cdot\left(1-\cos\left(\pi\frac{k}{n_{\mathrm{b}}}\right)\right)
$$

The value of $\hat{a}$ is chosen so that the maximum value of the signal is set to $\max \boldsymbol{r} = 2$.

The weights used for tuning $Q_{\mathrm{W}} = 1, S_{\mathrm{W}} = 0.5, R_{\mathrm{W}} = 0.001$.

The output is limited to range $\boldsymbol{y} \in [-2, 1]$, therefore the output trajectory is not feasible.

In Figure A-18 the results are presented in the time domain. The output trajectories are shown for selected iterations. The figure presents the prediction and the confidence interval and the actual observed output. The confidence interval touches the constraint value. Also, the observed output remains in the confidence interval. The confidence interval shrinks as the iterations proceed. Therefore, the algorithm serves its basic requirement that it must be able to narrow up to the constraints very close. In the time interval from the 50th and the 90th sample the algorithm also realizes perfect tracking in the 100th iteration. Thus, the algorithm fullfills the requirements.

For the matter of analysis an regression of the error evolution was done. As underlying function a potential decay with an offset was assumed. The offset was obtained by the final error observed.

Figure A-19 presents the error evolution over in the iteration domain. The error norm value is presented in the top frame. To analyze the convergence behaviour the bottom frame presents the

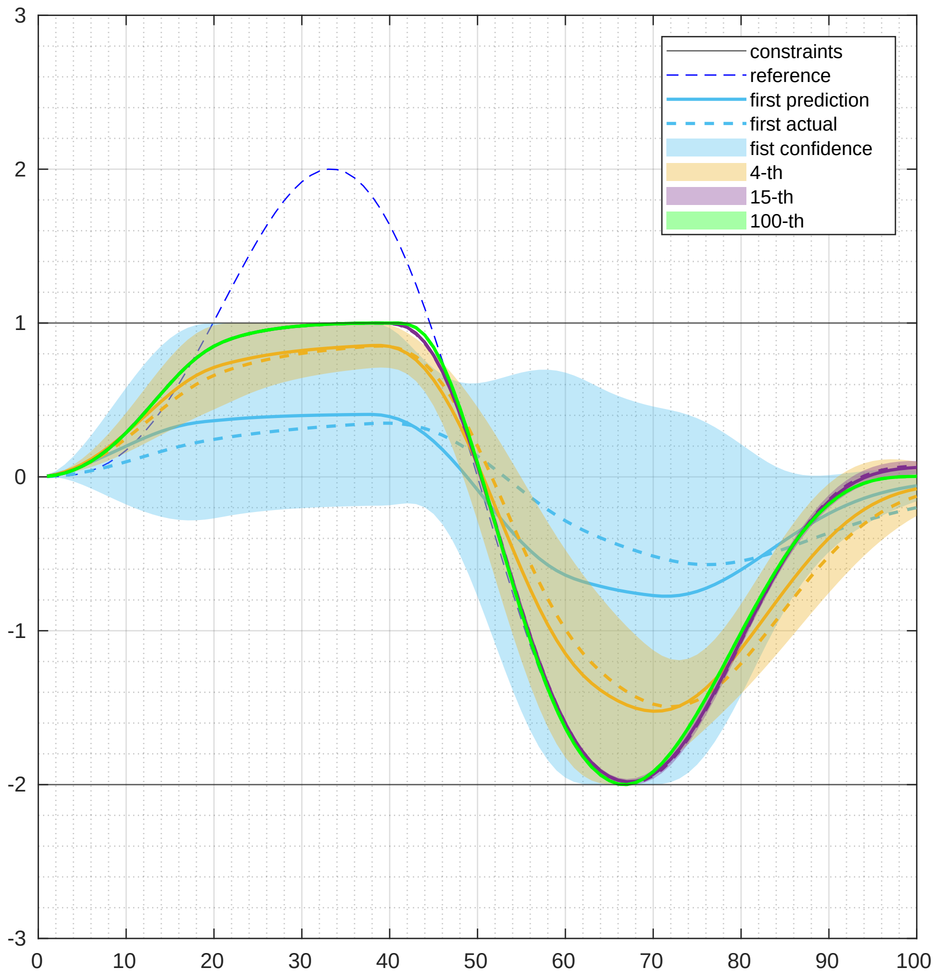


Figure A-18. ILC scheme with robust constraint satisfaction.

error norm difference to the final value in a logarithmic scale. In the early phase the constraints appear to play a minor role, since the error evolution is linear in the logarithmic scale it decays exponentially in the linear scale characteristic for norm optimal approaches. In the very end of the time scale the convergence speed appears to rise again. This effect is due to the approximation of the limit of the error norm by the last sample $||\boldsymbol{e}_\infty|| = \lim\limits_{j\to\infty} \boldsymbol{e}_j \approx \boldsymbol{e}_{n_{\text{ILC}}}$.

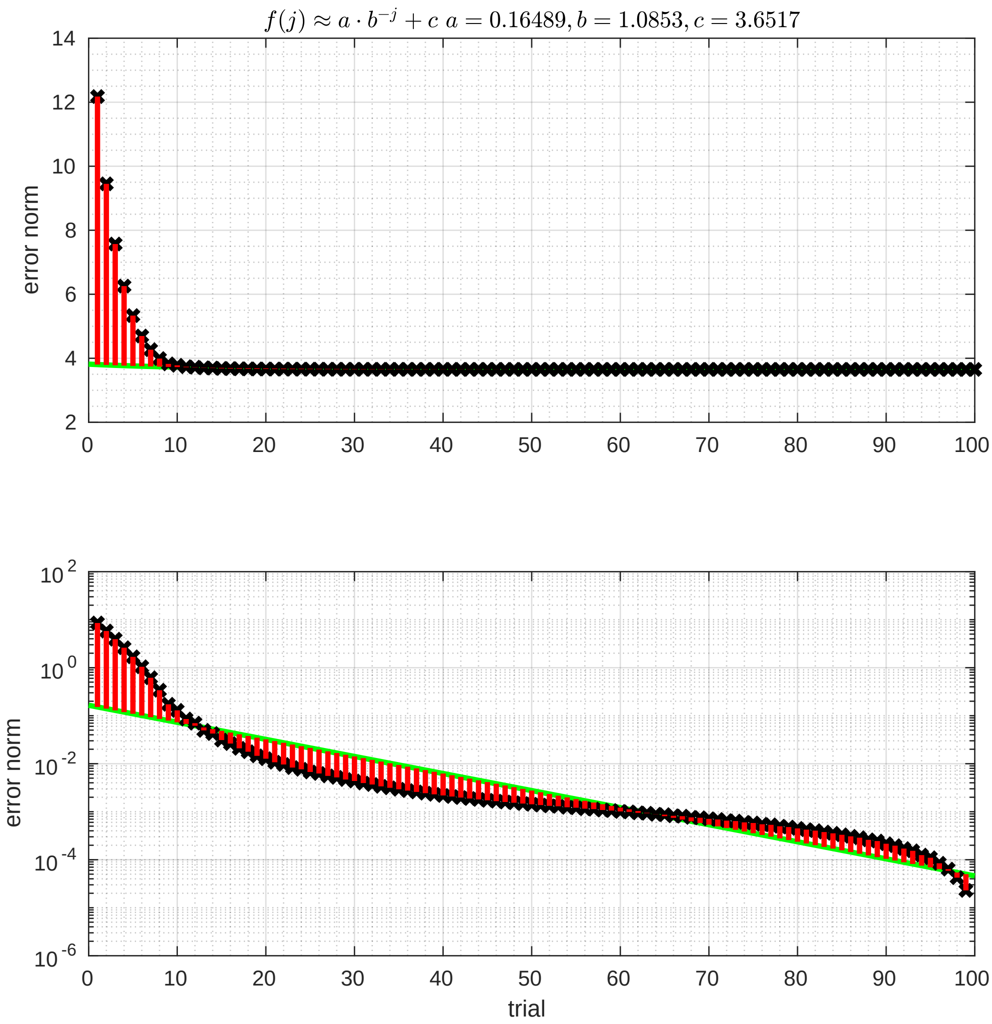
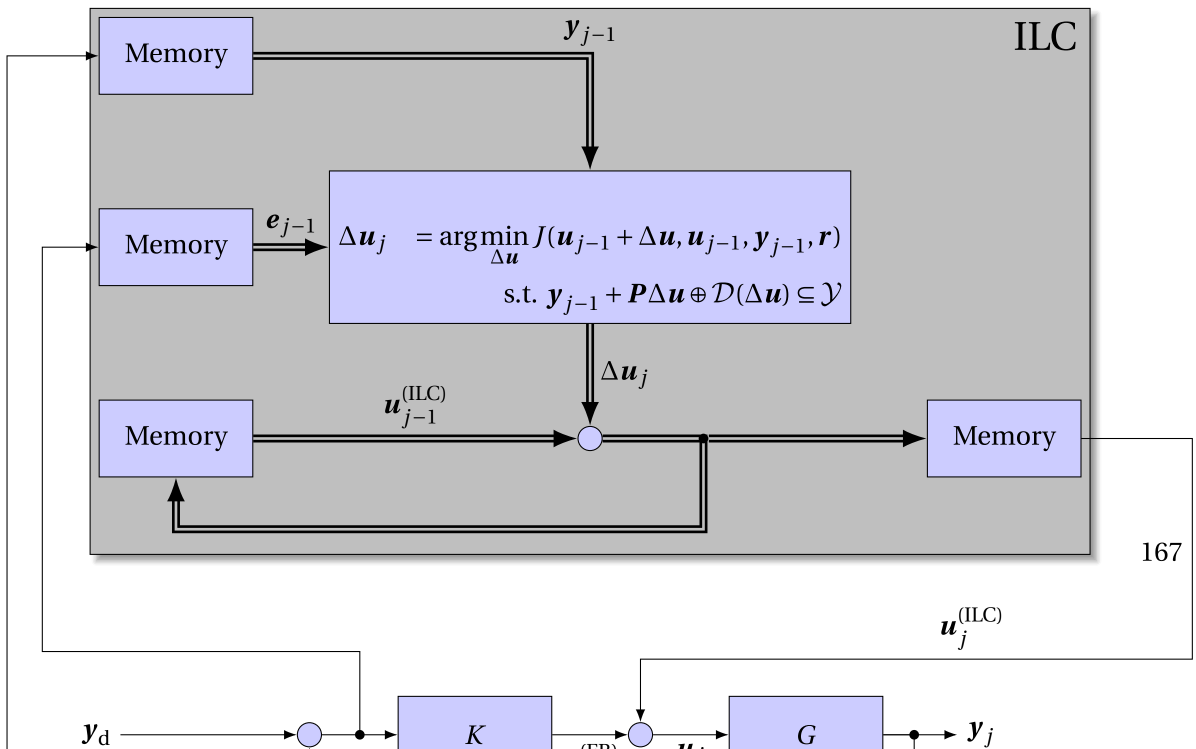


Figure A-19. Robust constraint error evolution.



A.7.4 System properties

In this section the property of a holonomy of constraints is done more precisely on the basis of literature. In the scientific discussion in the domain of motion planning and in mechanics the terminology appears to the author to be not unique, thus many problems that are holonomic in fundamental mechanics are called non-holonomic.

Holonomic in the kinematic sense

In the context of motion planing and controls the word holonomic systems often occurs. In fact this property is well known from the context of mechanics and describes the absence of a holonomic constraint representation. The rigorous definition is taken from the standard textbook [45, p. E10]. Therfore, a holonomic constraint is defined as

Definition 21 (Holonomic constraint)**.** *A holonomic kinetic constraint* c_h *can be represented by*

$$f_{c_\mathrm{h}}(q_1, \ldots, q_n, t) = 0. \tag{A.85}$$

If the function f *is not dependent on the time it is called skelernom and rheonom otherwise.*

Definition 22 (Non-holonomic constraint)**.** *Kinetic constraints that are not holonomic are non-holonomic.*

A non-holonomic c_nh constraint can only be presented as a function of the generalized velocities

$$f_{c_\mathrm{nh}}(\dot{q}_1, \ldots, \dot{q}_n, q_1, \ldots, q_n, t) = 0. \tag{A.86}$$

The effect of non-holonomic constraints is the loss of independence of the virtual displacements. Precisely, the number of degrees of freedom reduces in the infinitesimal view. The standard example is the car model that can maneuver in every direction and position in finite time. Contrarily, in infinitesimal time is only able to translate in a subspace of the plane and rotate around the up axis. A holonomic constraint leads to a loss of Degrees of freedom (DoF) in both the finite and infinitesimal time view.

In the case of the Two Wheeled Inverted Pendulum Robot (TWIPR) we may choose the holonomic constraint set of generalized coordinates as $(x, y, \vartheta, \varphi)$ namely the coordinates in the plane, the heading and the pitch. Thus we need to introduce the non-holonomic constraint

$$f_{c_\mathrm{nh}}(\dot{x}, \dot{y}, \vartheta) = \dot{x} \cdot \sin(\vartheta) + \dot{y} \cdot \cos(\vartheta) = 0. \tag{A.87}$$

Thus the non-holonomic character is exactly the same as for each other road vehicle. If we review the system given a unique orientation the constraint becomes holonomic as

$$f_{c_\mathrm{nh}}(\dot{x}, \dot{y}, \vartheta(t)) = \dot{x} \cdot \sin(\vartheta) + \dot{y} \cdot \cos(\vartheta) = \frac{\partial \dot{f}_{c_\mathrm{h}}}{\partial x}(x, y, \vartheta(t))\dot{x} + \frac{\partial \dot{f}_{c_\mathrm{h}}}{\partial y}(x, y, \vartheta(t))\dot{y}$$

$$f_{c_\mathrm{h}}(x, y, \vartheta(t)) = x \cdot \sin(\vartheta) + y \cdot \cos(\vartheta) = 0$$

Holonomic in the dynamic sense

In the context of controls the holonomy property is defined in an other way. In contrast to the former definition that assumes the kinetics and leads to implications on the dynamics i. e. a loss of dynamic degrees of freedom. The control sense definition relies on a dynamic model and presents implications on the solution space that can be reached using a controller. Many authors refere to the non-holonomic constraints eventhough not mentioning the precise definition or a meaningful reference. Non the less Oriolo and Nakamura published a paper with usefull and rigorous definitions [50] in 1991. So, assuming the general dynamic model

$$\boldsymbol{M}(q)\ddot{q} + \boldsymbol{c}(q,\dot{q}) + \boldsymbol{g}(q) = \boldsymbol{\tau}. \tag{A.88}$$

In the case of underactuation i.e. the generized forces can not be chosen independently we can assume without loss of generality (w. l. o. g.) that the system is portioned in actuated and underactuated coordinates $q = \begin{bmatrix} q_{\mathrm{a}} & q_{\mathrm{u}} \end{bmatrix}^{\mathrm{T}}$.

$$\begin{bmatrix} \boldsymbol{M}_{\mathrm{a}}(q) \\ \boldsymbol{M}_{\mathrm{u}}(q) \end{bmatrix} \cdot \begin{bmatrix} \ddot{q}_{\mathrm{a}} \\ \ddot{q}_{\mathrm{u}} \end{bmatrix} + \begin{bmatrix} \boldsymbol{c}_{\mathrm{a}}(q,\dot{q}) \\ \boldsymbol{c}_{\mathrm{u}}(q,\dot{q}) \end{bmatrix} + \begin{bmatrix} \boldsymbol{g}_{\mathrm{a}} \\ \boldsymbol{g}_{\mathrm{u}} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\tau}_{\mathrm{a}} \\ 0 \end{bmatrix} \tag{A.89}$$

Thus the second row forms a dynamic constraint of the form.

$$\boldsymbol{M}_{\mathrm{u}} \cdot \ddot{q} + \boldsymbol{c}_{\mathrm{u}}(q,\dot{q}) + \boldsymbol{g}_{\mathrm{u}} = 0 \tag{A.90}$$

This constraint can be either partially integrable if it could be integrated to $\tilde{f}(q,\dot{q},t) = 0$ or be fully integrable if a representation $f(q,t) = 0$ exists.

Theorem 1. *The dynamic constraint (A.90) of the underactuated system under actuated system (A.89) is partaialy integrable if and only if*

- *the $\boldsymbol{M}(q_{\mathrm{a}}, \dot{q}_{\mathrm{a}})$ does not depend on the under actuated generalized coordinates*
- *and the influence of the static conservative forces $\boldsymbol{g}_{\mathrm{u}}$ is constant.*

Proof. Refer to [50]. □

If a constraint is partially integrable it the consraint also take the form

$$\boldsymbol{M}_{\mathrm{u}}(q)\dot{q} + \boldsymbol{g}_{\mathrm{u}} t + k_2 = 0. \tag{A.91}$$

Consider the part $\boldsymbol{M}_{\mathrm{u}}(q)\dot{q}$ and define a mapping to the null space for a given configuration as $\Delta : \mathcal{C} \to \mathcal{P}(\mathcal{C}); q \mapsto \mathrm{null}(q)$.

Theorem 2. *The dynamic constraint (A.90) of the underactuated system under actuated system (A.89) is totally integrable and thus holonomic, if and only if*

- *it is partially integrable*
- *and the mapping Δ is involute.*

Proof. Refer to [50]. □

Remark 11. *Systems with two DoF that are partially integrable are always totally integrable [50].*

Remark 12. *The definition from mechanics does not conflict with the definition concerned in controls.*

Conclusion

Finally, the properties and their implications are summarized. In case of a partially integrable dynamic constraint we find a kinetic constraint. Thus, one may arrive in any point in the state space but the local direction in the statespace is bound to subspace. Conversely, a non-holonomic dynamic constraint behaves like a kinetec but also relies on the current acceleration. A holonomic constraint simply binds one DoF and thus the state representation should be changed. In the sense of motion planning the non holonomic constraint leads to the need of a Local Motion Planning (LMP) to take care of the possible motions that can be performed [15, pp. 788]. For a control system engineer the definition seems a little confusing since the most systems reviewed in teaching are non-holonomic and mentioning that is usually obmitted [43, 57].

A.8 Partially Differentially Flat (PDF)

The Two Wheeled Inverted Pendulum Robot (TWIPR) is PDF as defined in [49]. Partial differential flatness is a system property relating to flatness property known in nonlinear control [43, p. 209]. Flattness enables motion planning for systems to done easily. Unfortunately, in presence of friction the property is lost. The major advantage of this observation is that solving the boundary value problem only relies on the integration of a single equation of motion. In this section it is explained why the TWIPR is not flat but PDF and how this property relaxes requirements for motion planning. The property has be shown for the Ball-Bot in [49], while in this work the property is proven for the 3D TWIPR model.

Definition 23 (Differential Flatness.)**.** *[49] A system* $\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x},\boldsymbol{u}), \boldsymbol{x} \in \mathbb{R}^n, \boldsymbol{u} \in \mathbb{R}^m$ *is differentially flat if there exists an output* $\boldsymbol{y} \in \mathbb{R}^m$ *of the form* $\boldsymbol{y} = \boldsymbol{y}(\boldsymbol{x},\boldsymbol{u},\dot{\boldsymbol{u}},\dots,\boldsymbol{u}^{(p)})$ *such that the states and the inputs can be expressed as* $\boldsymbol{x} = \boldsymbol{x}(\boldsymbol{y},\dot{\boldsymbol{y}},\dots,\boldsymbol{y}^{(q)})$, $\boldsymbol{u} = \boldsymbol{u}(\boldsymbol{y},\dot{\boldsymbol{y}},\dots,\boldsymbol{y}^{(q)})$, *where* p, q *are finite integers.*

Fully actuated stationary industrial robotic manipulators are proven flat with respect to the joint angles [43, p. 212]. I.e. the state is $\begin{bmatrix} q & \dot{q} \end{bmatrix}$ and the input is the vector of generalized forces while the Differential Algebraic Equations (DAE) (4.33) is the mapping to the input given the outputs derivative. As the TWIPR is underactuated this argument is not sufficient for flattness.

Definition 24 (Partial differential flattness.)**.** *[49] A system* $\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x},\boldsymbol{u}), \boldsymbol{x} \in \mathbb{R}^n, \boldsymbol{u} \in \mathbb{R}^m$ *is partially deferentially flat if there exists a parition of the states* $\boldsymbol{x} = \begin{bmatrix} \boldsymbol{s} & \boldsymbol{r} \end{bmatrix}, \boldsymbol{s} \in \mathbb{R}^i, \boldsymbol{r} \in \mathbb{R}^{n-i}, i \leq n$, *(called* $\boldsymbol{s}$*-states and* $\boldsymbol{r}$*-states) and outputs* $\boldsymbol{y} \in \mathbb{R}^m$ *of the form* $\boldsymbol{y} = \boldsymbol{y}(\boldsymbol{s},\boldsymbol{u},\dot{\boldsymbol{u}},\dots,\boldsymbol{u}^{(p)})$, *such that the state-partition* $\boldsymbol{s}$ *and input* $\boldsymbol{u}$ *can be expressed* $\boldsymbol{s} = \boldsymbol{s}(\boldsymbol{y},\dot{\boldsymbol{y}},\dots,\boldsymbol{y}^{(q)})$ *and* $\boldsymbol{u} = \boldsymbol{u}(\boldsymbol{y},\dot{\boldsymbol{y}},\dots,\boldsymbol{y}^{(q)})$ *respectively, where* p, q *are finite integers, and the* $\boldsymbol{r}$*-state dynamics are those of one or more double integrators, i. e.,*

$$\dot{\boldsymbol{r}} = \begin{bmatrix} \boldsymbol{A}_1 & \boldsymbol{0} & \dots & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{A}_2 & \dots & \boldsymbol{0} \\ \vdots & \vdots & \ddots & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} & \dots & \boldsymbol{A}_l \end{bmatrix} \boldsymbol{r} + \begin{bmatrix} \boldsymbol{b}_1 \\ \boldsymbol{b}_2 \\ \vdots \\ \boldsymbol{b}_l \end{bmatrix} \tag{A.92}$$

with $\boldsymbol{A}_j$ and $\boldsymbol{b}_j$ given as

$$\boldsymbol{A}_j = \begin{bmatrix} \mathbf{0} & \boldsymbol{E} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}, \quad \boldsymbol{b}_j = \begin{bmatrix} \mathbf{0} \\ \boldsymbol{h}_j(\boldsymbol{y}, \dot{\boldsymbol{y}}, \dots, \boldsymbol{y}^{(q)}) \end{bmatrix} \tag{A.93}$$

where $\boldsymbol{h}_j$ is a smooth scalar function, and the bold constants are either row/column vectors or matrices of appropriate size.

Choosing the pitch φ and the differential wheel angle ϑ_Δ as outputs (**??**), the first and third equation of motion can be solved for the second derivative of the $\ddot{\vartheta}$.

$$0 = \left(\alpha + 2\beta\cos(\varphi) + \gamma\right)\ddot{\varphi} + \left(\alpha + \beta\cos(\varphi)\right)\ddot{\vartheta} - \beta\sin(\varphi)\left(\dot{\varphi}^2 + \frac{4{r_\mathrm{w}}^2}{d^2}\dot{\vartheta}_\Delta^{\,2}\right) - 2\epsilon\sin(2\varphi)\dot{\vartheta}_\Delta^{\,2} + \nu\sin(\varphi)$$

$$\ddot{\vartheta} = \frac{\beta\sin(\varphi)\left(\dot{\varphi}^2 + \frac{4{r_\mathrm{w}}^2}{d^2}\dot{\vartheta}_\Delta^{\,2}\right) + 4\epsilon\cos(\varphi)\sin(\varphi)\dot{\vartheta}_\Delta^{\,2} - \nu\sin(\varphi) - \left(\alpha + 2\beta\cos(\varphi) + \gamma\right)\ddot{\varphi}}{\alpha + \beta\cos(\varphi)} \tag{A.94}$$

$$\tau_{\vartheta_\Delta} = \left(\delta + \delta_1\sin^2(\varphi) + \delta_2\cos^2(\varphi)\right)\ddot{\vartheta}_\Delta + \frac{4{r_\mathrm{w}}^2}{d^2}\beta\sin(\varphi)\left(\dot{\varphi} + \dot{\vartheta}\right)\dot{\vartheta}_\Delta + 4\epsilon\sin(2\varphi)\dot{\varphi}\dot{\vartheta}_\Delta \tag{A.95}$$

The full state can be obtained by intergration and the full input by the second equation of motion.

$$\dot{\vartheta} = \int_{t_0}^{t} \ddot{\vartheta} + \dot{\vartheta}_0 \tag{A.96}$$

$$\vartheta = \int_{t_0}^{t} \dot{\vartheta} + \vartheta_0 \tag{A.97}$$

$$\tau_\vartheta = \left(\alpha + \beta\cos(\varphi)\right)\ddot{\vartheta} + \left(\delta + \delta_1\sin^2(\varphi) + \delta_2\cos^2(\varphi)\right)\ddot{\varphi} - \beta\sin(\varphi)\left(\dot{\varphi}^2 + \frac{4{r_\mathrm{w}}^2}{d^2}\dot{\vartheta}_\Delta^{\,2}\right) \tag{A.98}$$

In this representation the inputs are the generalized forces. Those contain the actuator losses in the gears that are complicated to compensate. But, it becomes evident, that losses in the gears do not affect the coupled rigid body dynamics and, therefore, are not of interest for motion planning purposes if only the outputs are of interest. The friction losses between the wheels and the underground behaves differently though, as observed in the following. The friction introduces damping in the dynamics of the $\boldsymbol{r}$-states and therefore annuls the PDF property. This leads to the fact that for planning integration is no longer sufficient but an Ordinary Differential Equations (ODE) needs to be solved. Furthermore, the input trajectory can not be given explicitly.

$$\boldsymbol{\tau} = \begin{bmatrix} -F_\mathrm{FR}r_\mathrm{w} - F_\mathrm{FL}r_\mathrm{w} \\ \tau_\mathrm{R} + \tau_\mathrm{L} - F_\mathrm{FR}r_\mathrm{w} - F_\mathrm{FL}r_\mathrm{w} \\ \tau_\mathrm{R} - \tau_\mathrm{L} - F_\mathrm{FR}r_\mathrm{w} + F_\mathrm{FL}r_\mathrm{w} \end{bmatrix} \tag{A.99}$$

$$
\begin{aligned}
\tau_{\varphi}(\dot{\varphi},\dot{\vartheta},\dot{\vartheta_\Delta}) &= \left(\alpha+2\beta\cos(\varphi)+\gamma\right)\ddot{\varphi}+\left(\alpha+\beta\cos(\varphi)\right)\ddot{\vartheta}-\beta\sin(\varphi)\left(\dot{\varphi}^2+\frac{4{r_\mathrm{w}}^2}{d^2}\dot{\vartheta_\Delta}^2\right)\\
&\quad -2\epsilon\sin(2\varphi)\dot{\vartheta_\Delta}^2+\nu\sin(\varphi)\\
0 &= \alpha\left(\ddot{\varphi}+\ddot{\vartheta}\right)+\beta\left(2\cos(\varphi)\ddot{\varphi}+\cos(\varphi)\ddot{\vartheta}-\sin(\varphi)\left(\dot{\varphi}^2+\frac{4{r_\mathrm{w}}^2}{d^2}\dot{\vartheta_\Delta}^2\right)\right)\\
&\quad +\gamma\ddot{\varphi}+\epsilon\left(-2\sin(2\varphi)\dot{\vartheta_\Delta}^2\right)+\nu\sin(\varphi)\\
&\quad + \underbrace{\mathbf{\Gamma}\left(\mathbf{\Phi}(\dot{\varphi}+\dot{\vartheta}+\dot{\vartheta_\Delta})+\mathbf{\Phi}(\dot{\varphi}+\dot{\vartheta}-\dot{\vartheta_\Delta})\right)}_{\tau_{\varphi}(\dot{\varphi},\dot{\vartheta},\dot{\vartheta}_\Delta)\text{ e. g.simple ANN modelling the wheel friction.}} \qquad \text{(A.100)}\\
\ddot{\vartheta} &= \frac{\beta\sin(\varphi)\left(\dot{\varphi}^2+\frac{4{r_\mathrm{w}}^2}{d^2}\dot{\vartheta_\Delta}^2\right)+4\epsilon\cos(\varphi)\sin(\varphi)\dot{\vartheta_\Delta}^2}{\alpha+\beta\cos(\varphi)}\\
&\quad \frac{-\nu\sin(\varphi)-\left(\alpha+2\beta\cos(\varphi)+\gamma\right)\ddot{\varphi}+\tau_{\varphi}(\dot{\varphi},\dot{\vartheta},\dot{\vartheta_\Delta})}{\alpha+\beta\cos(\varphi)}+\beta\sin(\varphi)\left(\dot{\varphi}^2+\frac{4{r_\mathrm{w}}^2}{d^2}\dot{\vartheta_\Delta}^2\right) \qquad \text{(A.101)}\\
\tau_{\vartheta_\Delta} &= \left(\delta+\delta_1\sin^2(\varphi)+\delta_2\cos^2(\varphi)\right)\ddot{\vartheta_\Delta}+4\epsilon\sin(2\varphi)\dot{\varphi}\dot{\vartheta_\Delta}+\frac{4{r_\mathrm{w}}^2}{d^2}\beta\sin(\varphi)\left(\dot{\varphi}+\dot{\vartheta}\right)\dot{\vartheta_\Delta} \qquad \text{(A.102)}\\
\tau_{\vartheta} &= \left(\alpha+\beta\cos(\varphi)\right)\ddot{\vartheta}+\left(\delta+\delta_1\sin^2(\varphi)+\delta_2\cos^2(\varphi)\right)\ddot{\varphi}-\beta\sin(\varphi)\left(\dot{\varphi}^2+\frac{4{r_\mathrm{w}}^2}{d^2}\dot{\vartheta_\Delta}^2\right) \qquad \text{(A.103)}
\end{aligned}
$$

Equation (A.100) forms a lumped representation that may be used for a system identification in future projects as descirbed in [58, Ch. 18.2.2].

In the case where friction is taken into account the system does not have a property that is distinct from ordinary Lagrange-Euler as discussed in the previous section system. The ODE that leads to the solution of $\ddot{\vartheta}$ (A.101) can also be solved for $\ddot{\varphi}$ as the $\vartheta, \dot{\vartheta}, \ddot{\vartheta}$ is given.

$$
\ddot{\varphi} = \frac{\beta\sin(\varphi)\left(\dot{\varphi}^2+\frac{4{r_\mathrm{w}}^2}{d^2}\dot{\vartheta_\Delta}^2\right)+4\epsilon\cos(\varphi)\sin(\varphi)\dot{\vartheta_\Delta}^2-\nu\sin(\varphi)-\left(\alpha+2\beta\cos(\varphi)\right)\ddot{\vartheta}+\tau_{\varphi}(\dot{\varphi},\dot{\vartheta},\dot{\vartheta_\Delta})}{\alpha+\beta\cos(\varphi)+\gamma} \qquad \text{(A.104)}
$$

In contrast to the former case the ODE is not stable for to upper equilibrium. This issue might lead to problems when solving the optimization problem that consider this dynamics.

Concluding this section it is found that the PDF property that was found for the ballbot extends to the TWIPR and trajectories can be generated for a given pitch and yaw trajectory while the position can be obtained using integration. Furthermore, it is shown that as friction is taken into account as second order ODE (A.101) must be solved to obtain inputs and the full state. Furthermore, it is shown that friction looses in the gearbox do not affect the trajectories in configuration space.

A.8.1 Solving the Local motion Planning Problem using the Markov parameters

In the aim of the overall project it is beneficial to solve the planning problem with the markov parameters instead of a state space model (Definition 8) that only maps the inputs to the configuration coordinates. The markov parameters do not expose the information regarding the order of the underlying system in a obvious way. As discussed earlier the information is needed to choose the dimension to sample from. In this section it will be shown how to obtain the system form the markov paremters only. This part, therefore serves as an additional argument and is not part of the implemented solution. If, the reader likes to continue with results he can skip this section.

Therefore, one obtains only a partial observation of the full state vector instead of the full state.

$$\boldsymbol{y} = \boldsymbol{C}\boldsymbol{x} \tag{A.105}$$

In general one meight not be aware of the state dimension. It is assumed that the only model inside that is given is the sequence of markov paramters. To assure that the states of two consective locally planned trajectories coincide the observability properties are used. First it will be assumed that the models order is know.

If the model order is known, the observability property allows to infer the state information from output data, i. e.the folowing equations can be uniquely solved for the state $\boldsymbol{x}(k)$ for $j \geq n$.

$$\begin{aligned}
\boldsymbol{y}(k) &= \boldsymbol{C}\boldsymbol{x}(k) && \text{(A.106)}\\
\boldsymbol{y}(k+1) &= \boldsymbol{C}\boldsymbol{A}\boldsymbol{x}(k) + \boldsymbol{C}\boldsymbol{B}\boldsymbol{u}(k) && \text{(A.107)}\\
&\vdots \\
\boldsymbol{y}(k+j) &= \boldsymbol{C}\boldsymbol{A}^{j}\boldsymbol{x}(k) + \sum_{i=0}^{j-1} \boldsymbol{C}\boldsymbol{A}^{i}\boldsymbol{B}\boldsymbol{u}(k+j-i) && \text{(A.108)}
\end{aligned}$$

Using this property one can also construct trajectories that coincide in one state without determinig it expicitly. If one assumes that the system matrix $\boldsymbol{A}$ is regular, arbitarily chosen consecutive $j_1, \ldots, j_n$ samples are sufficient to construct trajectories that have the same state $\boldsymbol{x}(k)$.

Furthermore, assumeing controlability yields to the property that n consecutive controlled inputs are sufficient to uniquely stear the linear system to a desired state as it is used in the last section.

$$\boldsymbol{x}(k) = \begin{bmatrix} \boldsymbol{A}^{n-1}\boldsymbol{B} & \ldots & \boldsymbol{A}\boldsymbol{B} & \boldsymbol{B} \end{bmatrix} \begin{bmatrix} \boldsymbol{u}(k-n) \\ \boldsymbol{u}(k-n+1) \\ \vdots \\ \boldsymbol{u}(k-1) \end{bmatrix} + \boldsymbol{A}^{n}\boldsymbol{x}(k-n) \tag{A.109}$$

Thus, one can write.

$$
\begin{bmatrix} \boldsymbol{y}(k) \\ \boldsymbol{y}(k+1) \\ \vdots \\ \boldsymbol{y}(k+j) \end{bmatrix} = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \dots & \mathbf{0} \\ \boldsymbol{CB} & \mathbf{0} & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{CA}^{j-2}\boldsymbol{B} & \boldsymbol{CA}^{j-3} & \dots & \mathbf{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{u}(k) \\ \boldsymbol{u}(k+1) \\ \vdots \\ k(k+j) \end{bmatrix}
+ \begin{bmatrix} \boldsymbol{C} \\ \boldsymbol{CA} \\ \vdots \\ \boldsymbol{CA}^j \end{bmatrix} \begin{bmatrix} \boldsymbol{A}^{n-1}\boldsymbol{B} & \dots & \boldsymbol{AB} & \boldsymbol{B} \end{bmatrix} \begin{bmatrix} \boldsymbol{u}(k-n) \\ \boldsymbol{u}(k-n+1) \\ \vdots \\ \boldsymbol{u}(k-1) \end{bmatrix} + \boldsymbol{A}^n \boldsymbol{x}(k-n)
$$

$$
\begin{bmatrix} \boldsymbol{y}(k) \\ \boldsymbol{y}(k+1) \\ \vdots \\ \boldsymbol{y}(k+j) \end{bmatrix} = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \dots & \mathbf{0} \\ \boldsymbol{CB} & \mathbf{0} & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{CA}^{j-2}\boldsymbol{B} & \boldsymbol{CA}^{j-3} & \dots & \mathbf{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{u}(k) \\ \boldsymbol{u}(k+1) \\ \vdots \\ k(k+j) \end{bmatrix}
+ \underbrace{\begin{bmatrix} \boldsymbol{CA}^{n-1}\boldsymbol{B} & \boldsymbol{CA}^{n-2}\boldsymbol{B} & \dots & \boldsymbol{CB} \\ \boldsymbol{CA}^{n}\boldsymbol{B} & \boldsymbol{CA}^{n-1}\boldsymbol{B} & \dots & \boldsymbol{CAB} \\ \vdots & \vdots & & \vdots \\ \boldsymbol{CA}^{n+j-1}\boldsymbol{B} & \boldsymbol{CA}^{n+j-2}\boldsymbol{B} & \dots & \boldsymbol{CA}^{j-1}\boldsymbol{B} \end{bmatrix}}_{\bar{\boldsymbol{P}}} \begin{bmatrix} \boldsymbol{u}(k-n) \\ \boldsymbol{u}(k-n+1) \\ \vdots \\ \boldsymbol{u}(k-1) \end{bmatrix}
+ \begin{bmatrix} \boldsymbol{CA}^n \\ \boldsymbol{CA}^{n+1} \\ \vdots \\ \boldsymbol{CA}^{n+j} \end{bmatrix} \boldsymbol{x}(k-n)
$$

One observes that $\bar{\boldsymbol{P}}$ is a slice of the systems lifted dynamics representation whose rank is limited to the system order. This leads to the a method to choose a set of outputs $j_1, \dots, j_{\tilde{n}} | \tilde{n} \leq n$ by analysing the lifted dynamics. The set of outputs is valid as slice of $\boldsymbol{\Phi}\bar{\boldsymbol{P}}$ does not loose rank.

$$
\operatorname{rank}\left[\boldsymbol{\Phi}\bar{\boldsymbol{P}}\right] = \operatorname{rank}\left[\bar{\boldsymbol{P}}\right] \tag{A.110}
$$

This observation is useful since the system can be designed so that does not need a state space representation of the system, but can determine a number of points in the configuration space that must coincied along the trajectory, so that the state must coincide in one known waypoint along the trajectory without knowing an representation. Furthermore, it is shown that they do not need to be consecutive, but can be chosen almost arbitrarily and the secelction can be proven valid by the inspection of the markov parameter only. But it is important to note that this approach, due to literature research [59], will only work if the markov parameters are generated by a model and not obtained in an experiment. In the field of subspace identification van der Veen [59, p. 1345] used the matrix $\bar{\boldsymbol{P}}$, that is the product of controlability and observabilty matrix, to inferre the system order using a Singular Value Decomposition (SVD). Thus, it is desirable that the order can be determined as proper lifteddynamics are given. Never the less it needs to be pointed out that system identification is task that needs background knowledge.

In the remainder of this section it is discussed how trajectories can be continued using the markov parameter representation.

If planning is done in the forward way, one could simply assume all past input fixed and only solve for the future ones.

$$\begin{bmatrix} \boldsymbol{y}(k_\text{s}) \\ \vdots \\ \boldsymbol{y}(k_\text{g}) \end{bmatrix} = \begin{bmatrix} \mathbf{0} & \boldsymbol{E} \end{bmatrix} \left(\underline{\boldsymbol{P}} \begin{bmatrix} \boldsymbol{u}(0) \\ \vdots \\ \boldsymbol{u}(k_\text{s}-1) \end{bmatrix} + \bar{\boldsymbol{P}} \begin{bmatrix} \boldsymbol{u}(k_\text{s}) \\ \vdots \\ \boldsymbol{u}(k_\text{g}) \end{bmatrix} + \boldsymbol{d} \right) \tag{A.111}$$

$$= \begin{bmatrix} \mathbf{0} & \boldsymbol{E} \end{bmatrix} \bar{\boldsymbol{P}} \begin{bmatrix} \boldsymbol{u}(k_\text{s}) \\ \vdots \\ \boldsymbol{u}(k_\text{g}) \end{bmatrix} + \begin{bmatrix} \mathbf{0} & \boldsymbol{E} \end{bmatrix} \bar{\boldsymbol{d}} \tag{A.112}$$

$$= \tilde{\boldsymbol{P}} \begin{bmatrix} \boldsymbol{u}(k_\text{s}) \\ \vdots \\ \boldsymbol{u}(k_\text{g}) \end{bmatrix} + \tilde{\boldsymbol{d}} \tag{A.113}$$

This method does only work for forward planned trajectories as the whole past input trajectory must be known. It is shown in the following that the lifted disturbance can also be calculated using the lifted dynamics representation. The crucial part leading to this restriction is the disturbance sequence $\tilde{\boldsymbol{d}}$. As it encodes the influence of past inputs and the initial state on future outputs it can also be computed using the hand over zone. Therefore, the system is assumed to be in the equilibrium in the beginning $\boldsymbol{y}(k-n) = \mathbf{0}$ and a virtual input for the handover setup is computed so that using the input of the handover zone leads to the same state in time instant k. The input of the handover zone must be applied as well leading to the same state in $k + j_{\bar{n}}$.

$$\begin{bmatrix} \boldsymbol{y}(k_\text{s}) \\ \vdots \\ \boldsymbol{y}(k_\text{g}) \end{bmatrix} = \begin{bmatrix} \mathbf{0} & \boldsymbol{E} \end{bmatrix} \left(\underline{\boldsymbol{P}} \begin{bmatrix} \boldsymbol{u}_\text{v}(k-j_{\bar{n}}) \\ \vdots \\ \boldsymbol{u}_\text{v}(k-1) \\ \boldsymbol{u}(k) \\ \vdots \\ \boldsymbol{u}(k_\text{s}-1) \end{bmatrix} + \bar{\boldsymbol{P}} \begin{bmatrix} \boldsymbol{u}(k_\text{s}) \\ \vdots \\ \boldsymbol{u}(k_\text{g}) \end{bmatrix} + \boldsymbol{d} \right) \tag{A.114}$$

$$= \begin{bmatrix} \mathbf{0} & \boldsymbol{E} \end{bmatrix} \bar{\boldsymbol{P}} \begin{bmatrix} \boldsymbol{u}(k_\text{s}) \\ \vdots \\ \boldsymbol{u}(k_\text{g}) \end{bmatrix} + \begin{bmatrix} \mathbf{0} & \boldsymbol{E} \end{bmatrix} \bar{\boldsymbol{d}} \tag{A.115}$$

$$= \tilde{\boldsymbol{P}} \begin{bmatrix} \boldsymbol{u}(k_\text{s}) \\ \vdots \\ \boldsymbol{u}(k_\text{g}) \end{bmatrix} + \tilde{\boldsymbol{d}} \tag{A.116}$$

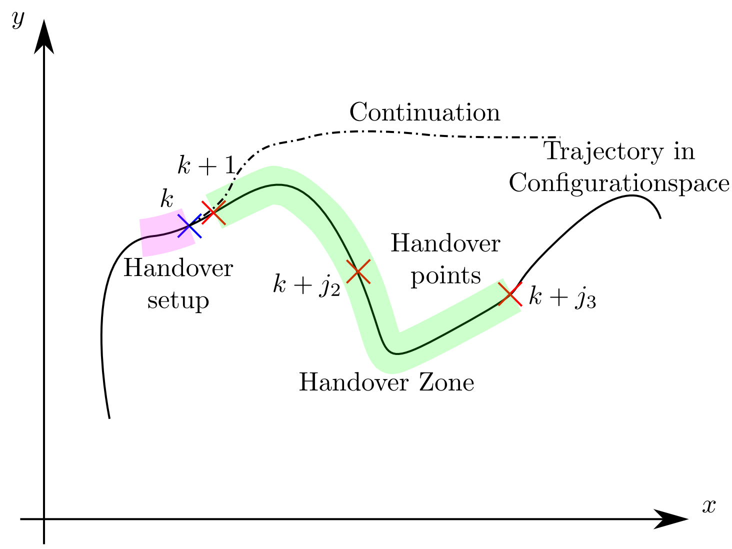


Figure A-21. Handover using multiple points in the Configuration space.

Bibliography

[1] D. Ioan, I. Prodan, S. Olaru, F. Stoican, and S.-I. Niculescu, "Mixed-integer programming in motion planning," Annual Reviews in Control, vol. 51, pp. 65–87, 2021. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1367578820300754

[2] D. Bristow, M. Tharayil, and A. Alleyne, "A survey of iterative learning control," IEEE Control Systems Magazine, vol. 26, no. 3, pp. 96–114, 2006. [Online]. Available: https://doi.org/10.1109/MCS.2006.1636313

[3] C. T. Freeman and Y. Tan, "Iterative Learning Control With Mixed Constraints for Point-to-Point Tracking," IEEE Transactions on Control Systems Technology, vol. 21, no. 3, pp. 604–616, 2013.

[4] M. Meindl, "Development and Iterative Learning Control of a Two-Wheeled Inverted-Pendulum-Robot," Master's thesis, Hochschule Karlsruhe, Karlsruhe, Germany, Jan. 2020.

[5] J. Raisch, Mehrgroeßenregelung im Frequenzbereich. Technische Universität Berlin Fak.IV - Elektrotechnik und Informatik Fachgebiet Regelungssysteme, 2010.

[6] S. H. Collins, M. Wisse, and A. Ruina, "A Three-Dimensional Passive-Dynamic Walking Robot with Two Legs and Knees," The International Journal of Robotics Research, vol. 20, no. 7, pp. 607–615, Jun. 2001, publisher: SAGE Publications.

[7] B. Henze, "Whole-Body Control for Multi-contact Balancing," in Whole-Body Control for Multi-Contact Balancing of Humanoid Robots: Design and Experiments. Cham: Springer International Publishing, 2022, pp. 65–128. [Online]. Available: https://doi.org/10.1007/978-3-030-87212-0_6

[8] E. Pall and O. Brock, "Analysis of Open-Loop Grasping From Piles," 2021 IEEE International Conference on Robotics and Automation (ICRA), pp. 2591–2597, Oct. 2021. [Online]. Available: http://www.robotics.tu-berlin.de/fileadmin/fg170/Publikationen_pdf/Pall-21-ICRA.pdf

[9] J. Raisch, "Skript zur Vorlesung - Grundlagen der Regelungstechnik," 2021. [Online]. Available: http://www.control.tu-berlin.de/wiki/images/8/81/grtSkript.pdf

[10] J. Moore, R. Cory, and R. Tedrake, "Robust post-stall perching with a simple fixed-wing glider using LQR-Trees," Bioinspiration & Biomimetics, vol. 9, no. 2, p. 025013, May 2014, publisher: IOP Publishing. [Online]. Available: https://doi.org/10.1088/1748-3182/9/2/025013

[11] R. Tedrake, Z. Jackowski, R. Cory, J. W. Roberts, and W. Hoburg, "Learning to fly like a bird," in 14th International symposium on robotics research. Lucerne, Switzerland. Citeseer, 2009.

[12] T. Seel, "Learning control and inertial realtime gait analysis in biomedical applications : improving diagnosis and treatment by automatic adaption and feedback control," Doctoral Thesis, Technische Universität Berlin, Berlin, 2016. [Online]. Available: http://dx.doi.org/10.14279/depositonce-4973

[13] M. Meindl, D. Lehmann, and T. Seel, "Bridging Reinforcement Learning and Iterative Learning Control: Autonomous Reference Tracking for Unknown, Nonlinear Dynamics," Jul. 2021. [Online]. Available: https://www.techrxiv.org/articles/preprint/Bridging_Reinforcement_Learning_and_Iterative_Learning_Control_Autonomous_Reference_Tracking_for_Unknown_Nonlinear_Dynamics/15022497

[14] M. Meindl, F. Molinari, D. Lehmann, and T. Seel, "Collective Iterative Learning Control: Exploiting Diversity in Multi-Agent Systems for Reference Tracking Tasks," IEEE Transactions on Control Systems Technology, pp. 1–13, 2021, publisher: Institute of Electrical and Electronics Engineers (IEEE). [Online]. Available: http://dx.doi.org/10.1109/TCST.2021.3109646

[15] S. M. LaValle, Planning Algorithms. Cambridge, U.K.: Cambridge University Press, 2006.

[16] T. George Thuruthel, Y. Ansari, E. Falotico, and C. Laschi, "Control Strategies for Soft Robotic Manipulators: A Survey," Soft Robotics, vol. 5, no. 2, pp. 149–163, Apr. 2018. [Online]. Available: https://www.liebertpub.com/doi/10.1089/soro.2017.0007

[17] R. Deimel, M. Radke, and O. Brock, "Mass control of pneumatic soft continuum actuators with commodity components," in 2016 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2016, pp. 774–779.

[18] V. Wall, G. Zöller, and O. Brock, "A method for sensorizing soft actuators and its application to the RBO hand 2," in 2017 IEEE International Conference on Robotics and Automation (ICRA), 2017, pp. 4965–4970.

[19] M. A. Mousa, M. Soliman, M. A. Saleh, and A. G. Radwan, "Biohybrid Soft Robots, E-Skin, and Bioimpedance Potential to Build Up Their Applications: A Review," IEEE Access, vol. 8, pp. 184 524–184 539, 2020.

[20] Z. Milosevic, R. A. S. Fernandez, S. Dominguez, and C. Rossi, "Guidance for Autonomous Underwater Vehicles in Confined Semistructured Environments," Sensors, vol. 20, no. 24, 2020. [Online]. Available: https://www.mdpi.com/1424-8220/20/24/7237

[21] J. Mainprice, N. Ratliff, M. Toussaint, and S. Schaal, "An Interior Point Method Solving Motion Planning Problems with Narrow Passages," in 2020 29th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), 2020, pp. 547–552.

[22] J. Mainprice, "(pyrieef) Python Riemannian Electrical Fields," Feb. 2022. [Online]. Available: https://github.com/humans-to-robots-motion/pyrieef

[23] D. H. Owens, Iterative Learning Control, ser. Advances in Industrial Control. London: Springer London, 2016. [Online]. Available: http://link.springer.com/10.1007/978-1-4471-6772-3

[24] J.-X. Xu and Y. Tan, Linear and Nonlinear Iterative Learning Control, ser. Lecture Notes in Control and Information Sciences. Berlin: Springer, 2003, no. 291.

[25] Y. Chen and C. Wen, Iterative Learning Control: Convergence, Robustness, and Applications., ser. Lecture Notes in Control and Information Sciences. London: Springer, 1999, no. 248.

[26] D. Owens, E. Rogers, and K. L. Moore, “Analysis Of Linear Iterative Learning Control Schemes Using Repetitive Process Theory,” Asian Journal of Control, vol. 4, no. 1, pp. 68–89, Oct. 2008. [Online]. Available: https://onlinelibrary.wiley.com/doi/10.1111/j.1934-6093.2002.tb00334.x

[27] R. King, “Vorlesungsskript - Grundlagen der Mess- und Relungstechnik,” 2021.

[28] P. Müller, C. Balligand, T. Seel, and T. Schauer, “Iterative Learning Control and System Identification of the Antagonistic Knee Muscle Complex During Gait Using Functional Electrical Stimulation,” IFAC-PapersOnLine, vol. 50, no. 1, pp. 8786–8791, 2017. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2405896317323509

[29] H. Wu, Y. He, X. Lin, and H. Li, “Research on the application of feedforward + high-order iterative learning of permanent magnet linear motor in wire bonding machine,” in 2021 22nd International Conference on Electronic Packaging Technology (ICEPT), 2021, pp. 1–6.

[30] M. Luo, Z. Wan, Y. Sun, E. H. Skorina, W. Tao, F. Chen, L. Gopalka, H. Yang, and C. D. Onal, “Motion Planning and Iterative Learning Control of a Modular Soft Robotic Snake,” Frontiers in Robotics and AI, vol. 7, p. 191, 2020. [Online]. Available: https://www.frontiersin.org/article/10.3389/frobt.2020.599242

[31] J.-C. Latombe, “Motion Planning: A Journey of Robots, Molecules, Digital Actors, and Other Artifacts,” International Journal of Robotics Research, vol. 18, no. 11, pp. 1119–1128, 1999.

[32] J. Pan, S. Chitta, and D. Manocha, “FCL: A general purpose library for collision and proximity queries,” in 2012 IEEE International Conference on Robotics and Automation, 2012, pp. 3859–3866.

[33] “MoveIt Motion Planning Framework.” [Online]. Available: https://moveit.ros.org/

[34] S. M. LaValle and P. Konkimalla, “Algorithms for Computing Numerical Optimal Feedback Motion Strategies,” International Journal of Robotics Research, vol. 20, no. 9, pp. 729–752, Sep. 2001.

[35] Y. Chen, B. Chu, and C. T. Freeman, “A coordinate descent approach to optimal tracking time allocation in point-to-point ILC,” Mechatronics, vol. 59, pp. 25–34, 2019. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0957415819300170

[36] “Convert model from continuous to discrete time - MATLAB c2d - MathWorks Deutschland.” [Online]. Available: https://de.mathworks.com/help/control/ref/lti.c2d.html

[37] B. Katz, J. D. Carlo, and S. Kim, “Mini Cheetah: A Platform for Pushing the Limits of Dynamic Quadruped Control,” in 2019 International Conference on Robotics and Automation (ICRA). Montreal, QC, Canada: IEEE, May 2019, pp. 6295–6301. [Online]. Available: https://ieeexplore.ieee.org/document/8793865/

[38] R. King, “Vorlesungsskript Lernende Regelung,” Feb. 2021. [Online]. Available: https://isis.tu-berlin.de/pluginfile.php/2130065/mod_resource/content/1/ILRG.pdf

[39] "Iterative Learning Control (ILC) · ControlSystems.jl." [Online]. Available: https://juliacontrol.github.io/ControlSystems.jl/latest/examples/ilc/

[40] K. B. Petersen and M. S. Pedersen, "The Matrix Cookbook," p. 72, Nov. 2012. [Online]. Available: https://www.math.uwaterloo.ca/~hwolkowi/matrixcookbook.pdf

[41] T. Strutz, A practical introduction to weighted least squares and beyond, 1st ed. Springer Vieweg Wiesbaden, 2016.

[42] "Constrained Nonlinear Optimization Algorithms - MATLAB & Simulink - MathWorks Deutschland." [Online]. Available: https://de.mathworks.com/help/optim/ug/constrained-nonlinear-optimization-algorithms.html#f26622

[43] J. Adamy, Nichtlineare Systeme und Regelungen. Berlin, Heidelberg: Springer Berlin Heidelberg, 2018. [Online]. Available: http://link.springer.com/10.1007/978-3-662-55685-6

[44] "Root-mean-square error between arrays - MATLAB rmse - MathWorks Deutschland." [Online]. Available: https://de.mathworks.com/help/matlab/ref/rmse.html

[45] J. Wittenburg, H.-A. Richard, J. Zierep, and K. Bühler, "E Technische Mechanik," in HÜTTE - Das Ingenieurwissen, H. Czichos and M. Hennecke, Eds. Berlin, Heidelberg: Springer Berlin Heidelberg, 2012, pp. 699–923. [Online]. Available: https://doi.org/10.1007/978-3-642-22850-6_5

[46] W. Nolting, Grundkurs theoretische Physik. 2: Analytische Mechanik / Wolfgang Nolting, 8th ed. Berlin Heidelberg: Springer, 2011.

[47] A. Meurer, C. P. Smith, M. Paprocki, O. Čertík, S. B. Kirpichev, M. Rocklin, A. Kumar, S. Ivanov, J. K. Moore, S. Singh, T. Rathnayake, S. Vig, B. E. Granger, R. P. Muller, F. Bonazzi, H. Gupta, S. Vats, F. Johansson, F. Pedregosa, M. J. Curry, A. R. Terrel, S. Roučka, A. Saboo, I. Fernando, S. Kulal, R. Cimrman, and A. Scopatz, "SymPy: symbolic computing in Python," PeerJ Computer Science, vol. 3, p. e103, Jan. 2017. [Online]. Available: https://doi.org/10.7717/peerj-cs.103

[48] S. Kim and S. Kwon, "Dynamic modeling of a two-wheeled inverted pendulum balancing mobile robot," International Journal of Control, Automation and Systems, vol. 13, no. 4, pp. 926–933, Aug. 2015. [Online]. Available: https://doi.org/10.1007/s12555-014-0564-8

[49] S. Ramasamy, G. Wu, and K. Sreenath, "Dynamically Feasible Motion Planning through Partial Differential Flatness." in Robotics: Science and Systems X. Robotics: Science and Systems Foundation, Jul. 2014. [Online]. Available: http://www.roboticsproceedings.org/rss10/p53.pdf

[50] G. Oriolo and Y. Nakamura, "Control of mechanical systems with second-order nonholonomic constraints: Underactuated manipulators," in Proceedings of the 30th IEEE Conference on Decision and Control, vol. 3. Brighton, UK: IEEE, 1991, pp. 2398–2403.

[51] "Linear-Quadratic Regulator (LQR) design - MATLAB lqr." [Online]. Available: https://www.mathworks.com/help/control/ref/lti.lqr.html

[52] "Systemtheorie Online: Minimalphasige und nichtminimalphasige Systeme." [Online]. Available: https://www.eit.hs-karlsruhe.de/mesysto/teil-a-zeitkontinuierliche-signale-und-systeme/uebertragungsglieder-der-regelungstechnik/minimalphasige-systeme-und-allpaesse/minimalphasige-und-nichtminimalphasige-systeme.html

[53] "Window-based FIR filter design - MATLAB fir1 - MathWorks Deutschland." [Online]. Available: https://de.mathworks.com/help/signal/ref/fir1.html

[54] K. J. Aström and B. Wittenmark, Computer Controlled Systems: Theory and Design. Prentice Hall, 1984.

[55] M. Meindl, F. Molinari, J. Raisch, and T. Seel, "Overcoming Output Constraints in Iterative Learning Control Systems by Reference Adaptation," IFAC-PapersOnLine, vol. 53, no. 2, pp. 1480–1486, 2020. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2405896320325647

[56] P. Ruge, C. Birk, and M. Wermuth, "A Mathematik und Statistik," in HÜTTE - Das Ingenieurwissen, H. Czichos and M. Hennecke, Eds. Berlin, Heidelberg: Springer Berlin Heidelberg, 2012, pp. 1–184. [Online]. Available: https://doi.org/10.1007/978-3-642-22850-6_1

[57] J. Lunze, "Reglerentwurf zur Polzuweisung," in Regelungstechnik 2: Mehrgrößensysteme Digitale Regelung. Berlin, Heidelberg: Springer Berlin Heidelberg, 2006, pp. 227–271. [Online]. Available: https://doi.org/10.1007/978-3-540-32511-6_6

[58] R. Tedrake, Underactuated Robotics: Algorithms for Walking, Running, Swimming, Flying, and Manipulation, 2022, published: Course Notes for MIT 6.832. [Online]. Available: http://underactuated.mit.edu

[59] G. van der Veen, J. van Wingerden, M. Bergamasco, M. Lovera, and M. Verhaegen, "Closed-loop subspace identification methods: an overview," IET Control Theory and Applications, vol. 7, no. 10, pp. 1339–1358, 2013, publisher: Institution of Engineering and Technology. [Online]. Available: https://research.tudelft.nl/en/publications/closed-loop-subspace-identification-methods-an-overview